

Post-Quantum Cryptography beyond Encryption: Design and Cryptanalysis

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par

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To Ayman, I wish you had the chance to write yours . . .

We've got five years, my brain hurts a lot
Five years, that's all we've got
— David Bowie, Five years



Abstract

We explore the design and analysis of post-quantum cryptographic primitives with an emphasis on two angles. First, diversity of assumptions, by building and analysing cryptography that does not rely on structured lattices. Second, diversity of functionalities, focusing on various primitives that extend beyond standard encryption. Motivated by the rapid development of quantum computing and the consequent threat to classical public-key cryptography, we therefore consider two families of post-quantum assumptions: isogeny-based and code-equivalence problems from which we build and analyse various primitives.

In the first part, we focus on isogeny-based cryptography. We start our study with digital signatures derived from SIDH squares and investigate their security by analysing a range of attack vectors, including higher-dimensional attack strategies (known as SIDH attacks), and consequently propose suitable parameters. Building on the powerful constructive tools that the SIDH attacks became, we design an updatable public-key encryption scheme from a variant of FESTA, a public-key encryption scheme by Basso, Maino and Pope, enabling an efficient instantiation of this primitive that allows an unbounded number of updates. Finally, we investigate isogeny-based group actions and propose a framework for general-purpose zero-knowledge proofs.

The second part focuses on code-based cryptography, particularly tensor group actions and code-equivalence problems. We first analyze and break a commitment scheme based on a structured tensor isomorphism problem, and we propose a secure fix. Then, we study variants of code-equivalence problems that underlie the security of two blind signature schemes. For one variant, we provide a reduction to a standard problem, whilst for another one we show that its security has been largely over-estimated.

Overall, this thesis contributes to the development of a diverse suite of post-quantum primitives by providing new constructions, security analyses, and insights into the use of alternative assumptions beyond lattice-based systems.

Key words: post-quantum cryptography, isogenies, codes

Résumé

Nous étudions la conception et l'analyse de primitives cryptographiques post-quantiques en mettant l'accent sur deux axes. Premièrement, la diversité des hypothèses, en construisant et en analysant des systèmes cryptographiques qui ne reposent pas sur les réseaux structurés. Deuxièmement, la diversité des fonctionnalités, en nous concentrant sur diverses primitives qui vont au-delà du chiffrement standard. Motivés par le développement rapide de l'informatique quantique et la menace qui en découle pour la cryptographie classique à clé publique, nous considérons ainsi deux familles d'hypothèses post-quantiques : les problèmes fondés sur les isogénies et ceux basés sur l'équivalence de codes, à partir desquels nous construisons et analysons différentes primitives.

Dans la première partie, nous nous intéressons à la cryptographie fondée sur les isogénies. Nous commençons par étudier des signatures digitales dérivées de "*SIDH squares*" et examinons leur sécurité en analysant un ensemble de vecteurs d'attaque, incluant les stratégies d'attaque de dimension supérieure (connues sous le nom d'attaques SIDH), ce qui nous permet de proposer des paramètres adaptés. En nous appuyant sur les puissants outils constructifs issus de ces attaques SIDH, nous concevons un schéma de chiffrement à clé publique actualisable ("*Updatable public key encryption*") à partir d'une variante de FESTA, un schéma proposé par Basso, Maino et Pope, permettant une instanciation efficace de cette primitive et autorisant un nombre illimité de mises à jour des paires de clés. Enfin, nous étudions les actions de groupe basées sur les isogénies et proposons un cadre pour des preuves à divulgation nulle de connaissance à usage général basées sur ce problème.

La seconde partie porte sur la cryptographie fondée sur les codes, en particulier les actions de groupe tensorielles et les problèmes d'équivalence de codes. Nous analysons d'abord un schéma d'engagement ("*commitment scheme*") basé sur un problème d'isomorphisme tensoriel structuré, que nous parvenons à rompre, puis nous proposons une correction jugée sûre. Ensuite, nous étudions plusieurs variantes des problèmes d'équivalence de codes qui sous-tendent la sécurité de deux schémas de signatures aveugles. Pour l'une de ces variantes, nous fournissons une réduction à un problème standard, tandis que pour une autre nous montrons que sa sécurité a été largement surestimée.

Chapter 0

Dans l'ensemble, cette thèse contribue au développement d'un ensemble diversifié de primitives post-quantiques en proposant de nouvelles constructions, des analyses de sécurité et des pistes de réflexions sur l'utilisation d'hypothèses cryptographiques alternatives au delà des systèmes fondés sur les réseaux.

Mots clefs: cryptographie post-quantique, isogénies, codes

Acknowledgements

Why does a thesis need acknowledgements?

I promised myself to give an honest overview of the thesis when writing my acknowledgements. Most times, when someone reads a thesis' acknowledgements it seems like it was 3,4,5+ years of smooth sailing, interesting research and fun with friends. Undoubtedly, there was some of that, but the truth is that a thesis is a lot more. There are moments of doubt, of stress, of loneliness, of imposter syndrome, of frustration. I think it matters to offer a transparent picture, because we all go through the same ups and downs. Acknowledging the hardships does not diminish the achievements, it makes them real.

EPFL is no doubt a great school, with amazing facilities, financial support and a stone's throw away from an amazing view of the lake and mountains, and it is filled with extremely smart people. But it is also competitive, it cares about its image (maybe a bit too much), it crushes you under imposter syndrome, and some days you wonder "What am I even doing here?".

It is also filled (at least in computer science) with a vast majority of men. Standing for 4 and half years as the only woman in most rooms is not a comfortable feeling, regardless of how nice the people in the room are. It is lonely. Always having to fight a little harder to earn your place, to fight the stereotypes and to make yourself heard gets exhausting. Diversity matters, in all its dimensions, and we would all benefit from more of it.

Finally, for every finished project, there are countless abandoned ones, discarded ideas and failed attempts. For every paper published, there were numerous rejections beforehand, and puzzling reviews. Doing a PhD somehow really requires believing in yourself and in your ideas. Even when Reviewer C says "I enjoyed reading the paper and I also like the idea" and rates it as "reject" (true story), or when you've been stuck for months on a project that ends up leading nowhere. Truth is, I think I found believing in myself to be the hardest part of the journey. But all these things only make the people whom I'm thanking next even more important (and the list is long!). So onto more positive things.

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Serge, you taught me rigour, attention to details, and I have lost count of the number of times

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This brings me to **LASEC**. There is a lot to say about this group, and the people that form it. Summarizing it in a few lines is not easy, and I won't try. Instead, I point to the whiteboard in our hallway which is filled with messages from past members of the lab, which always made me smile whenever I walked past it. I speak for my "generation" of LASEC, but I think that this group that Serge created throughout the years is truly special.

My start at LASEC was rather intimidating. I remember my first day, walking into the lab and seeing all these people knowing so much, and knowing each other so well. Not gonna lie, I was a little scared to not fit in. Thankfully, I started along **Dalia**, who made me feel a little less alone, and with whom I shared all the stress of joining a new lab, starting a PhD and doing these cryptosec homework. Dalia, I couldn't have done it without you and I really am so grateful that I had you by my side during this first semester. I hope the path you took brings you joy and fulfillment.

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Abdullah, sharing an office with you for all these years was amazing. You're the kindest, and

your passion for cryptography as well as the endless patience you showed for all my questions will make you an amazing teacher. Writing this thesis without our little random chats to distract me was hard. I'm so happy to see your life so fulfilled and I'm looking forward to seeing you shine as a professor!

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Finally, the youngest generation: an era finished and another one started. I finally convinced everyone to bring their lunch box!

Max, I loved the moments we shared with Boris and sometimes Sylvie in the cafeteria, discussing life and ranting about things. I'm looking forward to reading your papers! Well, if I'm being honest, I probably won't read more than the intro, but looking forward nonetheless. Keep chasing rabbits!

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LASEC has some bright years ahead with the three of you.

Then, because I had a co-supervisor, I also had a lab away from home. I am thankful to everyone in Christophe's group, and the broader isogeny community. It's such a welcoming community (some might have called us a sect because of our cool matching merch!), and at every workshop or conference I attended I was always extremely happy to meet up with such friendly and passionate people. In such a competitive and fast-paced environment as crypto, having such an open and positive community was a real breath of fresh air. I'm super grateful for all the dinners, drinks and karaokes we all shared together at conferences: after the work is over, the fun continues. More specifically, I am super grateful for the little group I found

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Then, I almost have a third lab thanks to my visit to Steven. First, let me say that every single person at the Auckland Uni maths department was absolutely lovely, welcoming and made me feel at home even though I was halfway across the world. Thanks to all the PhD students I shared lunches with, to all the people who joined for Friday drinks at OGH and, a special shoutout to Gab and Chris for going the extra mile at Christmas. I also have to say that this is the most diverse maths department I've ever been at (and I've travelled quite a bit!), and it is worth praising it. I wish it'd be the norm and not the exception.

Now onto specifics, **Pabasara**, thank you for welcoming me to the group, it was so nice to chat with you every morning. You're such a kind and hard-working person and you deserve a world of opportunities. I hope our paths will cross again in the future!

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Maddie, how lucky I am to have met a friend that shares my love for ballet, maths, skiing AND cheese! You're so talented and hardworking, both when it comes to ballet performances or being the best maths researcher, lecturer and person, and I look up to you so much. I'm looking forward to sharing a ballet class and a cheese board with you again.

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This is a good segway, into the fact that work-life balance matters to me, and I have a lot of hobbies. Particularly, ballet, which has acted as both a sport, a form of artistic expression and a necessary bit of therapy throughout the years. So I'm also super grateful for the Tuesday, Wednesday and Thursday classes at Igokat, the amazing teachers that taught me, particularly **Laura Smeak** and all of my fun classmates. Thanks for offering such a positive space where we can practice ballet. Also kudos to my tennis partner **Edlyn** along the years. Sometimes hitting a ball really hard feels great after a long day in the lab.

Another activity I particularly enjoy is eating cheese, and I was lucky enough to meet a perfect set of friends to do it with: **Lucas, Freya, Florian and Hussein**. We met on the first week of orientation, our PhD areas did not overlap whatsoever, so for 4+ years we shared weekly lunches and cheesy dinners forced to talk about anything but research (unless it was for complaining), and it was the best thing ever.

Similarly, I was lucky to have very fun lab neighbours: I met another **Lucas** at orientation who turned out to be an amazing cook and baker, and who actually was on the same floor. He fed us incredible food, until he had to leave for better things. Thank you Lucas for your generosity and kindness, and all the cookies.

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I strongly believe into creating supporting environments for everyone to thrive in. So a big shout-out to **EPIC**, and all its invested members, for creating so many events for our PhD students, I was very happy to take a small part in it. But most importantly, I'm very proud of the **Women+ in IC** committee. What started as a small idea that I pitched to **Vini** in 2022 (thanks for the support!), turned into a real association, with the help of so many dedicated people. **Mai and Mathilde**, we were an incredible trio, thank you both for all the time and energy you invested. I'm so proud of everything we achieved (the BC stairs will remember us). And a big shout-out to all the committee members that followed, for keeping the momentum and dedicating a bit of your time and energy to making IC a more inclusive place for everyone. Not everyone understands how much investment it takes to run an association, offer impactful and regular events, and do it all alongside studies or a PhD, and I'm very grateful to all of you for your dedication. This association would also not be the same without the support of **Marta, Annalisa and Cecilia** within the EDIC team. A big thank you for supporting our initiative year after year, and also for all the work you do behind the scene for all of us PhD students.

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I won't go around and thank all my remaining friends one by one, but I am grateful for every single one of them. Friendship is what carries me through life, whether they live close or far, whether I talk to them daily or once in a blue moon I feel privileged to have my friends in my life. Le fidèle quatuor qui a survécu au lycée: **Mathilde**, **Pauline** et nos chouettes réunions en dépit de la distance et surtout **Maëva** (trop fière de ma personal lawyer) merci d'être la pour moi et d'être toujours partante pour faire les 400 coups; l'équipe **Maths&mateam** pour être mon pilier à Genève et pour les aprem jeux, les dîners et toutes les chouettes activités; the Cambridge homies, especially **Radu**, and the Lausanne visit, **Sam** for the good ski and endless chats, **Oliver** for long calls and book recs, **Laeti** for a kind soul and upholding us to such strong values and **Jean** for pushing me to do a PhD and being so real with me always; as well as everyone else scattered around that I'm always happy to meet up with, I am truly grateful for your support. Gracias a **David** para darme la oportunidad de refrescar mis nociones de español. Joke aside, thanks for steadily supporting me through my ups and downs throughout this last final stretch, patiently listening to my rants, scratching my head (even when wearing a ski helmet), and cursing me with an elite taste in tortilla and a love for expensive bikes. I'm excited for the future.

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Laurane.

Lausanne, 19 Mars 2026

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1 Introduction

Cryptography, derived from the Greek words *kryptós* (“hidden”) and *gráphein* (“to write”), has historically been understood as the practice of concealing information. For much of its history, it served a single, crucial purpose: ensuring the confidentiality of communications. In early military and diplomatic contexts, messages were transformed into unintelligible forms through encryption, so that only the intended recipient, who possessed some secret information shared with the sender, could recover their meaning. This process of encryption and decryption formed the basis of symmetric cryptography, which relied on pre-established shared secrets between communicating parties. As a result, the management and secure exchange of these shared keys posed significant practical challenges, especially with the increase of the number of participants. With the advent of computers and the rise of digital communication, cryptography rapidly expanded beyond its original goal of secrecy. Modern cryptography now encompasses a broader set of objectives including ensuring the integrity of data, verifying its authenticity, and enabling complex forms of secure computation and interaction. All of these have become essential nowadays, redefining the way in which security and trust are achieved in the digital age.

Public-Key Cryptography. A major milestone in modern cryptography came with the introduction of public-key cryptography by Whitfield Diffie and Martin Hellman in 1976 [DH76]. They proposed a method that allowed two parties to establish a shared secret over an insecure communication channel, thereby opening an entirely new branch of cryptography. Unlike symmetric systems, where parties had to agree on a shared secret key in advance, their proposal provided an efficient way to derive a shared secret from public information alone. Instead of requiring each user to store a unique shared secret with every potential correspondent, the public-key paradigm provides each user with a pair of keys: one public and one private. These key pairs can be used in two complementary ways to construct two fundamental cryptographic primitives: encryption and digital signatures. For encryption, the sender uses the recipient’s public key to encrypt a message, which can then be decrypted only with the recipient’s private key. For digital signatures, the roles are reversed: the signer uses their

private key to sign a document, and anyone can verify the signature using the signer's public key.

Public-key cryptography quickly became a cornerstone of modern cryptography, enabling not only classical encryption and digital signatures but also a wide range of advanced primitives such as commitment schemes, updatable public-key encryption, threshold, group, and blind signatures, zero-knowledge proofs, and many others. The security of all these public-key primitives is usually based on two main families of computational assumptions: integer factorization (given a large integer, find its prime factors) and the discrete logarithm assumption (given an element in a finite group, find its logarithm with respect to a given base). Both assumptions are derived from long-standing and well-studied mathematical problems that have withstood decades of scrutiny by the mathematical and cryptographic communities, thereby providing strong confidence in their computational hardness.

One cannot overstate the importance of public-key cryptography, not only as a rich and elegant area of theoretical research but also, and perhaps more crucially, as the foundation of our digital infrastructure. In practice, it underpins secure messaging applications (such as Signal and WhatsApp), website authentication via HTTPS certificates, blockchain technologies and cryptocurrencies, secure email (PGP), online banking and payment systems, software update mechanisms, and even digital identity frameworks such as electronic passports and government e-services.

Post-quantum cryptography The next milestone of modern cryptography was the publication of Shor's algorithm in the 90's ([Sho97]). This algorithm, given access to a sufficiently powerful quantum computer, can efficiently solve the discrete logarithm and integer factorization problems. While in those times, when even classical computers were still limited, this threat might have seemed distant, the rapid progress of quantum technologies in the last decade has brought this threat to the forefront of cryptographic research. It led to the rise of a new area of research: post-quantum cryptography, a branch of public-key cryptography that seeks to design cryptographic systems which can be implemented on classical computers and remain secure even in the presence of quantum adversaries. The transition to post-quantum cryptography has become not only necessary but essential, due to the fast development of quantum technologies combined with the strong dependence of modern digital infrastructure on public-key cryptography.

In recognition of this, the U.S. National Institute of Standards and Technology (NIST) launched its Post-Quantum Cryptography Standardization Project in 2016.¹ In this initial call for proposals, NIST aimed to standardize both public-key encryption (or key-encapsulation mechanisms, a closely related variant) and digital signatures. After several rounds of evaluation, during which proposals based on a variety of mathematical problems were examined by the cryptographic community, several candidates were selected for standardization. However, NIST

¹<https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization>

was not entirely satisfied with the initial digital signature proposals, which all relied on lattice-based mathematical problems. As a result, in 2022 it issued an additional call for digital signature proposals, explicitly emphasizing the need for *diversity in the underlying assumptions*, quoting "NIST is primarily interested in additional general-purpose signature schemes that are not based on structured lattices." ^{II} Indeed, diversity in assumptions is particularly relevant in post-quantum cryptography, as we establish next. ^{III}

Build and break: the life-cycle of cryptographic assumptions. The security of public-key cryptography relies on the hardness of some mathematical assumptions which the community studies until they reach some level of confidence. To quantify the confidence in a certain cryptographic assumption, it is interesting to look at its maturity withing the assumptions's life cycle, which we describe next.

When an assumption is first introduced, researchers start constructing schemes directly based on it. Some are quickly broken, some need to be refined and others perfectly withstand initial scrutiny. Over time, as the community studies an assumption more thoroughly, confidence in its hardness grows, and new variants of the assumption emerge to enable more advanced constructions. Eventually, when an assumption has been tested for long enough without being broken, confidence in it solidifies and as the community moves its attention to a different series of problems, less and less results on its security appear. Classic examples of mature assumptions include the aforementioned factoring and discrete logarithm problems, which underpin the classical public-key ecosystem.

Post-quantum assumptions, on the other hand, are generally younger and less established. Some, such as lattice-based problems, have been studied since the 1980s and have accumulated a relatively high degree of confidence. Others, such as isogeny-based or code-equivalence problems, are less well understood, and the community continues to explore their limits in an active back-and-forth between protocol designers and cryptanalysts. These younger assumptions present both risks and opportunities: we are, for now, less confident in them, but they offer interesting trade-offs. For example, isogeny-based primitives are very compact, at the cost of being less efficient than lattice-based ones. Code-equivalence assumptions are easy to state, analyze and implement. In any case, it is essential to develop a range of suitable primitives relying on different assumptions for both security and functionality reasons. Security, because if one assumption is broken, we can easily fall back on alternatives. Functionality, because some primitives might be more easily and efficiently built from one assumption rather than the other, and also because the efficiency constraints depend on the final application: some applications, such as the Bluetooth protocol, might have harsh compactness requirements, while other might care more about speed.

With this in mind, this thesis studies the **design and analysis of post-quantum primitives** bearing in mind the notion of **diversity**, which we consider at two different levels. First, *diversity of*

^{II}<https://csrc.nist.gov/Projects/pqc-dig-sig/standardization>

^{III}This author would even argue that diversity is an essential part of any field, at every level.

assumptions: rather than concentrating on a single family of post-quantum assumptions, this thesis explores two different ones, each with its own advantages and limitations. We deliberately do not study lattice-based assumptions, which are already proposed for standardization, but instead turn our attention to isogeny-based cryptography, for its compactness, and code-equivalence problems, as an alternative candidate for post-quantum group actions. Second, *diversity of primitives*: while encryption and digital signatures are the textbook applications of public-key cryptography, our digital ecosystem relies on a much wider variety of primitives that must also transition to the post-quantum setting. This thesis therefore looks into the design and analysis of a broad range of primitives, including commitment schemes, digital signatures, blind signatures, proof systems, and updatable public-key encryption.

1.1 Overview and Contributions

This thesis is divided into two parts. The first part focuses on isogeny-based cryptography, and the second part focuses on code-based cryptography.

In the first part we consider three chapters, looking at three different primitives each built from a different sub-family of assumptions within isogeny based cryptography.

We start in Chapter 3 by looking at digital signatures based on the Supersingular Isogeny Diffie-Hellman (SIDH) problem. Barely a year into this thesis, in the summer of 2022, the SIDH problem was broken by a series of attacks proposed by Castryck and Déru [CD23], Maino, Martindale, Panny, Pope and Wesolowski [MMP⁺23] as well as Robert [Rob23] that exploit the additional torsion point information present in the SIDH protocol. In this chapter, we ask ourselves whether there exists any isogeny-based protocol that remains secure in a model where we give even more information to the adversary, i.e. in which we assume they know the endomorphism ring of all curves involved. Perhaps counterintuitively, we show that digital signatures based on SIDH squares remain secure within this framework under a careful choice of parameters.

While devastating for SIDH itself, the isogeny community quickly turned this series of attacks into a powerful constructive tool to build new isogeny-based primitives. This leads us to Chapter 4, where we design an updatable public-key encryption scheme by adapting a variant of FESTA [BMP23], an isogeny-based public-key encryption scheme that was born from the SIDH countermeasures. Updatable public-key encryption extends the notion of public-key encryption with an additional asynchronous key update mechanism. While lattice-based public-key encryption schemes are already proposed for standardization, it is notoriously difficult to adapt them to build efficient updatable public-key encryption schemes. We therefore explore the feasibility of such a primitive from isogeny-based assumptions, and propose a concrete construction with associated parameters, that is both practically efficient and allows for unbounded key updates.

Finally, in Chapter 5 we explore the design of zero-knowledge proofs from isogeny-based group actions. The framework of cryptographic group actions offers a simple and elegant

way to extend discrete logarithm style problems to the post-quantum setting. It is most often instantiated from isogeny-based group actions and has been used to build efficient encryption schemes [CLM⁺18] as well as digital signatures [BKV19, DG19]. However, the design of zero-knowledge proofs from isogeny-based group actions remains largely unexplored. In this chapter, we propose a framework for general purpose zero-knowledge proofs from isogeny-based group actions.

While isogeny-based group actions are undoubtedly the most popular choice to instantiate cryptographic group actions, they are not the only one. Code-based group actions have been proposed recently as an alternative instantiation ([SS13]) mostly for the design of digital signature schemes such as LESS [BMPS20], and then MEDS [CNP⁺23b]. While these group actions lack the commutativity offered by isogenies, they are conceptually simpler, generally faster, and more straightforward to implement. This motivates the second part of this thesis, which focuses on code-based cryptography. In particular, we study tensor-based group actions and code-equivalence problems, which are at the core of code-based group actions.

In Chapter 6, we analyze the security of a commitment scheme proposed at Asiacrypt 2023 by D’Alconzo, Flamini and Gangemi [DFG23]. They based the security of their scheme on a structured variant of the tensor isomorphism problem, which we show to be insecure. We propose two different attacks that completely break this underlying assumption. Leveraging the insights coming from these attacks, we propose a fix to their commitment scheme which we believe to be secure.

Finally in Chapter 7, we analyze the security of two code-equivalence problems that have been used to build blind signature schemes. Both problems are variants of standard problems pertaining to two different family of codes, namely linear codes and matrix codes. We give a reduction from the first problem to the standard linear code equivalence problem. We show that the security of the second problem was vastly overestimated by the authors who proposed it, and study its complexity using tools from algebraic cryptanalysis. These two chapters contribute to a better understanding of variants of tensor-based group actions and code-equivalence problems and their security, giving necessary insights for future protocol builders wishing to leverage these problems.

1.2 List of works

The author has contributed to the works listed below in chronological order of publication (most recent first). For every article, the authors are listed in alphabetical order.^{IV} The title of the papers included in this thesis are highlighted in bold.

1. **Updatable Public-Key Encryption from FESTA** - Andrea Basso, Tako Boris Fouotsa,

^{IV}<https://www.ams.org/learning-careers/leaders/CultureStatement04.pdf>

- Fatna Kouider, Péter Kutas, Luciano Maino, Laurane Marco. In submission. (Chapter 4)
2. **On the security of two blind signatures from code equivalence problems** - Valerie Gilchrist, *Laurane Marco*, Christophe Petit, Gang Tang. IACR Communications in Cryptology, Volume 2, Issue 4. ([GMPT25], Chapter 7)
 3. Non-Transferable Anonymous Tokens by Secret Binding - F.Betül Durak, *Laurane Marco*, Abdullah Talayhan, Serge Vaudenay. Published at ACM CCS 2024 ([DMTV24]).
 4. **Solving the Tensor Isomorphism Problem for special orbits with low rank points: Cryptanalysis and repair of an Asiacrypt 2023 commitment scheme.** Valerie Gilchrist, *Laurane Marco*, Christophe Petit, Gang Tang. Published at CRYPTO 2024 ([GMPT24], Chapter 6).
 5. **Exploring SIDH-based Signature Parameters.** Andrea Basso, Mingjie Chen, Tako Boris Fouotsa, Péter Kutas, Abel Laval, *Laurane Marco*, Gustave Tchoffo Saah. Published at ACNS 2024. ([BCF⁺24], Chapter 3)
 6. **Malleable Commitments from Group Actions and Zero-Knowledge Proofs for Circuits based on Isogenies.** Mingjie Chen, Yi Fu Lai, Abel Laval, *Laurane Marco*, Christophe Petit. Published at INDOCRYPT 2023. ([CLL⁺23], Chapter 5)
 7. Making Classical (Threshold) Signatures Post-Quantum for Single Use on a Public Ledger. *Laurane Marco*, Abdullah Talayhan, Serge Vaudenay. Published at IWSEC 2023. ([MTV23])

2 Preliminaries

The goal of this chapter is to set the ground for the rest of this thesis. We start in Section 2.1 by defining the primitives that will be used in this work, going from the most basic ones such as hash functions and commitment schemes to the most advanced ones such as digital signatures, proof systems or updatable public-key encryption.

We then introduce some hard problems which can be used to instantiate these primitives. First, we introduce a generic framework for cryptographic group actions in Section 2.2. This framework abstracts away the well-known setting of discrete logarithm type problems.

We then turn ourselves to two families of hardness assumptions. We first introduce isogeny-based cryptography and its associated problems in Section 2.3. We start by recalling the basics about elliptic curves, isogenies and their computational aspects. We then explain how to build post-quantum group actions from isogenies, and introduce two additional isogeny-based protocols SIDH and FESTA.

Finally, we turn our attention to tensor and code-based cryptography in Section 2.4, which offer an alternative to instantiate cryptographic group actions. We first introduce the general setting of tensor based cryptography, and then turn our attention to code equivalence type problems and some of their variants.

General notations

In this work, we denote by $\mathbf{1}_A(x)$ the function taking value 1 if $x \in A$ and 0 otherwise. We write PPT to stand for Probabilistic Polynomial Time. We let λ denote the security parameter. We call a function *negligible*, written $\text{negl}(\lambda)$, if its absolute value is asymptotically dominated by $O(x^{-\lambda})$ for all $\lambda > 0$. Given two integers a and b , $[a, b)$ denotes the set of integers x such that $a \leq x < b$ and $a||b$ denotes the concatenation of the bitstrings a and b .

2.1 Cryptographic primitives

In this section, we give a formal exposition to a wide range of cryptographic primitives. We start from the most basic building blocks, namely hash function and commitment scheme, all the way to proof systems and updatable public key encryption, giving a formal exposition to the various primitives studied in this thesis.

2.1.1 Hash functions

We start our exposition with a basic, yet essential cryptographic primitive: *hash functions*. Hash functions are a cornerstone of cryptography, and are used in a wide variety of protocols. The goal of a cryptographic hash function is to act as a compact, yet unique fingerprint of an object. The properties we wish to have from the resulting fingerprint depend in the cryptographic use case and we formalize several of them below in Definition 2.1

Definition 2.1 (2nd pre-image resistance, collision resistance, one-wayness). *A hash function $H : \mathcal{D} \rightarrow \{0, 1\}^{\tau(\lambda)}$, with message domain $\mathcal{D}$, output domain $\{0, 1\}^{\tau(\lambda)}$ and security parameter λ is:*

- *2nd pre-image resistant if for any probabilistic polynomial time algorithm $\mathcal{A}$, we have*

$$\Pr[2\text{Prelm}(\mathcal{A}) \rightarrow 1] \leq \text{negl}(\lambda)$$

- *one-way if for any probabilistic polynomial time algorithm $\mathcal{A}$, we have*

$$\Pr[\text{OW}(\mathcal{A}) \rightarrow 1] \leq \text{negl}(\lambda)$$

Futhermore, we say a keyed hash function $H : \mathcal{D} \times \mathcal{K}(\lambda) \rightarrow \{0, 1\}^{\tau(\lambda)}$ is collision resistant if for any probabilistic polynomial time algorithm $\mathcal{A}$, we have

$$\Pr[\text{CR}(\mathcal{A}) \rightarrow 1] \leq \text{negl}(\lambda)$$

The 2Prelm, CR and OW games are described in Figure 2.1. For simplicity, the parameter λ will be omitted from the notation of $\mathcal{D}$, τ and $\mathcal{K}$.

$\underline{2\text{Prelm}(\mathcal{A})}$	$\underline{\text{CR}(\mathcal{A})}$	$\underline{\text{OW}(\mathcal{A})}$
1: $x \xleftarrow{\$} \mathcal{D}$	1: $\text{hk} \xleftarrow{\$} \mathcal{K}$	1: $x \xleftarrow{\$} \mathcal{D}$
2: $\mathcal{A}(x) \rightarrow x'$	2: $\mathcal{A}(\text{hk}) \rightarrow x, x'$	2: $y \leftarrow H(x)$
3: return $\mathbb{1}_{H(x')=H(x) \wedge x \neq x'}$	3: return $\mathbb{1}_{H(x, \text{hk})=H(x', \text{hk}) \wedge x \neq x'}$	3: $\mathcal{A}(y) \rightarrow x'$
		4: return $\mathbb{1}_{H(x')=y}$

Figure 2.1: 2Prelm, CR, and OW games for hash function security.

2.1.2 Commitment schemes

Commitment schemes have many real-world use cases, finding applications in multiparty computation, zero-knowledge proofs, and coin tossing [LNS21, OPV09, JLO97, FPY18, PBF⁺18].

A commitment scheme is used when a sender wants to commit to some value, m , without initially revealing it to the receiver. Instead, the sender will commit to m by computing a commitment, c , that depends on m , and send it to the receiver. At a later time, the sender will reveal m , and the receiver should be able to verify that c was computed using m . The key security properties here are *hiding*, which means c should reveal nothing about m ; and *binding*, which means no other value $m' \neq m$ should be able to open c . We proceed to give formal definitions for a commitment scheme and its security properties.

Definition 2.2 (Commitment scheme). *A commitment scheme is defined by a message space, $\mathcal{M}(\lambda)$, randomness space, $\mathcal{R}(\lambda)$, commitment space, $\mathcal{C}(\lambda)$, and a tuple of algorithms (Setup, Commit, Verify), defined in the following way :*

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: This PPT algorithm takes as input a security parameter λ in unitary form and returns some public parameters pp which will be implicitly given as inputs to the next two algorithms.
- $\text{Commit}(m, r) \rightarrow c$: This PPT algorithm takes as input a message $m \in \mathcal{M}(\lambda)$ and randomness $r \in \mathcal{R}(\lambda)$ and outputs a commitment value $c \in \mathcal{C}(\lambda)$.
- $\text{Verify}(c, (m, r)) \rightarrow 0/1$: This is a deterministic algorithm which, given as input $c \in \mathcal{C}(\lambda)$, $(m, r) \in \mathcal{M}(\lambda) \times \mathcal{R}(\lambda)$ returns 1 if $\text{Commit}(m, r) = c$ and 0 otherwise.

For simplicity, we drop the λ notation from the message, randomness and commitment space.

In the case of a *bit commitment scheme*, the message space is restricted to two possible values: 0 or 1.

Definition 2.3 (Hiding Security). *A commitment scheme (Setup, Commit, Verify) is hiding if for any adversary, $\mathcal{A}$, the advantage, Adv , is negligible, where Adv is defined as*

$$\text{Adv} = |\Pr[\text{Hiding}(\mathcal{A}) \rightarrow 1] - 1/2|$$

and the Hiding game is defined in Figure 2.2.

Remark 2.4. Note that in the case of a bit commitment scheme, we have $\mathcal{M} = \{0, 1\}$, and in the Hiding game the adversary does not choose m_0, m_1 and does not need to provide a state st , rather the bit committed is the one sampled by the game, i.e. one can remove Line 2 in Figure 2.2 and set $m_b = b$, as depicted on the right-hand side of Figure 2.2 by the $\text{Hiding}_{\text{Bit}}$ game.

Hiding($\mathcal{A}$)

- 1: $\text{pp} \leftarrow \text{Setup}(1^\lambda)$
- 2: $m_0, m_1, st \leftarrow \mathcal{A}(\text{pp})$
- 3: $b \xleftarrow{\$} \{0, 1\}, r \xleftarrow{\$} \mathcal{R}$
- 4: $c \leftarrow \text{Commit}(m_b, r)$
- 5: $b' \leftarrow \mathcal{A}(c, st)$
- 6: **return** $b' = b$

Hiding_{Bit}($\mathcal{A}$)

- 1: $\text{pp} \leftarrow \text{Setup}(1^\lambda)$
- 2: $b \xleftarrow{\$} \{0, 1\}, r \xleftarrow{\$} \mathcal{R}$
- 3: $c \leftarrow \text{Commit}(b, r)$
- 4: $b' \leftarrow \mathcal{A}(c)$
- 5: **return** $b' = b$

Figure 2.2: The Hiding and Hiding_{Bit} game

Definition 2.5 (Binding Security). *A commitment scheme (Setup, Commit, Verify) is binding if for any adversary $\mathcal{A}$ the advantage Adv is negligible, where Adv is defined as*

$$\text{Adv} = \Pr[\text{Binding}(\mathcal{A}) \rightarrow 1]$$

and the Binding game is as defined in Figure 2.3.

Binding($\mathcal{A}$)

- 1: $\text{pp} \leftarrow \text{Setup}(1^\lambda)$
- 2: $m_0, m_1, r_0, r_1 \leftarrow \mathcal{A}(\text{pp})$
- 3: **return** $(m_0 \neq m_1) \wedge \text{Commit}(m_0, r_0) = \text{Commit}(m_1, r_1)$

Figure 2.3: The Binding game

Remark 2.6. We can make the hiding and binding definition more precise by quantifying the complexity of the adversary as well as its advantage. When $\mathcal{A}$ has unbounded complexity and $\text{Adv} = 0$, the commitment scheme is said to be *perfectly* binding/hiding. When $\mathcal{A}$ has unbounded complexity and $\text{Adv} = \text{negl}(\lambda)$, it is called *statistically* binding/hiding. When $\mathcal{A}$ is PPT, and $\text{Adv} = \text{negl}(\lambda)$ it is called *computationally* binding/hiding.

Remark 2.7. Note that a basic commitment scheme can be constructed from a hash function in the following way: $\text{Commit}(m; r) = H(m||r)$, where $||$ denotes concatenation, and the randomness r is of fixed length.

2.1.3 Digital Signatures

We now turn our attention to digital signatures. As their name indicates, they act as the digital counterpart of real-world signatures. They aim to certify the authenticity of a message as well as its integrity. They find a plethora of use-cases such as signing web certificates, software updates, blockchain transactions... We give a formal definition (Definition 2.1.3) together with a common security property (Definition 2.10).

Definition 2.8 (Digital Signature). *A digital signature scheme is a tuple of algorithms (KeyGen, Sign, Verify) together with a message space $\mathcal{M}$, such that*

- $\text{KeyGen}(1^\lambda) \rightarrow (\text{sk}, \text{pk})$ is a probabilistic polynomial time algorithm, that takes as input

$\text{EUF-CMA}(\mathcal{A})$	$\text{Osign}(m)$
1: $\text{KeyGen}(1^\lambda) \rightarrow (\text{sk}, \text{pk})$	1: $\sigma \leftarrow \text{Sign}(m, \text{sk})$
2: $\mathcal{Q} \leftarrow \emptyset$	2: $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{m\}$
3: $\mathcal{A}^{\text{OSign}(\text{pk})} \rightarrow m, \sigma$	3: return σ
4: if $m \in \mathcal{Q}$ then	
5: return 0	
6: return 1 if $\text{Verify}(\sigma, m, \text{pk})$, else 0	

Figure 2.4: EUF-CMA game for digital signature schemes.

the security parameter and outputs a secret key sk and public key pk .

- $\text{Sign}(m, \text{sk}) \rightarrow \sigma$ is a probabilistic polynomial time algorithm that takes as input a message $m \in \mathcal{M}$ and a secret key sk and outputs a signature σ .
- $\text{Verify}(\sigma, m, \text{pk}) \rightarrow 0, 1$ is a deterministic algorithm that takes as input a signature σ , a message m and a public key pk and either accepts (1) or rejects (0).

Definition 2.9 (Correctness of digital signatures). A digital signature scheme $(\text{KeyGen}, \text{Sign}, \text{Verify})$ with message space $\mathcal{M}$ is correct if

$$\forall m \in \mathcal{M}, \Pr[\text{Verify}(\text{Sign}(m, \text{sk}), m, \text{pk}) = 1] = 1$$

where $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$, and the probability is taken over the randomness used in KeyGen and Sign .

Definition 2.10 (EUF-CMA security). A digital signature scheme $(\text{KeyGen}, \text{Sign}, \text{Verify})$ with message space $\mathcal{M}$ is secure against existential forgery under chosen message attack if for all probabilistic polynomial time adversaries $\mathcal{A}$ the advantage of $\mathcal{A}$ in the EUF-CMA game is negligible, i.e.

$$\text{Adv}_{\mathcal{A}}^{\text{EUF-CMA}} = \Pr[\text{EUF-CMA}(\mathcal{A}) \rightarrow 1] \leq \text{negl}(\lambda)$$

where the EUF-CMA game is defined in Figure 2.4.

2.1.4 Sigma protocols

We now turn our attention to cryptographic proof systems, and more specifically *sigma protocols*. They give a structured framework for designing proofs of knowledge, i.e. proofs that a secret witness wt and a public statement st satisfy a given relation $\mathcal{R}$. We give below a definition of a sigma protocol that will be of use in Chapter 5, as well as in Chapter 6, Section 6.3.2.

Definition 2.11 (Sigma protocol). Given a language $L \in \text{NP}$ over an alphabet Z defined by a relation $\mathcal{R}$, and challenge space E a sigma protocol for L is a pair of interactive machines $\mathcal{P}(\text{st}, \text{wt})$ and $\mathcal{V}(\text{st})$ such that

- $\mathcal{V}$ is polynomially bounded.
- $\mathcal{P}$ starts with a message, com , $\mathcal{V}$ answers with a challenge, $\text{chall} \in E$, $\mathcal{P}$ terminates with a response, resp . $\mathcal{V}$ accepts (returns 1) or rejects (returns 0) only depending on $(\text{st}, \text{com}, \text{chall}, \text{resp})$.
- *Special soundness*: there exists a polynomially bounded algorithm $\mathcal{E}$ called extractor such that for any st , if $(\text{st}, \text{com}, \text{resp}; r)$ and $(\text{st}, \text{com}, \text{resp}'; r')$ are two accepting views for $\mathcal{V}$ such that $\text{chall} \neq \text{chall}'$ where $\text{chall} = \mathcal{V}(\text{st}, \text{com}; r)$ and $\text{chall}' = \mathcal{V}(\text{st}, \text{com}; r')$, then $\mathcal{E}(\text{st}, \text{com}, \text{chall}, \text{resp}, \text{chall}', \text{resp}')$ yields wt such that $\mathcal{R}(\text{st}, \text{wt})$.
- *Special honest-verifier zero-knowledge*: there exists a polynomially bounded algorithm Sim called simulator such that for any $\text{st} \in L$, wt such that $(\text{st}, \text{wt}) \in \mathcal{R}$ and chall , the transcript $(\text{com}, \text{chall}, \text{resp})$ of the interaction $\mathcal{P}(\text{st}, \text{wt}) \leftrightarrow \mathcal{V}(\text{st}, r_V)$ conditioned to chall has the same distribution as $\text{Sim}(\text{st}, \text{chall}; r)$. Note that here r_V denotes the randomness used by the verifier $\mathcal{V}$.

We call st the statement and wt the witness.

In more informal words, a sigma protocol is a three-move protocol between a prover and a verifier.

Remark 2.12. In some circumstances, we can relax the relation for special soundness to R' where $R \subseteq R'$. That is, we allow the extractor to output wt such that $(\text{st}, \text{wt}) \in R'$ but $(\text{st}, \text{wt}) \notin R$. As long as given st , finding wt such that $(\text{st}, \text{wt}) \in R'$ is hard, the sigma protocol can still serve as a proof system for some applications. For instance, [BKP20, BDK⁺22] both relax the relation by including the collision of the hash function, and call it *relaxed special-soundness* (see [BKP20], Definition 2.1 for a formal exposition).

Remark 2.13. The notion above is actually also referred to as 2-special soundness, since given **two** accepting transcripts with the same initial message (com) but different challenges, one can efficiently extract the witness. This can be generalized to n -special soundness for an integer n : given n accepting transcripts with the same commitment but n distinct challenges, one can efficiently extract a witness.

We illustrate the typical flow of a sigma protocol in Figure 2.5.

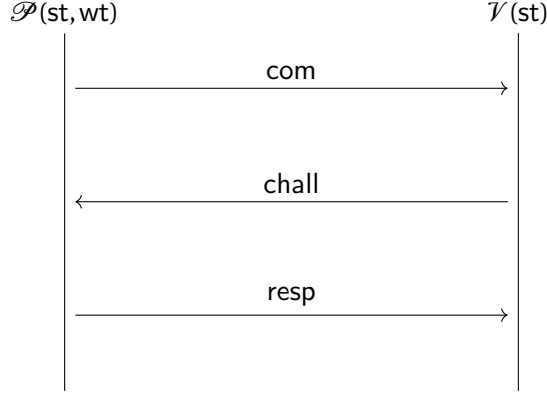


Figure 2.5: Illustration of the flow of a sigma protocol (Definition 2.11)

Definition 2.14 (Correctness). *A sigma protocol Π_Σ is said to be correct if for all $\lambda \in \mathbb{N}$, $(st, wt) \in R$, given that the prover and the verifier both follow the protocol specifications, then the verifier always accepts.*

Definition 2.15 (High Min-Entropy). *We say a sigma protocol Π_Σ has $\alpha(\lambda)$ min-entropy if for any $\lambda \in \mathbb{N}$, $(st, wt) \in R$, and a possibly computationally-unbounded adversary $\mathcal{A}$, we have*

$$\Pr \left[\text{com} = \text{com}' \mid \text{com} \leftarrow \mathcal{P}_1^\mathcal{O}(st, wt), \text{com}' \leftarrow \mathcal{A}^\mathcal{O}(st, wt) \right] \leq 2^{-\alpha},$$

where the probability is taken over the randomness used by P_1 and by the random oracle $\mathcal{O}$, and $\mathcal{P}_1$ denotes the prover in the commitment phase of the sigma protocol. We say Π_Σ has high min-entropy if $2^{-\alpha}$ is negligible in λ .

A sigma-protocol can be turned into a non-interactive zero-knowledge proof, as defined in Definition 2.1.5 using the Fiat-Shamir transform [FS86].

2.1.5 Proof Systems

We consider *non-interactive zero-knowledge proof of knowledge* (NIZKPoK) over the programmable random oracle model with statistical zero-knowledge and simulation-extractability.

Definition 2.16. *A NIZKPoK for a relation $\mathcal{R}$ is a tuple of PPT algorithms $\Pi = (\text{Prove}, \text{Verify})$ with access to the random oracle $\mathcal{O}$ such that for any valid public parameter of the scheme pp and the security parameter $\lambda \in \mathbb{N}$, the scheme satisfies the following properties:*

- *Completeness: for every $(st, wt) \in \mathcal{R}$,*

$$\Pr[\text{Verify}^\mathcal{O}(st, \pi) = 1 \mid \pi \leftarrow \text{Prove}^\mathcal{O}(st, wt)] = 1.$$

- *Statistical zero-knowledge: there exists a simulator Sim such that for any $(st, wt) \in \mathcal{R}$ and any possibly unbounded adversary $\mathcal{A}$, $\mathcal{A}$ can distinguish the distributions: $\{\pi \mid \pi \leftarrow$*

$\text{Prove}^{\mathcal{O}}(\text{st}, \text{wt})$, $\{\pi | \pi \leftarrow \text{Sim}^{\mathcal{O}}(\text{st})\}$ with only negligible advantage where the probability is taken over the randomness used in the experiment and the scheme.

- *Simulation-extractability* ([GM17]): for any non-uniform PPT adversary $\mathcal{A}$ there exists a non-uniform PPT extractor $\mathcal{E}_{\mathcal{A}}$ such that

$$\Pr \left[\begin{array}{c} \text{Verify}(\text{st}, \pi) = 1 \\ (\text{st}, \text{wt}) \notin \mathcal{R} \wedge (\text{st}, \pi) \notin \mathcal{Q} \end{array} \middle| \begin{array}{c} (\text{st}, \pi) \leftarrow \mathcal{A}^{\text{SimProve}, \mathcal{O}}(\text{pp}) \\ \text{wt} \leftarrow \mathcal{E}_{\mathcal{A}}(\text{st}, \pi) \end{array} \right] \leq \text{negl}(\lambda)$$

where $\text{SimProve}(\text{st}', \text{wt}')$ is an oracle that returns a simulated proof using $\pi' \leftarrow \text{Sim}(\text{st}')$ when $(\text{st}', \text{wt}') \in \mathcal{R}$, and stores its queries (st, π) in a set $\mathcal{Q}$. Otherwise, SimProve returns $\perp$.

Intuitively, the zero-knowledge property ensures that the proof does not leak any information about the witness to the adversary while the simulation extractability property ensure that an adversary should not be able to create a valid proof, unless they know the witness, regardless of how many proofs they have observed. In particular, simulation extractability only guarantees extraction for adversarially generated proofs that were not produced by the simulator oracle, and the extractor itself is adversary-specific.

2.1.6 Updatable Public Key Encryption

In this section, we introduce the notion of updatable public-key encryption (UPKE) schemes and some of their security notions following the descriptions given in [ABL25, DKW21, JD25]. Updatable public-key encryption (UPKE) can be seen as an advanced form of public-key encryption that allows for asynchronous key updates. It finds its main applications in secure messaging. Note that it is often mistaken with updatable encryption, which is a different primitive that allows to update ciphertexts to be decryptable under new keys.

To formally define UPKE, we take a necessary detour to first recall the definition of standard public-key encryption (PKE) schemes and their basic security.

Definition 2.17. A public-key encryption scheme (PKE) is a tuple $(\text{KeyGen}, \text{Enc}, \text{Dec})$ of algorithms together with a message space $\mathcal{M}$ satisfying the following:

- $\text{KeyGen}(1^\lambda) \rightarrow (\text{sk}, \text{pk})$ is a PPT algorithm that takes as input the security parameter and outputs a secret-public key pair (sk, pk) ;
- $\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$ is a PPT algorithm which on input a message $m \in \mathcal{M}$ and a public key pk , returns a ciphertext ct ;
- $\text{Dec}(\text{sk}, \text{ct}) \rightarrow m$ is a deterministic algorithm which on input the secret key sk and the ciphertext ct , outputs a message m .

Definition 2.18 (PKE Correctness). *A public-key encryption scheme $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is correct if for all security parameters λ and all secret-public key pairs (pk, sk) generated by KeyGen algorithm, and all $m \in \mathcal{M}$, the following condition is satisfied:*

$$\text{Dec}(\text{sk}, \text{Enc}(\text{pk}, m)) = m$$

Definition 2.19 (OW-CPA security). *Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a public key encryption scheme with message space $\mathcal{M}$. A PKE scheme is OW-CPA secure if for all PPT adversaries $\mathcal{A}$, its advantage $\text{Adv}_{\mathcal{A}}^{\text{OW-CPA}}$ is negligible. where the advantage is defined as*

$$\text{Adv}_{\mathcal{A}}^{\text{OW-CPA}} = \Pr[\text{OW-CPA}^{\mathcal{A}} \rightarrow 1].$$

and the OW-CPA game is defined in Figure 2.6.

OW-CPA($\mathcal{A}$)
 1: $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$
 2: $m^* \xleftarrow{\$} \mathcal{M}$
 3: $\text{ct}^* \leftarrow \text{Enc}(\text{pk}, m^*)$
 4: $m' \leftarrow \mathcal{A}(\text{ct}^*, \text{pk})$
 5: **return** $(m' = m^*)$

Figure 2.6: The OW-CPA game for public-key encryption.

This allows us to now define an updatable public-key encryption scheme. Roughly speaking, an UPKE scheme extends a standard PKE scheme with two additional algorithms that allow to update the public and secret keys respectively.

Definition 2.20. *An updatable public-key encryption (UPKE) scheme for a message space $\mathcal{M}$ is a tuple $(\text{KeyGen}, \text{Enc}, \text{Dec}, \text{UpdPk}, \text{UpdSk})$ of algorithms such that:*

- $\text{KeyGen}(\lambda) \rightarrow (\text{sk}, \text{pk})$ is a PPT key generation algorithm that outputs a secret key sk and a public key pk for a given security parameter λ ;
- $\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$ is a PPT algorithm that takes as input a public key pk and a message $m \in \mathcal{M}$, and outputs a ciphertext ct ;
- $\text{Dec}(\text{sk}, \text{ct}) \rightarrow m$ is a deterministic algorithm that takes as input a secret key sk and a ciphertext ct , and outputs a message m ;
- $\text{UpdPk}(\text{pk}, \rho) \rightarrow (\text{pk}', \text{up})$ is a PPT algorithm that takes as input a public key pk and explicit randomness $\rho \in \mathcal{R}$, and produces an updated public key pk' along with an update token up ;
- $\text{UpdSk}(\text{sk}, \text{up}) \rightarrow \text{sk}'$ is a deterministic algorithm that takes as input a secret key sk together with an update token up , and outputs an updated secret key sk' .

Definition 2.21 (UPKE Correctness [DKW21]). An UPKE scheme $(\text{KeyGen}, \text{Enc}, \text{Dec}, \text{UpdPk}, \text{UpdSk})$ with message space $\mathcal{M}$ is correct if:

- $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is a correct PKE with message space $\mathcal{M}$;
- for any $t \in \mathbb{N}$, any $(\text{sk}_0, \text{pk}_0) \leftarrow \text{KeyGen}(1^\lambda)$, any $(\text{pk}_{i+1}, \text{up}_{i+1}) \leftarrow \text{UpdPk}(\text{pk}_i, \rho_i)$, any $\rho_i \in \mathcal{R}$ $\text{sk}_{i+1} \leftarrow \text{UpdSk}(\text{sk}_i, \text{up}_{i+1})$ for any $i \in [0, t)$ and any message $m \in \mathcal{M}$, we have

$$\Pr[\text{Dec}(\text{sk}_{i+1}, \text{Enc}(\text{pk}_{i+1}, m)) = m] \geq 1 - \text{negl}(\lambda)$$

where the probability is taken over the randomness of Enc .

The various security notions for updatable public-key encryption, as established in [DKW21] follow closely the hierarchy established for the security of public-key encryption, but they include an additional key update oracle. Namely, we distinguish one-way (OW) security, indistinguishability under chosen-plaintext (IND-CPA) security and indistinguishability under chosen ciphertext (IND-CCA) security. Furthermore, each of these notions comes with two additional variants, which depend on the modeling of the key update oracle, namely chosen randomness (CR) where the adversary controls only the randomness used by the key update oracle, and chosen updates (CU) where the adversary can create entirely malicious public keys and update tokens.

Note that IND-CR-CPA is the standard notion that is usually proven for post-quantum updatable public key encryption schemes, such as in the constructions [JD25] and [ABL25]. Several works have shown generic transforms from IND-CR-CPA to IND-CU-CCA security, or from OW-CR-CPA to IND-CU-CCA, such as [DKW21, AW25]. This stronger security notion is therefore achievable from any IND-CR-CPA or OW-CR-CPA secure scheme, by using the generic transforms. In this thesis, in Chapter 4, we will use the transform introduced by [AW25] to turn an OW-CR-CPA secure UPKE into an IND-CU-CCA secure one.

In the following, we therefore define the notion of OW-CR-CPA secure UPKE (Definition 2.22). We further formalize an additional property needed for the transform to be valid, namely γ -spreadness (Definition 2.23), and we also explicitly recall the security notion of IND-CU-CCA security (Definition 2.24).

Definition 2.22 (OW-CR-CPA security). Let $\Pi = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{UpdPk}, \text{UpdSk})$ be an updatable public-key encryption scheme. We say it is OW-CR-CPA secure if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\mathcal{A}}^{\text{OW-CR-CPA}} = \left| \Pr[\text{OW-CR-CPA}_{\Pi}^{\mathcal{A}} \rightarrow 1] \right| \quad (2.1)$$

is negligible, where the OW-CR-CPA game is described in Figure 2.7.

Definition 2.23 (γ -spread). A UPKE scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec}, \text{UpdPk}, \text{UpdSk})$ is γ -spread if

OW-CR-CPA($\mathcal{A}$)

- 1: $i \leftarrow 0$
- 2: $(pk_0, sk_0) \leftarrow \text{KeyGen}(1^\lambda)$
- 3: $\mathcal{A}^{\text{UpdO}}(pk_0) \rightarrow \text{state}$
- 4: $q \leftarrow i$
- 5: $m^* \xleftarrow{\$} \mathcal{M}$
- 6: $ct^* \leftarrow \text{Enc}(pk_q, m^*)$
- 7: $\mathcal{A}^{\text{UpdO}}(ct^*, \text{state}) \rightarrow \text{state}$
- 8: $q' \leftarrow i$
- 9: $\rho^* \xleftarrow{\$} \mathcal{R}$
- 10: $(pk^*, up^*) \leftarrow \text{UpdPk}(pk_{q'}, \rho^*)$
- 11: $sk^* \leftarrow \text{UpdSk}(sk_{q'}, up^*)$
- 12: $\mathcal{A}(ct^*, pk^*, sk^*, up^*, \text{state}) \rightarrow m$
- 13: **return** $(m = m^*)$

UpdO(ρ)

- 1: $(pk_{i+1}, up_{i+1}) \leftarrow \text{UpdPk}(pk_i, \rho)$
- 2: $sk_{i+1} \leftarrow \text{UpdSk}(sk_i, up_{i+1})$
- 3: $i \leftarrow i + 1$

Figure 2.7: The OW-CR-CPA game for a UPKE scheme Π .

for all $\lambda, \ell \in \mathbb{N}$, all $(pk_0, sk_0) \leftarrow \text{KeyGen}(1^\lambda)$, all $m \in \mathcal{M}$, and all $i \in \{0, \dots, \ell\}$, we have:

$$-\log \left(\max_{ct \in \{0,1\}^*} \Pr[ct = \text{Enc}(pk_i, m)] \right) \geq \gamma,$$

where $(pk_j, up_j) \leftarrow \text{UpdPk}(pk_{j-1}, \rho)$, $sk_j \leftarrow \text{UpdSk}(sk_{j-1}, up_j)$ for $j = 1, \dots, \ell$, and the probability is taken over the randomness used in Enc .

This property aims to ensure that, given a message, the ciphertexts output over some encryption randomness are well spread. In other words, it characterizes how uniformly distributed are the ciphertexts, given messages and randomness.

Definition 2.24 (IND-CU-CCA security). *The IND-CU-CCA security of an UPKE scheme is defined by a game between an adversary $\mathcal{A}$ and a challenger $\mathcal{C}$ defined in Figure 2.8. $\mathcal{A}$'s advantage is defined by:*

$$\text{Adv}_{\Pi}^{\text{IND-CU-CCA}}(\mathcal{A}) = \left| \Pr[\text{IND-CU-CCA}_{\Pi}^{\mathcal{A}} \rightarrow 1] - \frac{1}{2} \right| \quad (2.2)$$

A UPKE scheme is IND-CU-CCA secure if for all PPT adversaries $\mathcal{A}$, its advantage $\text{Adv}_{\Pi}^{\text{IND-CU-CCA}}(\mathcal{A})$ is negligible.

Proposition 2.25. *Given an OW-CR-CPA secure UPKE that is also γ -spread with γ at least $O(\lambda)$, a symmetric encryption scheme that is OT-CPA secure and four random oracles, the transform of [AW25, Section 4.2] gives an IND-CU-CCA secure UPKE scheme.*

Remark 2.26. Note that any commonly used symmetric encryption scheme (e.g., AES in an appropriate mode of operation) is OT-CPA secure. Thus, the main requirement for obtaining IND-CU-CCA security via the above transform is to have a UPKE scheme that is both OW-CR-CPA secure and γ -spread with γ at least $O(\lambda)$.

<u>IND-CU-CCA($\mathcal{A}$)</u> 1: $i \leftarrow 0$ 2: $(pk_0, sk_0) \leftarrow \text{KeyGen}(1^\lambda)$ 3: $\mathcal{A}^{\text{UpdO}, \text{DecO}}(pk_0) \rightarrow (\text{state}, m_0, m_1)$ 4: $q \leftarrow i$ 5: $b \xleftarrow{\$} \{0, 1\}$ 6: $ct^* \leftarrow \text{Enc}(pk_q, m_b)$ 7: $\mathcal{A}^{\text{UpdO}, \text{DecO}^*}(ct^*, \text{state}) \rightarrow \text{state}$ 8: $q' \leftarrow i$ 9: $\rho^* \xleftarrow{\$} \mathcal{R}$ 10: $(pk^*, up^*) \leftarrow \text{UpdPk}(pk_{q'}, \rho^*)$ 11: $sk^* \leftarrow \text{UpdSk}(sk_{q'}, up^*)$ 12: $\mathcal{A}(ct^*, pk^*, sk^*, up^*, \text{state}) \rightarrow b'$ 13: return $(b = b')$	<u>UpdO(pk', up')</u> 1: Check validity of up' 2: $sk_{i+1} \leftarrow \text{UpdSk}(sk_i, up')$ 3: $pk_{i+1} \leftarrow pk'$ 4: $i \leftarrow i + 1$ <u>DecO(ct)</u> 1: return $\text{Dec}(sk_i, ct)$ <u>DecO*(ct)</u> 1: if $(i, ct) = (q, ct^*)$ then 2: abort 3: else 4: return $\text{Dec}(sk_i, ct)$
--	--

Figure 2.8: The IND-CU-CCA game for a UPKE scheme Π with chosen-update and decryption oracles.

For completeness, we recall below the definition of OT-CPA security in Definition ??.

Definition 2.27 (OT-CPA security). *Let $\Gamma = (\text{SKE.Enc}, \text{SKE.Dec})$ be a symmetric-key encryption scheme with key domain SK , and plaintext domain $\mathcal{M}$. We say it is OT-CPA secure if for all PPT adversaries $\mathcal{A}$, the advantage*

$$\text{Adv}_{\mathcal{A}}^{\text{OT-CPA}} = \left| \Pr[\text{OT-CPA}_{\Gamma}^{\mathcal{A}} \rightarrow 1] - \frac{1}{2} \right| \quad (2.3)$$

is negligible, where the OT-CPA game is defined below.

OT-CPA($\mathcal{A}$)
1: $k \xleftarrow{\$} \text{SK}$
2: $(m_0, m_1) \leftarrow \mathcal{A}$
3: $b \xleftarrow{\$} \{0, 1\}$
4: $ct^* \leftarrow \text{SKE.Enc}(k, m_b)$
5: $b' \leftarrow \mathcal{A}(ct^*)$
6: **return** $(b' = b)$

Figure 2.9: The OT-CPA game for a symmetric-key encryption scheme Γ .

We now leave the realm of protocols and turn our attention to the problems necessary to instantiate them. Several chapters of this thesis look at post-quantum group actions. We start by introducing a framework for these. We then turn our attention to two families of post-quantum assumptions: isogeny-based cryptography, and tensor and code problems.

2.2 Cryptographic group actions

Group actions have been of interest in cryptography because they provide a general framework to transition classical cryptosystems to post-quantum ones. The first well-studied instance of a post-quantum group action used in cryptography was in works from Couveignes [Cou06], and Rostovstev and Stolbunov [RS06], where they developed a Diffie-Hellman-like key exchange scheme from an isogeny group action on ordinary elliptic curves.

The seminal work from Alamiati, De Feo, Montgomery, and Patranabis in [ADMP20] generalizes the hard problems taken from isogeny group actions to use with any group action. In doing so, they establish standard definitions and notations that have since been upheld in the field, providing a valuable common ground for proceeding works. To begin, we recall some of these definitions that will be relevant to this thesis.

Definition 2.28 (Group action). *We say that a group G acts on a set X (by the left) if there exists a map $\star : G \times X \rightarrow X$ such that :*

- *Identity: If e is the identity element of G , then for any $x \in X$, $e \star x = x$;*
- *Compatibility: For any $g, h \in G$, and any $x \in X$, $(gh) \star x = g \star (h \star x)$.*

Such a group action can be denoted $(G, X, \star)$.

Definition 2.29 (Properties of group actions). *We call a group action*

- *commutative if for any $g, h \in G$ and any $x \in X$, $g \star (h \star x) = h \star (g \star x)$.*
- *faithful if for $g \in G$, $g \star x = x$ for all $x \in X$ implies that g is the identity element.*
- *free if for each $g \in G$, g is the identity element if and only if there exists an element $x \in X$ such that $x = g \star x$.*
- *transitive if for any two elements $x, y \in X$, there exists a group element $g \in G$ mapping x to y , i.e. $y = g \star x$.*
- *regular if it is both transitive and free.*

Definition 2.30 (Orbit). *The orbit, O_x , of an element $x \in X$, is the set of all $y \in X$ for which there exists an element mapping x to y i.e $O_x = \{y \in X | \exists g \in G, y = g \star x\} = \{g \star x | g \in G\}$.*

We can reformulate transitivity in terms of orbits : an action is transitive if for all $x, y \in X$ we get that $O_x = O_y$, i.e. there is only a single orbit.

Since we are considering group actions in the context of cryptography, we will only be concerned with *effective group actions*.

Definition 2.31 (Effective group action (EGA)). *We call a group action $\star : G \times X \rightarrow X$ effective if the following properties hold :*

- *G is finite and there exists a PPT algorithm for the following operations : group operation, computation of inverses, membership testing, equality testing and sampling.*
- *X is finite and there exists a PPT algorithm for membership testing and for computing a unique representation of elements in X .*
- *There exists a distinguished element $x_0 \in X$ such that its bit-string representation is known. We call this point the origin.*
- *$\star$ can be computed efficiently for any $g \in G, x \in X$.*

The framework can be refined to define a *known-order effective group action* (KO-EGA) as introduced in [ADMP20]. This notion is stronger because it contains additional information about the group structure and a generator set.

Definition 2.32 (Known-order Effective Group Action (KO-EGA)). *Let $(G, X, \star)$ be a group action. For an $x_0 \in X$, we say $(G, X, \star, x_0)$ forms a known-order effective group action if the following properties are satisfied:*

1. *$(G, X, \star, x_0)$ is an EGA.*
2. *The group G is finite, abelian and both the structure of $G \cong \bigoplus_i \mathbb{Z}/m_i\mathbb{Z}$ and a minimal generating set $\langle g_i \rangle = G$ are known.*
3. *There exists a PPT algorithm to evaluate $g_i^n \star x$ for any natural number $n \in \{1, \dots, \prod_i m_i\}$, $x \in X$ and g_i from the generating set.*

Since the group order of G is given, one can sample uniformly at random from G . Similarly, by multiplying by a proper factor, one can obtain a generating set for any subgroup of G and sample uniformly at random from the subgroup.

Post-quantum instantiations of KO-EGA are, at the moment, known only from isogenies. However, the less strict notion of EGA can be built from more diverse cryptographic assumptions, such as tensors or linear codes.

Hard problems in the context of group actions extend naturally from their classical counterpart, namely we have an extension of the discrete logarithm problem to group actions called group-action inverse problem (GAIP), its decisional version (dGA-IP), as well as extensions of the computational and decisional Diffie-Hellman problems to group actions.

Definition 2.33 (Group Action Inverse Problem). *Let $(G, X, \star, x_0)$ be a group action with a distinguished element $x_0 \in X$. Given x sampled uniformly in X , the Group Action Inverse Problem (GAIP) consists in finding an element $g \in G$ such that $g \star x_0 = x$.*

Remark 2.34. Note that this definition either requires X to have a single orbit (i.e. the action must be transitive), or that x is sampled uniformly from the orbit of x_0 .

Definition 2.35 (Decisional Group Action Inverse Problem [DFG23]). *Let $(G, X, \star)$ be a group action, and $x_0, x_1 \in X$ lie in distinct orbits. The decisional Group Action Inverse Problem (dGA-IP) game for $(G, X, \star, x_0, x_1)$ is described in Figure 2.10. For an adversary $\mathcal{A}$ against the dGA-IP game, we define the advantage Adv as follows :*

$$\text{Adv}(\mathcal{A}) = |\Pr[\text{dGA-IP}(\mathcal{A}) \rightarrow 1] - \frac{1}{2}|.$$

dGA-IP($\mathcal{A}$)

```

1:  $c, b \xleftarrow{\$} \{0, 1\}$ 
2:  $g, g' \xleftarrow{\$} G$ 
3:  $y \leftarrow g \star x_c$ 
4: if  $b = 1$  then
5:    $x \leftarrow g' \star y$ 
6: if  $b = 0$  then
7:    $x \leftarrow g' \star x_{1-c}$ 
8:  $b' \leftarrow \mathcal{A}(y, x)$ 
9: return  $b' = b$ 
```

Figure 2.10: The dGA-IP game

Remark 2.36. Note that the dGA-IP problem is only interesting when the group action is not transitive, i.e., when there exist at least two distinct orbits. In fact, if the group action is transitive, then for any $x \in X$, there exists a $g \in G$ such that $g \star x_0 = x$. In this case, the dGA-IP problem is trivial in this case since there is only one orbit, and thus any two elements in X lie in the same orbit.

Definition 2.37 (Group-Action Decisional Diffie-Hellman Problem). *Let $(G, X, \star, x_0)$ be a group action. Let the DDH game be defined in Figure 2.11. We define the advantage of $\mathcal{A}$ to be $\text{Adv}^{\text{DDH}}(\mathcal{A}) := |\Pr[\text{DDH}(\mathcal{A}) \rightarrow 1] - 1/2|$.*

Remark 2.38. The group-action decisional Diffie-Hellman has been shown in [CSV20] to be broken in the context of the action of ideal class groups of imaginary quadratic orders on the set of elliptic curves through isogenies (such as the CSIDH [CLM⁺18] group action, see Definition 2.60), as soon as there exists a quadratic character of small modulus .i.e. the discriminant has a small prime factor. In particular, the attack is very efficient when considering supersingular elliptic curves defined over $\mathbb{F}_p$ with $p \equiv 1 \pmod{4}$. However, one can still use this problem if restricting elements to a subgroup.

Definition 2.39 (Group-Action Computational Diffie-Hellman Problem). *Let $(G, X, \star, x_0)$ be a group action.*

Let the CDH game be defined in Figure 2.12. We define the advantage of $\mathcal{A}$ in solving CDH as

$$\text{Adv}^{\text{CDH}}(\mathcal{A}) := \Pr[\text{CDH}(\mathcal{A}) \rightarrow 1],$$

DDH($\mathcal{A}$)

```

1:  $b \xleftarrow{\$} \{0, 1\}$ 
2:  $g_1, g_2, g_3 \xleftarrow{\$} G$ 
3:  $x_1 \leftarrow g_1 \star x_0, x_2 \leftarrow g_2 \star x_0$ 
4: if  $b = 1$  then
5:    $y \leftarrow g_3 \star x_0$ 
6: if  $b = 0$  then
7:    $y \leftarrow g_1 g_2 \star x_0$ 
8:  $b' \leftarrow \mathcal{A}(y, x_1, x_2)$ 
9: return  $b' = b$ 

```

Figure 2.11: The DDH game

CDH($\mathcal{A}$)

```

1:  $g_1, g_2 \xleftarrow{\$} G$ 
2:  $x_1 \leftarrow g_1 \star x_0, x_2 \leftarrow g_2 \star x_0$ 
3:  $y \leftarrow \mathcal{A}(x_1, x_2)$ 
4: return  $y = g_1 g_2 \star x_0$ 

```

Figure 2.12: The CDH game

Remark 2.40. Note that it has been shown in a line of work ([GPSV21], [MZ22], [GLM24]) that in the context of abelian group action, discrete logarithm and CDH are quantumly equivalent.

2.3 Isogenies

As Chapters 3,4 and 5 focus on isogeny-based cryptography, we introduce in this section the necessary background on isogenies.

As each chapter covers a distinct type of isogeny-based problems, in the interest of conciseness we restrict ourselves to introduce only the minimal necessary background to understand them and refer the reader to relevant sources for further details. Let us start by recalling some general facts about elliptic curves and isogenies.

2.3.1 Elliptic curves and isogenies

In this thesis we are interested in elliptic curves over finite fields. A standard reference to study the basics of the theory of elliptic curves is [Sil09]. We assume the reader is already familiar with the definition of an elliptic curve and its group law, and we will focus more on the aspects relevant to isogenies.

Definition 2.41 (Isogeny). *Let E_1 and E_2 be elliptic curves defined over a finite field $\mathbb{F}_q$. An isogeny is a non-constant rational map*

$$\varphi : E_1 \rightarrow E_2$$

that is also a group homomorphism with respect to the group law on the elliptic curves.

Remark 2.42. An example of isogeny is the multiplication-by- m map which sends a point $P \in E$ to $[m]P$. It is in fact also an endomorphism (Defintion 2.45). The kernel of this map forms the m -torsion subgroup of an elliptic curve, and a point such that $[m]P = \mathcal{O}$, where $\mathcal{O}$ is the neutral element, is called an m -torsion point.

Definition 2.43 (Degree of an Isogeny). *When dealing with isogenies between supersingular elliptic curves, the degree of an isogeny φ , denoted $\deg(\varphi)$, is its degree as a rational map. It can be written as*

$$\deg(\varphi) = p^r d',$$

where d' is an integer coprime to p , the characteristic of the base field.

- If $r = 0$, the isogeny is called separable, and in this case

$$\deg(\varphi) = \# \ker(\varphi).$$

- Otherwise, the isogeny is said to be inseparable.

Unless otherwise stated, all isogenies considered in this work are assumed to be separable.

Definition 2.44 (Dual Isogeny). *For any isogeny $\varphi : E_1 \rightarrow E_2$, there exists a unique isogeny*

$$\hat{\varphi} : E_2 \rightarrow E_1$$

such that

$$\hat{\varphi} \circ \varphi = [\deg(\varphi)]_{E_1} \quad \text{and} \quad \varphi \circ \hat{\varphi} = [\deg(\varphi)]_{E_2}.$$

The isogeny $\hat{\varphi}$ is called the dual isogeny of φ .

Definition 2.45 (Endomorphism and Endomorphism Ring). *An endomorphism of an elliptic curve E is an isogeny*

$$\varphi : E \rightarrow E.$$

The set of all endomorphisms of E , denoted by $\text{End}(E)$ together with the zero map, forms a ring under addition and composition, called the endomorphism ring of E .

Theorem 2.46 (Structure of Endomorphism Rings over Finite Fields). *Let E be an elliptic curve defined over a finite field. Then:*

- If E is ordinary, the ring $\text{End}(E)$ is an order in an imaginary quadratic field.
- If E is supersingular, the ring $\text{End}(E)$ is a maximal order in a quaternion algebra.

In this thesis, only supersingular elliptic curves are considered.

With these ingredients we can introduce some central problems in isogeny-based cryptography.

Problem 2.47 (Endomorphism ring problem). *Given an elliptic E_1 over a finite field $\mathbb{F}_q$, compute (a basis of) its endomorphism ring.*

Problem 2.48 (General isogeny problem). *Given two isogenous supersingular elliptic curves E_1, E_2 over a finite field $\mathbb{F}_q$, find an isogeny $\varphi : E_1 \rightarrow E_2$.*

Remark 2.49. Note that the decisional version of this problem is easy since supersingular elliptic curves form a single isogeny class.

Problem 2.50 (d -isogeny problem). *Given two supersingular elliptic curves E_1, E_2 over a finite field $\mathbb{F}_q$, and a positive integer d find an isogeny $\varphi : E_1 \rightarrow E_2$ of degree d if it exists.*

Let us comment briefly on the relations between these problems. Recent work by Herlédan le Merdy and Wesolowski [MW25] shows unconditional polynomial time equivalence between Problem 2.47 and Problem 2.48. Note also that if one can solve Problem 2.50 then one can trivially solve Problem 2.48. The converse is not known.

2.3.2 Quaternion Algebras

As highlighted in definition 2.46, the endomorphism ring of a supersingular elliptic curve is isomorphic to a maximal order in a quaternion algebra. We now give the necessary background on quaternion algebras. The world of quaternion algebras is a powerful tool to study supersingular elliptic curves and their isogenies due to this correspondence between them.

Definition 2.51. *Let p be a prime. We write $\mathcal{B}_{p,\infty}$ for the quaternion algebra ramified only at p and ∞ , which is defined by*

$$\mathcal{B}_{p,\infty} = \left(\frac{-q, -p}{\mathbb{Q}} \right) = \mathbb{Q} + i\mathbb{Q} + j\mathbb{Q} + k\mathbb{Q}$$

where $0 \neq q \in \mathbb{N}$, $i^2 = -q$, $j^2 = -p$, and $k = ij = -ji$.

Definition 2.52 (Norm in a Quaternion Algebra). *Let B be a quaternion algebra over a number field $\mathbb{F}$. For an element*

$$x \in B, \quad x = a + bi + cj + dk, \quad a, b, c, d \in \mathbb{F},$$

the reduced norm of x , denoted $\text{Nrd}(x)$, is defined by

$$\text{Nrd}(x) := x \cdot \bar{x} = a^2 - b^2 i^2 - c^2 j^2 - d^2 k^2,$$

where $\bar{x} = a - bi - cj - dk$ is the conjugate of x in B , and the multiplication rules of B determine i^2, j^2, k^2 and $ij = -ji = k$, etc.

Equivalently, the reduced norm is a quadratic form on B satisfying

$$\text{Nrd}(xy) = \text{Nrd}(x)\text{Nrd}(y) \quad \text{for all } x, y \in B.$$

Definition 2.53 (Ring of integers). *Let $\mathbb{F}$ be an algebraic number field, then its ring of integers denoted $\mathcal{O}_{\mathbb{F}}$ is the ring of all algebraic integers contained of $\mathbb{F}$, i.e. the ring containing all the roots of the monic polynomials with integer coefficients. In particular, for any $\mathbb{F}$, we have $\mathbb{Z} \subset \mathcal{O}_{\mathbb{F}}$.*

Definition 2.54 (Order, Maximal order). *Let B be a quaternion algebra over a field $\mathbb{F}$. An order $\mathcal{O} \subset B$ is a subring of B which is a finitely generated module over the ring of integers $\mathcal{O}_{\mathbb{F}}$ of $\mathbb{F}$ and satisfies*

$$\mathcal{O} \otimes_{\mathcal{O}_{\mathbb{F}}} \mathbb{F} = B.$$

An order $\mathcal{O} \subset B$ is called maximal if it is not properly contained in any other order of B , i.e., if there is no order $\mathcal{O}' \subset B$ such that $\mathcal{O} \subsetneq \mathcal{O}'$.

As we will focus on $\mathcal{B}_{p,\infty}$, we further define a p -extremal maximal order as a maximal order containing j .

Examples of p -extremal maximal orders are those containing $\mathbb{Z}\langle i, j \rangle = \mathbb{Z} + i\mathbb{Z} + j\mathbb{Z} + k\mathbb{Z}$ as subring. For such a maximal order $\mathcal{O}$, if $R = \mathcal{O} \cap \mathbb{Q}[i]$ is the ring of integers $\mathbb{Z}[\omega]$ of $\mathbb{Q}[i]$, then the restriction of the norm to $R + Rj$ is given by

$$\text{Nrd}(x_1 + y_1\omega + (x_2 + y_2\omega)j) = f(x_1, y_1) + pf(x_2, y_2)$$

where f is a principal quadratic form of discriminant $\text{disc}(R)$ [KLPT14]. We have

$$f(x, y) = x^2 + \text{Trd}(\omega)xy + \text{Nrd}(\omega)y^2$$

We give below a few examples of the structure of $\mathcal{B}_{p,\infty}$, together with a p -extremal order $\mathcal{O}$, R and $f(x, y)$ as defined above for different values of p .

Example 2.55. 1. For $p \equiv 3 \pmod{4}$: $\mathcal{B}_{p,\infty} = \left(\frac{-1, -p}{\mathbb{Q}}\right)$; $\mathcal{O} = \langle 1, i, \frac{1+k}{2}, \frac{i+j}{2} \rangle$; $R = \mathbb{Z}[i]$; $f(x, y) = x^2 + y^2$.

2. For $p \equiv 5 \pmod{8}$: $\mathcal{B}_{p,\infty} = \left(\frac{-2, -p}{\mathbb{Q}}\right)$; $\mathcal{O} = \langle 1, i, \frac{1+j+k}{2}, \frac{i+2j+k}{4} \rangle$; $R = \mathbb{Z}[i]$; $f(x, y) = x^2 + 2y^2$.

3. For $p \equiv 1 \pmod{8}$: $\mathcal{B}_{p,\infty} = \left(\frac{-q, -p}{\mathbb{Q}}\right)$, where $q \equiv 1 \pmod{4}$ is a prime such that $\left(\frac{-p}{q}\right) = 1$; $\mathcal{O} = \langle 1, \frac{1+i}{2}, j, \frac{ci+k}{q} \rangle$, where $c^2 \equiv -p \pmod{q}$; $R = \mathbb{Z}[\frac{1+i}{2}]$; $f(x, y) = x^2 - xy + \frac{1+q}{4}y^2$.

2.3.3 Isogenies and class group action

In this section, we define the necessary background to interpret the action of the class-group on the supersingular isogeny graph.

Definition 2.56 (Ideal Class Group). *Let $\mathcal{O}$ be an order in an imaginary quadratic field. The ideal class group $\text{Cl}(\mathcal{O})$ is the group of invertible fractional ideals of $\mathcal{O}$, modulo principal ideals. It is a finite abelian group.*

Definition 2.57 (Supersingular Isogeny Graph). *Let $S = \{\ell_1, \dots, \ell_n\}$ be a set of small odd primes. The supersingular isogeny graph over $\mathbb{F}_p$ has:*

- nodes corresponding to $\mathbb{F}_p$ -isomorphism classes of supersingular elliptic curves, and
- edges corresponding to separable isogenies of degree $\ell \in S$.

Let E be a supersingular elliptic curve. We denote by $\text{End}_p(E)$ the subring of $\text{End}(E)$ that consists of endomorphisms defined over $\mathbb{F}_p$.

Definition 2.58 (Class Group Action on Supersingular Curves). *Let E be a supersingular elliptic curve over $\mathbb{F}_p$ with $\text{End}_p(E) \simeq \mathcal{O}$, and let $[\mathfrak{a}] \in \text{Cl}(\mathcal{O})$ be an ideal class. The action of $[\mathfrak{a}]$ on E is*

$$[\mathfrak{a}] \cdot E := E/E[\mathfrak{a}],$$

where $E[\mathfrak{a}]$ is the subgroup of points annihilated by $\mathfrak{a}$. This action is well-defined, transitive, and free on the set of supersingular curves with endomorphism ring $\mathcal{O}$.

Remark 2.59. The action of an ideal class $[\mathfrak{a}] \in \text{Cl}(\mathcal{O})$ on a supersingular elliptic curve E can be interpreted as a *walk in the supersingular isogeny graph*. Consider a base prime $p = \prod_i \ell_i \pm 1$. Each small prime ℓ_i dividing $p \pm 1$ corresponds to a set of edges in the graph representing separable ℓ_i -isogenies. The ideal $\mathfrak{a}$ above can be factored into prime ideals $\mathfrak{l}_i^{e_i}$ when the its norm is smooth, and its action is realized by successively applying the corresponding isogenies $\varphi_{\ell_i}^{e_i}$, we will denote the full isogenies corresponding to $\mathfrak{a}$ as $f_{\mathfrak{a}}$, and its codomain as $f_{\mathfrak{a}}(E)$. Algorithmically, computing $[\mathfrak{a}] \cdot E$ amounts to following a path in the graph, taking e_i steps along edges of small degree ℓ_i for each prime in the factorization. The isogeny at each of these steps can be computed as a combination of smooth-degree isogenies. The *commutativity of the class group action* ensures that the order of these steps does not affect the final curve (up to isomorphism), making the path well-defined.

We are now ready to present the specific CSIDH group action.

Definition 2.60 (CSIDH Group action). *Let $p \geq 5$ such that $p \equiv 3 \pmod{8}$, then $G = \text{Cl}(\mathbb{Z}[\sqrt{-p}])$ acts on $\text{Ell}_p(\mathbb{Z}[\sqrt{-p}], \pi)$, the set of supersingular elliptic curves defined over $\mathbb{F}_p$ with $\text{End}_p(E) \cong \mathbb{Z}[\sqrt{-p}]$, and π is the $\mathbb{F}_p$ -Frobenius endomorphism of E . The action is the following*

$$\begin{aligned} \text{Cl}(\mathbb{Z}[\sqrt{-p}]) \times \text{Ell}_p(\mathbb{Z}[\sqrt{-p}], \pi) &\rightarrow \text{Ell}_p(\mathbb{Z}[\sqrt{-p}], \pi) \\ ([\mathfrak{a}], E) &\mapsto E_{\mathfrak{a}} := f_{\mathfrak{a}}(E) \end{aligned}$$

This group action has been used to instantiate a non-interactive key exchange CSIDH [CLM⁺18]. We can use it to instantiate the GAIP Problem (Definition 2.33), the group action DDH prob-

lem (Definition 2.37), and the group action computational Diffie-Hellman problem (Definition 2.39). However, it is claimed that the original proposed parameters do not achieve their claimed quantum security [Pei20]. Obtaining a stronger parameter set from isogenies is an active research area [BKV19, DFK⁺23]. CSIDH can be used to instantiate the Known-Order Group Action Model, under a careful selection of parameters. Indeed, asymptotically the classical cost of computing the order of the class group is still subexponential. In this model, there exists various constructions, such as logarithmic (linkable) ring signatures [BKP20], threshold signatures [DM20], logarithmic (and tightly secure) group signatures [BDK⁺22], and compact blind signatures [KLLQ23].

2.3.4 Representing isogenies

After introducing a theoretical background on isogenies, we now discuss how to represent them efficiently for algorithmic purposes.

Definition 2.61. *An efficient representation of an isogeny $\varphi : E \rightarrow E'$ is any data of polynomial size that allows to efficiently evaluate the isogeny φ on any (accessible) point of E . We call a point accessible if it is defined over an extension of poly-logarithmic size.*

When giving $\varphi : E \rightarrow E'$ as an input or returning it as an output of an algorithm, we intend an efficient representation of φ . We list below two commonly used isogeny representations.

Kernel representation When the degree of φ is smooth and $\ker \varphi$ is accessible, then any representation of generator(s) of $\ker \varphi$ is an efficient representation of φ as one can compute/evaluate φ using Vélu's formulas [Vél71] or the square-root Vélu formulas [BDFLS20]. When φ is a composition of smooth degree isogenies, the isogeny φ can be represented by concatenating the representations of its factors. When the degree of φ is not smooth, the kernel representation becomes inefficient.

HD representation If one knows an evaluation of φ on some accessible torsion points of smooth order $N > \deg \varphi$, then as an application of Kani's lemma [Kan97], Robert [Rob22, Rob23] describes an algorithm that embeds φ into a higher dimensional (2, 4 or 8) isogeny Φ of degree N in such a way that from the evaluation of Φ on torsion points, one can retrieve the evaluation of φ on torsion points. This hence allows the *HD (higher dimensional) representation*, where one represents an isogeny φ using its evaluation on accessible torsion points of smooth order $N > 2\sqrt{\deg \varphi}$ (where the factor 2 comes from the post-composition of the HD isogenies by isomorphisms). In practice N is often chosen to be a power of 2.

2.3.5 Computing isogenies

We now discuss some existing algorithms for computing isogenies that will be of interest in Chapter 3. The main isogeny problem (Problem 2.48) is believed to be hard. Nevertheless, it may not be the case when more information is provided: the endomorphism rings of the curves and/or some torsion point information and/or the degree of the isogeny.

When the endomorphism rings of the curves are public, then a result of [GPST16] shows that the secret isogeny $\varphi : E_1 \rightarrow E_2$ can be recovered whenever it is the shortest isogeny connecting E_1 to E_2 . This result is formally given by the following theorem.

Theorem 2.62. *Let E_1 and E_2 be two supersingular curves, and let $\varphi : E_1 \rightarrow E_2$ be the shortest isogeny connecting E_1 and E_2 . Given a description of the endomorphism rings $\mathcal{O}_1 \simeq \text{End}(E_1)$ and $\mathcal{O}_2 \simeq \text{End}(E_2)$, there exists an efficient algorithm that computes the isogeny φ .*

Note that Theorem 2.62 is a straightforward generalization of [GPST16, Theorem 1] where the degree of the isogeny φ is a prime power to the case where there is no restriction on the degree of the isogeny. Two uniformly random supersingular curves are always connected by an isogeny of degree at most $O(\sqrt{p})$. Hence the attack in Theorem 2.62 does not help to recover isogenies of fixed/given degree $d \gg \sqrt{p}$. In [FKMT22], it is shown that if some reasonable amount of torsion point information is provided besides the endomorphism rings, then the secret isogeny can be efficiently recovered. More precisely, we have the following theorem which can be found in [FKMT22, Theorem 3.8].

Theorem 2.63. *Let E_1 and E_2 be two supersingular curves, and let d be the degree of the shortest isogeny connecting E_1 and E_2 . Let $\varphi : E_1 \rightarrow E_2$ be an isogeny of degree $N_1 \geq d$. Let N_2 be a smooth integer, set $E_1[N_2] = \langle P, Q \rangle$. Given $\varphi(P)$, $\varphi(Q)$ and a description the endomorphism rings $\mathcal{O}_1 \simeq \text{End}(E_1)$ and $\mathcal{O}_2 \simeq \text{End}(E_2)$, there exists an efficient algorithm that computes the isogeny φ provided that $N_1 \leq \frac{dN_2}{16}$.*

In practice, the endomorphism ring of the co-domain curve of the isogeny is not provided. In 2017, Petit [Pet17] described an attack that only requires the knowledge of a special endomorphism on the starting curve, and a large amount of torsion point information. This attack was later improved in [dQKL⁺21] but still required torsion point images of large order. In 2022, a series of three papers [CD23, MMP⁺23, Rob23] consecutively improved the state of art to reach a point where no known endomorphism is required and the amount of torsion point needed is way smaller than the degree of the isogeny: a supersingular isogeny of degree N_1 can be efficiently recovered from its action on torsion points of smooth order N_2 where $4N_1 < N_2^2$. These results led to a complete break of SIDH and are summarized in Theorem 2.64, which is based on [Rob23, Theorem 1].

Theorem 2.64. *Let E_1 and E_2 be two supersingular curves, let N_2 be a smooth integer and let $E_1[N_2] = \langle P, Q \rangle$. Let $\varphi : E_1 \rightarrow E_2$ be an isogeny of degree N_1 . Given $\varphi(P)$, $\varphi(Q)$ there exists an efficient algorithm that computes the isogeny φ provided that $4N_1 < N_2^2$.*

$$\begin{array}{ccc}
(E_0, P_a, Q_a, P_b, Q_b) & \xrightarrow{\varphi_a} & (E_a, \varphi_a(P_b), \varphi_a(Q_b)) \\
\downarrow \varphi_b & & \downarrow \varphi'_b \\
(E_b, \varphi_b(P_a), \varphi_b(Q_a)) & \xrightarrow{\varphi'_a} & E_{ab} \cong E_{ba}
\end{array}$$

Figure 2.13: The SIDH key exchange protocol

It may happen that when attempting to recover the secret isogeny, one does not directly have access to torsion point images, but only to images of some cyclic groups of the same order N . In [BKM⁺21, FP22], it is proven that if N has $O(\log \log p)$ prime factors, then one can efficiently recover the torsion point information from the images of three disjoint cyclic groups of order N . This implies that the secret isogeny can in fact be recovered from the images of three disjoint cyclic groups of order N .

In the next two subsections, we will introduce two isogeny-based protocols, SIDH and FESTA, which will be used as the basis of other primitives in Chapters 3 and 4 respectively.

2.3.6 SIDH

The Supersingular Isogeny Diffie-Hellman (SIDH) protocol was introduced in 2011 by Jao and De Feo [JD11]. It is a Diffie-Hellman type key exchange that uses supersingular isogenies. Supersingular isogenies do not commute in general. In order to get a commutative diagram that will help compute the shared secret in SIDH, the images of some torsion point basis through the secret isogenies are included in the public keys (see Figure 2.13). Moreover, in order to achieve the best possible efficiency, one uses isogenies of degree 2^a or 3^b between supersingular elliptic curves defined over $\mathbb{F}_{p^2}$ where the characteristic p is of the form $p = 2^a 3^b f - 1$, with f being a small co-factor. Primes of the form $p = 2^a 3^b f - 1$ (or $p = \ell_1^{e_1} \ell_2^{e_2} f - 1$ more generally) are usually referred to as *SIDH primes*. As mentioned before, it is now known that SIDH is completely broken [CD23, MMP⁺23, Rob23] due to the advances in isogeny reconstruction from torsion point information described in Theorem 2.64. However, we still describe the protocol here as it is the basis for an isogeny-based signature scheme considered in Chapter 3.

The detailed description of SIDH is as follows.

Setup. Let $p = 2^a 3^b f - 1$ be an SIDH prime and let E_0 be a supersingular elliptic curve defined over $\mathbb{F}_{p^2}$. Let $E_0[2^a] = \langle P_a, Q_a \rangle$ and $E_0[3^b] = \langle P_b, Q_b \rangle$.

Key generation. Alice samples a random scalar $k_a \in \mathbb{Z}/2^a\mathbb{Z}$, and computes the isogeny $\varphi_a : E_0 \rightarrow E_a$ whose kernel is $\langle P_a + [k_a]Q_a \rangle$. Her secret key is k_a and her public key is

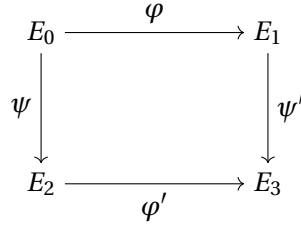


Figure 2.14: SIDH Square

$(E_a, \varphi_a(P_b), \varphi_a(Q_b))$. Similarly, Bob samples a random scalar $k_b \in \mathbb{Z}/3^b\mathbb{Z}$, and computes the isogeny $\varphi_b : E_0 \rightarrow E_B$ whose kernel is $\langle P_b + [k_b]Q_b \rangle$. His secret key is k_b and his public key is $(E_b, \varphi_b(P_a), \varphi_b(Q_a))$.

Shared secret. Given Bob's public key $(E_b, \varphi_b(P_a), \varphi_b(Q_a))$, Alice computes the isogeny $\varphi'_a : E_b \rightarrow E_{ba}$ whose kernel is generated by $\varphi_b(P_a) + [k_a]\varphi_b(Q_a)$. Given Alice's public key $(E_a, \varphi_a(P_b), \varphi_a(Q_b))$, Bob computes the isogeny $\varphi'_b : E_a \rightarrow E_{ab}$ whose kernel is generated by $\varphi_a(P_b) + [k_b]\varphi_a(Q_b)$. The shared secret is $j(E_{ab}) = j(E_{ba})$.

In SIDH, the isogenies φ_A and φ'_A (resp. φ_B and φ'_B) are said to be *parallel isogenies*. In general, two isogenies $\varphi : E_0 \rightarrow E_1$ and $\varphi' : E_2 \rightarrow E_3$ of the same degree are said to be *parallel* if there exists an isogeny $\psi : E_0 \rightarrow E_2$ such that $\ker \varphi' = \psi(\ker \varphi)$. Note that if $\varphi : E_0 \rightarrow E_1$ and $\varphi' : E_2 \rightarrow E_3$ are parallel, then $\widehat{\varphi} : E_1 \rightarrow E_0$ and $\widehat{\varphi}' : E_3 \rightarrow E_2$ are also parallel since $\ker \widehat{\varphi}' = \psi'(\ker \widehat{\varphi})$ where ψ' is the isogeny whose kernel is given by $\ker \psi' = \varphi(\ker \psi)$. We hence obtain a square (Figure 2.14) which is called an *SIDH square*.

2.3.7 FESTA

FESTA [BMP23] is an IND-CCA public-key encryption mechanism obtained by applying the OAEP transform to a trapdoor function. It was designed by constructively using the attacks against SIDH. We proceed to describe the corresponding trapdoor function below.

We define E_0 to be a supersingular elliptic curve over $\mathbb{F}_{p^2}$, where $p = 3 \pmod{4}$ and $2^a \mid (p+1)$, for some large enough $a > 0$. Additionally, we fix (P_0, Q_0) to be a basis of $E_0[2^a]$.

The key generation works as follows. We compute an isogeny $\varphi_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ and then define $(R_{\text{pk}}, S_{\text{pk}}) = ([\alpha_{\text{sk}}]\varphi_{\text{sk}}(P_0), [\beta_{\text{sk}}]\varphi_{\text{sk}}(Q_0))$, for some odd $\alpha_{\text{sk}}, \beta_{\text{sk}}$. The secret key sk is equal to $(\varphi_{\text{sk}}, \alpha_{\text{sk}}, \beta_{\text{sk}})$, whereas the public key pk consists in $(E_{\text{pk}}, R_{\text{pk}}, S_{\text{pk}})$.

In the evaluation phase, the trapdoor function takes as input two isogenies, the isogenies $\varphi_{\text{Enc}_1} : E_0 \rightarrow E_1$ and $\varphi_{\text{Enc}_2} : E_{\text{pk}} \rightarrow E_2$ such that $\deg(\varphi_{\text{sk}})$, $\deg(\varphi_{\text{Enc}_1})$ and $\deg(\varphi_{\text{Enc}_2})$ are pairwise coprime. The output is the tuple

$$(E_1, R_1, S_1, E_2, R_2, S_2),$$

where

$$(R_1, S_1) = ([\alpha_{\text{eval}}]\varphi_{\text{Enc}_1}(P_0), [\beta_{\text{eval}}]\varphi_{\text{Enc}_1}(Q_0)) \quad \text{and} \\ (R_2, S_2) = ([\alpha_{\text{eval}}]\varphi_{\text{Enc}_2}(R_{\text{pk}}), [\beta_{\text{eval}}]\varphi_{\text{Enc}_2}(S_{\text{pk}})).$$

for some odd integers $\alpha_{\text{eval}}, \beta_{\text{eval}}$.

Now, let $\varphi_{\text{Dec}} := \varphi_{\text{Enc}_2} \circ \varphi_{\text{sk}} \circ \widehat{\varphi}_{\text{Enc}_1} : E_1 \rightarrow E_2$. To invert the trapdoor function, we observe that

$$\varphi_{\text{Dec}} \begin{pmatrix} R_1 \\ S_1 \end{pmatrix} = [\deg(\varphi_{\text{Enc}_1})] \begin{pmatrix} [\alpha_{\text{sk}}^{-1}]R_2 \\ [\beta_{\text{sk}}^{-1}]S_2 \end{pmatrix}.$$

This implies that, using the secret key sk , we can recover a higher-dimensional (HD) representation of φ_{Dec} , which in turn allows us to compute φ_{Enc_1} and φ_{Enc_2} . We give a visual description of the FESTA trapdoor function in Figure 2.15.

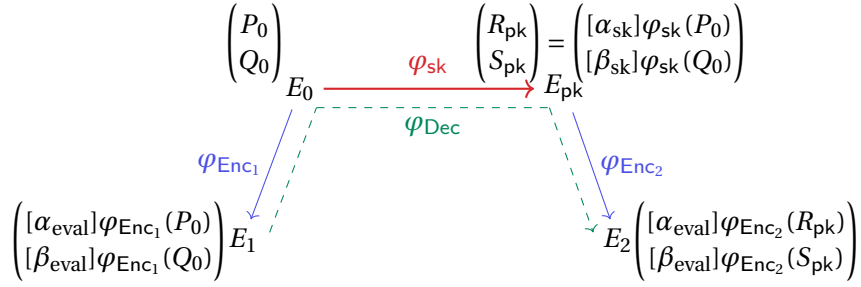


Figure 2.15: The FESTA trapdoor function.

2.4 Tensors and codes

In this section, we introduce the necessary background on tensors, codes and code-based cryptography that will be necessary for Chapters 6 and 7. We show how they can instantiate the group action framework introduced in Section 2.2, and we recall some known results on the hardness of related problems.

2.4.1 Tensors over finite fields

Tensors over finite fields allow to introduce a general framework to build group actions, and in Chapter 6, we will analyse a commitment scheme built on a specific instantiation of this framework, based on special orbits of the group action. In Chapter 7, we will rather focus on some instances of the code equivalence problem itself, and their application to blind signatures. Note that, as we will highlight later, code equivalence problems can be naturally expressed in terms of 3-tensors (or trilinear forms), as a restricted case of the tensor-based group action.

We start by recalling the definition of a 3-tensor over a finite field.

Definition 2.65 (3-tensor). *Fix a finite field $\mathbb{F}_q$ for a prime q , and ℓ, m, n positive integers. Then given bases $\{e_i\}_{i=1}^\ell, \{f_i\}_{i=1}^m, \{g_i\}_{i=1}^n$ of $\mathbb{F}_q^\ell, \mathbb{F}_q^m, \mathbb{F}_q^n$, respectively, a 3-tensor $v \in \mathbb{F}_q^\ell \otimes \mathbb{F}_q^m \otimes \mathbb{F}_q^n$ is defined as follows*

$$v = \sum_{i=1}^{\ell} \sum_{j=1}^m \sum_{k=1}^n v(i, j, k) e_i \otimes f_j \otimes g_k$$

where $\otimes$ denotes the Kronecker product. Alternatively a 3-tensor, v , can be represented as a 3-way array of field elements

$$v = [[v(i, j, k)]]_{i,j,k=1}^{\ell,m,n} \in \mathbb{F}_q^{\ell \times m \times n}.$$

3-tensors also encode trilinear forms in the following way:

Definition 2.66 (Equivalence of 3-tensors and trilinear forms). *To a given 3-tensor v , one can associate the 3-linear form $L_v : \mathbb{F}_q^\ell \times \mathbb{F}_q^m \times \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ defined as*

$$L_v(x, y, z) = \sum_{i_1=1}^{\ell} \sum_{i_2=1}^m \sum_{i_3=1}^n v(i_1, i_2, i_3) x_{i_1} y_{i_2} z_{i_3}.$$

For the rest of this work, we restrict to the case where $\ell = m = n$. We focus only on the case of 3-tensors introduced above, even if not explicitly stated, since the work from [GQ21] gives an equivalence between isomorphism problems on 3-tensors and higher dimensional d -tensors. This makes our study of 3-tensors an optimal instance of these problems.

Let $\mathbf{V}$ now denote the tensor space $\mathbf{V} = \mathbb{F}_q^n \otimes \mathbb{F}_q^n \otimes \mathbb{F}_q^n$, for which we will use the canonical basis $\{e_i\}_{i=1}^n$ on $\mathbb{F}_q^n$ to construct tensors.

Tensor rank. An essential invariant of tensors that will be considered in Chapter 6 is their *rank*.

Definition 2.67 (Rank 1 tensor). *A tensor $v \in \mathbf{V}$ is said to have rank 1 if it can be written as $v = a \otimes b \otimes c$, for $a, b, c \in \mathbb{F}_q^n$.*

Definition 2.68 (Rank of a tensor). *The rank of a tensor, $v \in \mathbf{V}$, is the minimal r such that we can write v in the form $v = \sum_{i=1}^r w_i$, where $\{w_i\}_{i=1}^r \subset \mathbf{V}$ is a set of rank 1 tensors.*

Problem 2.69 (Computational Tensor Rank Problem). *Given a tensor $v \in \mathbf{V}$, compute $\text{rank}(v)$.*

It has been shown that Problem 2.69 is NP-hard [Hås90, SS18, HL13].

It is shown in [HL13] that other tensor related problems are NP-hard. These include the decisional and computational versions of problems investigating eigenvalues, singular values, and spectral norms of tensors. These problems, however, are beyond the scope of this thesis.

Group action on tensors. We can instantiate cryptographic group actions from 3-tensors. The tensors are being acted on by the group $G = \text{GL}(n, q) \times \text{GL}(n, q) \times \text{GL}(n, q)$, where $\text{GL}(n, q)$ denotes the general linear group of degree n over $\mathbb{F}_q$, in the following way:

$$\star : G \times \mathbf{V} \rightarrow \mathbf{V},$$

$$\left((A, B, C), \sum_{i,j,k} v(i, j, k) e_i \otimes e_j \otimes e_k \right) \mapsto \sum_{i,j,k} v(i, j, k) A e_i \otimes B e_j \otimes C e_k, \quad (2.4)$$

for $v(i, j, k) \in \mathbb{F}_q$.

Lemma 2.70. *Let $(A, B, C) \in G$, and $v \in \mathbf{V}$. Then $\text{rank}((A, B, C) \star v) = \text{rank}(v)$.*

In other words, $(\star)$ preserves rank.

Proof. We can decompose $w := (A, B, C) \star v$ into a sum as shown in Equation 2.4, giving us an upper bound on its rank. Thus $\text{rank}(w) \leq \text{rank}(v)$.

Now consider the tensor $(A^{-1}, B^{-1}, C^{-1}) \star w$, which again can be written as a sum whose index is bounded above by $\text{rank}(w)$. This gives us that $\text{rank}((A^{-1}, B^{-1}, C^{-1}) \star w) \leq \text{rank}(w)$. We notice, however, that $(A^{-1}, B^{-1}, C^{-1}) \star w = v$, thus we get that $\text{rank}(v) \leq \text{rank}(w)$ and from before, $\text{rank}(w) \leq \text{rank}(v)$. Hence $\text{rank}(v) = \text{rank}(w)$. $\square$

Remark 2.71. Let us comment on the properties of this group action.

- The action is efficiently computable, as it only requires matrix multiplications and additions.
- The action is not commutative, since G is not abelian.
- The action is faithful, as the only element of G that acts trivially is the identity.
- The action is not transitive. Different orbits can be distinguished by the rank of the tensors.
- The action is not free as the stabilizer of a tensor can be non-trivial.

Several of these properties will be explored more in depth in Chapter 6.

Given this group action on tensors, we can consider two problems.

Problem 2.72 (Decisional Tensor Isomorphism Problem). *Given two tensors $v_0, v_1 \in \mathbf{V}$, decide whether there exists $(A, B, C) \in G$ such that $(A, B, C) \star v_0 = v_1$.*

Problem 2.73 (Computational Tensor Isomorphism Problem). *Given two tensors $v_0, v_1 \in \mathbf{V}$, such that $(A, B, C) \star v_0 = v_1$ for some $(A, B, C) \in G$, compute (A, B, C) .*

We can rephrase these problems in terms of *orbits*. Problem 2.72 asks to determine whether two tensors belong to the same orbit and Problem 2.73 asks to determine the isomorphism mapping two elements from the same orbit.

2.4.2 Code-equivalence problems

In Chapter 7, the problems of interest will use two main families of codes: *linear codes* and *matrix codes*.

Definition 2.74 (Linear code). *An $[n, m]_q$ linear code $\mathcal{C}$ is an m -dimensional linear subspace of $\mathbb{F}_q^n$. The elements in the code are called codewords.*

Definition 2.75 (Matrix code). *A $[m \times n, k]_q$ matrix code $\mathcal{C}$ is a k -dimensional $\mathbb{F}_q$ -linear subspace of $\mathbb{F}_q^{m \times n}$.*

Let $\text{vec}(\cdot)$ be the map that concatenates the rows of a matrix together. That is,

$$\text{vec} : \mathbb{F}_q^{m \times n} \rightarrow \mathbb{F}_q^{mn}, \quad \begin{bmatrix} -\vec{v}_1- \\ \vdots \\ -\vec{v}_m- \end{bmatrix} \mapsto \vec{v}_1 || \cdots || \vec{v}_m$$

Under this map an $[m \times n, k]$ matrix code can also be viewed as an $[mn, k]$ linear code. One can represent a code by a *generator matrix*.

Definition 2.76 (Generator matrix). *Given a $[n, m]_q$ linear code $\mathcal{C}$, a matrix $G \in \mathbb{F}_q^{m \times n}$ such that $\mathcal{C} = \{G^T v : v \in \mathbb{F}_q^m\}$ is called a generator matrix for $\mathcal{C}$.*

Code equivalence problems share a common general framework: they study the action of a group G on a certain set of codes under some equivalence relation. The modularity of the problems resides on which group is acting, which type of code is used and how the equivalence relation is specified, and we survey relevant problems in the next paragraphs. We keep our discussion brief, but a more careful description of these topics can be found in [BMPS20] for linear codes, and in [CNP⁺23b] for matrix codes.

Linear codes

The Linear Code Equivalence (LCE) problem focuses on the action of $G = \text{Mono}(n, q)$, the group of monomial matrices (products of a diagonal matrix and a permutation matrix), on the set of $[n, m]_q$ linear codes under the equivalence relation $[\mathcal{C}_0]_{\sim} = \{AG_0 : A \in \text{GL}(m, q)\}$, where G_0 is the generator matrix of $\mathcal{C}_0$. Note that this equivalence relation is trivial on the code itself, since generator matrices are always defined up to a change of basis. Hence, we say that two codes $\mathcal{C}_0, \mathcal{C}_1$ are equivalent under the action of monomial matrices, if $[C_1]_{\sim} = [C_0 M]_{\sim}$ for some $M \in \text{Mono}(n, q)$.

Note that we can decompose a matrix $M \in \text{Mono}(n, q)$ as $M = DP$ for $D \in D(n, q), P \in S_n$. This decomposition will allow us to better highlight the links between the different code equivalence problems. We formalize the LCE problem below.

Problem 2.77 (Linear Code Equivalence (LCE)). *Let $\mathcal{C}_0, \mathcal{C}_1$ be two linear $[n, m]_q$ codes with generator matrices $G_0, G_1 \in \mathbb{F}_q^{m \times n}$ respectively. Decide if there exists $A \in \text{GL}(m, q), D \in D(n, q), P \in S_n$ such that $G_1 = AG_0DP$. If such matrices exist, compute some choice of A, D, P .*

Remark 2.78. Note that when D is fixed to be the identity matrix I_n , the induced action reduces to the permutation group S_n acting on the set of equivalence classes of linear codes. This corresponds to the *Permutation Code Equivalence* (PCE) Problem, which lies outside the scope of this work. However, solving PCE trivially gives a solution to LCE.

Remark 2.79. The Linear Code Equivalence problem can be expressed as an instance of tensor isomorphism. For a generator matrix $G \in \mathbb{F}_q^{m \times n}$, define the order-3 tensor $T_G \in \mathbb{F}_q^{m \times n \times 1}$ by

$$T_G(i, j, 1) = G_{ij}.$$

Then, for $G_0, G_1 \in \mathbb{F}_q^{m \times n}$, the following are equivalent:

1. $G_1 = AG_0M$ for some $A \in \text{GL}(m, q)$ and some monomial matrix $M \in D(n, q) \rtimes S_n$;
2. $T_{G_1} = (A, M^T, 1) \star T_{G_0}$.

Hence, Linear Code Equivalence is precisely the orbit problem for the tensors $\{T_G\}$ under the subgroup

$$\text{GL}(m, q) \times (D(n, q) \rtimes S_n) \times \text{GL}(1, q) \subseteq \text{GL}(m, q) \times \text{GL}(n, q) \times \text{GL}(1, q).$$

(Note: The second factor remains $D(n, q) \rtimes S_n$ because the set of monomial matrices is closed under transposition.)

Solving LCE is believed to be hard for both classical and quantum adversaries under the appropriate parameter choices [SS13, BBPS23]. As a result, LCE has been used to construct the digital signature LESS [BMPS20]. Moreover, several variants of LCE have also appeared in the literature to accommodate the specific requirements of advanced cryptographic protocols.

The 2-LCE and *Inverse Linear Code Equivalence* (ILCE) problem are two notable examples. The 2-LCE was first introduced in [BBMP24] in the context of threshold signatures, where the adversary is given two LCE instances sharing the same secret monomial M .

Problem 2.80 (2-LCE). *Let $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ be four $[n, m]_q$ linear codes with generator matrices $G_0, G_1, G_2, G_3 \in \mathbb{F}_q^{m \times n}$ such that $G_1 = AG_0M$ and $G_3 = A'G_2M$ for some matrices $A, A' \in \text{GL}(m, q)$ and $M \in \text{Mono}(n, q)$. The 2-LCE problem asks to compute such M .*

The Inverse Linear Code Equivalence problem can be viewed as a special case of 2-LCE where the two samples are even more strongly related. ILCE was first introduced in [BBN⁺22] in the context of ring signatures.

Problem 2.81 (Inverse Linear Code Equivalence (ILCE)). *Let $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$ be three linear $[n, m]_q$ codes with generator matrices $G_0, G_1, G_2 \in \mathbb{F}_q^{m \times n}$ such that $G_1 = AG_0M, G_2 = A^{-1}G_0M^{-1}$ for some matrices $A \in \text{GL}(m, q)$ and $M \in \text{Mono}(n, q)$. The ILCE problem asks to compute such M .*

The ILCE problem, as well as the 2-LCE, have been shown by [BCD⁺24] to be solvable in polynomial time when $m = n/2$. The *Diagonal-masked Inverse Linear Code Equivalence* problem was introduced in Versions 1 and 2 of [DKQ⁺25] in the context of blind signatures. This problem corresponds to a different group action as well as a different equivalence relation for linear codes. Namely, given a linear code $\mathcal{C}$ represented by its generator matrix G , then the equivalence class of G is now defined as $[G]_D := \{AGD : A \in \text{GL}(m, q), D \in D(n, q)\}$, and we study the action of the permutation group S_n on the set of linear codes under this new equivalence relation.

Problem 2.82 (Diagonal-masked Inverse Linear Code Equivalence (DmILCE)). *Given $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$ three linear $[n, m]_q$ codes with generator matrices $G_0, G_1, G_2 \in \mathbb{F}_q^{m \times n}$, and $A_0, A_1 \in \text{GL}(m, q), D_0, D_1 \in D(n, q), P \in S_n$ such that $G_1 = A_0G_0D_0P, G_2 = A_1G_1D_1P^{-1}$. The DmILCE problem asks to compute such P .*

In Section 7.1 we will show that DmILCE, as defined above, is actually polynomial time equivalent to LCE.

Matrix codes

The central hard problem associated with matrix codes is to determine whether two matrix codes are equivalent under the action of a specified group with respect to a given equivalence relation. In this case, we consider the action of $\text{GL}(m, q) \times \text{GL}(n, q)$ on the set of matrix codes under the standard notion of linear equivalence, although the involved parameters differ from the linear code setting. This setting is formalized by the *Matrix Code Equivalence* problem defined below.

In what follows, $A^T \otimes B^T$ denotes the Kronecker (tensor) product of the matrices $A^T \in \mathbb{F}_q^{m \times m}, B^T \in \mathbb{F}_q^{n \times n}$ which is itself a matrix in $\mathbb{F}_q^{mn \times mn}$.

Problem 2.83 (Matrix Code Equivalence (MCE)). *Given two matrix codes, $\mathcal{C}_0, \mathcal{C}_1$, where G_0 is the generator matrix of $\mathcal{C}_0$ and G_1 that of $\mathcal{C}_1$, determine if there exists $A \in \text{GL}(m, q), B \in \text{GL}(n, q)$ such that $\mathcal{C}_1 = A\mathcal{C}_0B^T$. In other words, such that $G_1 = SG_0(A^T \otimes B^T)$ for some matrix $S \in \text{GL}(k, q)$. If so, compute A, B .¹*

¹A reader familiar with the MCE problem might expect the notation $G_1 = SG_0(A \otimes B)$, we follow instead the exposition of [KLP25] which is equivalent up to renaming of the matrices.

This problem is at the basis of the signature scheme MEDS [CNP⁺23b].

Note that, a $[m \times n, k]$ matrix code $\mathcal{C}$ can be viewed as a k -dimensional subspace of $\mathbb{F}_q^{m \times n}$, generated by a basis $(C_1, \dots, C_k)$, where each $C_s \in \mathbb{F}_q^{m \times n}$. Let $\mathcal{C}_{ij}^{(s)}$ denote the (i, j) -entry of C_s . Then C naturally induces a trilinear form $\mathcal{C}(x, y, z) = \sum_{i,j,s=1}^{m,n,k} C_{ij}^{(s)} x_i y_j z_s$, where $x \in \mathbb{F}_q^m, y \in \mathbb{F}_q^n, z \in \mathbb{F}_q^k$. Intuitively, this form encodes the interaction between the row space, column space, and basis index of the code. Under this representation, two matrix codes $\mathcal{C}_0$ and $\mathcal{C}_1$ are equivalent if and only if their associated trilinear forms satisfy

$$\mathcal{C}_1(x, y, z) = \mathcal{C}_0(Ax, By, Sz)$$

for A, B, S as above.

This formulation of the MCE problem in terms of trilinear forms allows us to make the connection with the tensor isomorphism problem, and its trilinear form representation as highlighted in 2.66.

The *Inverse Matrix Code Equivalence Problem* (IMCE), also introduced in [CNP⁺23b], is the inverse problem that corresponds to MCE.

Problem 2.84 (Inverse Matrix Code Equivalence (IMCE)). *Given three matrix codes, $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$, such that $\mathcal{C}_1 = A\mathcal{C}_0B^T$ and $\mathcal{C}_2 = A^{-1}\mathcal{C}_0B^{-T}$, for $A \in \text{GL}(m, q), B \in \text{GL}(n, q)$, find A, B up to equivalence.*

In [BCD⁺24], Budroni et al. give an upper bound on the number of samples (i.e. fresh instances of the problem that share the same secret) IMCE (and MCE) can publish using the same secret. We will summarize this result later on in Section 7.3. The single sample version of IMCE remains unaffected by their attack.

The *Modified Inverse Matrix Code Equivalence Problem* (MIMCE) focuses on (anti)symmetric matrices and was introduced in [KLP25] to construct a post-quantum blind signature.

Problem 2.85 (Modified Inverse Matrix Code Equivalence (MIMCE)). *Given three matrix codes, $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$, such that $\mathcal{C}_1 = A\mathcal{C}_0B^T$ and $\mathcal{C}_2 = A^{-1}\mathcal{C}_0B^{-T}$, for $A \in \text{GL}(m, q), B \in \text{GL}(n, q)$ (anti)symmetric; find D, F (anti)symmetric such that $\mathcal{C}_2 = D\mathcal{C}_1F^T$.*

This problem looks very similar to IMCE under the condition that A, B are (anti)symmetric. The outputs, however, are slightly different. We formalize IMCE with (anti)symmetric keys in Problem 2.86.

Problem 2.86 (symIMCE). *Given three matrix codes, $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$, such that $\mathcal{C}_1 = A\mathcal{C}_0B^T$ and $\mathcal{C}_2 = A^{-1}\mathcal{C}_0B^{-T}$, for $A \in \text{GL}(m, q), B \in \text{GL}(n, q)$ (anti)symmetric, find A, B up to equivalence.*

It was pointed out in [KLP25, Remark 2.21] that MIMCE and symIMCE should be equivalent, assuming the codes have a trivial automorphism group, i.e. are *rigid*.

Definition 2.87. A code $\mathcal{C}$ with automorphism group $\text{Aut}(\mathcal{C}) = \{\alpha I_m : \alpha \in \mathbb{F}_q\} \times \{\beta I_n : \beta \in \mathbb{F}_q\}$ is called rigid.

In the following lemma, we highlight the relationship between the symIMCE and MIMCE problems. This result will be relevant to our security analysis in Section 7.3.

Lemma 2.88. MIMCE reduces to symIMCE in polynomial time.

Proof. Consider an adversary $\mathcal{A}$ against MIMCE that is given a MIMCE instance $(\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2)$, then this also forms a symIMCE instance. $\mathcal{A}$ calls the symIMCE solver on $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$ to recover some matrices $\hat{A}, \hat{B}$. It then defines $D = \hat{A}^{-2}, F = \hat{B}^{-2}$ (which are symmetric since $\hat{A}, \hat{B}$ are) and returns D, F . Suppose that the symIMCE solver returns a correct output $\hat{A}, \hat{B}$, i.e. we have $\mathcal{C}_1 = \hat{A}\mathcal{C}_0\hat{B}$ and $\mathcal{C}_2 = \hat{A}^{-1}\mathcal{C}_2\hat{B}^{-1}$. Then $D\mathcal{C}_1F^T = D\mathcal{C}_1F = \hat{A}^{-2}\mathcal{C}_1\hat{B}^{-2} = \hat{A}^{-1}\mathcal{C}_0\hat{B}^{-1} = \mathcal{C}_2$, hence D, F is a correct solution to the MIMCE problem. $\square$

Part I - Isogenies

Isogeny-based cryptography is based on the problem of computing an isogeny of a given degree between two elliptic curves. Similarly to classical elliptic curve cryptography, the main attractive feature of isogeny-based cryptography is the very compact size of the primitives you can build from it, especially when compared to other post-quantum proposals. Considering the widespread adoption of classical elliptic curve cryptography in today's public-key infrastructure, it is crucial to explore whether isogeny-based cryptography can achieve comparable efficiency and confidence in the hardness of its foundational problems.

The first instance of isogeny based cryptography is due to Couveignes in 1997, and was rediscovered independently by Rostovstev and Stolbunov in 2004. However, this proposal based on ordinary elliptic curves, is far from being efficient and it is also subject to a sub-exponential quantum attack. To address these issues, De Feo and Jao designed in 2011 the SIDH (Supersingular Isogeny Diffie-Hellman) protocol [JD11] which was submitted as SIKE ([DJ20]) in the NIST post-quantum standardization process. For a long time, SIDH/SIKE was the golden child of isogeny based cryptography and even made it as an alternate candidate to the 4th round of the NIST standardization project. ^{II}.

However, in the summer of 2022, a year into this thesis, an earthquake shook isogeny-based cryptography to its foundations. A series of paper, by Castryck and Décrú [CD23], Maino, Martindale, Panny, Pope and Wesolowski [MMP⁺23] as well as Robert [Rob23] gave polynomial time algorithms that completely broke SIDH for a wide range of parameters, including those of SIKE. We will refer to these as the SIDH attacks. Luckily for the field, and this author, the SIDH problem was only a variant of the pure isogeny problem, and despite their devastating impact, the SIDH attacks did not put an end to isogeny-based cryptography. If anything, they gave it a second youth: they were so efficient that they were soon turned into a powerful constructive tool and enabled a plethora of new isogeny-based protocols, targeting mainly public-key encryption [BMP23, BF23, NO24, BM25] and digital signatures [DLRW24, BDD⁺24, NOC⁺24, NO25, XLZO25, BBC⁺25, Fou25].

The first part of this thesis looks into three complementary aspects of isogeny-based cryptography, and explores in each case the design of specific cryptographic primitives.

^{II}<https://csrc.nist.gov/Projects/post-quantum-cryptography/round-4-submissions>

In Chapter 3, we investigate minimal assumptions under which we can build secure isogeny protocols and show that, perhaps surprisingly, under a careful choice of parameters SIDH-based signatures are a candidate. More precisely, we look into the problem of building isogeny based protocols where the endomorphism rings of all curves are public. SIDH attacks were so devastating because they exploited the additional torsion information provided along the public elliptic curves. Giving away the endomorphism rings of all curves gives even more information to the attacker, and it is an interesting problem to see which protocols can be built in this setting. On the one hand, it shows that even if one were to be able to efficiently compute endomorphism rings of supersingular elliptic curves (a problem that is so far believed to be hard), isogeny-based cryptography would still be possible. On the other hand, it allows for more compact protocols since the requirement of using a large prime to address endomorphism-recovering attacks can be sidestepped. We explore the paradigm offered by digital signatures, and show through an extensive study of various attack vectors, including the SIDH attacks, and under a careful choice of parameters, that it is possible to build secure signatures from SIDH squares in this setting.

In Chapter 4, we turn our attention to the constructive aspects of the SIDH attacks, and show how they can be used to efficiently build an updatable public-key encryption (UPKE) scheme. UPKEs are an enhanced version of public-key encryption mostly used in the setting of secure messaging, that allows for asynchronous key updates. It is known to be difficult to achieve efficient updatable public-key encryption schemes with unbounded amount of key updates from lattice-based assumptions. Our isogeny-based construction fills this gap and highlights the role that isogenies have to play in the design of advanced protocols. More precisely, we build our scheme upon a four-dimensional version of FESTA with uniform public-keys, this means a public key is a uniformly random supersingular elliptic curves and the secret key is an isogeny from some starting curve to the public key. To update a public key we randomize it through a long-enough isogeny walk. To compute the corresponding secret key we describe a method that “shortens” an isogeny walk and avoids having the degree of the secret isogeny growing with the number of updates.

Finally, another vast subfield of isogeny-based cryptography is isogeny-based group action, of which CSIDH [CLM⁺18] is the figurehead. It suddenly gained attractivity again after the SIDH attacks. Isogeny-based group actions were already known to build non-interactive key exchange [CLM⁺18] as well as (moderately efficient) digital signatures [BKV19, DG19]. Chapter 5 aims to explore the feasibility of (isogeny-based) group actions to build generic zero-knowledge proofs. We give a general framework to build non-interactive zero-knowledge proofs from known-order effective group actions. We do so by formalizing a relaxed version of an homomorphic commitment which we call a malleable commitment. Using these, we show how to build basic proofs for addition and multiplication gates, and we show how to use it to build proofs for arithmetic circuits and rank-1 constraint systems. We also consider a different representation of computations called matrix branching programs and build a proof system for them as well.

3 Exploring SIDH-based Signature Parameters

This chapter is based on [BCF⁺24], published at ACNS 2024.

At the core of isogeny-based cryptography, lies the fundamental assumption that it is computationally hard to find an isogeny between two isogenous elliptic curves. When the curves are supersingular, the setting of nearly all modern constructions [CLG09, JD11, CLM⁺18, DMPS19, GPS20, DKL⁺20, DDF⁺21, FMP23, BMP23], the general isogeny problem (Problem 2.48) is equivalent to the endomorphism ring problem (Problem 2.47). The former asks to compute an isogeny between two supersingular elliptic curves, while the latter asks to find a basis of the ring of all the endomorphisms of a supersingular elliptic curve, i.e. all the isogenies from the curve to itself. The problem of finding an isogeny between two elliptic curves reduces to the endomorphism-ring problem because given two curves and a representation of their endomorphism rings, it is possible to compute an isogeny connecting them in polynomial time [KLPT14, Wes22b]. The actual unconditional equivalence has been shown recently Herlédan le Merdy and Wesolowski [MW25].

Due to this connection, the endomorphism ring problem and its relationship to the security of many isogeny-based protocols have been extensively studied. The best known algorithm to compute endomorphism rings is due to Eisenträger, Hallgren, Leonardi, Morrison, and Park [EHL⁺20], and it runs in $\tilde{O}(p^{1/2})$ time, where p is the characteristic of the underlying finite field (the result relies on some heuristics which were removed in [FIK⁺23]). Given a curve E_0 with known endomorphism ring $\text{End}(E_0)$, and an isogeny $\varphi : E_0 \rightarrow E$, one can push the endomorphism ring $\text{End}(E_0)$ of E_0 through φ to recover $\text{End}(E)$ [KLPT14, Wes22b]. Thus, finding an isogeny between a curve E_0 with known endomorphism ring, and a given curve E solves the endomorphism ring problem for the curve E . Since the characteristic p is exponential in the security parameter in practice, the general endomorphism ring problem remains hard.

It remains of interest to understand how the security of isogeny-based protocols is affected when an attacker has knowledge of the endomorphism rings. In several protocols, such as

the GPS signature [GPS17], SÉTA [DDF⁺21], SQISign [DKL⁺20] and SQISignHD [DLRW24], the secret keys are directly linked to a description of the endomorphism ring. Thus, solving the endomorphism ring problem trivially breaks such protocols. In other schemes, such as SIDH [JD11], CSIDH [CLM⁺18] and SCALLOP [DFK⁺23], CSI-FiSh–[BKV19], SeaSign [DG19], the secret isogenies have specific properties: if the endomorphisms of all curves were known, a direct application of [KLPT14, Wes22b] would prevent obtaining the correct isogeny. Nonetheless, it has been shown that the additional information that such protocols reveal, such as short degrees, torsion images or orientations, is sufficient to recover the secret isogeny [GPST16, CPV20, Wes22a, FKMT22]. More recently proposed schemes, such as M-SIDH/MD-SIDH [FMP23], FESTA [BMP23] and binSIDH/terSIDH [BF23] compute isogenies of degree roughly $\sqrt{p}$ or even smaller, hence the attack in [GPST16] trivially extends to those cases when endomorphism rings of curves are public. Moreover, many other protocols are insecure when the starting curves have known endomorphism ring. This is the case, for instance, for the CGL hash function [CLG09, EHL⁺18], the CSIDH-based oblivious transfer protocols [LGD21, BMM⁺23], the commitment scheme by Sterner [Ste22], the SIDH-based oblivious pseudorandom functions [BKW20, BKM⁺21, Bas24], and the hash proof systems and dual-mode PKEs based on group actions [ADMP20].

The relevance of endomorphism rings in isogeny-based cryptography and the consequences of their knowledge on security raises the following natural question:

*Can we construct a secure cryptographic protocol
where the endomorphism rings of all curves are public?*

One has to remark that one of the most natural algorithmic problems, namely finding an isogeny of a fixed degree d between supersingular elliptic curves, is not known to be equivalent to the endomorphism ring problem. An efficient classical equivalence between finding fixed degree isogenies and computing endomorphism rings would have important consequences, e.g., a significant speed-up of SQISign. Understanding whether we can build protocols which are secure even if endomorphism rings of all curves is public has both theoretical and practical consequences. On the theoretical side, a protocol that remains secure when the endomorphism rings of all curves are known shows that, even if the endomorphism ring problem is efficiently solvable, some isogeny-based constructions are still possible, and retain some security. On the practical side, the complexity of the endomorphism-recovering attacks generally imposes primes p with $p > 2^{2\lambda}$; without requiring endomorphism rings to remain secret, it is possible to design protocols with smaller primes, leading to more efficient and more compact protocols.

Contributions.

In this paper, we develop two protocols that appear to be secure, even if the endomorphism rings of all elliptic curves are public. This suggests an affirmative response to the question set out in the introduction (even though further cryptanalysis is necessary).

Both protocols are digital signatures, based on a proof of isogeny knowledge built on top of SIDH squares. In this work, we focus on digital signatures since it is the primitive that is most likely to be secure when endomorphism rings are known: the SIDH-based constructions generally reveal little information besides the degrees and the end curves of secret isogenies. Indeed, it is possible to construct SIDH-based signatures that do not reveal any torsion information [DDGZ22] or that are statistically independent from the secret key [BCC⁺23].

In this work, we analyze existing constructions of proofs of isogeny knowledge and identify three main design choices (Section 3.1). We also propose two practical instantiations, which are plausibly secure despite the underlying prime field having characteristic smaller than $2^{2\lambda}$.

To analyze the security of the proposed constructions, we identify and study three main lines of attacks. The first approach relies on the knowledge of endomorphism rings and the KLPT algorithm [KLPT14]. We analyze these attacks extensively in Section 3.2. Moreover, the KLPT algorithm has always been considered for constructive applications, and thus its analysis in the literature is bounded to polynomial running times. In this work, we study the output of the KLPT algorithm when running in superpolynomial time, which may be of independent interest. The method used is a variation of [Mer23, Section 3.4.].

We also consider attacks based on the recent attacks on SIDH (Section 3.3.1) and based on the lack of zero-knowledge of the underlying sigma protocol (Section 3.3.2). The results of these analyses shows that it is possible to design signatures based on proofs of isogeny knowledge with binary challenges, which are more efficient and compact than those based on proofs with ternary challenges (see Section 3.4 for an estimate of the concrete sizes). Combined with the previous analysis of KLPT-based attacks, this provides an argument for the security of the proposed constructions.

3.1 Signatures based on SIDH squares

Let us start by recalling various constructions of proofs of isogeny knowledge from the literature and highlight the main design options. We also introduce the potential lines of attack against some constructions, which are analyzed in detail in the following sections. For a comprehensive survey of proofs of isogeny, we refer the reader to [BDGP23], [GPV24].

Let us assume that a prover wants to demonstrate knowledge of a cyclic isogeny $\varphi : E_0 \rightarrow E_1$ of smooth degree d . The main framework, on which the following proofs are based on, is a sigma protocol due to De Feo, Jao, and Plût [FJP14]. The prover generates the SIDH square in Figure 2.14 where ψ has degree ℓ^n for some ℓ coprime with d , and they commit to E_2 and E_3 . The verifier sends a challenge bit $c \in \{0, 1\}$: if $c = 0$, the prover responds with the horizontal isogeny φ' , and if $c = 1$, the prover reveals the vertical isogenies ψ and ψ' . The verifier accepts if the response isogenies have the correct domain and codomain. The protocol has soundness error of $1/2$, and thus it needs to be repeated λ times to obtain a negligible soundness error of $2^{-\lambda}$. As pointed out in [GKPV21, DDGZ22], a malicious prover may not necessarily know φ

(such an isogeny might not exist at all); the proof in [FJP14] is thus sound with respect to the weaker relation

$$\mathcal{R}_{\text{weak}} = \left\{ ((E_0, E_1), \varphi) \mid \begin{array}{l} \varphi : E_0 \rightarrow E_1 \text{ is a cyclic } \ell^{2i} d\text{-isogeny,} \\ \text{for some integer } i \text{ and } \ell \text{ coprime with } d \end{array} \right\}. \quad (3.1)$$

In the case of an honest prover, this proof also reveals the action of φ on the torsion $E[\deg \psi]$ since the isogenies ψ and ψ' are parallel. This makes it potentially vulnerable to the recent attacks on SIDH [CD23, MMP⁺23, Rob23].

The authors of [DDGZ22] showed that it is possible to have a proof that is sound with respect to the strong relation

$$\mathcal{R}_{\text{strg}} = \{((E_0, E_1), \varphi) \mid \varphi : E_0 \rightarrow E_1 \text{ is a cyclic } d\text{-isogeny}\} \quad (3.2)$$

by ensuring that ψ and ψ' are parallel. However, to avoid the SIDH attacks and a technical issue with zero-knowledge^I, they have to resort to a proof with ternary challenges. Thus, to prove parallelness, the prover constructs the same SIDH square as in Figure 2.14, but additionally commits to P_2, Q_2 ^{II}, a basis of $E_2[d]$, its image $P_3 := \varphi'(P_2), Q_3 := \varphi'(Q_2)$ on E_3 , and the coefficients a, b such that $\ker \hat{\psi} = \langle [a]P_2 + [b]Q_2 \rangle$ and $\ker \hat{\psi}' = \langle [a]P_3 + [b]Q_3 \rangle$. The curves E_2 and E_3 are also committed with a hiding commitment scheme. Then, the challenges are ternary, i.e. $c \in \{-1, 0, 1\}$. When $c = \pm 1$, the verifier reveals a, b and either $(E_2, (P_2, Q_2))$ or $(E_3, (P_3, Q_3))$; the verifier reconstructs ψ or ψ' and ensures they have the correct codomain. In the case of $c = 0$, the verifier receives $(\varphi', (E_2, P_2, Q_2), (E_3, P_3, Q_3))$ and checks that the points P_3, Q_3 are the images of P_2, Q_2 under φ' . In all cases, the verifier also ensures the revealed values match the previously committed ones.

More recently, [BCC⁺23] introduced the concept of an *SIDH ladder*, which is obtained by gluing multiple SIDH squares together. This removes the requirement on the prime being an SIDH prime. It is thus possible to prove knowledge of an isogeny φ in *any* characteristic and even if φ and ψ have kernels that are not defined over $\mathbb{F}_{p^2}$.

This historical overview suggests there are three main design choices that determine how a proof of isogeny knowledge works:

1. The soundness relation: strong vs weak,
2. The challenge space: binary vs ternary,
3. The characteristic p and the degrees of φ and ψ .

Note that not all combinations are possible. For instance, when the kernels of ψ and ψ' are

^IThe Σ protocol with binary challenges does not satisfy the common definitions of zero-knowledge. This, however, does not constitute a problem when it is transformed into a signature scheme, as shown in Section 3.3.2.

^{II}This basis can be generated canonically, which avoids the need of its commitment.

not rational over $\mathbb{F}_{p^2}$, which requires using the SIDH ladder method proposed by [BCC⁺23], there is no known technique to prove the strong relation.

Proposed constructions

We now discuss some promising combinations and study their securities in later sections. Let E_0 be a random supersingular elliptic curve, ℓ_1 be a small prime and e_1 be a positive integer. Our goal is to prove the knowledge of a secret isogeny $\varphi : E_0 \rightarrow E_1$ of degree $\ell_1^{e_1}$ in $\mathcal{R}_{\text{strg}}$.

Variant 1.

This variant proves the knowledge of the **strong** relation, and uses a **binary** challenge space. Let public parameters $pp = (p, \ell_1, \ell_2, e_1, e_2, E_0)$ be such that $\#E_0(\mathbb{F}_{p^2}) = (\ell_1 \ell_2^{e_2} f)^2$, for $\ell_2^{e_2}$ of roughly the same size as in SIDH ([ACC⁺19]) and $d = \ell_1^{e_1} \gg \ell_2^{e_2}$, or $d = \ell_1^{e_1} \approx 2^\lambda (\ell_2^{e_2})^2$. Note that in this setting, $p = \ell_1 \ell_2^{e_2} f - 1 \approx \ell_2^{e_2}$ and thus it is smaller than $2^{2\lambda}$. Note also that E_0 is not a special curve and can in fact be generated by taking a long enough walk from $j = 1728$. Intuitively, the protocol is as described earlier in this section and the rigorous version is given in [DDGZ22, Figure 2]. The only difference is, the horizontal isogeny φ' is represented by a sequence of isogenies of degree ℓ_1 instead of a kernel point of order $\ell_1^{e_1}$. As noted in [DDGZ22], this sigma protocol has 2-special soundness, but does not satisfy the zero-knowledge (ZK) property if the distinguisher used in the ZK definition has access to the witness. We explore in Section 3.3.2 how we can still retain the security of the derived signature.

Variant 2.

This variant also proves the knowledge of the **strong** relation, but uses a **ternary** challenge space. The requirements on the public parameters are similar to Variant 1, with $p \approx \ell_1^{e_1} \ell_2^{e_2}$ as in SIDH, so $p \approx 2^{216}$ for $\lambda = 128$. The description of the protocol is as given in [DDGZ22, Figure 3]. Note again that we represent φ' by a sequence of isogenies of degree ℓ_1 . This sigma protocol has 3-special soundness (Remark 2.13) and zero-knowledge. Note that in this variant, E_0 does not need to be a random supersingular elliptic curve, and rather can be taken to be a special curve with an extremal order as endomorphism ring.

3.2 Analysis of KLPT-based attacks

In this section, we analyze KLPT-based attacks that break the security of SIDH-based signatures and proofs of knowledge. The attacks follow two main approaches: they can either recover the secret key from the public information, or they can forge a valid signature even if they fail to recover the secret key.

In the first approach, the attacker recovers a d -isogeny $\varphi : E_0 \rightarrow E_1$, given the domain and

codomain curves E_0, E_1 and their endomorphism rings.

This problem is linked to the problem of finding an ideal of norm d , connecting the maximal orders $\mathcal{O}_0 \cong \text{End}(E_0)$ and $\mathcal{O}_1 \cong \text{End}(E_1)$ through the computational Deuring correspondence [GPS17, DLLW23]. In [KLPT14, GPS17], the authors propose polynomial algorithms to find such ideal of smooth norm. A strategy to find a witness for the above relations could then be as follows:

1. Find an ideal I connecting $\mathcal{O}_0$ to $\mathcal{O}_1$;
2. Use the KLPT algorithm to compute an ideal J of norm d equivalent to I ;
3. Use the computational Deuring correspondence to compute an isogeny corresponding to J .

The KLPT algorithm produces ideals of norm in $O(3 \log(p))$ if either $\mathcal{O}_0$ or $\mathcal{O}_1$ is a special extremal order [PS18, KLPT14, GPS17], and $O(\frac{9}{2} \log(p))$ in the general case [PS18, DKL⁺20]. Hence, this strategy potentially fails for some $d \leq p^3$ or $d \leq p^{\frac{9}{2}}$, depending on the curves E_0 and E_1 .

A second attack strategy sidesteps these limitations and can possibly break the security of the protocol even when the secret isogeny is shorter than p^3 . When the underlying sigma protocol is sound with respect to the weak relation $\mathcal{R}_{\text{weak}}$, the prover demonstrates knowledge of an isogeny between E_0 and E_1 of degree $\ell^{2i}d$, for some integer i and ℓ coprime with d . An attacker can thus attempt to forge a proof, even without knowing the witness, by using the KLPT-based approach described above to compute an isogeny of degree $\ell^{2i}d$. Such an isogeny can then be written as the composition $\hat{\psi}' \circ \varphi' \circ \psi$, where φ' is a d isogeny and ψ, ψ' have degree ℓ^i ; the attacker can then correctly reply to any challenge. This attack can be avoided if the composition isogeny is shorter than the shortest isogeny returned by KLPT, or if the Sigma protocol is sound with respect to $\mathcal{R}_{\text{strg}}$: in that case, this approach would fail to produce isogenies $\hat{\psi}'$ and ψ that are parallel, because the isogenies φ' and parallel isogenies $\hat{\psi}'$ and ψ uniquely determine φ , which is too short to be determined by a KLPT-based attack.

In this section, we analyze the minimal norm of the ideal that KLPT can return. We extend the previous results by studying exponential-time algorithms and showing that there is a trade-off between KLPT's running time and the norm of the smallest ideal it can produce.

3.2.1 The KLPT algorithm for extremal order

We recall the following lemma from [KLPT14].

Lemma 3.1 ([KLPT14]). *Let I be a left $\mathcal{O}$ -ideal and $\alpha \in I$. Then $I \frac{\bar{\alpha}}{\text{Nrd}(I)}$ is a left $\mathcal{O}$ -ideal of norm $\frac{\text{Nrd}(\alpha)}{\text{Nrd}(I)}$.*

As consequence of this lemma, finding an equivalent $\mathcal{O}$ -ideal of I which has a norm in a certain set $\mathcal{N}$ consists of finding an element in I of norm $NNrd(I)$, for some $N \in \mathcal{N}$. Let $\mathcal{O}$ be one of the special extremal maximal orders given in 2.55; I an $\mathcal{O}$ -left ideal and ℓ a small prime. KLPT algorithm can be summarized as follows:

1. Compute an ideal J of prime norm equivalent to I , such that ℓ is a quadratic non-residue modulo N ;
2. Find an element $\gamma \in \mathcal{O}$ of norm $N\ell^{e_1}$ for some $e_1 \in \mathbb{N}$;
3. Find an element $\alpha \in J$ such that $J = \mathcal{O}\alpha + N\mathcal{O}$;
4. Compute $\mu_0 \in Rj$ such that $\gamma\mu_0 \equiv \alpha \pmod{N\mathcal{O}}$;
5. Compute $\lambda \in \mathbb{Z}/N\mathbb{Z}^*$ and $\mu_1 \in \mathcal{O}$ such that $\mu = \lambda\mu_0 + N\mu_1$ has norm ℓ^{e_2} , for some $e_2 \in \mathbb{N}$;
6. Return $J\frac{\bar{\beta}}{N}$, where $\beta = \gamma\mu$.

In Step 1, one computes a reduced basis of I and generates a small set of short elements until an element of norm $NNrd(I)$ is found for which N is prime. Experimentally, this algorithm returns an ideal of prime norm N , where $N \simeq \sqrt{p}$.

In Step 2, one solves the norm equation

$$f(x_1, y_1) = N\ell^{e_1} - pf(x_2, y_2). \quad (3.3)$$

In Step 3, it is enough to find an element $\alpha \in J$ such that $\gcd(N^2, \text{Nrd}(\alpha)) = N$. For such α , we have $J = \mathcal{O}\alpha + N\mathcal{O}$ i.e $J/N\mathcal{O} = \mathcal{O}\alpha/N\mathcal{O}$.

The idea of Step 4 is that $\mathcal{O}/N\mathcal{O}$ is isomorphic to $M_2(\mathbb{Z}/N\mathbb{Z})$ (an explicit isomorphism can be computed using [Voi20, Proposition 7.6.2]) and thus every left ideal only differs by a quaternion whose reduced norm is coprime to N and such an element can actually be chosen from Rj . Step 5 consists of finding $\mu \equiv \lambda\mu_0 \pmod{N\mathcal{O}}$ of norm ℓ^{e_2} . One has to look for $\mu_1 = x + y\omega + (z + t\omega)j \in R + Rj$ and $\lambda \in \mathbb{Z}/N\mathbb{Z}$ such that $\text{Nrd}(\mu) = \ell^{e_2}$ where $\mu = \lambda\mu_0 + N\mu_1$. For such μ , we have

$$\mu = N(x + y\omega) + [Nz + \lambda C + (Nt + \lambda D)\omega]j.$$

Hence $\text{Nrd}(\mu) = \ell^{e_2}$ is equivalent to

$$N^2 f(x, y) + pf(Nz + \lambda C, Nt + \lambda D) = \ell^{e_2}. \quad (3.4)$$

Modulo N , the previous equation becomes

$$\lambda^2 pf(C, D) \equiv \ell^{e_2} \pmod{N}.$$

Since l is a quadratic non-residue modulo N , the parity of e_2 should be adjusted so that $\left(\frac{pf(C,D)}{N}\right) = \left(\frac{\ell^{e_2}}{N}\right)$. We then have $\lambda = \sqrt{\frac{\ell^{e_2}}{pf(C,D)}} \bmod N$.

Furthermore, we also have

$$f(Nz + \lambda C, Nt + \lambda D) = N^2 f(z, t) + \lambda^2 f(C, D) + N\lambda L((C, D), (z, t)),$$

where

$$L((C, D), (z, t)) = 2Cz + \text{Trd}(\omega)(Dz + Ct) + 2\text{Nrd}(\omega)Dt = \langle C + D\omega, z + t\omega \rangle.$$

Hence, 3.4 is equivalent to

$$\lambda p L((C, D), (z, t)) = \frac{\ell^{e_2} - \lambda^2 pf(C, D)}{N} - N(f(x, y) + pf(z, t)). \quad (3.5)$$

(We recall that λ is chosen so that N divides $\ell^{e_2} - \lambda^2 pf(C, D)$). Modulo N , Equation 3.5 yields the linear equation

$$\lambda p L((C, D), (z, t)) = \frac{\ell^{e_2} - \lambda^2 pf(C, D)}{N} \bmod N. \quad (3.6)$$

This linear equation has N solutions (z, t) [KLPT14].

To find (x, y) , one takes a random solution (z, t) and tries to solve the following equation, which is equivalent to Equation 3.4:

$$f(x, y) = \frac{\ell^{e_2} - pf(Nz + \lambda C, Nt + \lambda D)}{N^2} =: r \quad (3.7)$$

Remark 3.2. In this step, there are two ways to proceed:

1. Take e_2 large enough so that r is always positive, and randomly take (z, t) so that r is a norm in R (as done in [KLPT14]);
2. Adjust e_2 for each value of (z, t) so that Equation 3.7 has a solution. This method gives an exponential approach that is studied next.

We summarize our discussion about the KLPT algorithm in the following lemma. Note that this result is already implied in [PS18].

Lemma 3.3. *Let $\mathcal{O}_0$ be a special extremal maximal order in $\mathcal{B}_{p,\infty}$. Let I be a $\mathcal{O}_0$ -left ideal. Using KLPT algorithm, we can compute an ideal of smooth norm d equivalent to I , where $d = \ell^e \approx p^3$*

Proof. In Step 1, the ideal J will have norm N of size about $\sqrt{p}$. For such N , one can solve the equation of Step 2 using the RepresentInteger methods if $N\ell^{e_1} > p$, hence $e_1 \geq \log_\ell(N/p) \approx$

$\log_\ell(p^{1/2})$. Then using the strategy of [PS18], in Step 5, we have $e_2 \approx \log_\ell(N^3 p) \approx \frac{5}{2} \log_\ell p$. Thus, $e = e_1 + e_2 \approx 3 \log_\ell(p)$, and the result follows. $\square$

Superpolynomial-time KLPT

We now analyze the second strategy discussed in Section 3.2, with a particular focus on superpolynomial-time algorithms. Note that the ideas of the strategy we present here first appeared in [Mer23, Section 3.4], and we give a variant of it.

Given $C, D \in \mathbb{Z}$, we look for solutions $(z, t) \in \mathbb{Z}^2$ such that 3.6 holds. In [PS18], it was shown that the solutions (z, t) for 3.6 can be viewed as a translated lattice as follows. We let $\Phi = p\lambda(2C + \text{tr}(\omega)D)$, $\Psi = p\lambda(\text{tr}(\omega)C + 2n(\omega)D)$, and $\chi := \frac{l^{e_2} - \lambda^2 p f(C, D)}{N}$, then (z, t) satisfies

$$\Psi z + \Psi t \equiv \chi \pmod{N}.$$

Let (z_0, t_0) denote one solution of this equation, then all solutions $(z, t) \in \mathbb{Z}^2$ are contained in the translated lattice $\mathcal{L} = (z_0, t_0)^T + \mathcal{L}_0$, where $\mathcal{L}_0 = \mathbb{Z}(\Phi, -\Psi)^T + \mathbb{Z}(0, N)^T$. To reduce the output length of KLPT, we aim to reduce $pf(Nz + \lambda C, Nt + \lambda D)$. In [PS18], this is reduced to a closest-vector problem where the involved lattice $\mathcal{L}$ is a deformation of $\mathcal{L}_0$. For our purpose, we do not recall the concrete basis of $\mathcal{L}_0$ here, but only note that this lattice is determined by C and D , and it has volume $pN^3 \sqrt{\Delta_{\mathbb{Q}(\omega)}}$. By the Gaussian heuristic, we estimate that the lattice contains a basis of size $\sqrt{p} N^{\frac{3}{2}} \Delta_{\mathbb{Q}(\omega)}^{\frac{1}{4}}$. This gives rise to the estimation that one can find (z, t) such that $pf(Nz + \lambda C, Nt + \lambda D) \approx pN^3 \sqrt{\Delta_{\mathbb{Q}(\omega)}}$. Hence, $\ell^{e_2} \approx pN^3 \sqrt{\Delta_{\mathbb{Q}(\omega)}} \approx p^{\frac{5}{2}}$. Below is a theorem that estimates the expected shortest vector of n independent random matrices from [AEN, Section 4.1].

Theorem 3.4. *Let $Z_1, \dots, Z_n$ be the length of the shortest vectors in n independent random matrices of unit volume and $Z_{\min} := \min\{Z_1, \dots, Z_n\}$, then $\mathbb{E}(Z_{\min}) \leq O(\frac{1}{\sqrt{n}})$ for $n \geq 2$.*

If we can generate n pairs of C, D that gives rise to n independent random lattices, then according to Theorem 3.4, the expected shortest vector among these n lattices has length $\frac{\sqrt{p} N^{\frac{3}{2}} \Delta_{\mathbb{Q}(\omega)}^{\frac{1}{4}}}{\sqrt{n}}$.

Therefore, $pf(Nz + \lambda C, Nt + \lambda D)$ would be $\frac{pN^3 \sqrt{\Delta_{\mathbb{Q}(\omega)}}}{n}$. Let $n \approx N^{e_3}$, then $\frac{pN^3 \sqrt{\Delta_{\mathbb{Q}(\omega)}}}{n} \approx pN^{3-e_3} \sqrt{\Delta_{\mathbb{Q}(\omega)}} \approx p^{\frac{5}{2} - \frac{e_3}{2}}$.

Hence, the length of the path returned using this approach reaches $e = \frac{5-e_3}{2} \log(p)$. The number n is exactly the number of solutions provided by the modular constraint Step 4 that we want to analyze.

On the modular constraint

In a general context, the modular constraint step consists of finding an element $[\mu] \in (\mathcal{O}/N\mathcal{O})^* \equiv GL_2(\mathbb{Z}/N\mathbb{Z})$ such that $(\mathcal{O}\gamma/N\mathcal{O})[\mu] = J/N\mathcal{O}$. The existence of such element is justified by the

transitivity of the action of $GL_2(\mathbb{Z}/N\mathbb{Z})$ on $\mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$. We recall the following lemma:

Lemma 3.5 ([KLPT14]). *Let N be a prime and $A = M_2(\mathbb{Z}/N\mathbb{Z})$. The set of proper nontrivial left A -ideals is in bijection with the set*

$$\left\{ \mathbb{P}^1(\mathbb{Z}/N\mathbb{Z}) \times \{(x : y)\}; (x : y) \in \mathbb{P}^1(\mathbb{Z}/N\mathbb{Z}) \right\},$$

and the right action of $PGL_2(\mathbb{Z}/N\mathbb{Z})$ on left A -ideals is transitive and induced by the natural action on $\mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$.

We recall that the action of $PGL_2(\mathbb{Z}/N\mathbb{Z})$ on $\mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$ is induced by the action of $GL_2(\mathbb{Z}/N\mathbb{Z})$ which has kernel $(\mathbb{Z}/N\mathbb{Z})^*$. Using an explicit isomorphism between $\mathcal{O}/N\mathcal{O}$ and $M_2(\mathbb{Z}/N\mathbb{Z})$ we have an action of $(\mathcal{O}/N\mathcal{O})^*$ on the left $\mathcal{O}/N\mathcal{O}$ -ideals. In the context of the KLPT algorithm, this action is restricted to the action of $Rj/N\mathcal{O}$. This is why Step 4 just consists of finding a pair (C, D) . Hence, the number n is upper bounded by the number of $[\mu]$. The number of such $[\mu]$ is exactly $\#Stab([x : y])$ (where $Stab([x : y])$ denotes the stabilizer of $[x : y]$) for some $[x : y] \in \mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$, since the action is transitive. Furthermore, we have $\#Stab([x : y]) = \frac{\#PLG_2(\mathbb{Z}/N\mathbb{Z})}{\#\mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})} = N(N+1)$. Hence, we have $n \approx N^2$. We summarize the discussion into the following theorem.

Theorem 3.6. *Let I be a left $\mathcal{O}$ -ideal for an extremal order $\mathcal{O}$. Then applying the KLPT algorithm to I one could find an equivalent left $\mathcal{O}$ -ideal J such that $n(J)$ is an ℓ -power and is of length $\frac{7-e}{2} \log p$ in time $\tilde{O}(n)$ where n is any positive integer less than p and e is a rational number such that $(\sqrt{p})^e \approx n$.*

Proof. We set $n(J) = \ell^{e_1+e_2}$ where e_1 and e_2 are given respectively by Step 2 and Step 5. Following the arguments in the proof of Lemma 3.3, we have $e_1 = \frac{1}{2}$. Let e be such that $N^e \approx (\sqrt{p})^e$ is the number of solutions we generate in the modular constraint step, then based on the discussions above, we have that $e_2 \approx \frac{5-e}{2} \log_\ell(p)$. This approach then has complexity in $\tilde{O}(\sqrt{p}^e)$. The number of solutions that can be generated in the modular constraint step is bounded above by $N^2 \approx p$.

□

The shortest path returned by this approach when considering a complexity of $\tilde{O}(p)$ has length $\frac{5}{2} \log p$. Note that in both our variants p is a slightly smaller than $2^{2\lambda}$. The cost of this attack to generate a path of length $\frac{5}{2} \log p$ is thus far greater than the security parameter. On the other hand, if we choose the runtime to be $\sqrt{p} \approx 2^\lambda$, then the output path length from this approach is $3 \log p$.

3.2.2 KLPT algorithm for non-extremal order

Let E and E_1 be two supersingular elliptic curve defined over $\mathbb{F}_{p^2}$, of known endomorphism rings $\mathcal{O}$ and $\mathcal{O}_1$. Let I be a connecting ideal of $\mathcal{O}$ and $\mathcal{O}_1$. Let $\mathcal{O}_0$ be a special extremal order, and $I_0 = I(\mathcal{O}_0, \mathcal{O})$. The problem is to find an $\mathcal{O}$ -left ideal of smooth norm equivalent to I .

Approach from [KLPT14]

The idea in [KLPT14] is as follows:

1. Compute $I_1 = I_0 \frac{\tilde{\gamma}_1}{\text{Nrd}(I_0)}$ where $\text{Nrd}(\gamma_1) = n_1 \text{Nrd}(I_0)$;
2. Compute $I_2 = I_0 I \frac{\tilde{\gamma}_2}{\text{Nrd}(I_0 I)}$ where $\text{Nrd}(\gamma_2) = n_2 \text{Nrd}(I_0 I)$;
3. Return $I \frac{\tilde{\gamma}}{\text{Nrd}(I)}$, where $\gamma = \tilde{\gamma}_1 \gamma_2$.

With this approach, the length of the shortest path is greater than $4 \log(p)$.

The SQISign approach

The idea in SQISign [DKL⁺20] is to transfer the problem in the special case using pullback and push forward through I_0 .

$$\begin{array}{ccc}
 \mathcal{O}_0 & \xrightarrow{I_0} & \mathcal{O} \\
 \left(\begin{array}{c} \chi_K(\beta') \\ \parallel \\ \mathcal{O}_R(K) \end{array} \right) & \begin{array}{c} K = [I_0]^* I \\ \parallel \\ \mathcal{O}_R(K) \end{array} & \left(\begin{array}{c} I \\ \parallel \\ \mathcal{O}_1 \end{array} \right) \begin{array}{c} [I_0]_* \chi_K(\beta') \end{array}
 \end{array}$$

Where $\chi_K(\beta') = K \frac{\tilde{\beta}'}{\text{Nrd}(K)}$. To obtain this, one must have $\beta' \in K \cap \mathcal{D}$, where $\mathcal{D} = \mathcal{O}_0 \cap \mathcal{O} = \mathbb{Z} + I_0$ [DKL⁺20, Corollary 1]. For $n = \ell^e$, the algorithm can be described as follows: We suppose that $\text{Nrd}(I_0) = N_0$ is prime inert in R such that ℓ is a quadratic non residue modulo N_0 .

1. Compute $K = [I_0]^* I$;
2. Compute an ideal L of prime norm N equivalent to K , such that ℓ is a quadratic non-residue modulo N . Let δ such that $L = \chi_K(\delta)$;
3. Find an element $\gamma \in \mathcal{O}$ of norm $N \ell^{e_1}$ for some $e_1 \in \mathbb{N}$;
4. Find an element $\alpha \in \mathcal{O}$ such that $L = \mathcal{O} \alpha + N \mathcal{O}$;
5. Compute $\mu_0 = (C_0 + \omega D_0) j \in R j$ such that $\gamma \mu_0 \equiv \alpha \pmod{N \mathcal{O}}$;

6. Compute $\mu_1 = (C_1 + \omega D_1)j \in Rj$ such that $\gamma\mu_0\delta \in \mathcal{O} \cap \mathcal{O}_0$;
7. Compute $C = CRT_{N_0, N}(C_0, C_1)$ and $D = CRT_{N_0, N}(D_0, D_1)$ and let $\mu' = (C + \omega D)j$
8. Compute $\lambda \in \mathbb{Z}/NN_0\mathbb{Z}^*$ and $\mu'_1 \in \mathcal{O}_0$ such that $\mu = \lambda\mu' + NN_0\mu'_1$ has norm ℓ^{e_2} , for some $e_2 \in \mathbb{N}$.
9. Return $\chi_L(\beta)$, where $\beta = \gamma\mu$.

The main difference between this algorithm and the algorithm from [KLPT14] described in 3.2.1 is Step 8. Here the approximation is done modulo NN_0 and then the computation of λ becomes more delicate than what we have in Step 5, 3.2.1. In the present context, the approximation equation is 3.8, which corresponds to 3.4, replacing N by NN_0 .

$$N^2 N_0^2 f(x, y) + pf(NN_0 z + \lambda C, NN_0 t + \lambda D) = \ell^{e_2}. \quad (3.8)$$

Modulo NN_0 , this equation becomes

$$\lambda^2 pf(C, D) \equiv \ell^{e_2} \pmod{NN_0}.$$

For this equation to have a solution, we need the following equality:

$$\left(\frac{pf(C, D)}{N} \right) = \left(\frac{\ell^{e_2}}{N} \right) \text{ and } \left(\frac{pf(C, D)}{N_0} \right) = \left(\frac{\ell^{e_2}}{N_0} \right).$$

Since ℓ is a quadratic non residue modulo N and N_0 , we always have $\left(\frac{\ell^{e_2}}{N} \right) = \left(\frac{\ell^{e_2}}{N_0} \right)$. Hence, we need

$$\left(\frac{pf(C, D)}{N} \right) = \left(\frac{pf(C, D)}{N_0} \right). \quad (3.9)$$

This last equality has a probability $\frac{3}{4}$ to fail, for given γ from Step 3 and δ from Step 2 [DKL⁺20]. To minimize this failure probability, the authors of [DKL⁺20] take e_1 large enough so that there are many possibilities for γ (we recall that γ is computed by solving Equation 3.3). The advantage of this method is that it only modifies the parameters (C, D) , and N remains fixed.

Lemma 3.7. *Let $\mathcal{O}$ and $\mathcal{O}_1$ be two non extremal maximal orders in $\mathcal{B}_{p, \infty}$ where $p \equiv 3 \pmod{4}$ or $p \equiv 5 \pmod{8}$. Given a connecting ideal I of $\mathcal{O}$ and $\mathcal{O}_1$, there is a probabilistic polynomial time algorithm which find an equivalent ideal of norm ℓ^e , for some small prime ℓ and $e \approx \frac{9}{2} \log_\ell(p)$.*

Proof. Following the idea in Lemma 3.3, we can obtain $e_1 = \frac{1}{2}$ in step 3 Equation 3.9 has a success probability of $1 - \left(\frac{3}{4}\right)^4 \approx 68.4\%$. In the failure case we can either go back to Step 2 and compute another L , or compute another I_0 . In Step 8, we can use the strategy of [PS18] to obtain $e_2 \approx 4 \log_\ell p$ and the result follows. $\square$

Remark 3.8. The exponential approach of Section 3.2.1 can be applied here. Using similar analysis, we see that the output length is $(\frac{9}{2} - e) \log p$ in time $\tilde{O}(n)$ where n is any positive integer less than p and e is a rational number such that $(\sqrt{p})^e \approx n$, for $p \equiv 3 \pmod{4}$ or $p \equiv 5 \pmod{8}$. And in this case, if we bound the runtime by $\sqrt{p}$, then the shortest path returned is of length $\frac{7}{2} \log p$.

3.2.3 Parameters secure against KLPT-based attacks

Combining the analysis presented so far, we obtain the following limitations for an attacker of complexity 2^λ . Since $O(p) = O(2^\varepsilon)$ and $\lambda < \varepsilon \leq 2\lambda$, we use $O(\sqrt{p})$ as an upper bound for the runtime of exponential KLPT, and we summarize the output length here.

Takeaway 1. Consider an isogeny $\varphi : E_0 \rightarrow E_1$ of degree d . Given the endomorphism rings of E_0 and E_1 , KLPT-based methods cannot recover φ in time $\tilde{O}(\sqrt{p}) < \tilde{O}(2^\lambda)$ if:

1. E_0 is a special curve and $\log d < 3 \log p$;
2. E_0 is not a special curve and $\log d < \frac{7}{2} \log p$.

Given these results, we obtain that the two protocols proposed in Section 3.1 are secure against KLPT-based attacks. In both instances, E_0 is chosen to not be a special curve, hence the limit 2 applies. In the first variant, from Takeaway 2, the isogeny φ has degree $d \approx 2^\lambda p^2$ and ψ has degree $\approx p$, **Variant 1** is secure according to the summary above. Similarly, **Variant 2** relies on isogenies of degree $\approx p$, and thus KLPT-based attacks do not apply. In both instantiations, the signatures rely on the stronger relation, and thus the attack that recovers the composition $\hat{\psi} \circ \varphi' \circ \psi'$ cannot be used.

3.3 Analysis of other attacks

We now proceed to analyze further attack vectors against our proposal.

3.3.1 Attacks based on the SIDH attacks

The SIDH attacks [CD23, MMP⁺23, Rob23] recover an isogeny $\varphi : E_0 \rightarrow E_1$ of degree d given:

- The curves E_0 and E_1 ;
- The degree d ;
- The image of a torsion basis of smooth order n with $n \geq \sqrt{d}$.

However, it is possible to brute-force part of an unknown isogeny (which is always cheaper than brute-forcing torsion point information), thus we need $d \geq 2^\lambda \sqrt{\ell}$ to avoid the attacks.

Binary challenges. The proofs of isogeny knowledge with binary challenges are potentially vulnerable to the SIDH attacks.

Consider the following diagram:

$$\begin{array}{ccc}
 E_0 & \xrightarrow{\varphi} & E_1 \\
 \psi \downarrow & & \downarrow \psi' \\
 E_2 & \xrightarrow{\varphi'} & E_3
 \end{array}$$

When the proof has binary challenges, the isogenies ψ and ψ' are revealed together. If the prover is honest, we have

$$\ker \psi' = \varphi(\ker \psi),$$

which allows an attacker to recover the image of the subgroup $\ker \psi$ under the secret isogeny φ . After three such challenges, the attacker has recovered enough information to apply the SIDH attacks. In fact, as shown in [BKM⁺21, FP22] this allows the attacker to recover the image of a torsion basis on E_0 under φ , up to the same scalar. The square of this scalar can be computed through Weil pairing. If the degrees of ψ and ψ' are prime powers, as in SIDH, there are only two possible square roots. It is easy to apply the SIDH attacks each time $\deg \varphi < (\deg \psi)^2$, assuming the degree of ψ is not much smaller than that of φ , and recover φ . In practice, it is enough to have $2^\lambda (\deg \psi)^2 \leq \deg \varphi$, so that one needs to first guess an isogeny of degree at least 2^λ before being able to apply the SIDH attacks.

Takeaway 2. Binary challenges require $2^\lambda (\deg \psi)^2 \leq \deg \varphi$.

Remark 3.9. An alternative idea could be to use commitment isogenies ψ and ψ' with non-rational torsion as the torsion point images are defined over large extensions fields and thus the SIDH attacks do not apply directly. However, in our case we know endomorphism rings and can go on a different route. As described in [CII⁺23], one has that $(\text{End}(E)/N\text{End}(E))^*$ acts on the set of degree N isogenies. In general without any extra information this action is hard to compute when only the codomain of the isogeny is known. In our case however, this action is exactly provided by the parallel isogenies. Thus one can compute the stabilizer of this action as in [CII⁺23] using a polynomial-time quantum algorithm which essentially reveals the connecting ideal corresponding to the secret isogeny. Since the secret isogeny here is smooth, one can recover the isogeny itself step by step. Note that just knowing the codomain of the parallel isogeny would not always have been enough (as it does not determine the action precisely as one might have several degree N isogenies between two supersingular elliptic curves). However, knowing the second vertical isogeny already is enough information to evaluate the group action (assuming that the torsion is large enough to ensure unicity).

3.3.2 Attacks on the zero-knowledge property

In this section, we explore the zero-knowledge property of the underlying sigma protocol for the binary challenge variant as described in Section 3.1.

We set ourselves in the setting of **Variant 1**, from Section 3.1, since the others variants use ternary challenges and are provably (honest-verifier) zero-knowledge. The question we explore is whether we can address the zero-knowledge issue that arises in Variant 1 in the context where we only use the identification protocol for the purpose of turning it into a signature scheme.

This turns out to be a key observation. Since we turn the sigma protocol into a signature scheme via the Fiat-Shamir transform [FS87], a natural question that arises is whether we can allow a relaxation of the zero-knowledge property of the underlying sigma protocol whilst still retaining the security of the Fiat-Shamir transform. In other words, what are the minimal assumptions for the security of the Fiat-Shamir transform and can we achieve them? In [AABN02], they introduce a new notion of security for sigma-protocols, namely *security against impersonation under passive attacks*. They show that this is a minimal assumption for the Fiat-Shamir transform to be secure. We define this notion formally below:

Definition 3.10 ([AABN02]). *We say that a sigma protocol is secure against impersonation under passive attacks if for any polynomial time adversary $\mathcal{A}$ the advantage $\text{Adv}^{\text{Impersonate}}(\mathcal{A})$ of $\mathcal{A}$ in the Impersonate game is negligible, where*

$$\text{Adv}^{\text{Impersonate}}(\mathcal{A}) = \Pr[\text{Impersonate}(\mathcal{A}) \rightarrow I] = \text{negl}(\lambda)$$

The Impersonate game is described below.

$\text{Impersonate}(\mathcal{A})$	$\mathcal{O}^{\text{TG}}$
1: $\text{Setup}(1^\lambda) \rightarrow \text{pp}$	1: $R_p \xleftarrow{\$} \mathcal{R}$ (generate randomness)
2: $\text{KeyGen}(\text{pp}) \rightarrow (\text{sk}, \text{pk})$	2: $\text{com} \leftarrow \mathcal{P}(\text{sk}; R_p)$
3: $\mathcal{A}^{\mathcal{O}^{\text{TG}}}(\text{pk}) \rightarrow \text{com}, \text{st}$	3: $\text{ch} \xleftarrow{\$} \mathcal{C}$
4: $\text{ch} \xleftarrow{\$} \mathcal{C}$	4: $\text{resp} \leftarrow \mathcal{P}(\text{sk}, \text{com}, \text{ch}; R_p)$
5: $\mathcal{A}(\text{st}, \text{ch}) \rightarrow \text{resp}$	5: return (com, ch, resp)
6: return $\mathcal{V}(\text{pk}, \text{com}, \text{ch}, \text{resp})$	

Theorem 3.11 ([AABN02], Theorem 3.3). *Let Π_Σ be a non-trivial sigma-protocol, and $\mathcal{DS}_\Sigma$ be the associated digital signature scheme obtained via the Fiat-Shamir transform, then $\mathcal{DS}_\Sigma$ is secure against existential forgery under chosen-message attacks if and only if Π_Σ is secure against impersonation under passive attacks.*

Remark 3.12. Note that non-triviality here requires the challenge space to be super-polynomial. Otherwise, a trivial winning strategy would be to replay a transcript obtained from the oracle and then one gets a probably $1/|\mathcal{C}|$ of winning. So if the size of the challenge space is polynomial, we have a winning strategy with probability that is not negligible.

We claim that this relaxed security notion is achieved by the parallel repetition of the binary challenge version under some conditions, and we hence get a secure signature scheme by applying the Fiat-Shamir transform. Let us formulate this more formally.

Problem 3.13. Consider two supersingular elliptic curves E_0, E_1 and an isogeny $\varphi : E_0 \rightarrow E_1$ of degree $D = \ell_1^{e_1}$ in $\mathcal{R}_{\text{strg}}$ such that $\#E_0(\mathbb{F}_{p^2}) = (\ell_1 \ell_2^{e_2} f)^2$ as defined in **Variant 1**. Given access to an oracle that outputs either :

- two isogenies $E_0 \rightarrow E_2, E_1 \rightarrow E_3$ and the images of the torsion basis on the codomain curves;
- an isogeny $E_2 \rightarrow E_3$ of degree D ;

then the adversary must recover the secret isogeny φ .

Theorem 3.14. Under the hardness of Problem 3.13, the λ parallel repetition of the sigma protocol is secure against impersonation under passive attacks.

Proof. Problem 3.13 is the translation of the impersonate game to our setting. □

This problem has already been discussed in the previous subsections. Notably if we ensure that the takeaways from Sections 3.2 and 3.3.1 are respected then we can reasonably assume its hardness.

Takeaway 3. Under reasonable assumptions, a digital signature derived from a proof of isogeny knowledge with binary challenges via the Fiat-Shamir transform is secure.

3.4 Concrete instantiations and parameters size

In this section, we analyze the size of the signatures derived from the sigma protocols described in Section 3.1 using the Fiat-Shamir transform [FS87]. We rely on a hash function $\mathcal{H} : \{0, 1\}^* \rightarrow \{0, 1\}^{2\lambda}$, where λ is the security parameter.

Variant 1.

We first fix $\ell_1 = 2, \ell_2 = 3$ for efficiency reasons. Then, we choose the value e_2 such that isogenies of degree 3^{e_2} are hard to recover via meet-in-the-middle or van Oorschot-Wiener attacks [vW99] (e.g., we have $e_2 = 137$ for $\lambda = 128$). The exponent e_1 is chosen such that the isogenies of degree 2^{e_1} are hard to recover, even when their action on the 3^{e_2} torsion is known, i.e. $e_1 = \lambda + \lceil 2e_2 \log 3 \rceil$. We define the prime p to be of the form $p = 2 \cdot 3^{e_2} f - 1$, where f is a small cofactor. The public parameters of the signature are then $pp = (p, e_1, e_2, E_0, T_0, P_0, Q_0)$, where E_0 is a random supersingular elliptic curve defined over $\mathbb{F}_{p^2}$. The point $T_0 \in E_0$ is an

auxiliary point of order 2, used in the CGL hash function computations [CLG09], and $\{P_0, Q_0\}$ is a basis of $E_0[3^{e_2}]$.

By the definition of p , any point of order 2^{e_1} is defined over a large extension of $\mathbb{F}_{p^2}$. We then represent the secret isogeny $\varphi : E_0 \rightarrow E_1$ (of degree 2^{e_1}) by a sequence of 2-isogenies. To do that, we represent it as a seed $s \in \{0, 1\}^{e_1}$ that has to be hashed using CGL hash function [CLG09], with the first step chosen between the 2-torsion points that are not T_0 . The secret key can then be represented by $e_1 \approx \lambda + 2 \log p \approx 5\lambda$ bits. The public key is (E_1, P_1, Q_1) where $P_1 = \varphi(P_0)$ and $Q_1 = \varphi(Q_0)$. This requires $6 \log p \approx 12\lambda$ bits.

In this variant, the soundness error is $\frac{1}{2}$: this means the signature needs to repeat the sigma protocol $k = \lambda$ times to obtain a negligible soundness error. Since we are in the binary case, the response to both challenges includes all the committed values; thus, we rely on hashed commitments only for compression, but we do not require any hiding property. Using some of the compression techniques from [GPV24], the size of the response for the horizontal challenge is $3 \log p + 2\lambda$, while the vertical challenge is $\log p$. Repeating λ times and including the hashed commitment sizes of $2 \times 2\lambda$, we obtain an asymptotic size of $\lambda(5\lambda + 2 \log p) \approx 9\lambda^2$ bits. Note that this is asymptotic and based on the assumption that $e_3 \approx 2\lambda / \log 3$, but we can choose smaller parameters based on the cost of the van Oorschoot-Wiener attack, as done in SIDH. For $\lambda = 128$, this results in a 218-bit prime for which we obtain a signature of about 17 kB, and a public key of about 164B, which may be reduced even further by relying on seed trees [GPV24]. This resulting signature is much larger than the current SQISign NIST proposal, which at the same security level as a signature of 177B and a public key of 64B. However, SQISign's security crucially relies on the secrecy of the endomorphism ring, and thus would be insecure in the setting we explore.

Variant 2.

The second variant relies on ternary challenges, and thus torsion images under the secret isogeny is not revealed. The public key hence consists of a single curve, which then requires $2 \log p$ bits.

In this variant, we need to rely on a computationally binding and statistically hiding commitment scheme $\mathcal{C}$: we construct one from a hash function $\mathcal{H} : \{0, 1\}^* \rightarrow \{0, 1\}^{2\lambda}$ by defining $\mathcal{C}(m) = \mathcal{H}(m|r)$, for some random string r of λ bits. For each execution of the sigma protocol, we need 6λ bits to represent the three commitments (following the commitment algorithm in [DDGZ22, Figure 3]), $\lambda + \log p$ for the responses to the vertical isogeny challenges, and $\lambda + 3 \log p$ for the response to the horizontal isogeny challenge. The soundness error of the underlying sigma protocol is $\frac{2}{3}$, which means the signature needs to repeat the sigma protocol $k = \lceil -\lambda / \log 2/3 \rceil$ times to obtain a negligible soundness error. This results in an average signature of asymptotic size $\frac{31}{3} \lambda \lceil -\lambda / \log 2/3 \rceil$ bits. For $\lambda = 128$, we can take $\log p = 216$ and this results in a signature syze of about 114kB and a public key of 54B.

3.5 Conclusion

In this chapter, we explore the feasibility of SIDH-based signatures when the endomorphism ring of all curves are public. We identify two variants of the construction that are secure in this setting, where the difference resides in the use of binary or ternary challenges and give concrete parameters. We provide a thorough security analysis of our proposals notably in terms of attacks based on KLPT, with both a polynomial and superpolynomial adversary, attacks derived from the recent SIDH-attacks and analyze the zero-knowledge property of the binary challenge variant. Note that the results we derive from the KLPT attacks could be affected by improvements on the output size of KLPT but this would only require to adjust the parameters to retain the security.

4 Updatable Public-Key Encryption from FESTA

This chapter is based on an article of the same name currently in submission.

Updatable Public-Key Encryption (UPKE) is an extension of public-key encryption that was proposed in the setting of secure messaging. A necessary property of secure messaging is *forward secrecy*, which ensures that in a scenario where a secret key is exposed at a certain point of time, all messages exchanged before that key leakage remain secure.

This notion is easy to achieve from a primitive called a *ratchet*, which can be instantiated from both symmetric or Diffie-Hellman assumptions. However, a ratchet requires interaction at every update. An equivalent notion would be therefore desirable from public-key encryption, allowing any party to update a public key and removing the need for interaction. Forward-secure public-key encryption as a notion was proposed and realized in the form of a mechanism which created key pairs which could and should be synchronously updated. However, no efficient constructions have been proposed yet. Furthermore, requiring synchronicity does not seem to be a realistic assumption in a messaging context.

Updatable public-key encryption (UPKE) was therefore put forward as an alternative by Jost et al. [JMM19] and refined by Alwen et al. [ACDT20]. The idea is to introduce an asynchronous key update mechanism, where the sender can update the receiver's public key and send them an encrypted update token that will allow them to update their corresponding secret key. The security guarantee is slightly more relaxed than forward security in the sense that exposing a secret key preserves the security of previous messages, as long as an honest update was performed at some point before the exposure.

Efficient constructions of UPKEs were proposed by the original authors [JMM19, ACDT20] from discrete logarithms assumptions. However, achieving post-quantum security is now a necessary goal in messaging applications, not only in theory but also in practice, as shown for example with Signal's introduction of the post-quantum ratchet.^I Building an efficient,

^I<https://signal.org/blog/spqr/>

quantum-resistant UPKE is therefore a necessary step towards efficient and secure messaging protocols.

Related work

The first post-quantum updatable public-key encryption schemes were proposed from lattice-based assumptions, namely by Dodis et al. and Haidar et al. [DKW21, HPS23]. However, both of these schemes only support a bounded number of updates, due to the inherent noisy nature of lattice-based assumptions. This limit on the number of updates was lifted in [ABL25], by Albrecht et al. but the resulting scheme is still far from being practically efficient.

Therefore, a natural candidate for post-quantum UPKE lies in isogeny-based cryptography. In fact, Eaton et al. [EJKM22] first proposed two isogeny-based UPKE. One from SIDH, now broken [CD23, MMP⁺23, Rob23], and one from CSIDH. Both protocols unfortunately require synchronous updates. Additionally, selecting appropriate CSIDH parameters to guarantee post-quantum security is still subject to debate [Pei20]. These results however opened the door to a series of work aiming to build practical and secure isogeny-based UPKEs. Duparc, Fouotsa and Vaudenay [DFV24] proposed the first secure isogeny-based UPKE that was not based on group actions, leveraging so-called generalized lollipop attacks on MSIDH [FMP23] by Castryck and Vercauteren [CV23]. However, the scheme is far from being practically efficient. Most recently, Jana and Dutta [JD25] propose two isogeny-based UPKE, both based on CSIDH assumptions. Despite all this progress, the question of achieving a practically efficient post-quantum UPKE with unbounded updates (that does not rely on class group actions) remains open.

Contribution. In this chapter, we propose a practically efficient isogeny-based UPKE with unbounded updates. It relies on two key insights.

Our first ingredient is a generalization of the FESTA [BMP23] public-key encryption (PKE) scheme using dimension-four isogenies. This generalization allows us to relax constraints on the degree of the isogenies used compared to the original FESTA and implies a simpler security proof, relying only on the CIST^2 assumption [BMP23, Problem 8].

Using this new PKE as a basis, we build an update mechanism by randomizing the public-key through a long-enough isogeny walk. To compute the corresponding secret key, in Section 4.2, we explain how to “shorten” an isogeny walk. More specifically, we introduce ConstrainedIsogeny (Algorithm 1), an algorithm that, given the knowledge of the endomorphism ring of two curves, outputs the description of an isogeny of degree $d \leq 2^a \sqrt{p}$ between them matching some constraints on the 2^a -torsion. Crucially, this prevents the update mechanism from having secret isogenies whose degrees grow with the number of updates. This algorithm is therefore a key ingredient to be able to allow an unbounded amount of key updates.

Finally, we provide a formal proof of the security of the UPKE, relying on the CIST² assumption [BMP23, Problem 8] only. Note that for comparison the construction of [JD25], relies on both the standard CSIDH assumption, as well as a non-standard “knowledge of exponent” CSIDH assumption to prove the security of their proposed schemes.

This results in the first practically efficient post-quantum UPKE with unbounded updates that does not come from group actions. Moreover, its security is based on a well-known problem in the literature.

4.1 Known algorithms

For ease of reading, we gather here all the known, isogeny-related algorithms we will use in the rest of the chapter. Most of these algorithms are considered as standard in the literature, and we consider them as black-boxes giving adequate references.

- $(E', \varphi) \leftarrow \text{IsogenyWalk}_\ell(E, n, r)$: on input a curve E , an integer n and $r = (r_0, r_1, \dots, r_{n-1}) \in [0, \ell+1] \times [0, \ell]^{n-1}$, the algorithm returns (E', φ) , where $\varphi : E \rightarrow E'$ is the non-backtracking isogeny walk of length n defined by r in the ℓ -isogeny graph. The isogeny $\varphi = \varphi_{n-1} \circ \dots \circ \varphi_0$ is computed step-by-step, in a CGL-like fashion [CLG09]: the kernel of each isogeny φ_i is determined by r_i . When $n \geq \varepsilon \log_\ell(p)$, then the distribution of the curve E' is at distance $O(2^{-\frac{1}{2}(\varepsilon-1)\log_2(p)})$ from the stationary distribution on the ℓ -isogeny graph [BCC⁺23, Theorem 11].
- $(P, Q) \leftarrow \text{Gen2Tor}(E)$: on input a curve E , it outputs a deterministic basis (P, Q) of $E[2^a]$, where $a = \max\{n : 2^n \mid (p+1)\}$. In the literature, there exist several ways to do so; see for instance [AAA⁺24, §2.2.3] or [ZSP⁺19].
- $\text{End}(E') \leftarrow \text{IsogenyToEnd}(\varphi : E \rightarrow E', \text{End}(E))$: on input an isogeny $\varphi : E \rightarrow E'$ of degree 3^δ , where $\delta := \lceil 2 \log_3(p) + 2 \log_3(2)\lambda \rceil$, and $\text{End}(E)$, this algorithm returns $\text{End}(E')$. This algorithm is a straightforward generalization of [AAA⁺24, Algorithm 3.17]. We write the isogeny $\varphi = \varphi_r \circ \dots \circ \varphi_1 : E \rightarrow E_1 \rightarrow \dots \rightarrow E_r = E'$, where each φ_i has rational kernel generators. Then, each isogeny φ_i is translated to its corresponding ideal I_{φ_i} , where we have that $\text{End}(E_i)$ is the right order of I_{φ_i} . The output is $\text{End}(E_r)$.
- $r \leftarrow \text{KLPT}_\ell(E, \text{End}(E), E', \text{End}(E'))$: on input two supersingular elliptic curves E and E' together with their endomorphism rings $\text{End}(E)$ and $\text{End}(E')$, this algorithm returns $r = (r_0, r_1, \dots, r_{n-1}) \in [0, \ell+1] \times [0, \ell]^{n-1}$, such that $(E', \varphi) \leftarrow \text{IsogenyWalk}_\ell(E, n, r)$ where $\varphi : E \rightarrow E'$ is an isogeny of degree ℓ^n . It runs the KLPT [KLPT14] algorithm used in earlier versions of SQIsign [DKL⁺20, DLLW23, EPSV23, SEMR24] to get an isogeny $\varphi : E \rightarrow E'$ of degree $\ell^n \approx p^{15/4}$, then recovers the output $r = (r_0, r_1, \dots, r_{n-1}) \in [0, \ell+1] \times [0, \ell]^{n-1}$ by mapping the kernel of each degree ℓ factor φ_i of φ to the corresponding scalars $r_0 \in [0, \ell+1]$ and $r_i \in [0, \ell]$ for $i = 1, \dots, n-1$.

4.2 Computing isogenies matching constraints

In this section, we explain how to compute special isogenies between elliptic curves of known endomorphism rings. Such isogenies must act in a prescribed way on some torsion. Before specializing to our use case, we provide some simple structural results.

4.2.1 Structural results

It is well-known that isogenies between two supersingular elliptic curves form a 4-dimensional lattice. Let N be an integer coprime to p and let E_1, E_2 be supersingular elliptic curves. Also, fix two bases (P_1, Q_1) and (P_2, Q_2) of the N -torsion of E_1 and E_2 , respectively. We now introduce some notation that we will use to define a sublattice of $\text{Hom}(E_1, E_2)$.

Definition 4.1. *Let E_1, E_2 be supersingular elliptic curves and let P_1, Q_1 and P_2, Q_2 be fixed bases of the N -torsion of E_1 and E_2 , respectively. We denote by $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ the set of isogenies taking P_1 to the subgroup generated by P_2 and taking Q_1 to the subgroup generated by Q_2 , i.e.,*

$$\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)} := \left\{ \varphi \in \text{Hom}(E_1, E_2) \left| \begin{array}{l} \varphi(\langle P_1 \rangle) \leq \langle P_2 \rangle, \\ \varphi(\langle Q_1 \rangle) \leq \langle Q_2 \rangle \end{array} \right. \right\}.$$

where $\leq$ denotes a subgroup.

Our goal is to prove that $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is a 4-dimensional lattice. Before that, we recall that endomorphisms that fix the subgroup generated by a point P are of the form $\mathbb{Z} + I_\varphi$, where φ is the isogeny whose kernel is generated by P and I_φ is the corresponding left ideal. The set $\mathbb{Z} + I_\varphi$ is also the Eichler order obtained from intersecting $\text{End}(E_1)$ with the right order of I_φ . We will call this the *Eichler order associated to P* .

Proposition 4.2. *Let E_1, E_2 be supersingular elliptic curves and let (P_1, Q_1) and (P_2, Q_2) be fixed bases of the N -torsion of E_1 and E_2 respectively. Then the following hold:*

1. $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is not empty.
2. Let Λ_{P_1} and Λ_{Q_1} be the Eichler orders associated to P_1 and Q_1 . Then $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is a right $\Lambda_{P_1} \cap \Lambda_{Q_1}$ -module.
3. As a right $\Lambda_{P_1} \cap \Lambda_{Q_1}$ -module, $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is isomorphic to a right ideal in $\Lambda_{P_1} \cap \Lambda_{Q_1}$.

Proof. [FKMT22, Lemma 3.4] implies that there exist isogenies for any prescribed action on the N -torsion thus $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is not empty.

The second claim follows from the fact that elements of $\Lambda_{P_1} \cap \Lambda_{Q_1}$ fix both the subgroup generated by P_1 and Q_1 . Indeed, Λ_{P_1} consists of endomorphisms of the form $\mathbb{Z} + I_{P_1}$, where

$I_{P_1} = \{\alpha \in \text{End}(E_1) \mid \alpha(P_1) = 0\}$; hence the claim that elements of $\mathbb{Z} + I_{P_1}$ send P_1 to an integer multiple of P_1 . The same holds for Λ_{Q_1} and Q_1 .

As for the third claim, we show a concrete isomorphism. We take an isogeny $\psi : E_2 \rightarrow E_1$ that sends the subgroup generated by P_2 to the subgroup generated by P_1 and the subgroup generated by Q_2 to the subgroup generated by Q_1 ; this exists because of [FKMT22, Lemma 3.4]. Then, composing ψ with $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ yields a set J that is a collection of endomorphisms in $\Lambda_{P_1} \cap \Lambda_{Q_1}$. Furthermore, since $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is a right $\Lambda_{P_1} \cap \Lambda_{Q_1}$ -module, J is a right ideal in $\Lambda_{P_1} \cap \Lambda_{Q_1}$. $\square$

Proposition 4.2 implies that the lattice $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is a 4-dimensional lattice. We now prove that the determinant of $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is equal to $p^2 M^2$, where M is a divisor of N^2 . First, we recall the following easy result.

Lemma 4.3. *Let G be an abelian group and let H, K be subgroups of finite index in G . Then $H \cap K$ is also of finite index and $|G : (H \cap K)|$ divides $|G : H| |G : K|$.*

Proof. We define the following homomorphism of groups: $\theta : G/(H \cap K) \rightarrow G/H \times G/K, \theta(g) = (g \bmod H, g \bmod K)$. We show that θ is injective. Assume that $g_1 \equiv g_2 \pmod{H}$ and $g_1 \equiv g_2 \pmod{K}$. This implies that $g_1 g_2^{-1}$ is both in H and K , hence it lies in $H \cap K$ which proves injectivity. Now the statement follows from the first isomorphism theorem (or even Lagrange's theorem). $\square$

To prove that the determinant of $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ is equal to $p^2 M^2$ where M is a divisor of N^2 , we observe what follows. Let $\mathcal{O}_1, \mathcal{O}_2$ and $\mathcal{O}_3$ be three maximal orders where $\mathcal{O}_1 \cap \mathcal{O}_2$ and $\mathcal{O}_1 \cap \mathcal{O}_3$ are Eichler orders of level N ; that is $\mathcal{O}_1 \cap \mathcal{O}_2$ and $\mathcal{O}_1 \cap \mathcal{O}_3$ are both additive subgroups of $\mathcal{O}_1$ of index N . By Lemma 4.3, we have that the index of $\mathcal{O}_1 \cap \mathcal{O}_2 \cap \mathcal{O}_3$ in $\mathcal{O}_1$ is equal to M , a divisor of N^2 . Now the upper bound on the determinant $p^2 N^4$ follows from the fact that a sublattice of index M of an order $\mathcal{O}$ has determinant $p^2 M^2$.

4.2.2 Specializing to our use case

Equipped with the above theoretical tools, we now describe an algorithm to sample isogenies from $\text{Hom}(E_1, E_2)_{(P_1, Q_1), (P_2, Q_2)}$ of small norm. For our purposes we will assume that we have a prime of the form $p = f 2^a 3^b 5^c - 1$, where f is a small cofactor and $2^a > 3^b 5^c \sqrt{p} (2a)^3$; this technical condition will become clear in what follows. We also define another exponent c such that $5^c \approx 3^b$. We also fix $E_0 : y^2 = x^3 + x$, whose endomorphism ring $\text{End}(E_0)$ is given by all the integral combinations of the endomorphisms $[1, \iota, \frac{\iota + \pi}{2}]$ and $\frac{1 + \iota \pi}{2}$ (see Section 2.3.2), as well as (P_0, Q_0) , which is the output of $\text{Gen2Tor}(E_0)$.

We assume that we are given an elliptic curve E together with its endomorphism ring $\text{End}(E)$. Let $(P, Q) \leftarrow \text{Gen2Tor}(E)$. Our goal is to compute an isogeny representation of an isogeny

$\varphi : E_0 \rightarrow E$ such that $([\alpha]\varphi(P_0), [\beta]\varphi(Q_0))$ for some $\alpha, \beta \in \mathbb{Z}/N\mathbb{Z}$. To represent the isogeny φ , we will use the information coming from the evaluation of the 2^a -torsion. Similarly to what is done in FESTA, and as we will explain in Section 4.3.4, in our construction, we will need to provide a representation for an isogeny $\varphi_2 \circ \varphi \circ \widehat{\varphi}_1$, where $\deg(\varphi_1) = 3^b$ and $\deg(\varphi_2) = 5^c$. This translates to the condition $2^{2a} > 3^b 5^c \deg(\varphi)$.

We remark that the isogeny φ we are after is an isogeny in the lattice $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$. As we have already shown in the previous section, the discriminant of $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$ is upper bounded by $p^2 2^{4a}$. This means that the shortest isogeny (*i.e.*, the isogeny of smallest degree) in the lattice $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$ will have degree $\approx \sqrt{p} 2^a$. Hence, to represent $\varphi_2 \circ \varphi \circ \widehat{\varphi}_1$, we need $2^{2a} > 3^b 5^c \deg(\varphi) \approx \sqrt{p} 2^a 3^b 5^c$, which is satisfied since we imposed $2^a > 3^b 5^c \sqrt{p} (2a)^3$. We next explain how to sample such a short isogeny in $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$.

Given the information coming from $\text{End}(E_0)$ and $\text{End}(E)$, we can compute a quaternion representation of $\text{Hom}(E_0, E)$: we first compute an ideal connecting $\text{End}(E_0)$ and $\text{End}(E)$, and then, from this one, we can compute all the remaining ones. The lattice $\text{Hom}(E_0, E)$ is a 4-dimensional lattice; define (b_1, b_2, b_3, b_4) to be such a basis. Our next task is evaluate the isogenies $(\widetilde{\psi}_1, \widetilde{\psi}_2, \widetilde{\psi}_3, \widetilde{\psi}_4)$ corresponding to the basis (b_1, b_2, b_3, b_4) on the 2^a -torsion.

The conversion from ideals to isogenies is a problem that has been widely studied in isogeny-based cryptography (*e.g.* [KLPT14, DKL⁺20, DLLW23, EPSV23]). The fastest method to date is described in [BSE⁺25]. We remark that, while a straightforward application to the algorithm in [BSE⁺25] to our case is not applicable as we use a prime p of a different form, it is still possible to adapt it to our use case; namely, this will involve computing chains of (3,3)-isogenies and potentially a (5,5)-isogeny. Hence, we can assume that we can evaluate the isogenies $(\widetilde{\psi}_1, \widetilde{\psi}_2, \widetilde{\psi}_3, \widetilde{\psi}_4)$ on the 2^a -torsion efficiently.

Our next steps is to find a basis of $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$. Using discrete logarithm computations and solving a system of linear equations as in [FKMT22, Algorithm 2], we can find a basis $(\psi_1, \psi_2, \psi_3, \psi_4)$ of $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$, where the ψ_i 's are isogeny representations.

As shown in [Sil09, Ch. III, Corollary 6.3], the lattice $\text{Hom}(E_0, E)$ is endowed with a positive definite quadratic form given by the degree map. This means that also $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$ inherits such a quadratic form. Given the basis $(\psi_1, \psi_2, \psi_3, \psi_4)$, using the LLL reduction [LLL82], we find the shortest isogeny $\varphi \in \text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$. We define d to be its degree; in particular $d \lesssim 2^a \sqrt{p}$. Given the isogeny representation of φ , we compute $\alpha, \beta \in \mathbb{Z}/2^a\mathbb{Z}$ such that $([\alpha]P, [\beta]Q) = (\varphi(P_0), \varphi(Q_0))$.

As we mentioned earlier, in our construction, we will need to represent $\varphi_2 \circ \varphi \circ \widehat{\varphi}_1 : E_1 \rightarrow E_2$ for some isogenies φ_1 and φ_2 such that $\deg(\varphi_1) = 3^b$ and $\deg(\varphi_2) = 5^c$. In practice, this means that we need $2^{2a} - d 3^b 5^c$ to be a sum of two squares; this will allow us to represent the isogeny in dimension four. To have a sum of two squares, it would be desirable that $2^{2a} - d 3^b 5^c \equiv 1 \pmod{4}$. Additionally, for computational reasons, we need that $2^{2a} - d 3^b 5^c$ is a prime. Clearly, this cannot happen if d is even, so we first need to remove the even part of d .

Define n to be the largest integer such that 2^n divides n and call $d' = d/2^n$. Now, we need to check that $2^{2a-n} - d'3^b5^c = 1 \pmod{4}$. If $d'3^b5^c = 1 \pmod{4}$, it is not possible that $2^{2a-n} - d'3^b5^c = 1 \pmod{4}$. If this is the case, we need to compute an additional 7-isogeny $\chi : E_2 \rightarrow E'_2$, so we need to represent the isogeny $\chi \circ \varphi_2 \circ \varphi \circ \hat{\varphi}_1 : E_1 \rightarrow E_2 \rightarrow E'_2$. In particular, we will have that $2^{2a-n} - d'3^b5^c7 = 1 \pmod{4}$. If $2^{2a-n} - (d/2^n)3^b5^c = 3 \pmod{4}$, we will redefine d' to be equal to $7d/2^n$.

It can still happen that $2^{2a-n} - d'3^b5^c$ is not a prime. If this is the case, we check for increasing odd s when $2^{2a-n} - d'3^b5^c s^2$ is a prime. From the isogeny perspective, this means that we need to compose the isogeny we need to represent with a scalar multiplication by s . This does not incur any additional cost.

We assume that the quantity $2^{2a-n} - d'3^b5^c s^2$ behaves as a random integer, so we need to be able to try $\approx 2a$ times before we find a prime; this is due to the prime number theorem. Since we are only trying to add squares to the quantity $d'3^b5^c s^2$, we need that $2^{2a-n} > d'3^b5^c (2a)^2$. Since we do not have control on the exponent n and the fact that we may need an additional factor 7, we will assume that $2^{2a} > 2^a 3^b 5^c \sqrt{p} (2a)^3$.

Remark 4.4. We have tried to simulate experimentally this case with protocol-compatible parameters and did not observe any failure case. We leave a more thorough analysis of potential failures as future work.

It is also worth noting that we adopted a conservative approach: we decided to only use the shortest isogeny to avoid estimating the probability of encountering lattices $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$ where we only have one short isogeny and the remaining ones are all longer than the bound $2^a \sqrt{p}$. It is possible to combine the approach where we have more than one short-enough, basis generator with what we have explained to obtain a simpler algorithm at the cost of relying on some heuristics.

We collect what we have described so far in Algorithm 1, which we will denote as ConstrainedIsogeny in the rest of the paper.

Remark 4.5. Concurrent (and yet unpublished) work [BFLS26] has proposed and implemented techniques to compute isogenies with torsion constraints, including the constraints from Definition 4.1. While the concurrent work targets different applications than an UPKE, it shows the practical viability of these techniques, thus strengthening our claims of practicality.

4.3 Updatable FESTA

In this section, we describe a variant of FESTA we will use to introduce an Updatable public-key encryption. This version builds on top of the four-dimensional version of FESTA briefly described in [BMP23, §6.3] together with the PKE construction in [NO24, §3.2]. Our goal is to obtain uniformly distributed public keys.

Algorithm 1 ConstrainedIsogeny($E, \text{End}(E)$)**Require:** An elliptic curve E together with its endomorphism ring $\text{End}(E)$.**Ensure:** A tuple (α, β, d, s) describing an isogeny $\varphi : E_0 \rightarrow E$ such that $\deg(\varphi) = d$, $([\alpha]P, [\beta]Q) = (\varphi(P_0), \varphi(Q_0))$, the quantity $2^{2a-n} - d'3^b5^c s^2$ is prime $\equiv 1 \pmod{4}$, where

$$d' = \begin{cases} d/2^n & \text{if } ((d/2^n)3^b5^c) \equiv 3 \pmod{4} \\ 7d/2^n & \text{otherwise} \end{cases} \quad \text{and } n \text{ is the largest integer such that } 2^n \text{ divides } d, \text{ and} \\ (P, Q) \leftarrow \text{Gen2Tor}(E).$$

- 1: Compute a basis $(\widetilde{\psi}_1, \widetilde{\psi}_2, \widetilde{\psi}_3, \widetilde{\psi}_4)$ of $\text{Hom}(E_0, E)$.
- 2: Evaluate the basis $(\widetilde{\psi}_1, \widetilde{\psi}_2, \widetilde{\psi}_3, \widetilde{\psi}_4)$ on the 2^a -torsion.
- 3: Compute a basis of $(\psi_1, \psi_2, \psi_3, \psi_4)$ of $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$.
- 4: Find the shortest isogeny φ in $\text{Hom}(E_0, E)_{(P_0, Q_0), (P, Q)}$.
- 5: $d \leftarrow \deg(\varphi)$.
- 6: Compute the largest n such that $2^n \mid d$.
- 7: **if** $(d/2^n)3^b5^c \equiv 3 \pmod{4}$ **then**
- 8: $d' \leftarrow d/2^n$.
- 9: **else**
- 10: $d' \leftarrow 7d/2^n$.
- 11: Find a small s such that $2^{2a-n} - d'3^b5^c s^2$ is prime.
- 12: **return** (α, β, d, s) .

4.3.1 Parameters

We now fix the parameters we will use in the rest of the paper. Our base prime p is of the form $p = f \cdot 2^a \cdot 3^b \cdot 5 - 1$. With respect to the notations in Section 2.3.7, $3^b \approx 2^{2\lambda}$ ($b := \lceil 2\lambda / \log_3(2) \rceil$) will be the degree of φ_{Enc_1} and $5^c \approx 2^{2\lambda}$ ($c := \lceil 2\lambda / \log_2(5) \rceil$) will be the degree of φ_{Enc_2} . The (long) secret isogeny and the update isogeny will both have degree 3^δ where $\delta := \lceil \frac{15}{4} \log_3(p) \rceil$. There are two motivations for this choice of δ . The first is that the public keys are uniformly distributed in the supersingular isogeny graph (see Section 4.1). The second, which will be useful in our security proof, is that given two elliptic curves with known endomorphism ring, one can link them by an isogeny of such degree using KLPT (see Section 4.1).

As in FESTA, during decryption, we will have to compute the isogeny $\varphi_{\text{Dec}} := \varphi_{\text{Enc}_2} \circ \varphi_{\text{sk}} \circ \widehat{\varphi}_{\text{Enc}_1} : E_1 \rightarrow E_2$. We will use the 2^a -torsion to represent φ_{Dec} . Nevertheless, this time we will go for a dimension 4 representation which only requires $2^{2a} > \deg(\varphi_{\text{Dec}})$, as opposed to the dimension 2 representation which requires $2^a > \deg(\varphi_{\text{Dec}})$. Combining this with the explanation in Section 4.2.2, we obtain

$$2^{2a} > \deg(\varphi_{\text{Dec}})(2a)^3 \approx 2^a 3^b 5^c \sqrt{p}(2a)^3 \approx 2^{3a/2} 2^{5\lambda} (2a)^3,$$

Hence, we require $a - 6\log(2a) > 10\lambda$. We will make use of a hash function $H_{\text{up}} : \{0, 1\}^* \rightarrow [0, 4) \times [0, 3)^{\delta-1}$ modeled as a random oracle.

The starting curve is the supersingular elliptic curve $E_0 : y^2 = x^3 + x$ whose endomorphism ring is given by all the integral combinations of the endomorphisms $[1]$, ι , $\frac{\iota + \pi}{2}$ and $\frac{1 + \iota\pi}{2}$ (see

Section 2.3.2). We also fix (P_0, Q_0) to be the output of $\text{Gen2Tor}(E_0)$ and (R_3, S_3) to be a basis^{II} of $E_0[3^b]$.

Remark 4.6. From the previous estimates on the exponent a , we have that $p > 2^{12\lambda}$. We can now have a more precise estimate of the statistical distance between the distribution of the public keys in our UPKE and the uniform distribution on the supersingular isogeny graph. From Section 4.1, this distance is roughly

$$2^{-\frac{1}{2}(15/4-1)\log_2(p)} < 2^{-33/2\lambda} = 2^{-16.5\lambda}.$$

While this is unnecessarily small with regards to public-key distribution, it is useful in our security proof.

4.3.2 Key Generation

The first step in KeyGen is to generate a long isogeny $\tilde{\varphi}_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ originating from E_0 ; such an isogeny is generated by uniformly sampling ρ_{sk} from $\{0, 1\}^{2\lambda}$ and then applying $\text{IsogenyWalk}_3(E_0, \delta, H_{\text{up}}(\rho_{\text{sk}}))$. It is worth noting that, given the length of $r_{\text{sk}} = H_{\text{up}}(\rho_{\text{sk}})$, the curve E_{pk} is uniformly distributed in the supersingular isogeny graph.

Once the isogeny $\tilde{\varphi}_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ has been computed, we retrieve $\text{End}(E_{\text{pk}})$ via IsogenyToEnd . Leveraging the knowledge of $\text{End}(E_{\text{pk}})$, we compute the shorter secret isogeny $\varphi_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ by applying $\text{ConstrainedIsogeny}$. In particular, we obtain $(\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}})$, where $d_{\text{sk}} = \deg(\varphi_{\text{sk}})$, $([\alpha_{\text{sk}}]P_{\text{pk}}, [\beta_{\text{sk}}]Q_{\text{pk}}) = (\varphi_{\text{sk}}(P_0), \varphi_{\text{sk}}(Q_0))$ and $(P_{\text{pk}}, Q_{\text{pk}}) \leftarrow \text{Gen2Tor}(E_{\text{pk}})$.

The public key pk is set to be E_{pk} , whereas, the secret sk is set to be $(\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, \text{End}(E_{\text{pk}}))$. We summarize the procedure in Algorithm 2.

Algorithm 2 KeyGen

Require: The Setup information described in Section 4.3.1.

Ensure: The public key $\text{pk} = E_{\text{pk}}$ and the secret key $\text{sk} = (\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, \text{End}(E_{\text{pk}}))$.

- 1: $\rho_{\text{sk}} \leftarrow_{\$} \{0, 1\}^{2\lambda}$.
 - 2: $r_{\text{sk}} \leftarrow H_{\text{up}}(\rho_{\text{sk}})$.
 - 3: $E_{\text{pk}}, \tilde{\varphi}_{\text{sk}} \leftarrow \text{IsogenyWalk}_3(E_0, \delta, r_{\text{sk}})$.
 - 4: $\text{End}(E_{\text{pk}}) \leftarrow \text{IsogenyToEnd}(\text{End}(E_0), \tilde{\varphi}_{\text{sk}})$.
 - 5: $(\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}}) \leftarrow \text{ConstrainedIsogeny}(E_{\text{pk}}, \text{End}(E_{\text{pk}}))$.
 - 6: $\text{pk} \leftarrow E_{\text{pk}}$.
 - 7: $\text{sk} \leftarrow (\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}}, \text{End}(E_{\text{pk}}))$.
 - 8: **return** pk, sk .
-

^{II}The basis (P_0, Q_0) must not satisfy the conditions described in [CV23], i.e., there should not exist an endomorphism $\theta \in \text{End}(E_0)$ of small norm such that either P_0 or Q_0 is an eigenvector for θ . As explained in [CV23, §5.3], this happens with negligible probability when Gen2Tor is used.

4.3.3 Encryption

Following the PKE construction in [NO24, §3.2], we encode messages in the scalars used to mask torsion information: we introduce the algorithm `MexToScalars` that converts messages into scalars $(\bmod 2^a)$. A message $m \in \mathcal{M} = [0, 2^{a-2})$ is encoded into an odd scalar $\alpha = 2 \cdot m + 1$. On input m , the function `MexToScalars` returns (α, β) , where $\alpha = 2 \cdot m + 1$ and $\beta = \alpha^{-1} \pmod{2^a}$.

Similarly to the evaluation procedure in Section 2.3.7, during encryption we compute two isogenies $\varphi_{\text{Enc}_1} : E_0 \rightarrow E_1$ and $\varphi_{\text{Enc}_2} : E_{\text{pk}} \rightarrow E_2$; for efficiency reason $\deg(\varphi_{\text{Enc}_1}) = 3^b$ and $\deg(\varphi_{\text{Enc}_2}) = 5^c$. Since during decryption we will need to recover $\ker(\varphi_{\text{Enc}_1})$, we must have that $E_0[3^b]$ is defined over $\mathbb{F}_{p^2}$.

To compute $\varphi_{\text{Enc}_1} : E_0 \rightarrow E_1$, we sample a nonce $r_3 \leftarrow_{\$} [0, 3^b)$ and compute the isogeny with kernel $\langle R_3 + [r_3]S_3 \rangle$, where (R_3, S_3) is the basis of $E_0[3^b]$ we have fixed. We then define $(R_1, S_1) = ([\alpha_{\text{eval}}]\varphi_{\text{Enc}_1}(P_0), [\beta_{\text{eval}}]\varphi_{\text{Enc}_1}(Q_0))$.

Since we do not need to recover $\ker(\varphi_{\text{Enc}_2})$ during decryption, we compute $\varphi_{\text{Enc}_2} : E_{\text{pk}} \rightarrow E_2$ as a chain of c 5-isogenies. This is done by sampling r_5 uniformly at random from $[0, 6) \times [0, 5)^{c-1}$ and then applying `IsogenyWalk5`(E_{pk}, c, r_5). Similarly, we define $(R_2, S_2) = ([\alpha_{\text{eval}}]\varphi_{\text{Enc}_2}(P_{\text{pk}}), [\beta_{\text{eval}}]\varphi_{\text{Enc}_2}(Q_{\text{pk}}))$. The final ciphertext is then given as $\text{ct} = (E_1, R_1, S_1, E_2, R_2, S_2)$.

We describe `Enc` in Algorithm 3.

Algorithm 3 `Enc(pk, m)`

Require: A message $m \in \mathcal{M} = [0, 2^{a-2})$ and a public key $\text{pk} = E_{\text{pk}}$.

Ensure: A ciphertext $\text{ct} = (E_1, R_1, S_1, E_2, R_2, S_2)$.

- 1: $r_3 \leftarrow_{\$} [0, 3^b)$.
 - 2: $\alpha_{\text{eval}}, \beta_{\text{eval}} \leftarrow \text{MexToScalars}(m)$.
 - 3: $(P_{\text{pk}}, Q_{\text{pk}}) \leftarrow \text{Gen2Tor}(E_{\text{pk}})$.
 - 4: Compute the isogeny $\varphi_{\text{Enc}_1} : E_0 \rightarrow E_1$ with kernel $\langle R_3 + [r_3]S_3 \rangle$.
 - 5: $(R_1, S_1) \leftarrow ([\alpha_{\text{eval}}]\varphi_{\text{Enc}_1}(P_0), [\beta_{\text{eval}}]\varphi_{\text{Enc}_1}(Q_0))$.
 - 6: $r_5 \leftarrow_{\$} [0, 6) \times [0, 5)^{c-1}$.
 - 7: $E_2, \varphi_{\text{Enc}_2} \leftarrow \text{IsogenyWalk}_5(E_{\text{pk}}, c, r_5)$.
 - 8: $(R_2, S_2) \leftarrow ([\alpha_{\text{eval}}]\varphi_{\text{Enc}_2}(P_{\text{pk}}), [\beta_{\text{eval}}]\varphi_{\text{Enc}_2}(Q_{\text{pk}}))$.
 - 9: **return** $\text{ct} = (E_1, R_1, S_1, E_2, R_2, S_2)$.
-

4.3.4 Decryption

To decrypt ciphertexts, we first define a decoding function for `MexToScalars`; we name it `ScalarsToMex`. The algorithm `ScalarsToMex` recovers a message m from a tuple (α, β) . The first step is to check that $0 < \alpha, \beta < 2^a$ and that α and β are odd; if $\alpha > 2^{a-1}$, then the tuple

(α, β) is redefined to be equal to $(2^a - \alpha, 2^a - \beta)$. Then, the algorithm `ScalarsToMex` proceeds to check that $\alpha \cdot \beta = 1 \pmod{2^a}$ and outputs $m = \frac{\alpha-1}{2}$; if $\alpha \cdot \beta \neq 1 \pmod{2^a}$, it returns $\perp$.

Remark 4.7. The algorithm `ScalarsToMex` normalizes the scalars (α, β) so that $\alpha < 2^{a-1}$. This step is necessary because, during decryption, we will be able to recover the encryption isogeny φ_{Enc_1} up to automorphisms on the codomain curve. Unless the codomain of φ_{Enc_1} is the curve with j -invariant $\in \{0, 1728\}$, which happens with negligible probability, the only automorphisms on the codomain curve are $\pm \text{id}$.

We are now ready to introduce the decryption algorithm `Dec` (Algorithm 5). We recall that from a ciphertext $(E_1, R_1, S_1, E_2, R_2, S_2)$, we have the evaluation of the isogeny $\varphi_{\text{Dec}} := \varphi_{\text{Enc}_2} \circ \varphi_{\text{sk}} \circ \widehat{\varphi}_{\text{Enc}_1} : E_1 \rightarrow E_2$ on $E_1[2^a]$, namely

$$\varphi_{\text{Dec}} \begin{pmatrix} R_1 \\ S_1 \end{pmatrix} = [3^b] \begin{pmatrix} [\alpha_{\text{sk}}] R_2 \\ [\beta_{\text{sk}}] S_2 \end{pmatrix}.$$

Given the evaluation of φ_{Dec} on $E_1[2^a]$ and a secret $(\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}})$, we explain how to extract $\widehat{\varphi}_{\text{Enc}_1} : E_0 \rightarrow E_1$ in `Extract3b` (see Algorithm 4). We first factor out the even degree components of φ_{Dec} . Define $e_\alpha := \max\{n \mid 2^n \text{ divides } \alpha\}$ and $e_\beta := \max\{n \mid 2^n \text{ divides } \beta\}$, and take $e := \min\{e_\alpha, e_\beta\}$. This means that $\varphi_{\text{Dec}} = [2^e] \varphi'_{\text{Dec}}$.

If $e_\alpha \neq e_\beta$, this means that $\deg(\varphi'_{\text{Dec}})$ is even. We now try to factor φ'_{Dec} as $\varphi'_{\text{Dec}} = \varphi'_{\text{Dec,even}} \circ \varphi'_{\text{Dec,odd}}$. To do so, we notice the following relation

$$[2^{\max\{e_\alpha, e_\beta\}}] \varphi'_{\text{Dec,odd}} \begin{pmatrix} R_1 \\ S_1 \end{pmatrix} = \widehat{\varphi}'_{\text{Dec,even}} \begin{pmatrix} [\alpha_{\text{sk}}] R_2 \\ [\beta_{\text{sk}}] S_2 \end{pmatrix}.$$

In particular, if $e_\alpha > e_\beta$, we have that $\ker(\widehat{\varphi}'_{\text{Dec,even}}) = \langle [2^e] S_2 \rangle$; otherwise $\ker(\widehat{\varphi}'_{\text{Dec,even}}) = \langle [2^e] R_2 \rangle$. We now have a representation of an odd-degree isogeny, namely the HD representation of $\varphi'_{\text{Dec,odd}}$, this follows from the discussion in Section 4.2.2.

From the HD representation of $\varphi'_{\text{Dec,odd}}$, we compute n to be the largest power of 3 such that $[3^n]$ divides $\varphi'_{\text{Dec,odd}}$. If $n = 0$, then $\ker(\widehat{\varphi}_{\text{Enc}_1}) = \ker(\varphi_{\text{Dec}})[3^b]$, which directly gives a representation of $\widehat{\varphi}_{\text{Enc}_1}$ since $E_0[3^b]$ is defined over $\mathbb{F}_{p^2}$. Otherwise, from $(\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}})$, we compute $\mu : E_0 \rightarrow \widetilde{E}$ such that $\varphi_{\text{sk}} = \widetilde{\mu} \circ \mu$ and $\deg(\mu) = 3^n$. Then $\widehat{\varphi}_{\text{Enc}_1} = \widehat{\mu} \circ \widetilde{\varphi}_1$, where $\ker(\widetilde{\varphi}_1) = \ker(\varphi'_{\text{Dec,odd}})[3^{b-n}]$.

Finally, from $\widehat{\varphi}_{\text{Enc}_1}$ we recover the scalars (α, β) such that

$$(R_1, S_1) = ([\alpha] \varphi_{\text{Enc}_1}(P_0), [\beta] \varphi_{\text{Enc}_1}(Q_0));$$

such scalars encode the encrypted message.

Algorithm 4 $\text{Extract}_{3^b}(\varphi, (\alpha, \beta, d, s))$

Require: An HD representation on $\psi = \varphi_2 \circ \varphi \circ \varphi_1 : E_1 \rightarrow E_0 \rightarrow E \rightarrow E_2$ of degree $3^b 5^c d$, where $\deg(\varphi_1) = 3^b$, $\deg(\varphi) = d$ and $\deg(\varphi_2) = 5^c$, a tuple (α, β, d, s) describing a representation of $\varphi : E_0 \rightarrow E$, where $\deg(\varphi) = d$ and s is the auxiliary quantity defined in ConstrainedIsogeny (Algorithm 1).

Ensure: A representation of $\varphi_1 : E_1 \rightarrow E_0$.

- 1: $(R_1, S_1) \leftarrow \text{Gen2Tor}(E_1)$.
- 2: $(R_2, S_2) \leftarrow (\psi(R_1), \psi(S_1))$.
- 3: $e_\alpha, e_\beta \leftarrow \max\{n \mid 2^n \text{ divides } \alpha\}, \max\{n \mid 2^n \text{ divides } \beta\}$.
- 4: $e \leftarrow \min\{e_\alpha, e_\beta\}$.
- 5: **if** $e_\alpha > e_\beta$ **then**
- 6: Compute $\chi : E_2 \rightarrow E'_2$ with kernel $\langle [2^e] S_2 \rangle$.
- 7: **if** $e_\alpha < e_\beta$ **then**
- 8: Compute $\chi : E_2 \rightarrow E'_2$ with kernel $\langle [2^e] R_2 \rangle$.
- 9: $(R_2, S_2) \leftarrow (\chi(R_2), \chi(S_2))$.
- 10: Redefine $\psi : E_1 \rightarrow E'_2$ as the odd part of ψ . $\triangleright \psi = \hat{\chi} \circ \psi_{\text{odd}}$
- 11: $n \leftarrow \max\{s \mid [3^s] \text{ divides } \psi\}$.
- 12: **if** $n = 0$ **then**
- 13: **return** a representation of the isogeny with kernel $\ker(\psi)[3^b]$.
- 14: From (α, β, d, s) , compute μ such that $\varphi = \tilde{\mu} \circ \mu : E_0 \rightarrow \tilde{E} \rightarrow E$ and $\deg(\mu) = 3^n$.
- 15: Compute the isogeny $\tau : E_1 \rightarrow \tilde{E}$ with kernel $\ker(\psi)[3^{b-n}]$.
- 16: **return** A representation of $\hat{\mu} \circ \tau$.

Algorithm 5 $\text{Dec}(\text{sk}, \text{ct})$

Require: A ciphertext $\text{ct} = (E_1, R_1, S_1, E_2, R_2, S_2)$ and a secret key $\text{sk} = (\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}}, \text{End}(E_{\text{pk}}))$.

Ensure: A message m .

- 1: $(\tilde{R}_2, \tilde{S}_2) \leftarrow ([\alpha_{\text{sk}}^{-1}] R_2, [\beta_{\text{sk}}^{-1}] S_2)$.
- 2: Using $(R_1, S_1, \tilde{R}_2, \tilde{S}_2)$ compute the representation of an isogeny $\psi : E_1 \rightarrow E_2$ of degree $d_{\text{sk}} 3^b 5^c$.
- 3: $\tilde{\varphi}_1 : E_1 \rightarrow E \leftarrow \text{Extract}_{3^b}(\psi, (\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}}))$.
- 4: Check that E is isomorphic to E_0 ; if not abort.
- 5: Define $\varphi : E_0 \rightarrow E_1$ to be the dual of $\tilde{\varphi}_1 : E_1 \rightarrow E_0$.
- 6: Compute the scalars α, β such that $[\alpha]\varphi(P_0) = R_1$ and $[\beta]\varphi(Q_0) = S_1$.
- 7: **return** $\text{ScalarsToMex}(\alpha, \beta)$.

4.3.5 Public-key update

To update a public key pk , we simply compute a long isogeny $\varphi_{\text{up}} : E_{\text{pk}} \rightarrow E_{\text{pk}'}$; the updated public key is then given by $\text{pk}' = E_{\text{pk}'}$. The degree of φ_{up} is greater than $p \cdot 2^{2\lambda}$, so that the distribution of $E_{\text{pk}'}$ is at negligible distance from the uniform distribution. This is achieved by feeding some randomness $\rho \in \{0, 1\}^{2\lambda}$ to UpdPk (Algorithm 6) that is then digested by H_{up} , together with E_{pk} , to obtain an integer $r \in [0, 3^\delta)$, determining the isogeny walk $\text{IsogenyWalk}_3(E_{\text{pk}}, \delta, r)$. Finally, the update token up is obtained by encrypting the randomness ρ using Enc .

Algorithm 6 UpdPk(pk, ρ)

Require: Current public key $\text{pk} = E_{\text{pk}}$ and the randomness $\rho \in \{0, 1\}^{2\lambda}$.

Ensure: Updated public key $\text{pk}' = E_{\text{pk}'}$ and the update token up .

- 1: $r \leftarrow H_{\text{up}}(\rho || E_{\text{pk}})$.
- 2: $E_{\text{pk}', -} \leftarrow \text{IsogenyWalk}_3(E_{\text{pk}}, \delta, r)$.
- 3: $\text{up} \leftarrow \text{Enc}(\text{pk}, \rho)$.
- 4: **return** $E_{\text{pk}'}, \text{up}$.

4.3.6 Secret-key update

Given a secret key $\text{sk} = (\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s_{\text{sk}}, \text{End}(E_{\text{pk}}))$ and an update token up , we explain how to update sk in UpdSk (Algorithm 7). The first step is to recover the randomness used to update the public key; to do so, we compute $\tilde{\rho} \leftarrow \text{Dec}(\text{up}, \text{sk})$. This $\tilde{\rho}$ is then digested by H_{up} , which returns a scalar r determining the isogeny walk $\varphi_{\text{up}} : E_{\text{pk}} \rightarrow E'_{\text{pk}}$ via IsogenyWalk. Note that while the public key is required to be hashed by H_{up} , we do not give it as an explicit input to the algorithm as it can be recovered from the secret key.

Given the isogeny φ_{up} , we recover $\text{End}(E'_{\text{pk}})$ via IsogenyToEnd. Finally, we recover an isogeny representation $(\alpha'_{\text{sk}}, \beta'_{\text{sk}}, d'_{\text{sk}}, s'_{\text{sk}})$ of the updated secret isogeny $\varphi'_{\text{sk}} : E_0 \rightarrow E'_{\text{pk}}$ via ConstrainedIsogeny. The updated secret key is given as $(\alpha'_{\text{sk}}, \beta'_{\text{sk}}, d'_{\text{sk}}, s'_{\text{sk}}, \text{End}(E'_{\text{pk}}))$.

Algorithm 7 UpdSk(sk, up)

Require: Current secret key $\text{sk} = (\alpha_{\text{sk}}, \beta_{\text{sk}}, d_{\text{sk}}, s'_{\text{sk}}, \text{End}(E_{\text{pk}}))$, the updated public key $\text{pk}' = E_{\text{pk}'}$ and the update token up .

Ensure: Updated secret key $\text{sk}' = (\alpha'_{\text{sk}}, \beta'_{\text{sk}}, d'_{\text{sk}}, s'_{\text{sk}}, \text{End}(E_{\text{pk}'}))$

- 1: $\tilde{\rho} \leftarrow \text{Dec}(\text{up}, \text{sk})$.
- 2: $\tilde{r} \leftarrow H_{\text{up}}(\tilde{\rho}, E_{\text{pk}})$.
- 3: $-, \varphi_{\text{up}} \leftarrow \text{IsogenyWalk}_3(E_{\text{pk}}, \delta, \tilde{r})$.
- 4: $\text{End}(E_{\text{pk}'}) \leftarrow \text{IsogenyToEnd}(\text{End}(E_{\text{pk}}), \varphi_{\text{up}})$
- 5: $(\alpha'_{\text{sk}}, \beta'_{\text{sk}}, d'_{\text{sk}}, s'_{\text{sk}}) \leftarrow \text{ConstrainedIsogeny}(E_{\text{pk}'}, \text{End}(E_{\text{pk}}))$.
- 6: $\text{sk}' \leftarrow (\alpha'_{\text{sk}}, \beta'_{\text{sk}}, d'_{\text{sk}}, s'_{\text{sk}}, \text{End}(E_{\text{pk}'}))$.
- 7: **return** sk' .

4.4 Security

In this section, we will prove the security of our UPKE, which we refer to as Updatable FESTA.

To this extent, we start by proving a few preliminary results. Namely, we show in Proposition 4.9 that FESTA-PKE, the PKE obtained using Algorithm 2 as KeyGen, Algorithm 3 as Enc and Algorithm 5 as Dec is OW-CPA. We then move to Updatable FESTA, proving its correctness in Proposition 4.11, γ -spreadness property in Theorem 4.12 and its OW-CR-CPA security in Theorem 4.14. Putting these results together, we apply a transform from [AW25, §4.3; 4.4] to get Theorem 4.15 that provides an IND-CU-CCA UPKE.

4.4.1 OW-CPA security of FESTA-PKE

To prove that FESTA-PKE is OW-CPA, we recall the CIST^2 problem from [BMP23, Problem 8], tailoring it to our use case.

Problem 4.8 (CIST^2). Let $E_0 : y^2 = x^3 + x$ be a supersingular elliptic curve defined over $\mathbb{F}_{p^2}$, and let E'_0 be a random supersingular elliptic curve defined over the same field. Consider two isogenies $\varphi : E_0 \rightarrow E_1$ and $\varphi' : E'_0 \rightarrow E'_1$ of smooth degrees 3^b and 5^c , respectively, and let $\alpha, \beta \in (\mathbb{Z}/N\mathbb{Z})^*$. Given the curves E_0, E_1, E'_0, E'_1 and the points

$$(P_1, Q_1) = ([\alpha]\varphi(P_0), [\beta]\varphi(Q_0)) \quad \text{and} \quad (P'_1, Q'_1) = ([\alpha]\varphi'(P'_0), [\beta]\varphi'(Q'_0)),$$

where $(P_0, Q_0) \leftarrow \text{Gen2Tor}(E_0)$ and $(P'_0, Q'_0) \leftarrow \text{Gen2Tor}(E'_0)$, compute some representations for the isogenies φ and φ' .

Proposition 4.9. Under the CIST^2 assumption, FESTA-PKE is OW-CPA.

Proof. Let $\mathcal{A}$ be an adversary against OW-CPA of FESTA-PKE, and consider an adversary $\mathcal{B}$ against CIST^2 , with parameters $d = 3^b$, $d' = 5^c$ and $N = 2^a$.

On input a tuple $(E_0, E_1, E'_0, E'_1, P_1, Q_1, P'_1, Q'_1)$, the adversary $\mathcal{B}$ sets $\text{pk} = E'_0$. We highlight that, thanks to the use of isogeny walks of length δ guaranteeing uniform distribution of public curves (Section 4.3.1), the curve E'_0 is indeed an admissible public key.

Then, the adversary $\mathcal{B}$ defines the challenge ciphertext to be

$$c^* = (E_1, P_1, Q_1, E'_1, P'_1, Q'_1),$$

which has the correct distribution by direct construction, and calls $\mathcal{A}(\text{pk}, c^*)$. The adversary $\mathcal{B}$ finally collects the response m^* .

To recover φ, φ' , $\mathcal{B}$ runs $\text{MexToScalars}(m^*) = (\alpha^*, \beta^*)$, which it can use to unmask $(P_1, Q_1), (P'_1, Q'_1)$ to get $\varphi(P_0), \varphi(Q_0)$, and $\varphi'(P'_0), \varphi'(Q'_0)$. This information gives HD representations for φ, φ' , hence breaking CIST^2 . $\square$

Remark 4.10. Note that unlike FESTA [BMP23] and QFESTA [NO24], we solely rely on the CIST^2 to prove the one-way security of FESTA-PKE and do not need the additional DIST assumption.

The DIST assumption was assumed to argue that public keys are indistinguishable from random elliptic curves. In our construction, public keys are truly uniform, hence strengthen our security claims.

It is also worth noting that given a public curve E_{pk} , the basis $(P_{\text{pk}}, Q_{\text{pk}})$ is not necessarily the image of $\langle P_0 \rangle$ and $\langle Q_0 \rangle$ under the isogeny generated by IsogenyWalk_3 . Concretely, the degree of the isogeny matching the constraint $(\alpha_{\text{sk}}, \beta_{\text{sk}})$ is secret. This removes important information that was showed to be crucial to define weak instances [DFP24, CV23].

4.4.2 Correctness of Updatable FESTA

Proposition 4.11 (Correctness). *The UPKE described in Section 4.3 is correct.*

Proof. First, the correctness of FESTA-PKE follows directly from the correctness of FESTA ([BMP23, Section 3]), the discussion in Sections 4.3.2 to 4.3.4, and our choice of parameters (Section 4.3.1).

Second, we need to show that the correctness of description is preserved by updates. This comes as a direct consequence of the correctness of the subroutines used in UpdSk (Algorithm 7). Given a valid key pair (sk_i, pk_i) , and an update randomness $\rho_i \xleftarrow{\$} \{0, 1\}^{2\lambda}$, UpdPk sends $pk_i = E_{pk_i}$ to a new curve $pk_{i+1} = E_{pk_{i+1}}$ by performing a walk of length δ in the 3-isogeny graph described by $r_i \leftarrow H_{up}(\rho_i)$. By our choice of δ , the distribution of resulting curve will be close to uniform. Then given an honestly update key $sk_i = (\alpha_i, \beta_i, d_i, s_i, \text{End}(E_{pk_i}))$, and update token $up_i = \text{Enc}(\rho_i, pk_i)$, one first computes $\text{Dec}(up_i, sk_i)$. By the correctness of decryption for (sk_i, pk_i) this is exactly ρ_i . Then one recomputes $E_{pk_{i+1}}$ by performing the same isogeny walk φ_{up_i} described by $r_i \leftarrow H_{up}(\rho_i)$, and stores the corresponding isogeny, calls IsogenyToEnd on input $(\text{End}(pk_i), \varphi_{up_i})$ to get $\text{End}(E_{pk_{i+1}})$ and finally calls ConstrainedIsogeny to obtain $(\alpha_{i+1}, \beta_{i+1}, d_{i+1}, s_{i+1})$. By the discussion of Section 4.2.2, ConstrainedIsogeny is correct which ensures that $sk_{i+1} = (\alpha_{i+1}, \beta_{i+1}, d_{i+1}, s_{i+1}, \text{End}(E_{pk_{i+1}}))$ is well-formed. Thus (sk_{i+1}, pk_{i+1}) is a valid key pair, and by the correctness of decryption the statement is proven. $\square$

4.4.3 OW-CR-CPA security and γ spreadness of Updatable FESTA

Theorem 4.12 (γ -spreadness). *The Updatable-FESTA UPKE scheme described in Section 4.3 satisfies the γ -spreadness property.*

Proof. Let $ct^* = (E_1^*, R_1^*, S_1^*, E_2^*, R_2^*, S_2^*)$ be any fixed ciphertext, and let $ct = (E_1, R_1, S_1, E_2, R_2, S_2)$ a random ciphertext returned by the encryption algorithm of our UPKE. Then

$$\Pr[ct = ct^*] \leq \Pr[E_1 = E_1^*, E_2 = E_2^*] = \Pr[E_1 = E_1^*] \cdot \Pr[E_2 = E_2^*],$$

where the last equality comes from the fact that E_1 and E_2 are independently generated in the encryption algorithm.

First, the curve E_1 is obtained by computing a 3^b -isogeny $\varphi_{\text{Enc}_1} : E_0 \rightarrow E_1$ of kernel $\langle R_3 + [r_3]S_3 \rangle$ where r_3 is sampled uniformly from $[0, 3^b)$ and (R_3, S_3) is a fixed basis of $E_0[3^b]$. As in [OAT20, Theorem 1], one can prove that $E_0 : y^2 = x^3 + x$ admits no endomorphism of degree 3^{2b} whenever $3^{2b} < \frac{p+1}{4}$, which is the case for our parameters. Hence, each choice of r_3 yields a distinct curve E_1 , implying that

$$\Pr[E_1 = E_1^*] = \frac{1}{3^b} = 2^{-2\lambda}.$$

Second, $\varphi_{\text{Enc}_2} : E_{pk} \rightarrow E_2$ is a uniformly random isogeny of degree 5^c from E_{pk} , and there are $6 \cdot 5^{c-1} = 5^c + 5^{c-1} > 2^{2\lambda}$ such isogenies. From [LB20], the number of supersingular curves having an endomorphism of degree 5^{2c} is bounded by $O(5^{3c})$. Since E_{pk} is uniformly distributed in the supersingular isogeny graph by construction, then the probability that E_{pk} admits an endomorphism of degree 5^{2c} is bounded by $\frac{12 \cdot 5^{3c}}{p} \approx \frac{2^{6\lambda}}{2^{12\lambda}} \approx 2^{-6\lambda}$. We hence obtain

$$\begin{aligned} \Pr[E_2 = E_2^*] &= \Pr[E_2 = E_2^*, \varphi_{\text{Enc}_2} = \varphi_{\text{Enc}_2}^*] + \Pr[E_2 = E_2^*, \varphi_{\text{Enc}_2} \neq \varphi_{\text{Enc}_2}^*] \\ &\leq \frac{1}{5^c + 5^{c-1}} + \Pr[\exists \theta \in \text{End}(E_{pk}), \deg(\theta) = 3^{2b}] \\ &\leq 2^{-2\lambda} + 2^{-6\lambda}. \end{aligned}$$

Finally, we obtain

$$\begin{aligned} \Pr[\text{ct} = \text{ct}^*] &\leq \Pr[E_1 = E_1^*] \cdot \Pr[E_2 = E_2^*] \\ &\leq 2^{-2\lambda} \cdot (2^{-2\lambda} + 2^{-6\lambda}) \\ &= 2^{-4\lambda} + 2^{-8\lambda}, \end{aligned}$$

which means that our UPKE is γ -spread with $\gamma \approx 4\lambda$. $\square$

Proposition 4.13 (Key indistinguishability). *In Updatable FESTA, freshly generated key pairs, i.e. key pairs generated directly as $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$ are statistically indistinguishable from updated key pairs i.e. key pairs obtained as $((\text{pk}', \text{up}), \text{sk}') \leftarrow (\text{UpdPk}(\text{pk}, \rho), \text{UpdSk}(\text{sk}, \text{up}))$ for some $\rho \xleftarrow{\$} \{0, 1\}^{2\lambda}$.*

Proof. First of all, in both cases the secret keys are obtained as a deterministic procedure of the public key, so we focus on the public key distribution. During key generation, the public key is generated as a random isogeny walk of length $\delta := \left\lceil \frac{15}{4} \log_3(p) \right\rceil$ in the 3-isogeny graph. As pointed out in Remark 4.6, by our choice of parameters this means that the public distribution is $O(2^{-16.5\lambda})$ close to the uniform distribution on the supersingular isogeny graph. By a similar argument, updated public keys are also generated as a walk of length δ in the 3-isogeny graph but starting from the previous public key. Nonetheless, the resulting distribution is equally close to the stationary one. Putting together these two results we get that the distributions of the fresh and updated public keys are $O(2 \cdot 2^{-16.5\lambda})$ close. $\square$

Theorem 4.14 (OW-CR-CPA security). *In the random oracle model, assuming the hardness of the CIST^2 problem, the Updatable-FESTA UPKE scheme described in Section 4.3 is OW-CR-CPA.*

Proof. We preliminary remark that, under the assumption that the CIST^2 is hard, we have already proved that FESTA-PKE is OW-CPA in Proposition 4.9. Moreover, by Proposition 4.13, we have that fresh keys and updated keys are statistically indistinguishable.

Now, consider G_0 to be the original OW-CR-CPA game. We model H_{up} as a random oracle. So the adversary has access to both UpdO and RO . We also explicitly unroll the UpdPk procedure within the oracle.

We write out G_0 explicitly in Figure 4.1

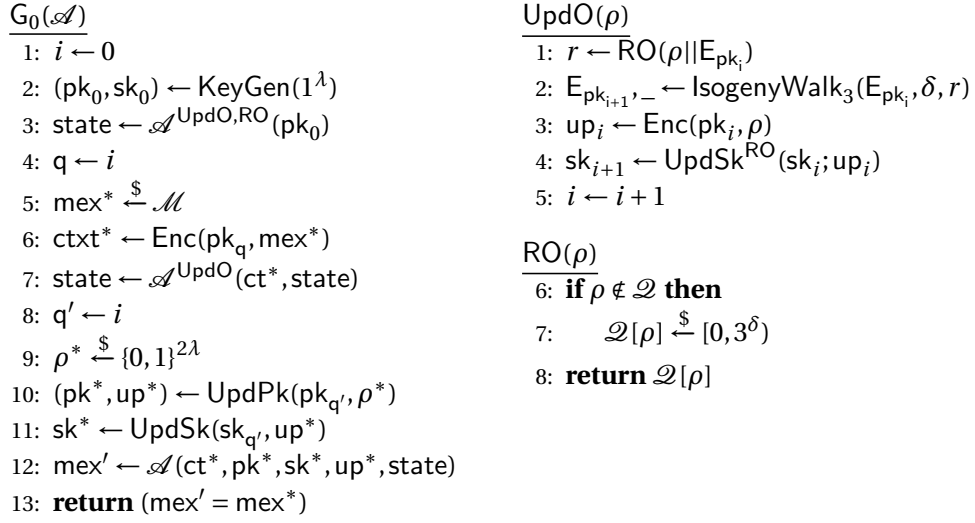


Figure 4.1: G_0

G_1 Our first goal is to reduce to a scenario where we can estimate the epoch q during which the adversary is going to make its challenge query for c^* . This is a standard guessing argument. We consider a first series of hybrid games H_i , where we keep the game essentially the same, except that the challenger in H_i makes a guess i in advance on the value of q for which the adversary is going to issue the first challenge query, and aborts and returns a random message if $q \neq i$. When $i = q$, the game is perfectly simulated. Otherwise, we output a random plaintext. Let q_1 be the maximal number of queries issued by $\mathcal{A}$ to UpdO in the first phase of the game (i.e on line 3.). Then, by a standard hybrid argument $\text{Adv}_{G_0}(\mathcal{A}) \leq \sum_{i=0}^{q_1} \text{Adv}_{H_i}(\mathcal{A})$. So in particular, there exists some i^* such that $\text{Adv}_{G_0}(\mathcal{A}) \leq (q_1 + 1) \text{Adv}_{H_{i^*}}(\mathcal{A})$. Let $G_1 = H_{i^*}$. So at this stage, we assume the challenger has correctly guessed q .

G_2 Now that we assume we have correctly guessed the value of q , we want to ensure that (sk_q, pk_q) is generated honestly, so that the challenge ciphertext comes from an honest key pair.

The idea is to let the adversary make their first $q - 1$ queries to UpdO freely, and then use the programmability of the random oracle to force the last update of the adversary to end up on a target, honestly generated public key $(sk_q, pk_q) \leftarrow \text{KeyGen}(1^\lambda)$. This is possible because the challenger has knowledge of the entire secret key chain $sk_0 \rightarrow \dots \rightarrow sk_{q-1}$ as well as sk_q . The secret keys contain the knowledge of the endomorphism ring of the corresponding curves, therefore, the challenger can use the KLPT algorithm, as described in Section 4.1, to compute an isogeny between $E_{pk_{q-1}}$ and E_{pk_q} of degree 3^δ ^{III}. Then, they encode that isogeny as the

^{III}This step justifies our choice of large value for δ .

corresponding CGL path value r_q which is given as input to IsogenyWalk , i.e. we have $E_{pk_q} \leftarrow \text{IsogenyWalk}(E_{pk}, \delta, r_q)$. Having knowledge of this r_q value, all is left to do is to embed it as the answer of the q^{th} query of $\mathcal{A}$ to UpdO , i.e. we program the random oracle such that $H(\rho_q || E_{pk_{q-1}}) = r_q$, where ρ_q denotes the value given as input to UpdO at the time of the adversary q^{th} query. However, the adversary also has access to the random oracle, and could have previously queried this same value, which prevents us to program it freely. To address this, we use a standard query-guessing argument. Let q_H denote the number of queries made by $\mathcal{A}$ to RO in this epoch. Then we need to make a guess on the index i^* on which of the q^H possible RO queries will also be made to UpdO and plant our value r_q at this i^* query. If we make a wrong guess or if the query to UpdO is actually fresh, we abort. The probability of correctly guessing i^* is therefore $\frac{1}{q_H+1}$, which leads to a security loss factor of $(q_H + 1)$.

So now at the end we arrive at an equivalent game G_2 where $(sk_q, pk_q) \leftarrow \text{KeyGen}$ directly, so we can ignore the first phase. We summarize it in Figure 4.2.

$G_2(\mathcal{A})$	$\text{UpdO}(\rho)$
1: $(sk_q, pk_q) \leftarrow \text{KeyGen}(1^\lambda)$	1: $r \leftarrow \text{RO}(\rho E_{pk_i})$
2: $m^* \xleftarrow{\$} \mathcal{M}$	2: $E_{pk_{i+1}, -} \leftarrow \text{IsogenyWalk}_3(E_{pk_i}, \delta, r)$
3: $ct^* \leftarrow \text{Enc}(pk_q, m^*)$	3: $up_i \leftarrow \text{Enc}(pk_i, \rho)$
4: $\text{state} \leftarrow \mathcal{A}^{\text{UpdO}, \text{RO}}(pk_q, ct^*)$	4: $sk_{i+1} \leftarrow \text{UpdSk}^{\text{RO}}(sk_i, up_i)$
5: $q' \leftarrow i$	5: $i \leftarrow i + 1$
6: $\rho^* \xleftarrow{\$} \{0, 1\}^{2\lambda}$	$\text{RO}(\rho)$
7: $(pk^*, up^*) \leftarrow \text{UpdPk}(pk_{q'}, \rho^*)$	6: if $\rho \notin \mathcal{Q}$ then
8: $sk^* \leftarrow \text{UpdSk}(sk_{q'}, up^*)$	7: $\mathcal{Q}[\rho] \xleftarrow{\$} [0, 3^\delta)$
9: $m' \leftarrow \mathcal{A}^{\text{RO}}(ct^*, pk^*, sk^*, up^*, \text{state})$	8: return $\mathcal{Q}[\rho]$
10: return $(m' = m^*)$	

Figure 4.2: G_2

G_3 This is the same as G_2 except that we compute (sk^*, pk^*) as an update from (sk_q, pk_q) directly, rather than from $(sk_{q'}, pk_{q'})$ using some randomness ρ^* which is sampled at the beginning of the game. The difference in the two games is bounded by the statistical distance between the distributions of the two curves which, by a similar reasoning as that of Proposition 4.4.3 bounded by $2^{-16.5\lambda+1}$. Hence, we get $|\text{Adv}_{G_3} - \text{Adv}_{G_2}| \leq 2^{-16.5\lambda+1}$. Furthermore, note that since the leaked key pair is computed directly from (sk_q, pk_q) the updates are no longer used in the game anymore.

G_4 This is the same as G_3 except that we remove the adversary's access to UpdO and instead we can perfectly simulate it using RO, and manually updating the public key. The advantage remain the same.

G_5 This is the same as G_4 except that if the adversary queries $RO(\rho^*)$ (recall ρ^* is used to update pk_q to pk_q^*), the game returns 1. We have $\text{Adv}(G_5) = \text{Adv}(G_4) + \frac{1}{2^{2\lambda}}$, accounting for the probability of the adversary guessing $\rho^* \in \{0, 1\}^{2\lambda}$.

G_6 This is the same as G_5 except that instead of having $r \leftarrow RO(\rho^*)$, we directly compute $r \xleftarrow{\$} [0, 3^\delta]$. Since in the random oracle model r is independent of ρ^* until ρ^* is queried, the advantage stays the same. (Indeed, we took care of the case where ρ^* is queried in the previous game).

G_7 So now, we are ready to replace (sk_*, pk_*) by a freshly generated key pair instead of an update from (sk_q, pk_q) . The difference in advantage between these two games is bounded by the statistical distance between the distributions of freshly generated keys and updated keys. By Proposition 4.4.3, and our choice of parameter, this distance is negligible. Hence, we have $|\text{Adv}(G_7) - \text{Adv}(G_6)| \leq 2 \cdot 2^{-16.5\lambda}$.

Let us summarize the game in Figure 4.3

$G_6(\mathcal{A})$	$RO(\rho)$
1: $\rho^* \xleftarrow{\$} \{0, 1\}^{2\lambda}$	1: if $\rho = \rho^*$ then
2: $(sk_q, pk_q) \leftarrow \text{KeyGen}(1^\lambda)$	2: FreeWin $\leftarrow 1$
3: $m^* \xleftarrow{\$} \mathcal{M}$	3: return $\perp$
4: $ct^* \leftarrow \text{Enc}(pk_q, m^*)$	4: if $\rho \notin \mathcal{Q}$ then
5: $\text{state} \leftarrow \mathcal{A}^{RO}(pk_q, ct^*)$	5: $\mathcal{Q}[\rho] \xleftarrow{\$} [0, 3^\delta]$
6: $(pk^*, sk^*) \leftarrow \text{KeyGen}(1^\lambda)$	6: return $\mathcal{Q}[\rho]$
7: $up^* \leftarrow \text{Enc}(pk_q, \rho^*)$	
8: $m' \leftarrow \mathcal{A}^{RO}(ct^*, pk^*, sk^*, up^*, \text{state})$	
9: return $(m' = m^*)$ or FreeWin	

Figure 4.3: G_7

Now we argue that G_7 is essentially the same as the OW-CPA game for FESTA. The challenge ciphertext is computed from an honest and freshly generated key pair, as in the OW-CPA game for FESTA-PKE. The random oracle can be simulated with lazy sampling and the exposed key pair also comes from a freshly generated key generation, which can be perfectly simulated. Finally, up^* is also the encryption of a random value which can be simulated. So, for any adversary $\mathcal{B}$ against OW-CPA game for FESTA-PKE, we have $\text{Adv}_{G_7}(\mathcal{A}) \leq \text{Adv}_{\text{FESTA-PKE}}^{\text{OW-CPA}}(\mathcal{B})$.

Thus, by unrolling this series of game, given an adversary $\mathcal{A}$ against OW-CR-CPA, and adversaries $\mathcal{B}$, against the OW-CPA of FESTA-PKE, we have

$$\text{Adv}_{\text{FESTA-UPKE}}^{\text{OW-CR-CPA}}(\mathcal{A}) \leq (q_1 + 1)(q_H + 1) \left(\text{Adv}_{\text{FESTA-PKE}}^{\text{OW-CPA}}(\mathcal{B}) + \frac{2}{2^{16.5\lambda+1}} + \frac{1}{2^{2\lambda}} \right)$$

By assumption q_1, q_H are polynomial in the security parameter and $\text{Adv}_{\text{FESTA-PKE}}^{\text{OW-CPA}}(\mathcal{B})$ is negligible, which proves the statement. $\square$

4.4.4 IND-CCA-CU security

We construct an IND-CCA-CU-secure FESTA-based UPKE scheme $\Sigma := (\text{KeyGen}', \text{Enc}', \text{Dec}', \text{UpdPk}', \text{UpdSk}')$ detailed in Figure 4.4 based on our OW-CR-CPA-secure UPKE scheme $\Pi := (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{UpdPk}, \text{UpdSk})$ and an OT-CPA-secure symmetric scheme $\Gamma := (\text{SEnc}, \text{SDec})$ together with four random oracles described as follows:

- $G : \{0, 1\}^* \rightarrow \mathcal{SK}_{\text{sym}}$,
- $H : \{0, 1\}^* \times \{0, 1\}^* \times \{0, 1\}^* \rightarrow \mathcal{R} = \{0, 1\}^{2\lambda}$,
- $\hat{G} : \{0, 1\}^* \rightarrow \mathcal{SK}_{\text{sym}}$,
- $\hat{H} : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \mathcal{R}$.

where $\mathcal{R}$, and $\mathcal{SK}_{\text{sym}}$ are the spaces of randomness of Π , and the secret keys of Γ , respectively.

Theorem 4.15. *Let Π be the UPKE scheme described in Section 4.3 and let $\Gamma := (\text{SEnc}, \text{SDec})$ be a symmetric key encryption scheme that satisfies OT-CPA security. Then the UPKE scheme Σ obtained by applying the transformation of [AW25, §4.2] satisfies IND-CU-CCA security.*

Proof. The result follows from putting together Theorem 4.12 and Theorem 4.14, and Proposition 2.25. The proof is detailed in [AW25, §4.3; 4.4]. $\square$

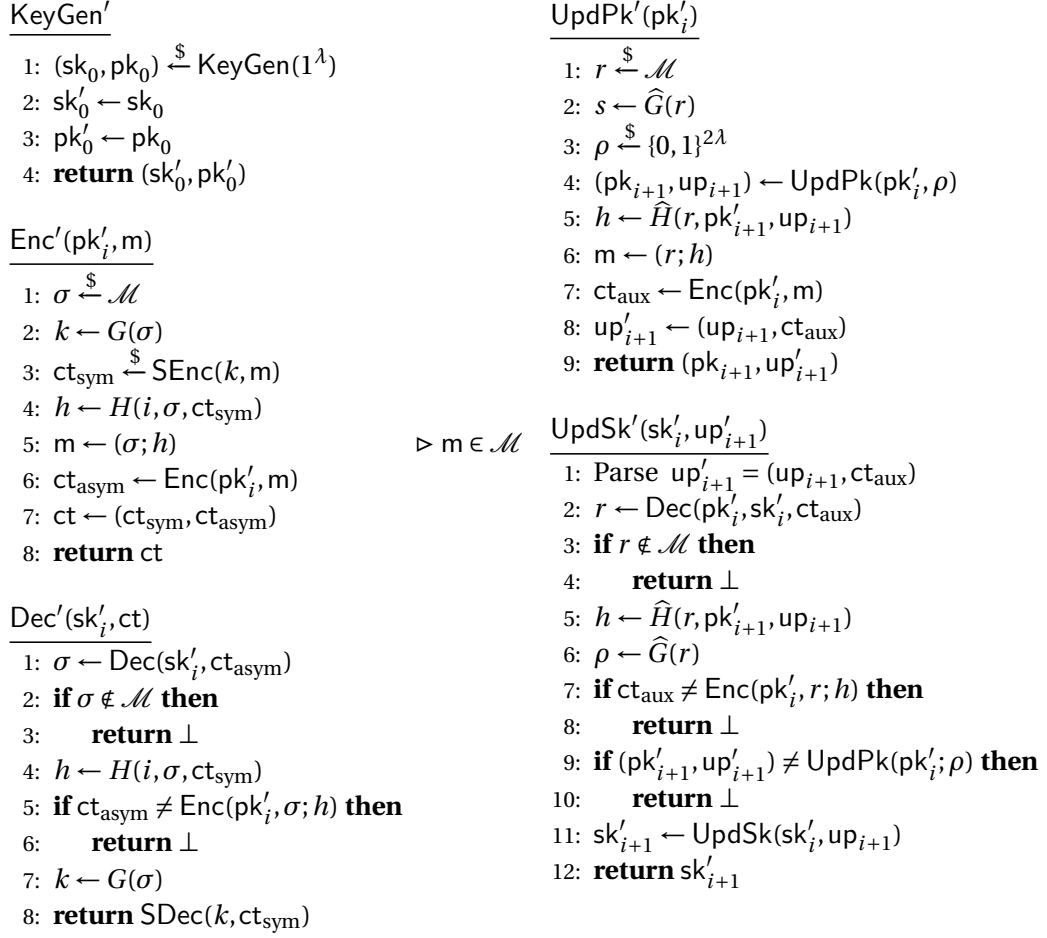
4.4.5 Results

We now instantiate our UPKE by providing actual parameters for the NIST security level I and discussing the key and ciphertext sizes. We use the 1631 bits prime $p_{1631} = f \cdot 2^a \cdot 3^b \cdot 5 - 1$ where

$$a = 1367, b = 162, f = 21.$$

The degree of the second encryption isogeny φ_{Enc_2} is 5^c where $c := 111$, and the degree of the update isogeny is 3^δ where $\delta = 3858$.

The public key is a supersingular elliptic curve E_{pk} defined over $\mathbb{F}_{p^2}$. It can be represented by its Montgomery coefficient using $2 \log_2(p)$ bits. The ciphertext is a tuple $(E_1, R_1, S_1, E_2, R_2, S_2)$ where E_1 and E_2 are supersingular elliptic curves defined over $\mathbb{F}_{p^2}$, R_i and S_i are points of order 2^a on E_i for $i \in \{1, 2\}$. As before, the curves E_i can be represented by their Montgomery coefficient using $2 \log_2(p)$ bits. The points R_i, S_i can be represented using their coordinates in the deterministic basis $(P_i, Q_i) \leftarrow \text{Gen2Tor}(E_i)$ of $E_i[2^a]$, at the cost of some discrete logarithm computations. Each point is hence represented by two a bits long scalars. The total size of the

Figure 4.4: IND-CU-CCA FESTA-UPKE Scheme Σ .

ciphertext is hence $2\log_2(p) + 8a$ bits. For our NIST level I prime p_{1631} , the public key is 408 bytes while the ciphertext is 1775 bytes.

As far as efficiency is concerned, the only non- standard subroutines in our UPKE are ConstrainedIsogeny (Algorithm 1) and the 4-dimensional isogeny computed during decryption.

- We believe that the algorithm ConstrainedIsogeny to be concretely efficient. All the steps in ConstrainedIsogeny are just minor variants of what has already appeared in the literature. While we have not provided a concrete implementation, we highlight that concurrent work has fully implemented a similar version of our algorithm, and it is practically efficient (see Remark 4.5).
- The 4-dimensional isogeny computed during decryption. During decryption, we are given the evaluation of $\varphi_{\text{Dec}} : E_1 \rightarrow E_2$ on the 2^a torsion. The degree of φ_{Dec} is about 2^{2a} , and the degree of the dimension 4 isogeny Φ into which φ_{Dec} is embedded is 2^{2a} . The idea here is that Φ is in fact an endomorphism of $E_1 \times E_1 \times E_2 \times E_2$. Writing Φ as $\Phi = \Phi_1 \circ \Phi_2$ where each Φ_i has degree 2^{2a} , one can recover the kernel of each Φ_i from $\varphi_{\text{Dec}}(E_1[2^a])$. One then evaluate $\Phi_1 : E_1 \times E_1 \times E_2 \times E_2 \rightarrow C_1$ and $\Phi_2 : E_1 \times E_1 \times E_2 \times E_2 \rightarrow C_2$ where C_1 and C_2 are two isomorphic dimension 4 principally polarized abelian varieties. To terminate the evaluation of Φ one needs to compute the isomorphism $C_1 \simeq C_2$, which can be derived from the evaluation of φ_{Dec} on the 2^a -torsion. We refer the reader to [Dar25] for more details on the computation of 4-dimensional isogenies.

4.5 Conclusion

In this work, we introduce a novel updatable public-key encryption scheme based on a 4-dimensional version of FESTA. One of its main advantages is that updated public keys are provably uniformly distributed among all supersingular elliptic curves. This feature allows us to prove its security by only assuming the hardness of a problem that has already appeared in the literature, namely [BMP23, Problem 8].

Our construction supports unbounded updates and does not rely on class group actions, avoiding dealing with the uncertainty surrounding the quantum security of this framework. We also provide evidence of its concrete practicality. All things considered, this results in the first practically efficient post-quantum updatable public-key encryption scheme that supports unbounded updates and has a full security proof.

5 Malleable Commitments from Group Actions and Zero-Knowledge Proofs for Circuits based on Isogenies

This chapter is based on [CLL⁺23], published at INDOCRYPT 2023.

Zero-knowledge proofs are a cryptographic tool that ensures the integrity of a computation, by providing a compact proof that the computation was carried out properly, while preserving the privacy of the inputs through the zero-knowledge property. They find many applications, such as multiparty computations, verifiable computations, cryptocurrencies and more... and are an essential building block in many cryptographic protocols.

In 1991, Goldreich, Micali and Wigderson [GMW91] introduced a construction of proof systems for all NP statements from one-way functions. However, such proofs remain inefficient in general and in practice, one usually relies on the structure given by *homomorphic* commitments to build efficient zero-knowledge proofs [Gro09b, BCC⁺16, BBB⁺18]. The field of zero-knowledge proofs has seen a tremendous amount of progress in the last decade. However, current homomorphic commitment constructions rely on classical hardness assumptions such as the hardness of the discrete logarithm problem [Ped92] and factorisation or pairing-based assumptions [Gro09a]. The security of these problems no longer holds in a post-quantum setting.

Lattice-based cryptography provides some constructions of post-quantum commitments with some homomorphic properties [GSW13, GVW15, BGG⁺14], it is currently the main candidate for the instantiation of general-purpose zero-knowledge proofs. Isogeny-based cryptography, and in particular isogeny-based group actions, whose main attractive feature is its compactness, is a promising candidate for compact zero-knowledge proofs. This is particularly interesting in applications where the size of the proof is crucial, such as in blockchain settings. However, due to the weaker algebraic structure offered by isogeny-based group actions, obtaining homomorphic commitments does not seem plausible. Hence, this brings us to the main question of this work:

Can we construct a zero-knowledge proof for generic NP statements from isogenies

better than from one-way functions?

Our contributions. In this work, we build a generic zero-knowledge proof of knowledge from isogeny-based assumptions. More specifically, we work under the known-order effective group action model [ADMP20] which can be instantiated from isogenies [BKV19, DFK⁺23]. We consider two representations for NP statements: arithmetic circuits over a small field converted to rank 1 constraint systems and matrix-based branching programs over a large field. We construct proof systems for proving the addition and multiplication gates for the arithmetic circuits over a small field or $\mathbb{Z}_h$ for some small $h \in \mathbb{N}$. Then, using these two components we construct a proof system for rank-1 constraints system over a small field or ring, which is a more efficient representation for an arithmetic circuit. Furthermore, we construct a proof system for a matrix branching relation. Using the technique given in [BDK⁺22], each of these constructions leads to an online-extractable zero-knowledge proof of knowledge for NP statements without trusted setup in the random oracle model.

We use the known-order effective group action (KO-EGA) model to develop our schemes, where we know the group order and have an efficient algorithm for computing the action of ng for any g within the (additive) group and $n \in \mathbb{N}$. At the time of writing this paper, the only known post-quantum instantiations of KO-EGA are from isogeny-based group actions, such as CSI-FiSh [BKV19] and SCALLOP [DFK⁺23]. The quantum security levels of these two instantiations are somewhat lower than the NIST-1 requirement [Pei20]. As such, our results remained predominantly theoretical. It is essential to highlight that the KO-EGA model implies the existence of numerous cryptographic primitives with significant relevance in real-world applications [BKP20, DM20, BDK⁺22, KLLQ23]. The challenge of instantiating a stronger KO-EGA remains an open problem and an active area of research, which is orthogonal to the focus of our work. In fact since the publication of the paper corresponding to this chapter, many variants of SCALLOP [CLP24, ABE⁺25, PPS25] have emerged to enable more practical isogeny-based group actions. Our primary focus centers on constructing schemes within this model and providing performance metrics in terms of the number of group actions, group elements, and set elements. We leave it up to future work to choose a suitable group action among the most recent candidates to instantiate it efficiently.

Another limitation of our proof systems is that their sizes grow linearly with the input length. We describe the obstacles and challenges towards obtaining succinct proofs in Section 5.4.4.

Overview. In this chapter, we first introduce *malleable commitments*, a generalization of homomorphic commitments. Intuitively, a malleable commitment scheme for a given relation must satisfy the following property: given a commitment com on a message m , there exists an efficient algorithm that can derive a commitment com' on m' such that m and m' are related. The idea is to settle for a middle point between the absence of structure given by commitments from one-way functions, and the constraints of homomorphic commitments.

We then provide a concrete instantiations of malleable commitments in the KO-EGA model. Our construction of malleable commitment is based on an adaptation of the public key encryption scheme of [BDK⁺22] (Section 5.2). It is computationally hiding and perfectly binding, both classically and in a post-quantum setting.

The original version of this work contained a construction based on pSIDH [Ler22] which is both computationally hiding and binding under isogeny assumptions, but only against classical adversaries due to the attacks of [CII⁺23]. As this thesis focuses on post-quantum cryptography, we do not include it.

Then, we proceed to develop ad hoc proof systems for proving the correctness of a given commitment, taking advantage of the malleability property for greater efficiency (Section 5.3).

Finally, we use these proof systems as building blocks to provide zero-knowledge proofs for NP statements (Section 5.4). More precisely, we derive proof systems for arithmetic circuits, rank-1 constraint systems and branching programs. The constructions for arithmetic circuits and rank-1 constraint systems impose some size restrictions on the underlying field as well as the message space, which can be lifted using the branching program approach.

5.1 Malleable commitments

In this section we introduce the notion of malleable commitments. We will use this notion to build commitments based on group actions and isogenies from which we can derive zero-knowledge proofs for general NP statements. We first give a general definition together with the associated security notion and then restrict the notion to a specific kind of malleability which we will use later on in our constructions. We recall that the necessary background on commitment schemes can be found in Section 2.1.2.

5.1.1 A generic notion of malleability

Intuitively, we say that a commitment scheme is *malleable* if given a commitment for an unknown value m , anyone can create a second commitment for some m' related to m . We formalise this notion below.

Definition 5.1 (Malleable Commitment). *Given a set $\mathcal{M}$ on which we have a relation R , a commitment scheme $(\mathcal{M}, \mathcal{R}, \mathcal{C}, \text{Setup}, \text{Commit}, \text{Verify})$ is said to be R -malleable if there exists a PPT algorithm $\mathcal{A}_m^R : \mathcal{C} \rightarrow \mathcal{C}$ with the following property: for any $\text{com}' = \mathcal{A}_m^R(\text{com})$, if there exists some $(m, r) \in \mathcal{M} \times \mathcal{R}$ such that $\text{com} = \text{Commit}(m, r)$, then there exists $(m', r') \in \mathcal{M} \times \mathcal{R}$ such that $\text{com}' = \text{Commit}(m', r')$ and $R(m, m')$.*

Remark 5.2. Depending on the application, r and r' might be equal. Additionally, the algorithm $\mathcal{A}_m$ does not have to be able to retrieve any of the values m, m' or r . However, in our constructions we only consider schemes for which whenever one party knows (m, r) , they can retrieve the message m' corresponding to the commitment com' .

Remark 5.3. One can easily check that homomorphic commitments are a special case of malleable commitments for which $\mathcal{M}, \mathcal{R}$ and $\mathcal{C}$ are all groups.

5.1.2 Admissible Group-Action-based Malleable Commitments

As is, malleable commitments *a priori* do not impose any structure on $\mathcal{M}, \mathcal{R}$ or $\mathcal{C}$ and provide a very generic framework. Within this framework, we instantiate a specific type of malleable commitments exploiting the structure provided by isogeny-based cryptography. Namely, we enforce a group structure on the message and randomness spaces $\mathcal{M}$ and $\mathcal{R}$, while the commitment space $\mathcal{C}$ is only assumed to be a generic set. This is motivated by the fact that messages and randomness will be encoded as isogenies, which form a group under composition, while commitments will be seen as (isomorphism classes of) elliptic curves, which possesses no built-in group structure. The malleability will then be obtained by defining a group action of $\mathcal{M} \times \mathcal{R}$ on $\mathcal{C}$. In particular, since $\mathcal{C}$ is not a group, what we obtain is *not* an homomorphic commitment.

We formalize this additional structure required through the notion of *Admissible Group-Action-based Malleable Commitment* (AGAMC).

Definition 5.4 (Admissible group-action-based malleable commitment (AGAMC)). *A commitment scheme $\Pi_{\text{Commit}} = (\mathcal{M}, \mathcal{R}, \mathcal{C}, \text{Setup}, \text{Commit}, \text{Verify})$ is said to be an admissible group-action based malleable commitment scheme with respect to a group action $(\mathcal{M} \times \mathcal{R}) \times \mathcal{C} \rightarrow \mathcal{C}$ if the following requirements are satisfied:*

1. $\mathcal{M} = G_M, \mathcal{R} = G_R$ are (additive) abelian groups with identities $0_M, 0_R$ respectively, and there is a distinguished element $X = \text{Commit}(0_M, 0_R) \in \mathcal{C}$.
2. The action $\star : (\mathcal{M} \times \mathcal{R}) \times \mathcal{C} \rightarrow \mathcal{C}$ is regular and can be computed in polynomial-time. Any commitment is computed as $\text{Commit}(m, r) := (m, r) \star X$.
3. The group action gives rise to the following malleability: for any $(m, r), (m', r') \in \mathcal{M} \times \mathcal{R}$, we have $(m', r') \star \text{Commit}(m, r) = \text{Commit}(m + m', r + r')$.
4. There are efficient uniform sampling methods for any subgroups of $\mathcal{M}, \mathcal{R}$.
5. There is an efficient algorithm to determine the membership of $\mathcal{C}$.
6. Every element in $\mathcal{M}, \mathcal{R}, \mathcal{C}$ has a unique representation.

Remark 5.5. We do not require the algorithm Commit specified by pp to be computed efficiently itself. Instead, one uses malleability to produce a commitment $\text{Commit}(m, r) = (m, r) \star \text{Commit}(0_{\mathcal{M}}, 0_{\mathcal{R}})$. We abuse the notation 0 to represent $0_{\mathcal{M}}, 0_{\mathcal{R}}$. Also, we simplify the commitment scheme notion to be $\Pi_{\text{Commit}} = (\text{Setup}, \text{Commit}, \text{Verify})$ and the public parameters pp are implicitly used when the context is clear.

5.2 Constructing malleable commitments from group actions

We present a general construction of an AGAMC scheme from any known-order effective group action (Theorem 2.32), and prove its security.

Construction. We need to specify $\Pi = (\mathcal{M}, \mathcal{R}, \mathcal{C}, \text{Setup}, \text{Commit}, \text{Verify})$. Suppose we have a known-order effective group action $(G, \mathcal{E}, E_0, \star)$ where $N = |G|$ is given. Let $\mathcal{M} = \mathcal{R} = G$ and $\mathcal{C} = \mathcal{E} \times \mathcal{E}$.

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: on input λ , output $\text{pp} = (G, G, \mathcal{E} \times \mathcal{E}, (E_0, E), \star')$ i.e. $\mathcal{M} = \mathcal{R} = G$, $\mathcal{C} = \mathcal{E} \times \mathcal{E}$ and $X = (E_0, E)$, where $E = s \star E_0$ for some $s \in G$ and $s \neq 0$. We define the malleability action $\star'$ as:

$$(m, r) \star' (Y_0, Y_1) := (r \star Y_0, (r + m) \star Y_1), \text{ for } (Y_0, Y_1) \in \mathcal{E} \times \mathcal{E}$$

- $\text{Commit}(m, r) \rightarrow \text{com}$: on input $(m, r) \in G^2$, returns $\text{com} = (m, r) \star' X$.
- $\text{Verify}(\text{com}, (m, r)) \rightarrow 0/1$: on input $(\text{com}, (m, r)) \in \mathcal{C} \times G^2$, checks whether $(m, r) \star' X = \text{com}$. If yes, then outputs 1 and outputs 0 otherwise.

The requirements of an AGAMC follow naturally from the known-order effective group action. The uniqueness and membership test of $\mathcal{C}$ and the feasibility of $\star$ come from Definition 2.32. The uniform sampling methods of any subgroups of $\mathcal{M}, \mathcal{R}$ come from the known order and minimal generating set of G . Since we know minimal generating sets, we have a unique representation of $\mathcal{M}, \mathcal{R}$.

Let us now comment on the security of this commitment scheme.

Theorem 5.6. *The commitment scheme described above is (computationally) hiding assuming the DDH problem over $(G, \mathcal{E}, E_0, \star)$ is hard.*

Proof. Consider a DDH setup $(G, \mathcal{E}, E_0, \star)$ (as in Definition 2.37). Given a DDH challenge (E_1, E_2, E_3) and access to a hiding adversary $\mathcal{A}$, we build an adversary $\mathcal{B}$ against DDH as follows.

1. Set $X = (E_0, E_1)$ and $\text{pp} = (G, G, \mathcal{E} \times \mathcal{E}, X, \star')$. This gives a valid setup for our commitment construction.
2. Invoke the adversary $\mathcal{A}$ on the the public parameter p .
3. Upon receiving $m_0, m_1 \in \mathcal{M}$ from $\mathcal{A}$, reply with $(E_2, m_b \star E_3)$ where $b \leftarrow \{0, 1\}$.
4. Output $(b = b')$ where b' is returned by $\mathcal{A}$.

We claim that this adversary $\mathcal{B}$ has half of the advantage of $\mathcal{A}$. Write $E_1 = s_1 \star E_0$ and $E_2 = s_2 \star E_0$ where s_1, s_2 are sampled uniformly from G by the DDH challenger. In the case that $E_3 = (s_1 + s_2) \star E_0$, i.e. $b = 1$, it is clear that $\mathcal{B}$ returns a well-formed commitment of m_b with respect to p . In the case that $E_3 \leftarrow \mathcal{E}$ by the challenger, the commitment is potentially malformed. Since the input $(E_2, m_i \star E_3)$ follows a uniform distribution over $\mathcal{E}^2$ regardless of m_b and b in the view of $\mathcal{A}$, the random bit b equals b' returned by $\mathcal{A}$ with probability exactly $1/2$ in this case. Hence, the reduction $\mathcal{B}$ can win with probability exactly $1/2$ in this case. Therefore,

$$\begin{aligned} |\Pr[\mathcal{B} \text{ wins}] - 1/2| &= |1/2 \Pr[\mathcal{B} \text{ wins} \mid E_3 = (s_1 + s_2) \star E_0] \\ &\quad + 1/2 \Pr[\mathcal{B} \text{ wins} \mid E_3 \neq (s_1 + s_2) \star E_0] - 1/2| \\ &= 1/2 |\Pr[\mathcal{B} \text{ wins} \mid E_3 = (s_1 + s_2) \star E_0] - 1/2| \\ &= 1/2 \text{Adv}_{\Pi}^{\text{Hide}}(\mathcal{A}). \end{aligned}$$

Hence, we have $\text{Adv}^{\text{DDH}}(\mathcal{B}) = \frac{1}{2} \text{Adv}_{\Pi}^{\text{Hide}}(\mathcal{A})$. $\square$

Theorem 5.7. *The commitment scheme described above is (perfectly) binding.*

Proof. Given $(m_0, r_0) \star' X = (m_1, r_1) \star' X$, we have $(r_0 \star Y_0, (r_0 + m_0) \star Y) = (r_1 \star Y_0, (r_1 + m_1) \star Y)$ for some $Y_0, Y \in \mathcal{E}$. Since the action is regular, we have $r_0 = r_1$ and $m_0 = m_1$, and the result follows. $\square$

Commitment products

We show that given two AGAMCs we can derive a new one via a direct product. Concretely, given two AGAMCs $\Pi_1 = (\text{Setup}_1, \text{Commit}_1, \text{Verify}_1)$ and $\Pi_2 = (\text{Setup}_2, \text{Commit}_2, \text{Verify}_2)$ we can define $\Pi = (\text{Setup}, \text{Commit}, \text{Verify})$ in the following way:

Given $\text{Setup}_1 \rightarrow (\mathcal{M}_1, \mathcal{R}_1, \mathcal{C}_1, X_1, \star_1)$, $\text{Setup}_2 \rightarrow (\mathcal{M}_2, \mathcal{R}_2, \mathcal{C}_2, X_2, \star_2)$ then $\text{Setup} \rightarrow (\mathcal{M}, \mathcal{R}, \mathcal{C}, X, \star)$ with $\mathcal{M} = \mathcal{M}_1 \times \mathcal{M}_2$, $\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2$, $\mathcal{C} = \mathcal{C}_1 \times \mathcal{C}_2$, $X = (X_1, X_2)$ and $((m_1, m_2), (r_1, r_2)) \star X = ((m_1, r_1) \star_1 X_1, (m_2, r_2) \star_2 X_2)$. The definitions of Commit and Verify follow accordingly.

Proposition 5.8 (Commitment Product). *Given two AGAMCs Π_1 and Π_2 , then their product Π defined above gives an AGAMC. Moreover, Π is perfectly (resp. computationally) hiding (resp. binding) when both Π_1, Π_2 are perfectly (resp. computationally) hiding (resp. binding).*

Proof. The uniqueness and membership test of $\mathcal{M}, \mathcal{R}, \mathcal{C}$, the feasibility of $\star$, and the uniform sampling immediately follow from the properties of Π_1 and Π_2 . Regarding the hiding property, by using a standard hybrid argument, for any hiding adversary $\mathcal{A}$ against Π , there exists B_1 against Π_1 and B_2 against Π_2 such that $\text{Adv}_{\Pi}^{\text{Hide}}(\mathcal{A}) \leq \text{Adv}_{\Pi_1}^{\text{Hide}}(\mathcal{B}_1) + \text{Adv}_{\Pi_2}^{\text{Hide}}(\mathcal{B}_2)$. Regarding the binding property, suppose $(m, r) \star X = (m', r') \star X$ where $m = (m_1, m_2)$, $m' = (m'_1, m'_2)$, $r = (r_1, r_2)$, $r' = (r'_1, r'_2)$. Then, we have $(m_1, r_1) \star_1 X_1 = (m'_1, r'_1) \star_1 X_1$ and $(m_2, r_2) \star_2 X_2 = (m'_2, r'_2) \star_2 X_2$. Therefore, Π is binding when both Π_1, Π_2 are binding. $\square$

Remark 5.9. Proposition 5.8 implies that the AGAMC is closed under the product. Alternatively, given an AGAMC with message and randomness space $\mathcal{M}$ and $\mathcal{R}$, we can extend both to vector spaces or matrix spaces $\mathcal{M}^{m \times n}, \mathcal{R}^{m \times n}$. As we will show in Section 5.4.2, this algebraic structure is sufficient to efficiently encode the information of circuits.

5.3 Proof systems for an admissible group-action based commitment

In this section, we build two proof systems for AGAMC that will serve as essential tools for constructing a generic proof system from this special commitment scheme. Both of them are proving the knowledge of the input of a commitment with a restricted message space. The nuance between the two protocols lies in the structure of the message space being a subgroup or not.

The first system (Section 5.3.1 and Fig. 5.1) will be needed to commit on the multiplicative gates of an arithmetic circuit (as $\mathcal{M}$ is not a *multiplicative* group) while the second (Section 5.3.2 and Fig. 5.2) will be used for the addition gates (and thus we can exploit the additive group structure of $\mathcal{M}$).

The additional structure available in the second construction makes it more efficient: it is therefore simpler to commit on additions than on multiplications.

Formally, let $\text{pp} = (\mathcal{M}, \mathcal{R}, \mathcal{C}, X, \star)$, the parameters of an AGAMC and M' a subset of $\mathcal{M}$. Consider the following relation $\mathcal{R}_{\text{pp}, M'} \subseteq \mathcal{C} \times \mathcal{M} \times \mathcal{R}$ where

$$\mathcal{R}_{\text{pp}, M'} = \{(\text{st} = c, \text{wt} = (m, r)) \mid c = (m, r) \star X \wedge m \in M'\}. \quad (5.1)$$

Equivalently, $\mathcal{R}_{\text{pp}, M'} = \{(\text{st} = c, \text{wt} = (m, r)) \mid c = \text{Commit}(m, r) \wedge m \in M'\}$ due to the malleability. Therefore, to have a proof system for the relation $\mathcal{R}_{\text{pp}, M'}$ implies to have a proof system for the commitment without revealing m or r .

We then turn it into non-interactive zero-knowledge proof (NIZK) using parallel repetition and the Fiat-Shamir transform ([FS87]). Looking ahead, in Sections 5.4.1 and 5.4.2 we will use this tool to build NIZK to prove satisfiability of arithmetic circuits and for rank-1 constraint systems (R1CS).

As mentioned earlier, commitment schemes from one-way functions can indeed be used to build generic zero-knowledge proof, but they result in highly inefficient ones. Homomorphic commitments (e.g. Pedersen commitments) have additional structure that makes them useful in many pre-quantum succinct zero-knowledge proofs, such as [BCC⁺16, ZGK⁺17, WTs⁺18]. Here, we demonstrate that a malleable commitment (AGAMC) can also provide a structure suitable for use in post-quantum generic proof systems for arithmetic circuits when the underlying ring is small.

5.3.1 Proof system for small message space

We start with a fundamental proof system of AGAMC for proving the knowledge of the input of a commitment with an arbitrary small message space.

We first consider the case where M' is polynomially small. A concrete construction is given in 5.1 using Merkle trees to make the proof size logarithmic in $|M'|$. The use of Merkle trees is standard except for one modification: two leaves are concatenated in alphabetical order, resulting in an index-hiding Merkle tree [BKP20].

To be more precise, the modification to the alphabetical order serves as a random (in the ROM) permutation of the input, ensuring that revealing a path in the tree does not reveal the location of the leaf. The construction is inspired by [BDK⁺22]. The high-level idea is that the prover uses the malleability property to create multiple new commitments $(-m', r')$ for each $m' \in M'$, and generates a Merkle root. The reason M' is required to be small is to make this possible. The prover is then challenged to either reveal all randomnesses used in the malleability, including the randomness used to generate the whole tree, or to reveal $(0, r + r')$ and the path of the tree that results in the same root. Intuitively, knowing the whole tree and the secret path allows for the determination of m along with r . Zero-knowledge follows from the uniform distribution of $r + r'$ (remember $\mathcal{R}$ has a group structure), and the use of an index-hiding Merkle tree to protect m .

Figure 5.1 gives a construction for the relation $\mathcal{R}_{pp, M'}$ (Eq. (5.1)) using an AGAMC with $pp = (\mathcal{M}, \mathcal{R}, \mathcal{C}, X, \star)$. The set M' is of cardinality t and can be publicly enumerated as $M' = \{m_1, \dots, m_t\}$. The hash function H is modeled as a random oracle with prefixes EXP (for expand) and COM, for a pseudo-random number generator (PRNG) and a commitment scheme respectively and is also implicitly used in $\text{MerkleTree}(\cdot)$.

Theorem 5.10. *The proof system described in Figure 5.1 is correct, 2-special sound (assuming collision resistance of the hash function H), and statistical honest-verifier zero-knowledge (HVZK) in the random oracle model.*

Correctness. For the challenge $\text{chall} = 0$, the correctness holds naturally. For the challenge $\text{chall} = 1$, for the response $\text{resp} = (r'', \text{path}, \text{bits})$, we have $r'' = r' + r$. Therefore, $\widetilde{\text{com}} = (0, r'') \star X = (0, r' + r) \star X$. Recall that $\text{com}_I = (-m_I, r') \star u$. Since $\text{bits}_I = \text{bits}$, we have $\widetilde{\text{com}} = \text{com}_I$ so that $\tilde{C} = C_I$. $\text{ReconstructRoot}(\tilde{C}, \text{path})$ will result in the same root and the correctness follows.

2-Special Soundness. Let $(\text{comm} = \text{root}, 0, \text{resp}_0 = \text{seed}), (\text{comm} = \text{root}, 1, \text{resp}_1 = (r'', \text{path}, \text{bits}))$ be two valid transcripts. We can either extract a collision of the hash function H or a witness for the statement (p, M', u) . Following $\mathcal{P}_1$ of Fig. 5.1, say seed generates r' , $\text{com}_i = (-m_i, r') \star u$ for each $i \in [t]$. Let $\widetilde{\text{com}} = (0, r'') \star X$. Since both $\text{resp}_0, \text{resp}_1$ result in the same root root , we assume $\widetilde{\text{com}} = \text{com}_{\tilde{I}}$ for some $\tilde{I} \in [t]$; otherwise, a collision of the hash function H can be found in the process of generating the root (tree, ReconstructRoot) due to the binding property of the Merkle tree [BKP20, Lemma 2.9]. We claim that $(m_{\tilde{I}}, r'' - r')$ is the witness. This is because we have $(0, r'') \star X = (-m_{\tilde{I}}, r') \star u$. Therefore, $(m_{\tilde{I}}, r'' - r') \star X = u$. The result follows.

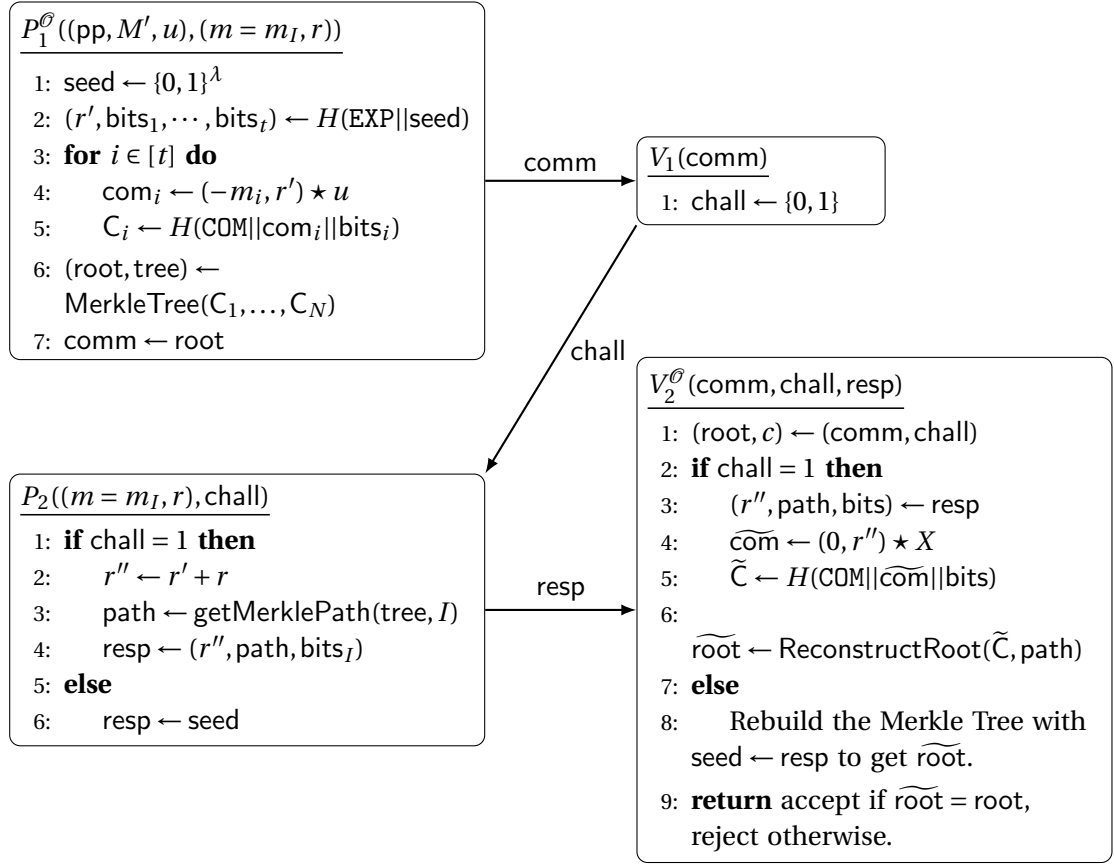


Figure 5.1: Protocol flow for the relation $\mathcal{R}_{pp, M'}$: prover P interacts with verifier V .

Statistical HVZK. We show the scheme is statistical HVZK by modeling the hash function H as a random oracle, which is essential to achieve statistical zero-knowledge. For the challenge $\text{chall} = 0$, the simulator follows the first round of Fig. 5.1 and generates $(\text{seed}, 0, \text{comm})$ as the transcript. For the challenge $\text{chall} = 1$, the simulator generates r'' from $\mathcal{R}$ uniformly at random and computes $\widetilde{\text{com}}_1 = (0, r'') \star X$. Then, the simulator generates uniformly r' from $\mathcal{R}$ and $\text{bits}_1, \dots, \text{bits}_t$ from $\{0, 1\}^\lambda$, computes $\widetilde{C}_1 = H(\text{COM} \parallel \widetilde{\text{com}}_1 \parallel \text{bits}_1)$ and $C_i = H(\text{COM} \parallel (-m_i, r') \star u \parallel \text{bits}_i)$ for $i \in \{2, \dots, t\}$, and generates $(\text{root}, \text{path}) \leftarrow \text{MerkleTree}(\widetilde{C}_1, C_2, \dots, C_N)$. Finally, the simulator outputs the transcript $(\text{root}, 1, (r'', \text{path}, \text{bits}_1))$. Even though the simulator simulates for $I = 1$, the distribution is independent of this particular choice due to the hiding property of the Merkle tree [BKP20, Lemma 2.10]. The only difference is that $r', \text{bits}_1, \dots, \text{bits}_t$ are not generated by $H(\text{EXP} \parallel \cdot)$. Since seed has λ bits of min-entropy and is information-theoretically hidden from the distinguisher, for any unbounded distinguisher with Q queries to H of the form $H(\text{EXP} \parallel \cdot)$ the statistical difference between the distributions is $Q_H/2^\lambda$.

Efficiency. Suppose it requires x bits to represent an element over the message space M' and y bits for the randomness space $\mathcal{R}$. The scheme in Fig. 5.1 has proof size $\frac{\lambda + (y + \log \lambda |M'| + \lambda)}{2}$ bits on average and requires $O(|M'|)$ malleability actions.

5.3.2 Proof system for message spaces with a subgroup structure

In Section 5.3.1, we require the message space to be small since we do not have any assumptions on its structure. Here we show that we can remove this restriction when the message space M' is a subgroup of $\mathcal{M}$, we obtain a more efficient NIZK based on the proof system shown in Fig. 5.2.

Theorem 5.11. *The proof system described in Fig. 5.2 is correct, 2-special sound (assuming collision resistance of the hash function H) and statistical honest verifier zero-knowledge.*

We give brief proofs for correctness, special soundness, statistical honest-verifier zero knowledge for the proof system in Fig. 5.2.

Proof. Correctness. For the challenge $\text{chall} = 0$, the correctness holds naturally. For $\text{chall} = 1$, for the response $\text{resp} = (r'', r')$, we have $r'' = r' + r$. Therefore, $(m'', r'') \star X = (m' + m, r' + r) \star X$. We have $\tilde{C} = C$ and the correctness follows.

Special Soundness. Let $(\text{comm} = C, 0, \text{resp}_0 = \text{seed})$, $(\text{comm} = C, 1, \text{resp}_1 = (m'', r''))$ be two valid transcripts. We show that can either extract a collision of the hash function H or a witness for the statement (p, M', u) .

Following **round 1** of Fig. 5.1, say seed generates m', r' . Due to the validity of both transcripts, we have $C = H(\text{COM} || (m'', r'') \star X) = H(\text{COM} || (m', r') \star u)$. We assume $(m'', r'') \star X = (m', r') \star u$; otherwise, a collision of the hash function H can be found. It is clear that $(m'' - m, r'' - r)$ is the witness since $(m'' - m, r'' - r) \star X = u$. We therefore have the extractor.

Statistical HVZK. For the challenge $\text{chall} = 0$, the simulator follows the first round of Fig. 5.2 and generates $(\text{seed}, 0, \text{comm})$ as the transcript. For the challenge $\text{chall} = 1$, the simulator generates m'', r'' from $M', \mathcal{R}$ uniformly at random and computes $\tilde{C} = H(\text{COM} || (m'', r'') \star X)$. The distributions $\{(m', r') \star u | (m', r') \leftarrow M' \times \mathcal{R}\}$ and $\{(m'', r'') \star X | (m'', r'') \leftarrow M' \times \mathcal{R}\}$ are identical for all u , hence the scheme is statistical HVZK. Concretely, since seed has λ bits of min-entropy and is information-theoretically hidden from the distinguisher, for any unbounded distinguisher with Q queries to H of the form $H(\text{EXP} || \cdot)$ the statistical difference between the distributions is $Q_H/2^\lambda$. $\square$

Efficiency. Say it requires x bits to represent an element over the message space M' and y bits for the randomness space $\mathcal{R}$. The scheme in Fig. 5.2 has proof size $\frac{\lambda + (x+y)}{2}$ bits on average and requires 2 malleability actions.

5.3.3 NIZK via the Fiat-Shamir Transform

By applying the Fiat-Shamir transform [FS87], we obtain NIZKs for the relation $\mathcal{R}_{\text{pp}, M'}$ as shown in Fig. 5.3. Using Theorems 5.10 and 5.11 respectively, and the properties of the Fiat-

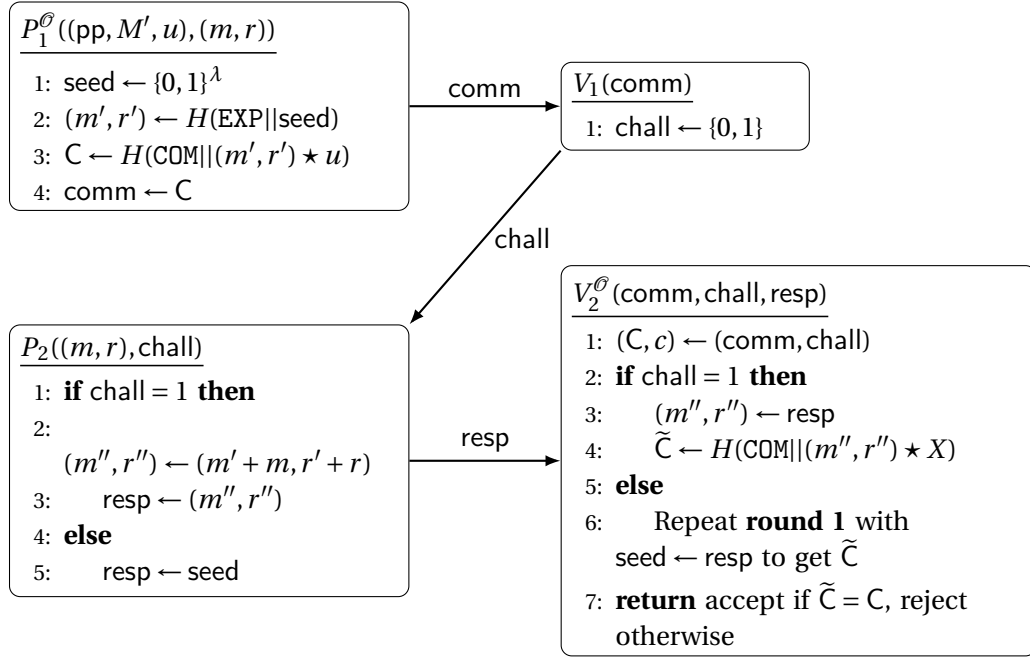


Figure 5.2: Our construction for the relation $\mathcal{R}_{\text{pp}, M'}$ (Eq. (5.1)) using an AGAMC with $\text{pp} = (\mathcal{M}, \mathcal{R}, \mathcal{C}, X, \star)$ where M' is a subgroup of $\mathcal{M}$. The hash function H is modeled as a random oracle with prefixes EXP and COM as a PRNG and a commitment scheme respectively.

Shamir transform we may prove that the NIZKs are complete and statistical zero-knowledge with standard proofs.

In order to demonstrate simulation-extractability, we can reuse the proof of Thm. 6.1 in [BDK⁺22] and show that our NIZKs are also (multi-proof) online-extractable (see Def. 2.10 of [BDK⁺22] for the definition), which implies the simulation extractability and has online-extractability. From a high level, the scheme is online-extractable because we use the PRNG to generate the randomness in the commitment in the sigma protocol. In the security proof, the PRNG is modeled as a random oracle and the challenge is binary. Alternatively, a simulator can extract the response for the challenge 0 by checking the random oracle queries regardless of the challenge $c \in \{0, 1\}$. In other words, since we use the random oracle as PRNG to generate the randomness for the commitment and the challenge is binary, the proof of Theorem 6.1 in [BDK⁺22] is agnostic to the underlying sigma protocol by replacing Step 2.(b) with the 2-special soundness extractor. We, therefore, omit the proof for simplicity.

Theorem 5.12. *The NIZKs of Figs. 5.1 and 5.2 compiled by Fig. 5.3 is complete, simulation-extractable and statistically zero-knowledge in the random oracle model.*

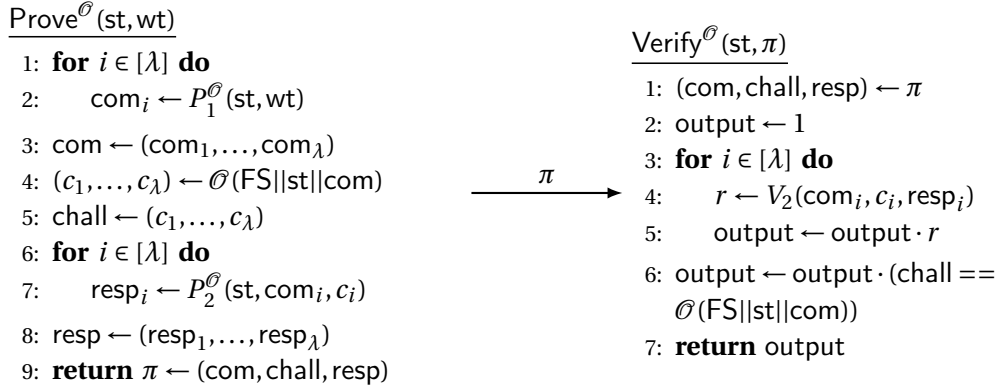


Figure 5.3: NIZK for the relation by applying the Fiat–Shamir transform to $\Pi = (P = (P_1, P_2), V = (V_1, V_2))$ from either Fig. 5.1 or Fig. 5.2 with λ repetitions.

5.4 Proof Systems for NP Statements

This section presents proof systems for an arithmetic circuit over a finite field or ring constructed from an AGAMC. We first consider two cases: first the arithmetic circuit itself with its addition and multiplication gates in Section 5.4.1, and then the representation using rank-1 constraint system in Section 5.4.2. In both cases, the underlying finite field or ring must be small or embedded into a small subgroup of the message space. One can find a reminder on arithmetic circuits and RICS in Appendix A.

In Section 5.4.3, we consider yet another model to represent computations, namely matrix branching programs and we construct a proof system for the branching program representation without size restrictions.

5.4.1 Arithmetic circuits over a small ring

This subsection gives a NIZK for the satisfiability of an arithmetic circuit over a small ring. We encode this ring as $M' \subseteq \mathcal{M}$ of a given AGAMC $\Pi = (\mathcal{M}, \mathcal{M}, \mathcal{C}, X, \star)$ (e.g. via a dictionary map).

The idea is fairly straightforward. For each gate of the circuit, the prover uses an AGAMC to make a commitment to the inputs and the output. Then, the prover uses Figure 5.1 to generate a proof based on the commitment product (Proposition 5.8). Concretely, for an input (m_1, m_2) and an output m_3 for a certain gate, the prover computes $\text{Commit}(m_1, r_1)$, $\text{Commit}(m_2, r_2)$, and $\text{Commit}(m_3, r_3)$. Suppose the ring of the circuit is encoded as $M' \subseteq \mathcal{M}$, then the prover uses Section 5.1 for the triple commitment $\Pi \times \Pi \times \Pi$ with message space $M'' \subseteq (M')^3$ by collating the valid input-output tuple corresponding to the gate. For example, consider an addition gate with inputs m_1, m_2 and output m_3 . We need to prove that $\text{Commit}(m_1, r_1)$, $\text{Commit}(m_2, r_2)$, and $\text{Commit}(m_3, r_3)$ are all in Π and that $m_1 + m_2 = m_3$. Regarding a multiplication gate with inputs m_1, m_2 and output m_3 . We need to prove that $\text{Commit}(m_1, r_1)$, $\text{Commit}(m_2, r_2)$, and $\text{Commit}(m_3, r_3)$ are all in Π and that $m_1 * m_2 = m_3$.

Efficiency. Say it requires x bits to represent an element over the message space M' and y bits for the randomness space $\mathcal{M}$. Let the number of parallel repetitions of the base schemes (Figures 5.1 and 5.2) be λ in Fig. 5.3. The scheme in Fig. 5.1 has proof size $\frac{\lambda^2 + \lambda(3y + x \log \lambda + 3\lambda)}{2}x$ bits on average and requires $O(|M'|^3)$ malleability actions. The equality test can also be done in the same way. Say the prover commits to $\text{Commit}(m_1, r_1)$, $\text{Commit}(m_1, r_2)$. Then, the prover uses Fig. 5.1 for the commitment product $\Pi \times \Pi$ with the message space $M'' \subseteq M'^2$ by collecting $(m, m) \in M'^2$.

Optimization for a group M' . When the ring of the circuit can be embedded as a subgroup $M' \leq \mathcal{M}$, the proofs for a variety of gates can be significantly improved, both with respect to proof size and efficiency.

Concretely, we can improve the proofs of the gates of equalities, additions, addition-by-a-constant, or multiply-by-a-constant. This is because the tuple of valid input-output tuples will form a subgroup of $\mathcal{M}^3$ by Proposition 5.13. In this case we can use Figure 5.2 to generate the proofs. In contrast, the proof size is improved by a factor of $\log_2(M'^2)$ and the number of actions is reduced by a factor of M' .

However, the main bottleneck of our technique is rooted in the multiplication gate where the input-output tuples cannot form a subgroup over $\mathcal{M}^3$. Therefore, multiplication gates still requires Fig. 5.1 to generate a proof for a multiplication gate and the proof size is of $O(\log(M'))$. This is also the reason that our result cannot be extended to a larger ring in a computationally feasible manner.

5.4.2 Proof System for Rank-1 Constraint System Over a Small Ring

Let $\mathbb{F}$ be a finite field (or a ring), the rank-1 Constraint System (R1CS) relation consists of statement-witness pairs $((A, B, C, v), w)$ where $A, B, C \in \mathbb{F}^{m \times (n+1)}$ and v, w are matrices such that $(Az) \circ (Bz) = Cz$ for $z := (1, v, w) \in \mathbb{F}^{n+1}$ and $\circ$ denotes the entry-wise product (i.e. the Hadamard product).

R1CS capture arithmetic circuit satisfaction : A, B, C encode the circuit's gates, the circuit's public input v , and w is the circuit's private input and wire values. We assume that $\mathbb{F}$ is small.

Recall from Remark 5.9 that we can assume the message space of an AGAMC to be a vector. Therefore, by using an AGAMC, a prover generates proofs for the rowcheck and lincheck problems as follows.

1. Commit to $z := (1, v, w)$, Az, Bz, Cz using Commit. Let C_z, C_A, C_B, C_C be the resulting commitment vectors.
2. **Rowcheck:** Generate the proof for C_z, C_A, C_B, C_C using Fig. 5.2 for the relation

$$\left\{ \begin{array}{l|l} \text{st} = (A, B, C, C_z, C_A, C_B, C_C), & C_z = \text{Commit}(z; r_z), \\ \text{wt} = (z, r_z, r_A, r_B, r_C) & C_A = \text{Commit}(Az; r_A), \\ & C_B = \text{Commit}(Bz; r_B), \\ & C_C = \text{Commit}(Cz; r_C). \end{array} \right\}.$$

3. **Lincheck:** Generate the proof for C_A, C_B, C_C using Fig. 5.1 for the relation

$$\left\{ \begin{array}{l|l} \text{st} = (C_A, C_B, C_C), & C_A = \text{Commit}(y_A; r_A), \\ \text{wt} = (y_A, y_B, y_C, r_A, r_B, r_C) & C_B = \text{Commit}(y_B; r_B), \\ & C_C = \text{Commit}(y_C; r_C), \\ & y_C = y_A \circ y_B. \end{array} \right\}.$$

4. Output C_z, C_A, C_B, C_C and the proofs.

Note that we are allowed to apply Figure 5.2 to the rowcheck because if M' is a subgroup of $\mathcal{M}$, where $\mathcal{M}$ is of dimensional $n + 1$, then $\{(m, Am, Bm, Cm) | m \in M'\}$ is also a subgroup of $\mathcal{M}^4$, which can be formalized as Proposition 5.13. In contrast, Figure 5.2 is not applicable to the lincheck due to the non-subgroup structure of the restriction $y_C = y_A \circ y_B$. At first sight, it might seem like the message space blows up exponentially. However, recall that $\mathbb{F}$ is polynomially small, and we use the product version of our AGAMC. In particular this means that $\mathcal{C}_A = (\mathcal{C}_A^0, \dots, \mathcal{C}_A^n)$, i.e. the commitment is done component-wise. This allow us to run the proof component-wise to avoid an exponential message space blow up at the cost of a proof linear in n .

Proposition 5.13. *Let M_1, M_2 be subgroups of a finite abelian group $(\mathcal{M}, +)$. Then,*

- (i) $M_1 \times M_2$ is also a subgroup of $\mathcal{M}^2 := \mathcal{M} \times \mathcal{M}$.
- (ii) the set $\{(x, x) \in M_1^2\}$ is a subgroup of $\mathcal{M}^2$.
- (iii) for any integer $a, b, c \in \mathbb{Z}$, the set $\{(x, y, z) \in M_1 \times M_2 \times \langle M_1, M_2 \rangle \mid ax + by = cz\}$ is a subgroup of $\mathcal{M}^3$.

For a given instantiation of an AGAMC scheme, we give in Table 5.1 an efficiency and proof size estimate with $A, B, C \in \mathbb{F}^{m \times (n+1)}$ and $z \in \mathbb{F}^{1+n}$ for an arithmetic circuit over a finite field $\mathbb{F}$ which can be embedded as a subgroup into $\mathcal{M}$. The parameters η, γ are the numbers of bits to represent the elements of $\mathcal{C}$ and $\max\{|\mathcal{M}|, |\mathcal{R}|\}$, respectively. The density $\rho \in (0, 1)$ is the Hamming weight of A, B, C divided by $3m(n + 1)$. The computational cost is evaluated for both parties in terms of the malleability operations.

For example, when $\lambda = 128$ with an instantiation of AGAMC using CSIDH-2048 (assuming the existence), we have $\eta \approx \gamma^2 \approx 2048$.

Item	Cost
pp	$\frac{\eta}{4} B$
Commitment	$(3m + n + 1) \frac{\eta}{8} B$
Rowcheck Proof	$\frac{\lambda \rho}{16} (\lambda + (3m + n + 1) \log_2(\mathbb{F}) + \gamma(3m + n + 1) B$
Lincheck Proof	$\frac{\lambda \rho}{16} (\lambda + \eta(m + \lambda \log_2(\mathbb{F}))) B$
Computational Cost	$\mathcal{O}(mn\lambda \mathbb{F}^2)$

Table 5.1: Efficiency estimate of R1CS over $\mathbb{F}$ where $\mathbb{F}$ can be embedded as a subgroup using Figs. 5.1 and 5.2.

5.4.3 Zero-knowledge proofs for branching programs

Barrington's theorem is a fundamental result in computational complexity theory that has important implications for circuit design. It states that any circuit of depth d and fan-in 2 can be represented by a matrix branching program of length at most 4^d and width 5 [Bar86].

In more formal terms, a matrix branching program over a field $\mathbb{F}$ of depth d and width w for ℓ -bit inputs is defined by a sequence $\text{BP} = (i, A_{i,0}, A_{i,1})_{i \in [d]}$, where $A_{i,b}$ is a square matrix of $\mathbb{F}$ of dimension w for $i \in [d]$ and $b \in \{0, 1\}$. The evaluation function e maps each index $i \in [d]$ to a bit position $e(i) \in [\ell]$.

Given an input $x \in \{0, 1\}^\ell$, the matrix branching program evaluates by computing the product of matrices $\prod_{i=1}^d A_{i, x_{e(i)}}$. The program outputs 1 if and only if the resulting product equals the identity matrix.

In this section, we show how to prove the satisfiability of a circuit in the form of a branching program (over a ring) using AGAMCs. We start by introducing the high-level idea. Let N be a natural number and $(i, A_{j,0}, A_{j,1})_{j \in [d]}$ represent a matrix branching program defined over $\mathbb{Z}_N$, with depth d and width w and let square matrices $M_0 = I_w, (M_j)_{j \in [d]}$ be a transcript of the program where $M_1 = A_{1,0}M_0$ or $= A_{1,1}M_0$, depending on the input x and the evaluation function e , and M_d is the final product.

Our proof system of the matrix branching program will proceed as follows.

1. Commit to M_j via $u_j = \text{Commit}(M_j, R_j)$.
2. Compute $M_{j+1} = A_{j,b}M_j$ depending on the secret $b \in \{0, 1\}$.
3. Commit to M_{j+1} via $u_{j+1} = \text{Commit}(M_{j+1}, R_{j+1})$.
4. Generate a proof that the values committed in u_{j+1} and u_j are related by either $A_{j,0}$ or $A_{j,1}$ (i.e. $M_{j+1} = A_{j,0}M_j$ or $A_{j,1}M_j$).

It suffices to build a proof system for [Item 4](#), as described in the following.

Base proof system

This subsection introduces a fundamental tool using AGAMC for our proof system of a circuit in the form of a branching program. Concretely, for a given parameter $\text{pp} = (\mathcal{M}^2, \mathcal{R}^2, \mathcal{C}^2, X, \star)$, where $\mathcal{M}, \mathcal{R}, \mathcal{C}$ correspond to the message space, the randomness space and the commitment space in the square matrix variant as explained in [Remark 5.9](#), we consider the following relation

$$\mathcal{R}_{\text{pp}, A_0, A_1} = \left\{ \begin{array}{l} (\text{st} = (u_1, u_2), \\ \text{wt} = ((M_1, M_2), (R_1, R_2))) \\ \in \mathcal{C}^2 \times \mathcal{M}^2 \times \mathcal{R}^2 \end{array} \middle| \begin{array}{l} M_1, M_2 \in \mathcal{M} \\ u_1 = (M_1, R_1) \star X \wedge \\ u_2 = (M_2, R_2) \star X \wedge \\ A_0 M_1 = M_2 \vee A_1 M_1 = M_2 \end{array} \right\}.$$

Note that a standard OR-proof [\[CDS94a\]](#) does not seem feasible here. An OR-proof requires statements for both relations and proves the knowledge of a witness for one of the two statements. Therefore, using OR-proofs for a branching program would induce a statement and proof size growing exponentially in the length because a prover would need to provide all possible statements in advances.

In contrast, we construct a proof system with linear growth in [Figure 5.4](#). The high-level idea is that the prover generates and commits to $(M', A_0 M', R'), (M', A_1 M', R')$, then reveals either $(M_b + M', R + R')$ or (M', R') . However, this would disclose the information of b when the response is $(M_b + M', R + R')$ and the witness is given. To this end, the prover additionally commits to $(M_1 + M', A_{1-b}(M_1 + M'), R + R')$ and shuffles with the previous commitments in a specific way to hide the information of b .

Notations. When the context is clear, we abuse notation by writing $(M'_1, M'_2, R'_1, R'_2) \star (x_1, x_2)$ to represent $((M'_1, R'_1) \star x_1, (M'_2, R'_2) \star x_2)$.

Theorem 5.14. *In the random oracle model, by assuming MerkleTree to be collision-resistant and $H(\text{COM}||\cdot)$ to be binding, the construction $(P^\mathcal{O} = (P_1^\mathcal{O}, P_2), V = (V_1, V_2^\mathcal{O}))$ in [Fig. 5.4](#) is correct, 2-special sound, and HVZK for the relation $\mathcal{R}_{\text{pp}, A_0, A_1}$ with Q_H queries to H .*

Proof. Correctness clearly holds when the challenge bit is 0. When the challenge is 1 the verifier have $\text{resp} = (M'', R'', \text{bits}_0, \text{bits}_2, C_1)$ where $M'' = M_1 + M'$ and $R'' = R + R'$. Then, we have either $\{\tilde{C}, \tilde{C}'\} = \{C_0, C_2\}$ where $\tilde{C} = H(\text{COM}|| (M'', A_0 M'', R'') \star (X, X) || \text{bits}_0)$ and $\tilde{C}' \leftarrow H(\text{COM}|| (M'', A_1 M'', R'') \star (X, X) || \text{bits}_2)$. Therefore, the Merkle tree on input $(\tilde{C}||C_1||\tilde{C}')$, $(\tilde{C}'||\tilde{C}||C_1)$ and $(C_0||C_1||C_2), (C_2||C_0||C_1)$ will result in the same root since the entries of the input will be ordered alphabetically.

2-Special Soundness. Let u be the statement and $(\text{root}, 1, (M'', R'', \text{bits}_0, \text{bits}_2, C_1))$, $(\text{root}, 0, (\text{seed}, \text{bits}_{b+1}, \text{bits}_{b+2}, C_2))$ be the two valid transcripts. We claim that we are able to extract a relation witness of the statement u for $\mathcal{R}_{\text{pp}, A_0, A_0}$ or $\mathcal{R}_{\text{pp}, A_1, A_1}$, a collision witness for $\text{MerkleTree}(\cdot)$ or a binding witness for $H(\text{COM}||\cdot)$.

Due to the validness of the proof, we have

$$\text{root} = \text{MerkleTree}((\tilde{C}||C_1||\tilde{C}'), (\tilde{C}'||\tilde{C}||C_1)) = \text{MerkleTree}((\hat{C}||\hat{C}'||C_2), (C_2||\hat{C}||\hat{C}')),$$

where

$$\begin{aligned}\tilde{C} &\leftarrow H(\text{COM}||(M'', A_0 M'', R'') \star (X, X)||\text{bits}_0), \\ \tilde{C}' &\leftarrow H(\text{COM}||(M'', A_1 M'', R'') \star (X, X)||\text{bits}_2), \\ (M', R') &\leftarrow H(\text{EXP}||\text{seed}), \\ \hat{C} &\leftarrow H(\text{COM}||(M', A_0 M', R') \star u||\widehat{\text{bits}}), \\ \hat{C}' &\leftarrow H(\text{COM}||(M', A_1 M', R') \star u||\widehat{\text{bits}}').\end{aligned}$$

If the alphabetically ordered set $\{(\tilde{C}||C_1||\tilde{C}'), (\tilde{C}'||\tilde{C}||C_1)\}$ is not equal to $\{(\hat{C}||\hat{C}'||C_2), (C_2||\hat{C}||\hat{C}')\}$, then these two ordered set constitute a witness of collision of $\text{MerkleTree}(\cdot)$. Therefore, we may assume

$$\{(\tilde{C}||C_1||\tilde{C}'), (\tilde{C}'||\tilde{C}||C_1)\} = \{(\hat{C}||\hat{C}'||C_2), (C_2||\hat{C}||\hat{C}')\}.$$

We consider two cases: $(\tilde{C}||C_1||\tilde{C}') = (\hat{C}||\hat{C}'||C_2)$ or $(\tilde{C}||C_1||\tilde{C}') = (C_2||\hat{C}||\hat{C}')$.

For each case, we will extract a witness for either the relations or the binding commitment $H(\text{COM}||\cdot)$.

- **Case** $(\tilde{C}||C_1||\tilde{C}') = (\hat{C}||\hat{C}'||C_2)$: Due to $\tilde{C} = \hat{C}$, if the equality $((M'', A_0 M'', R'') \star (X, X)||\text{bits}_0) = ((M', A_0 M', R') \star u||\widehat{\text{bits}})$ does not hold, then the pairs serve as a binding witness for $H(\text{COM}||\cdot)$. If the equality holds then we have $(u, ((M, A_0 M), R)) \in \mathcal{R}_{\text{pp}, A_0, A_0}$ where $M = M'' - M'$ and $R = R'' - R'$.
- **Case** $(\tilde{C}||C_1||\tilde{C}') = (C_2||\hat{C}||\hat{C}')$: Due to $\tilde{C}' = \hat{C}'$, if the equality $((M'', A_1 M'', R'') \star (X, X)||\text{bits}_2) = ((M', A_1 M', R') \star u||\widehat{\text{bits}}')$ does not hold, then the pairs serve as a binding witness for $H(\text{COM}||\cdot)$. If the equality holds then we have $(u, ((M, A_1 M), R)) \in \mathcal{R}_{\text{pp}, A_1, A_1}$ where $M = M'' - M'$ and $R = R'' - R'$.

Therefore, the special soundness follows.

HVZK. Given a parameter pp and a statement u , a simulator $\mathcal{S}$ on input $(\text{pp}, u, \text{chall})$ proceeds as Fig. 5.5.

We claim that for the transcripts generated by the simulator $\mathcal{S}$ are statistically indistinguishable from those generated by a real prover.

Let $((M_1, M_2), R)$ be the witness for u . When $\text{chall} = 1$, two transcript distributions are identical except for two differences. First, there exist no $\text{seed} \in \{0, 1\}^\lambda$ has been queried to $H(\text{EXP}||\cdot)$ such that $(M'', R'') = (M_1 + M', R + R')$ where M', R' are generated by the oracle. Also, there exists no $\text{bits}_1 \in \{0, 1\}^\lambda$ has been queried to $H(\text{COM}||\cdot)$ such that $C_1 = H(\text{COM}||u'|||\text{bits}_1)$ for some set element $u' \in \mathcal{C}'^2$. Since we model H as a random oracle, the statistical difference in this case is at most

$$\left| \frac{2Q_H}{2^\lambda} + \frac{Q_H}{|\mathcal{R}'^2|} \right|.$$

Similarly, when $\text{chall} = 0$, two transcript distributions are identical except that there exist no $\text{bits}_2 \in \{0, 1\}^\lambda$ has been queried to $H(\text{COM}||\cdot)$ such that $C_2 = H(\text{COM}||u'|||\text{bits}_2)$ for some set element $u' \in \mathcal{C}'^2$. Since we model H as a random oracle, the statistical difference in this case is at most

$$\left| \frac{Q_H}{2^\lambda} \right|.$$

It follows that the statistical difference of the transcripts generated by $\mathcal{S}$ and a real prover is at most

$$\left| \frac{2Q_H}{2^{2\lambda}} + \frac{Q_H}{|\mathcal{R}'^2|} \right|.$$

□

Remark 5.15. The Merkle tree only has two leaves in **round 1** so the prover does not need to store additional information for the tree to generate the response.

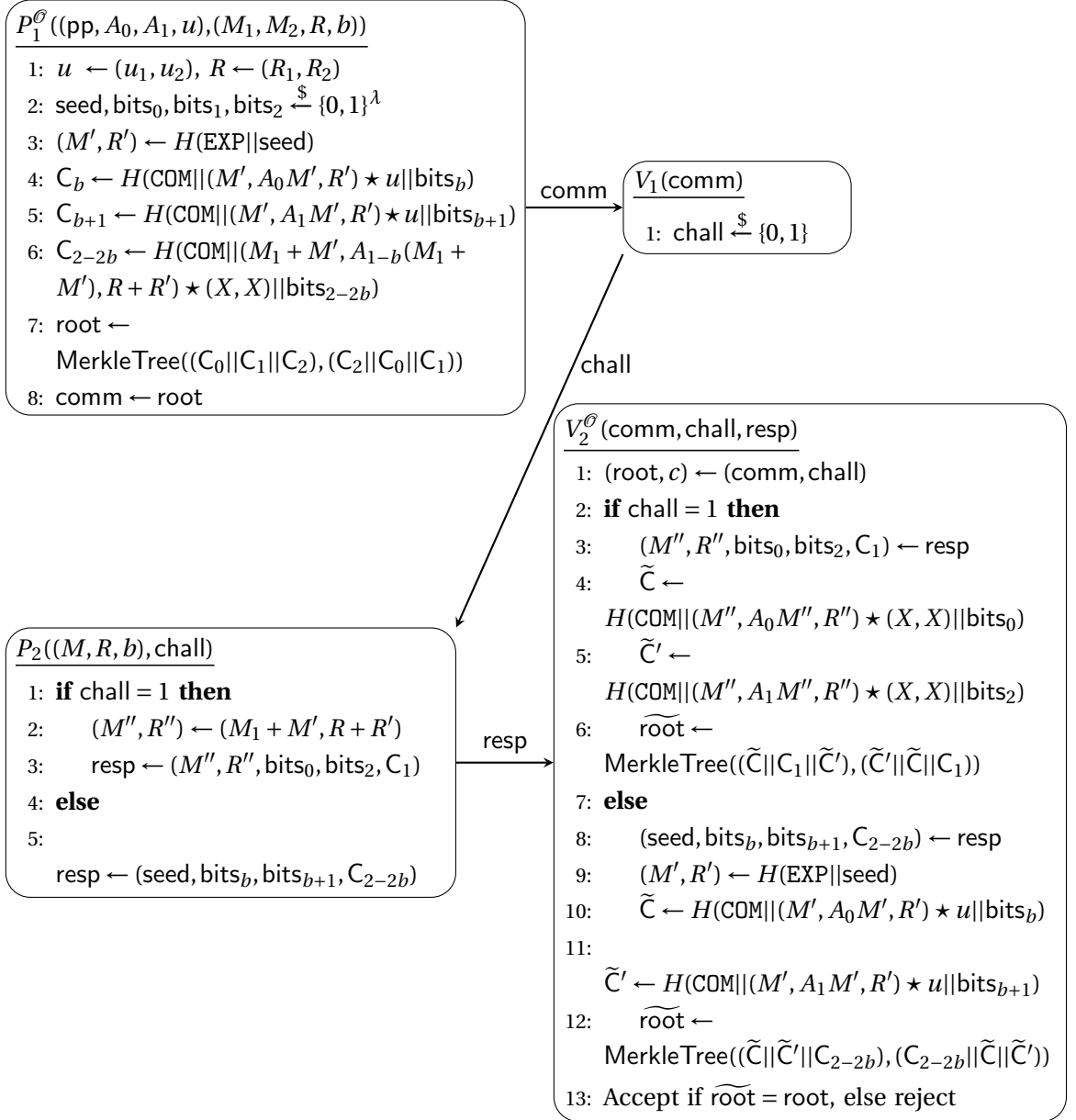


Figure 5.4: A three-move interactive protocol $(P^{\mathcal{O}} = (P_1^{\mathcal{O}}, P_2), V = (V_1, V_2^{\mathcal{O}}))$ to prove possession of a witness (M_1, M_2, R_1, R_2, b) for a statement $u = (u_1, u_2)$ such that $u = (M_1, M_2, R_1, R_2) \star (X, X)$ and $M_2 = A_b M_1$.

```

 $\mathcal{S}^{\mathcal{O}}(\text{pp}, \text{st} = u, \text{chall})$ 
1: if chall = 1 then
2:    $(M'', R'', \text{bits}_0, \text{bits}_2, C_1) \leftarrow \mathcal{M}' \times \mathcal{R}'^2 \times \{0, 1\}^\lambda \times \{0, 1\}^\lambda \times \{0, 1\}^{2\lambda}$ 
3:    $C_0 \leftarrow H(\text{COM} \parallel (M'', A_0 M'', R'') \star (X, X) \parallel \text{bits}_0)$ 
4:    $C_2 \leftarrow H(\text{COM} \parallel (M'', A_1 M'', R'') \star (X, X) \parallel \text{bits}_2)$ 
5:    $\widetilde{\text{root}} \leftarrow \text{MerkleTree}((C_0 \parallel C_1 \parallel C_2), (C_2 \parallel C_0 \parallel C_1))$  ▷ Index-hiding Merkle Tree
6: else
7:    $(\text{seed}, \text{bits}_0, \text{bits}_2, C_1) \leftarrow \{0, 1\}^\lambda \times \{0, 1\}^\lambda \times \{0, 1\}^\lambda \times \{0, 1\}^{2\lambda}$ 
8:    $(M', R') \leftarrow H(\text{EXP} \parallel \text{seed})$ 
9:    $C_0 \leftarrow H(\text{COM} \parallel (M', A_0 M', R') \star u \parallel \text{bits}_0)$ 
10:   $C_1 \leftarrow H(\text{COM} \parallel (M', A_1 M', R') \star u \parallel \text{bits}_1)$ 
11:   $\widetilde{\text{root}} \leftarrow \text{MerkleTree}((C_0 \parallel C_1 \parallel C_2), (C_2 \parallel C_0 \parallel C_1))$ 
12:  resp  $\leftarrow$  (root, chall, resp)
13: return  $\pi \leftarrow$  (com, chall, resp)

```

Figure 5.5: The simulator for the protocol described in Fig. 5.4.

Proof Size and Computational Cost.

We turn Figure 5.4 into a NIZK via Figure 5.3, giving an essential tool for our proof system for a branching program.

Let N be a natural number and $(i, A_{j,0}, A_{j,1})_{j \in [d]}$ represent a matrix branching program defined over $\mathbb{Z}_N$, with depth d and width w . For a given instantiation of an AGAMC scheme, We give in Table 5.2 an estimation of the branching program proof system computational cost and proof size, using the technique described in Remark 5.9 to produce the commitment scheme in the matrix form. We consider a matrix branching problem over $\mathbb{Z}_N$ and of length d and width w and N is also the number of bits to represent elements of M . The parameters η, γ are the numbers of bits to represent the elements of $\mathcal{C}$ and $\mathcal{M}$, respectively. Note that $(\mathcal{C}', \mathcal{M}', \mathcal{R}') = (\mathcal{C}^{w \times w}, \mathcal{M}^{w \times w}, \mathcal{R}^{w \times w})$. The computational cost is evaluated for both parties given in terms of the malleability operations. For example, regarding the proof size, when $\lambda = 128$ with an instantiation of AGAMC using CSIDH-2048 (assuming the existence), we have $\eta \approx \gamma^2 \approx 2048$.

Regarding the computational complexity, as shown in Table 5.2, we measure the cost in terms of the malleability operation, which is the group action operation that gives malleability. Within each layer $j \in [d]$, both the prover and verifier need to perform $O(\lambda)$ malleability operation. Therefore, the proof takes both parties $O(d\lambda w^2)$ CSIDH group actions regardless of the size of the underlying ring. This cost estimate applies to each layer of the proof and takes into account the size of the matrices used in the computation.

Item	Size (Bytes)
pp	2η
Commitment Size	$dw^2 \frac{\eta}{8}$
Proof Size	$\frac{\lambda d}{16} ((N + 2\gamma)w^2 + 9\lambda)$
Computational Cost	$\mathcal{O}(\lambda dw^2)$

Table 5.2: An evaluation of the commitment size and the proof size [Fig. 5.4](#) for a matrix branching problem over $\mathbb{Z}_N$ and of length d and width w .

5.4.4 Discussion and further work

Compared to the classical setting, there are three major challenges in our current context that require further attention.

The first bottleneck, concerning our proof systems for generic arithmetic circuits over a ring, described in [Sections 5.4.1 and 5.4.2](#), is that to prove the multiplication relation of the committed messages, we employ an OR-proof-type proof system. Although the proof remains logarithmic in the ring size (of the arithmetic circuit), both the prover and the verifier must compute the actions of all possible combinations of $(M_1, M_2, M_1 M_2)$. This limits the ring size as well as the message space, as they cannot both be large. Developing a more efficient proof system for the multiplication relation of the committed messages would improve the capability of the proof systems presented. Note however, that this restriction *does not affect* the proof system for branching program, which relies on an affine relation, as discussed in [Section 5.4.3](#).

Another restriction is the lack of an efficient commitment scheme that allows a prover to commit to a vector of messages using **only** one element. In classical cryptography, the generalized Pedersen commitment provides such a functionality. In our setting, a naive approach is to use randomly generated elements $y, g_1, \dots, g_n$, and Y_0, Y from the group, and commit to $(y^r \star Y_0, (r \prod_i^n g_i^{m_i}) \star Y)$ for the messages $m_1, \dots, m_n$ (using multiplicative notation for the group operation). However, this commitment is not quantumly binding since the relations between the g_i 's can be recovered using a quantum period-finding algorithm. This is a challenge that is worth investigating in future research.

Finally, the folding technique, which enables proof compression [[BCC⁺16](#), [BBB⁺18](#)] does not extend from homomorphic to malleable commitment or the Naor-Reingold-type VRF [[Lai23](#)], resulting in an argument with square root communication complexity. A variant of that technique making use of the malleability rather than the homomorphic property is an interesting question to explore.

However, our construction does offer several advantages. First, we do not require any form of trusted set-up. Second, we achieve statistical, rather than computational, zero-knowledge, as well as simulation extractability, meaning that an adversary cannot simulate a new proof unless they know a witness, regardless of the number of simulated proofs they have seen.

This is a stronger notion than soundness or even knowledge soundness, and highly desirable in situations where the non-malleability of the proofs is required. Finally, we obtain linear efficiency in the size of circuit. Despite being less efficient than state of the art post-quantum zero-knowledge proofs, such as [AHIV17, BCR⁺19, BS23] whose proofs sizes are sublinear or even polylogarithmic in the size of the witness, our approach still improves on the naive construction from one-way functions and provides a first general purpose zero-knowledge proof from isogenies.

5.5 Conclusion

In this chapter, we introduce and formalize the notion of malleable commitment, which extends the notion of homomorphic commitments but enforces less structure on the underlying commitment scheme. We show how to instantiate such malleable commitments from an isogeny-based group action. Subsequently, we build different flavors of generic proofs for NP statements. Our first two approaches, based on arithmetic circuits and rank-1 constraint systems impose restrictions on the size of the underlying ring and message space. However, this restriction is lifted by our third approach based on branching programs. Regardless, these three approaches offer the first feasibility results towards post-quantum zero-knowledge proofs based on isogeny assumptions, improving on the naive construction from one-way functions.

Part II - Codes

Group actions are one of the cornerstones of classical public-key cryptography. The Discrete Logarithm Problem and its elliptic curve analogous are such examples of classical group actions and they have been used to instantiate a wide range of primitives from the most basic ones to the most advanced. The abstract framework of cryptographic group actions allows us to design cryptographic protocols in a generic manner, independently of the underlying group action. This is particularly useful when considering post-quantum group actions, as the landscape of post-quantum assumptions is still very much in movement, and it is important to be able to quickly adapt existing protocols to new assumptions as they arise. However, this framework does not directly allow to translate all discrete logarithms protocol to the post-quantum setting, since discrete logarithm problems benefit from extra structure. Indeed, in the discrete logarithm setting, we have a group acting not just on a set, but on another group. This rich algebraic structure is what allows for its versatility in cryptographic protocol design, but it is also the weakness that quantum algorithms such as Shor's algorithm can exploit. Therefore, to be able to replicate the functionalities offered by discrete logarithm-based protocols while retaining post-quantum security, post-quantum group actions must rely on extra structure or extra assumptions.

In this second part of the thesis, we will explore how one can instantiate post-quantum group actions from equivariance problems coming from codes and tensors. While isogenies have long been considered as the main candidate for post-quantum group actions, tensors and codes offer a promising alternative. They are generally more efficient, easier to implement and it is easier to select appropriate parameters for a given security level. On the other hand, they are less compact and less versatile since the resulting group action is not commutative. To overcome these limitations, protocol designers have come up with variants of the classical code-based group actions, by either imposing more constraints on the set being acted upon, or by adding more structure on the group that is acting. However, introducing more structure often means introducing more attack vectors, unless a reduction to a well-known problem is available.

In Chapter 6, we analyze a variant of the Tensor Isomorphism Problem, which considers the action of tuples of invertible matrices on random 3-tensors, in specific structured orbits. This problem was used to design a commitment scheme. We show that this extra structure actually allows to completely break the protocol and give two polynomial-time attacks on the

decisional and computational variants of the problem. The attacks relies on the existence of low rank point for the special tensors considered, which we combine with an analysis of the structure of the stabilizer subgroups allowing us to recover the isomorphism in polynomial-time. Leveraging the insights gained from the attacks, we propose an adaptation of the commitment scheme whose security relies on the pure Tensor-Isomorphism Problem.

In Chapter 7, we turn our attention to variants of the Linear Code Equivalence (LCE) problem and Matrix Code Equivalence (MCE) problem, which studies the action of pairs of invertible matrices on linear and matrix codes respectively. Both variants were introduced for the design of blind signature schemes. In both cases, additional structure is added, either by considering a new equivalence class of linear codes, or by adding extra constraints on the acting group for matrix codes, namely by considering (anti)symmetric matrices only. We first show the equivalence between the Linear Code Equivalence problem and its variant, giving confidence in the security of this newly introduced problem. We show however that the extra structure imposed by the symmetry of the matrices in the Matrix Code Equivalence problem strongly decreases the security of the problem, making it challenging to select appropriate security parameters while retaining the efficiency of the scheme.

These two chapters highlight the delicate balance to strike when designing cryptographic protocols based on post-quantum group actions, and the need for further cryptanalysis research to establish well-studied yet versatile problems on which we can rely to design advanced primitives.

6 Solving the Tensor Isomorphism Problem for special orbits with low rank points

This chapter is based on [GMPT24], published at CRYPTO 2024.

Isogenies have historically been the most well-studied example of post-quantum group actions [BKV19, CLM⁺18]. However, cryptographic group actions can be obtained from other hard problems, some of which have already been proposed for use in post-quantum cryptography. Such group actions include alternating trilinear forms [TDJ⁺22], lattices [DvW22], polynomials [Pat96], linear codes [CNP⁺23b, BBPS23], and tensors [JQSY19].

Grochow and Qiao [GQ21] defined the tensor isomorphism class TI, that is, if a problem is polynomially equivalent to tensor isomorphism, it is TI-complete. In particular, the alternating trilinear form equivalence problem is TI-complete [GQT21]. Based on this problem, a signature scheme ALTEQ [BDN⁺23] has recently been submitted to NIST as a round 1 candidate. The work of [GQ21] shows that three problems, the matrix code equivalence problem, the trilinear form equivalence problem, and the tensor isomorphism problem (TIP) are all equivalent. Particularly given a 3-tensor with dimensions l, m, n , we can naturally represent it as a $[l \times m, n]$ -matrix code. Thus, though our focus in this work will be on TIP, we will be inherently studying the security of all three of these problems. Two recent independent works [JQSY19, CNP⁺23b] proposed using tensor isomorphism (or matrix code equivalence) for cryptographic purposes from the different algebraic structures of tensors and codes. Very recently, a digital signature scheme, MEDS [CNP⁺23a, CNP⁺23b], based on matrix code equivalence was submitted to the NIST standardization competition for post-quantum signatures as a round 1 candidate. The security of MEDS, that is, the algorithm of matrix code equivalence, has been studied in several works [CNP⁺23b, CNP⁺23a, CDG20, RST24, NQT24]. We proceed to briefly review the state-of-the-art algorithms for matrix code equivalence, and hence TIP.

Related work

There are three types of algorithms for matrix code equivalence. The first one use algebraic methods, whereby matrix code equivalence is translated into solving systems of polynomial equations. Gröbner basis techniques are often employed for this approach, albeit with efficiency contingent upon the values of l , m and n . Notably, efficiency diminishes significantly when these dimensions become slightly large. The second is the graph-theoretic algorithm [RST24], which involves transforming matrix codes into quadratic mappings, thereby facilitating the adaptation of algorithms designed for Quadratic Maps Linear Equivalence (QMLE) [BFV13]. Nevertheless, this approach does not perform well when $l = m = n$. The last one is the Leon-like algorithm, which is an adaptation of the code equivalence problem algorithm on the Hamming metric. The basic idea comes from the observation that equivalence preserves the Hamming weight as well as the weight distribution of the codewords. Leon's [Leo82] algorithm entails finding the set of codewords with minimal Hamming weight in two given codes, which can reveal enough information to recover the equivalence. Beullens [Beu20] recently improved this algorithm by building two lists of codewords with a particular weight and then searching for collisions between them to recover the equivalence. This approach naturally translated to matrix code [CNP⁺23b]: by creating two lists of matrices with low rank in the two given matrix codes one can then find the collision to recover the equivalence. Very recently, Narayanan, Qiao and Tang [NQT24] further improved Beullens' algorithm and introduced a new invariant called "co-rank1 associated invariant" that avoids finding collisions by Gröbner basis.

In this work, we will consider the Tensor Isomorphism Problem, but on special orbits. This problem was first introduced at ASIACRYPT 2023 by D'Alconzo, Flamini and Gangemi [DFG23] in the context of a commitment scheme construction. While the general Tensor Isomorphism Problem considers tensors that are generated randomly, which we call *random tensors*, the work of [DFG23] restricts itself to the use of very particular tensors with a lot of structure, which we will call *unit tensors*. Their commitment scheme uses the orbits of two of these unit tensors, which they conjecture is still secure.

Overview

In this paper, we propose polynomial-time algorithms for the decisional and computational versions of TIP on special orbits. This implies that the commitment scheme proposed by D'Alconzo, Flamini and Gangemi in Asiacypt 2023 is broken. We also propose a countermeasure to fix the commitment scheme. We summarize our contributions in what follows.

Algorithms for tensor isomorphism on special orbits. We present a polynomial-time algorithm for the TIP on special orbits. One observation is that some low-rank points exist for the tensors on these special orbits. This allows us to distinguish tensors on two different orbits,

which implies that the decisional TIP is broken. Another observation is that the automorphism groups of these unit tensors are not trivial, that is, they have stabilizers. Therefore, we can avoid finding collisions and quickly recover an equivalent isomorphism which solves the computational TIP.

Regarding the commitment scheme proposed by D’Alconzo, Flamini and Gangemi, its security is based on the hardness of decisional TIP on these unit tensors. We implement our algorithms and use them to break the hiding property of the scheme, as well as recover the random values used when creating the commitment. These algorithms render the scheme completely insecure; see more details in Section 6.2.

Repairing Asiacrypt 2023 scheme. We provide a countermeasure to repair the commitment scheme from [DFG23] using *random* (instead of unit) tensors. Our new scheme fills the gap of a post-quantum, non-interactive commitment scheme from non-transitive group actions, which was the aim of the Asiacrypt 2023 commitment scheme [DFG23], and which is still missing in the literature. We stress that our new scheme makes no new assumptions, but only depends on already existing hard problems from random tensors. The scheme we propose is statistically binding and computationally hiding. We conclude by proposing a zero-knowledge proof of opening for our new commitment scheme.

Organization. In Section 6.1 we will start by recalling the protocol we will be attacking from [DFG23], and in 6.1.2, we present a key tool for our attacks related to the MinRank problem. In Sections 6.2.1 and 6.2.2 we discuss low rank points and our attack on the hiding property of [DFG23], and in Sections 6.2.3, 6.2.4, and 6.2.5, we compute stabilizers and give an attack that recovers the full secret. Finally, in Section 6.3, we give a repair on their commitment scheme that preserves the non-interactive structure and still has statistical binding, with a discussion on relevant zero-knowledge proofs in Section 6.3.2.

6.1 D’Alconzo et al.’s commitment scheme from tensors

While there is already literature about commitment schemes [Nao91, Ped91], developing post-quantum equivalents is necessary in order to curb the risk of future quantum attacks. Many post-quantum commitments exist from lattice-based assumptions with recent constructions such as [FMN24, BDL⁺18, AFLN24], code-based assumptions [MRS17, JKPT12], isogeny-based assumptions [Ste22] and *transitive* group actions [CLL⁺23, BY90, JQSY19]. *Non-transitive* group actions, however, are less restrictive and arise naturally, making them interesting for use in commitment schemes. In [DFG23], at Asiacrypt 2023, D’Alconzo, Flamini, and Gangemi describe a general bit commitment framework from post-quantum non-transitive group actions that they call GACE (Group Action with Canonical Element), and then give an instance using tensors in Section 6.2 of their work.

In particular, they take inspiration from the fact that it is widely believed to be hard to compute the rank of a random tensor (see Section 2.4.1), and proven NP-hard [Hås90, SS18, HL13]. In the paper they provide a method of constructing a distinguished element for any rank, such that given the rank and some additional data it is easy to verify the correctness of the original claim. Let $b \in \{0, \dots, n-1\}$, then their distinguished element is

$$t_b := \sum_{i=1}^{n-b} e_i \otimes e_i \otimes e_i. \quad (6.1)$$

We will refer to the t_b as *unit tensors*, following the terminology from [BI10].

In their commitment scheme they use t_b to create either an n -rank tensor or an $(n-1)$ -rank tensor, thereby encoding a bit $b \in \{0, 1\}$. They then sample an element $(A, B, C) \in G = \text{GL}(n, q) \times \text{GL}(n, q) \times \text{GL}(n, q)$, and compute the commitment $c = (A, B, C) \star t_b$ via the group action described in Equation 2.4. The sender sends the commitment c to the receiver. When the sender is ready, they will reveal $(A, B, C), b$. The receiver can then easily check that $c = (A, B, C) \star t_b$, and so has rank $n - b$.

This is summarized in Figure 6.1.

Setup(1^λ)	Commit(b)	Verify($c, (b, (A, B, C))$)
1: $t_0 \leftarrow \sum_{i=1}^n e_i \otimes e_i \otimes e_i$	1: $A, B, C \xleftarrow{\$} (\text{GL}(n, q))^3$	1: return $(c = (A, B, C) \star t_b)$
2: $t_1 \leftarrow \sum_{i=1}^{n-1} e_i \otimes e_i \otimes e_i$	2: $c \leftarrow (A, B, C) \star t_b$	
3: return t_0, t_1	3: return c	

Figure 6.1: Tensor-based commitment scheme from [DFG23]

6.1.1 Parameters

There are no parameters given for the commitment scheme from [DFG23], and it does not rely on a pure form of the tensor isomorphism problem, so parameter choice for this scheme is not obvious. Though our attacks run in polynomial time complexity, we would like a concrete example instance of the commitment scheme on which to test the algorithms outlined in Section 6.2. Thus, we will consider the parameters outlined in MEDS [CNP⁺23a, CNP⁺23b], a post-quantum signature from matrix code equivalences, submitted to the NIST competition for post-quantum signatures. Their proposed choices of vector dimension and field size, (n, q) , are $(14, 4093), (22, 4093), (30, 2039)$.

The matrix code equivalence problem is polynomially equivalent to the general trilinear form equivalence problem and the tensor isomorphism problem as noted in [GQ21], meaning parameter sizes should be the same across all of these hard problems.

6.1.2 The MinRank problem

In Section 6.2 we will be exploring how low rank points can affect the security of some tensor-based hard problems. For this, we will need a way of solving a MinRank instance. While some more straightforward approaches to MinRank, such as [KS98], would suffice for computing low rank points, we choose to use a more general algorithm to eliminate the chance of using different rank tensors as a countermeasure to our attacks later on. Here we summarize the relevant results from [BBC⁺20], the current state-of-the-art for this problem. The authors of this work revise the algebraic approach to MinRank, and thus completely avoid the use of a Gröbner basis computation for some parameters, keeping all the equations linear. When applying this approach to three NIST candidate key exchange schemes, they were able to improve upon or match the state-of-the-art attacks.

The MinRank problem is as follows:

Problem 6.1 (MinRank problem). *Given an integer $r \in \mathbb{N}$, and k matrices $M_1, \dots, M_k \in \mathbb{F}_q^{m \times n}$, determine field elements $x_1, \dots, x_k$ not all zero, such that*

$$\text{rank}\left(\sum_{i=1}^k x_i M_i\right) \leq r.$$

In previous works [AGHT18], the MinRank problem was solved by creating equations using *coefficient* variables and *support* variables depending on maximal minor equations. This system of equations would then be solved using a Gröbner basis algorithm. In [BBC⁺20], they instead define even more bilinear equations using the coefficient and support variables. So many, in fact, that they are sufficient for solving the entire problem, thereby avoiding the Gröbner basis computation in many cases.

The algorithm. We are given n matrices, $M_1, \dots, M_n$, of size $n \times n$, and a target rank r . We want to find $\{x_i\}_{i=1}^n$ such that $\mathbf{M} := \sum x_i M_i$ has rank r . So the entries of $\mathbf{M}$ are linear expressions in the variables x_i . When $\mathbf{M}$ has rank r , we can write it as $\mathbf{M} = \mathbf{S}\mathbf{R}$, where columns of the $n \times r$ matrix $\mathbf{S}$ form a basis for the column-space of $\mathbf{M}$, and the $r \times n$ matrix $\mathbf{R}$ contains a basis for the row-space of $\mathbf{M}$. The entries in $\mathbf{S}$ and $\mathbf{R}$ are referred to as the *support* and *coefficient* variables, respectively. The matrix $\mathbf{R}$ will necessarily have full rank r . Hence, if we add a row from $\mathbf{M}$ to it, all of its maximal minors will be vanishing. These vanishing maximal minors give us algebraic equations depending on the x_i and the entries of $\mathbf{R}$. We can do this for each row of $\mathbf{M}$, obtaining enough equations to linearize and solve for the x_i .

6.2 Solving Tensor Isomorphism Problems in t_b orbits

In this section we focus on attacking the commitment scheme from [DFG23], which uses special cases of tensor hard problems. We begin in Section 6.2.1 by establishing some background

on the rank of points in tensors. This will help us in Section 6.2.2, where we focus on attacking a variant of the Decisional Tensor Isomorphism Problem (DTI, see Problem 2.72) present in their commitment scheme, and provide a polynomial time attack on it. We then describe some elements of the stabilizer group of the family of tensors $\{t_b : b \in \mathbb{Z}_n\}$ in Section 6.2.3. Lastly, in Sections 6.2.4 and 6.2.5, we use these findings to attack their variant of the Computational Tensor Isomorphism Problem (CTI, see Problem 2.73).

All the algorithms and experiments described in the proceeding sections were coded in Magma, and can be found at

<https://github.com/vgilchri/tensor-group-action>.

6.2.1 Computing the rank of points

In the case of tensors, we can first choose to view a 3-tensor, $g \in \mathbf{V}$, as a list of n matrices $G_1, \dots, G_n \in \mathbb{F}_q^{n \times n}$. Given $u, v, w \in \mathbb{F}_q^n$, we represent $u \otimes v \otimes w$ as $[G_1, \dots, G_n]$ where $G_i = u_i \cdot (v \cdot w^T)$. Note, there are two additional ways to do this, which essentially changes along which axis you are “slicing”. While the choice of this axis is not important, choosing one and staying consistent is. We highlight our chosen representation in the following example.

Example 6.2. Suppose we are working over $\mathbb{F}_3^3$. We would like to compute the tensor $u \otimes v \otimes w$, where $u = [4, 1, 2]$, $v = [3, 0, 1]$, $w = [2, 4, 0]$.

We proceed by first expanding the matrix $v \cdot w^T$:

$$v \cdot w^T = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 2 & 4 & 0 \end{bmatrix}.$$

Now we multiply this matrix by each entry of u , storing them in a list as we go and we obtain the following:

$$\begin{bmatrix} 4 & 3 & 0 \\ 0 & 0 & 0 \\ 3 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 2 & 4 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 4 & 0 \\ 0 & 0 & 0 \\ 4 & 3 & 0 \end{bmatrix}.$$

Using this representation of tensors, we can define the rank of a *point* in a tensor.

Definition 6.3 (Rank of a point). *Let $g \in \mathbf{V}$ be a tensor, written in the form $g = [G_1, \dots, G_n]$. Then the rank of a point $u := [u_1, \dots, u_n] \in \mathbb{F}_q^n$ in g is exactly the matrix rank of $u_1 G_1 + \dots u_n G_n$.*

Lemma 6.4. *The set of rank 0 points of a tensor, $v \in \mathbf{V}$, is a vector space.*

Proof. Associativity, commutativity, and distributivity are all inherited from $\mathbb{F}_q^n$. The zero-vector in $\mathbb{F}_q^n$ is trivially a rank 0 point in any tensor, and so serves as the additive identity

element for the vector space. Furthermore, we have closure under additive inverses and scalar multiplication since multiplication by a non-zero scalar does not affect the rank of a matrix. Lastly, given two rank 0 points u and w , we get that $u + w$ is a rank 0 point, since the sum of rank 0 matrices will have rank 0 as well. $\square$

Computing the rank of points in a tensor will be useful to us later on. In the rest of this paper, when talking about the rank of points, we will consider points up to scalar multiplication, unless stated otherwise.

Lemma 6.5. *Let $v \in \mathbf{V}$ be a tensor. Denote $L_k(v) := \{u \in \mathbb{F}_q^n : \text{rank}_v(u) = k\}$ to be the set of rank k points in v . Now let $(A, B, C) \in G$ be an isomorphism, and $w = (A, B, C) \star v$. Then $|L_k(v)| = |L_k(w)|$.*

In particular, we get a bijection that sends $u \in L_k(v)$ to $uA^{-1} \in L_k(w)$.

Proof. We observe that, given a tensor v , the action of B and C does not affect the rank of a point, meaning that a point u of rank r in v will also be of rank r for $(I, B, I) \star v$, and $(I, I, C) \star v$. This follows straightforwardly from the fact that given $v = [T_1, \dots, T_n]$, we can write the action of (I, B, I) as $(I, B, I) \star v = [BT_1, \dots, BT_n]$ and that of (I, I, C) as $(I, I, C) \star v = [T_1 C^T, \dots, T_n C^T]$ with T denoting the transpose.

We observe that only the action of A has some influence on the rank of points, and A sends a rank r point u in v to the rank r point $u' = uA^{-1}$ in $(A, I, I) \star v$, where $\star$ is as defined in Equation 6.2.3.

Combining these results, we obtain a one-to-one correspondence between the rank r points in v , and the rank r points in $w = (A, B, C) \star v$, with an explicit mapping, which proves the statement. $\square$

When considering tensors, Lemma 6.5 tells us that the number of points of fixed rank in a tensor is preserved via the group action ($\star$) from Equation 6.2.3. Computing a group action in this case amounts to transforming the bases of $\mathbb{F}_q^n$ being used to compute the tensor, to new bases, via linear transformations (recall we require three bases of $\mathbb{F}_q^n$). Hence, the overall structure or rank of the tensor does not change.

Rank 0 points in tensors, when considering a tensor as a list of matrices, means that there exists some linear dependency between these matrices. Transforming the bases of $\mathbb{F}_q^n$ preserves this dependency, regardless of the exact changes of bases being used. Solving for these points can be done with Gaussian elimination, as we will see in Section 6.2.2. Solving for higher rank points will amount to a MinRank problem, described in Section 6.1.2.

6.2.2 Hiding attack

We now describe an attack that breaks the hiding property of the commitment scheme described in [DFG23]. That is, given a committed value $c = \text{Commit}(b, r)$ for a given bit b , we recover b by solving a system of linear equations, therefore disproving the security claims made in [DFG23].

Let us recall the problem we are attacking:

Problem 6.6. *Given $c = (A, B, C) \star t_b$ with $t_b = \sum_{i=1}^{n-b} e_i \otimes e_i \otimes e_i$, $b \in \{0, \dots, n-1\}$, $(A, B, C) \in G$ sampled randomly, and $\star$ is as defined in Equation 6.2.3, recover b .*

Note, this problem is slightly more general than the commitment scheme from [DFG23], since we are considering any $b \in \{0, 1, \dots, n-1\}$. From Problem 6.6, we see that c is precisely the tensor t_b with an isomorphism acting on it, as described in Lemma 6.5.

Lemma 6.7. *Let $t_b = \sum_{i=1}^{n-b} e_i \otimes e_i \otimes e_i$, then the vector space of rank 0 points has dimension b .*

Proof. Lemma 6.4 immediately tells us that the set of rank 0 points is a vector space. It remains to prove the dimension.

Let us write t_b as an n list of $n \times n$ matrices, which we denote $T_b^1, \dots, T_b^n$. These matrices can be described in the following way :

$$(T_b^k)_{ij} = \begin{cases} 1 & \text{if } i = j = k \text{ and } k \leq n - b \\ 0 & \text{otherwise} \end{cases}$$

We know that the rank of a point $u \in \mathbb{F}_q^n$ of t_b is the rank of the matrix $U_b = u_1 T_b^1 + \dots + u_n T_b^n$. Using the above definition of the T_b^i , $i = 1, \dots, n$ we observe that U_b will be a diagonal matrix with entries

$$(U_b)_{ii} = \begin{cases} u_i, & \text{for } i = 1, \dots, n - b \\ 0 & \text{otherwise} \end{cases}.$$

Suppose $u \neq 0 \in \mathbb{F}_q^n$, then U_b will consist of a diagonal matrix of dimension $(n - b) \times (n - b)$ with the remaining b rows/columns being completely zeros. This upper bounds the rank to be at most $n - b$. For example, U_0 will have rank at least 1 when u is non-zero, and therefore no non-trivial rank 0 points and thus the subspace has dimension 0. Similarly U_1 will have one zero column so rank at most $n - 1$ and will always have a rank 0 point (up to scalar multiplication), in the form $u_1 = \dots = u_{n-1} = 0$ and u_n any non-zero element of $\mathbb{F}_q$.

More generally, for t_b where $b > 0$, letting e_i denote the canonical basis element of $\mathbb{F}_q^n$, we can define the basis B generating the vector space of rank 0 points in t_b as

$$B := \{e_n, \dots, e_{n-b+1}\}.$$

□

We focus for a moment on the case $b \in \{0, 1\}$. Using Lemma 6.7, we have that t_1 has exactly one rank 0 point, and that t_0 has no rank 0 points. Combining this with Lemma 6.5 shows that we can use the existence of a rank 0 point as a distinguisher on c (as defined in Problem 6.6). We can find the rank 0 points of c by solving the system of n^2 linear equations

$$\alpha_1 G_1 + \cdots + \alpha_n G_n = 0, \quad (6.2)$$

for $\alpha_i \in \mathbb{F}_q$, $i = 1, \dots, n$. If such a solution exists, then $b = 1$, else $b = 0$. For the more general case of $b \in \{0, 1, \dots, n-1\}$, the dimension of the solution space will be exactly b .

While more complex algorithms could be used to solve this system, Gaussian elimination is sufficient for our purposes. The challenge here is selecting a set of n linearly independent equations from the n^2 possible choices. In the worst case this gives an upper bound on the number of operations of $O(n^4)$. This proves Theorem 6.8.

Theorem 6.8. *There exists an algorithm that solves Problem 6.6 using $O(n^4)$ operations.*

For all parameters highlighted in Section 6.1.1 the attack runs in under a second on a laptop. While running these experiments we found that sampling the first n equations was sufficient for correctly solving the system. While we would not expect this to be true for a random set of n^2 equations, this could be due to the structure of the tensor which is not random. So heuristically, we expect a complexity of $O(n^3)$. Note that Theorem 6.8 always returns a correct answer to Problem 6.6.

To this end, we give a second probabilistic algorithm that also requires $O(n^3)$ operations but uses a slightly different approach to random sampling.

Theorem 6.9. *There exists a probabilistic algorithm that solves Problem 6.6 using $O(n^3)$ operations.*

Proof. Recall from Problem 6.6, given $c = (A, B, C) \star t_b$, we would like to recover b .

We begin by restricting to the case $b \in \{0, 1\}$. Consider a random point $u \in \mathbb{F}_q^n$. Then u has rank n in t_0 and rank $n-1$ in t_1 with very high probability. Namely u will have rank n in t_0 with probability $(q-1)^n/q^n$ since there can be no zeros in this point. Similarly, in t_1 , one of the entries can be free but the rest must be non-zero, so we get probability $(q-1)^{n-1}/q^{n-1}$. By Lemma 6.5, these probabilities will translate to c . Thus, it suffices to compute the rank of u in the commitment, c , to recover b with overwhelming probability, for large q . This approach generalizes to $b \in \{0, 1, \dots, n-1\}$, in the obvious way.

This algorithm consists of computing a random linear combination of n square matrices, and computing the sum's rank over the finite field $\mathbb{F}_q$. Using Gaussian elimination, we can do this in $O(n^3)$ complexity.

□

Remark 6.10. This attack method works because of how structured t_b is. For a random tensor we do not expect to have any rank 0 point (this will be shown in Lemma 6.25). For this reason, our distinguishing attack is not expected to have any impact on the general case of Problem 2.72 (the Decisional Tensor Isomorphism Problem), which asks an attacker to determine whether two given tensors are isomorphic. For the same reason, we also do not claim any improvement on Problem 2.73 (the Computational Tensor Isomorphism Problem). To date, both of these problems are believed to be cryptographically hard in the general setting, and we will use them in Section 6.3 to build a new commitment scheme from random tensors.

6.2.3 Stabilizer subgroup of t_b

We now investigate some elements belonging to the stabilizer group of tensors, and construct the entire stabilizer group of t_0 . These findings will be of use to us in an attack in later sections.

Stabilizers for all tensors. Recall, we define $v \in \mathbf{V}$ as

$$v := \sum_{i,j,k=1}^n v(i,j,k) e_i \otimes e_j \otimes e_k,$$

and we are considering isomorphisms $(A, B, C) \in G$ that act on v via the group action, $(\star)$, defined as

$$(A, B, C) \star \sum_{i,j,k} v(i,j,k) e_i \otimes e_j \otimes e_k = \sum_{i,j,k} v(i,j,k) A e_i \otimes B e_j \otimes C e_k.$$

Lemma 6.11. *Let $\lambda_a, \lambda_b, \lambda_c \in \mathbb{F}_q$ be such that $\lambda_a \lambda_b \lambda_c = 1$. Then for all $v \in \mathbf{V}$, we have that $(\lambda_a I_n, \lambda_b I_n, \lambda_c I_n) \star v = v$.*

The proof of this lemma can be done straightforwardly using the definition of the group action from Equation 6.2.3.

We reserve the details for Appendix B.

Stabilizer subgroup of t_b . Now we focus our attention to t_b , which can be written as

$$t_b := \sum_{i=1}^{n-b} e_i \otimes e_i \otimes e_i.$$

In the following two lemmas we formalize how diagonal matrices and permutation matrices can be used to create stabilizing elements for t_b , and more specifically, for t_0 .

Lemma 6.12. Suppose $\Lambda_A, \Lambda_B, \Lambda_C$ are three diagonal $n \times n$ matrices over $\mathbb{F}_q$ such that $\Lambda_A \Lambda_B \Lambda_C = I_n$. Then $(\Lambda_A, \Lambda_B, \Lambda_C) \star t_b = t_b$, i.e. $(\Lambda_A, \Lambda_B, \Lambda_C)$ is in the stabilizer subgroup of t_b .

Proof. We claim that $(\Lambda_A, \Lambda_B, \Lambda_C)$ is an element of the stabilizing group of t_b . We can see this explicitly as

$$\begin{aligned} (\Lambda_A, \Lambda_B, \Lambda_C) \star t_b &= \sum_{i,j,k=1}^{n-b} \sum_{s=1}^{n-b} \Lambda_{is}^A \Lambda_{js}^B \Lambda_{ks}^C e_i \otimes e_j \otimes e_k \text{ (the matrices are diagonal)} \\ &= \sum_{i=1}^{n-b} \Lambda_{ii}^A \Lambda_{ii}^B \Lambda_{ii}^C e_i \otimes e_i \otimes e_i \\ &= \sum_{i=1}^{n-b} e_i \otimes e_i \otimes e_i = t_b. \end{aligned}$$

□

Additional stabilizer elements for t_0 . To explore how permutation matrices can be used to build stabilizing elements, we first focus on the case of $b = 0$. In what follows, we use S_n to denote the symmetric group of integers $\{1, \dots, n\}$.

Lemma 6.13. Let $\sigma \in S_n$, and let P_σ be the associated permutation matrix given by $P_\sigma = (p_{ij})_{i,j=1}^n$ and $p_{ij} = \mathbf{1}_{i=\sigma(j)}$. Then $(P_\sigma, P_\sigma, P_\sigma) \star t_0 = t_0$, i.e. $(P_\sigma, P_\sigma, P_\sigma)$ is in the stabilizer subgroup of t_0 .

Proof. Consider any permutation matrix P_σ . When applied consistently to the bases being used to construct $t_0 = \sum_{s=1}^n e_s \otimes e_s \otimes e_s$, it simply permutes the order in which the summation is performed, resulting in the same tensor t_0 . We reserve the explicit calculations showing this equality for Appendix B. □

In [BI10, Proposition 4.1], they state that the entire stabilizer group of t_0 , defined over the complex numbers, is exactly the semi-direct product of the diagonal matrices from Lemma 6.12 and the set of permutation matrices P_σ for $\sigma \in S_n$. This is consistent with our findings over finite fields, and was further corroborated via our experiments.

Additional stabilizer elements for t_1 . We now turn to the study of the stabilizer elements specific to t_1 .

Lemma 6.14. Let $\sigma \in S_{n-1}$, and let P_σ be the associated permutation matrix given by $P_\sigma = (p_{ij})_{i,j=1}^{n-1}$, $p_{ij} = \mathbf{1}_{i=\sigma(j)}$ and M_σ be the $n \times n$ matrix obtained by adding an additional zero row

and zero column to P_σ , in the following way :

$$M_\sigma := \begin{pmatrix} P_\sigma & \mathbf{0}_v \\ \mathbf{0}_h & 0 \end{pmatrix}$$

where $\mathbf{0}_v$ and $\mathbf{0}_h$ are zero column and zero row vectors respectively.

Then $(M_\sigma, M_\sigma, M_\sigma) \star t_1 = t_1$, i.e. $(M_\sigma, M_\sigma, M_\sigma)$ is in the stabilizer subgroup of t_1 .

Proof. As in Lemma 6.13, the action of the permutation matrices is permuting the order of the bases being used to construct $t_1 = \sum_{i=1}^{n-1} e_i \otimes e_i \otimes e_i$. In this case, however, we risk mapping one of the basis vectors $\{e_1, \dots, e_{n-1}\}$ to e_n , which should not be included in our summation. Hence it is crucial to exclude the permutations that do so by forcing the final row/column to be zeros. This leaves us with the permutations from S_{n-1} .

□

Lemma 6.15. Let I be the $(n-1) \times (n-1)$ identity matrix, $\mathbf{v} = (v_1, \dots, v_{n-1}) \in \mathbb{F}_q^{n-1}$, $v_n \in \mathbb{F}_q$ and let M be a matrix of the following form:

$$M := \begin{pmatrix} I & \mathbf{v} \\ \mathbf{0}_h & v_n \end{pmatrix}$$

where $\mathbf{0}_h$ is a zero row vector. Then $(M, I, I) \star t_1 = t_1$, $(I, M, I) \star t_1 = t_1$, and $(I, I, M) \star t_1 = t_1$. i.e. these three families of isomorphisms are in the stabilizer subgroup of t_1 .

Proof. Recall, applying an isomorphism (A, B, C) to t_1 means

$$(A, B, C) \star t_1 = \sum_{i=1}^{n-1} A e_i \otimes B e_i \otimes C e_i.$$

Since we are restricted to $i < n$, every vector contains a zero in the n^{th} component. Thus the n^{th} columns of A, B, C will always be multiplied by zeros. Therefore the families of isomorphisms $\{(M, I, I), (I, M, I), (I, I, M)\}$ are contained in the stabilizer subgroup of t_1 .

□

Equivalence classes on G . The study of stabilizer groups leads us to the definition of an equivalence relation on $G = \text{GL}(n, q) \times \text{GL}(n, q) \times \text{GL}(n, q)$. This observation will allow us to add more constraints to the solutions we are looking for in CTI (Computational Tensor Isomorphism Problem), and thus accelerate our attack in the proceeding section.

Definition 6.16. Let $(A_1, B_1, C_1), (A_2, B_2, C_2)$ be two elements from G , and $t \in \mathbf{V}$ a tensor. We say that (A_1, B_1, C_1) and (A_2, B_2, C_2) belong to the same equivalence class via t , denoted $(A_1, B_1, C_1) \equiv_t (A_2, B_2, C_2)$, if and only if they are equal up to right-multiplication by an element in the stabilizer subgroup of t .

6.2.4 Solving CTI in orbits of t_0

In this section we focus on the $b = 0$ case from the commitment scheme, and outline an approach to recovering an element from the isomorphism class via t_0 of the secret (A, B, C) . Recall our problem is as follows:

Problem 6.17. Let $t_0, g \in \mathbf{V}$ be two 3-tensors such that

$$t_0 := \sum_{i=1}^n e_i \otimes e_i \otimes e_i, \text{ and } g := \sum_{i,j,k=1}^n g_{i,j,k} e_i \otimes e_j \otimes e_k$$

for $\{g_{i,j,k}\} \subset \mathbb{F}_q$. Assuming it exists, find an isomorphism $(A, B, C) \in G$ such that

$$(A, B, C) \star t_0 = g, \tag{6.3}$$

where $\star$ is as defined in Equation 6.2.3.

We begin by noticing that the tensor t_0 has exactly n rank 1 points (up to scalars), which we will prove in Lemma 6.18. Then from Lemma 6.5 this tells us that g will also have n rank 1 points. Denote a set of rank 1 points of t_0 as $\{e_1, \dots, e_n\}$. Suppose we could find the rank 1 points in g , denoted $\{a_1, \dots, a_n\}$. Then, we can try to match the e_i to the a_i to recover part of the isomorphism, namely A^{-1} . We formalize this idea in Lemma 6.19.

Lemma 6.18. Let $t_0 = \sum_{i=1}^n e_i \otimes e_i \otimes e_i$ then t_0 has exactly n rank 1 points (up to scalars).

Proof. Let $u := [u_1, \dots, u_n]$ be a rank 1 point in t_0 , define matrices $T_0^1, \dots, T_0^n$ as

$$(T_b^k)_{ij} = \begin{cases} 1 & \text{if } i = j = k \text{ and } k \leq n - b \\ 0 & \text{otherwise} \end{cases}.$$

Since u has rank 1 in t_0 , we expect the following diagonal matrix to have rank 1 (by definition of rank of a point, see Definition 6.3).

$$U_0 := \begin{pmatrix} u_1 & & 0 \\ & \ddots & \\ 0 & & u_n \end{pmatrix}$$

Thus, for U_0 to have rank 1, we see that u can have only one non-zero entry. This gives n distinct points up to scalars. □

Lemma 6.19. Consider Problem 6.17. Let $\{a_1, \dots, a_n\}$ be the set of rank 1 points of g , and $\{e_1, \dots, e_n\}$ be the set of rank 1 points in t_0 (up to scalar multiplication). Then there exists an ordering, σ , of $\{a_1, \dots, a_n\}$ and a matrix A such that for each $i \in [1, \dots, n]$, $a_{\sigma(i)} = e_i A^{-1}$.

Proof. As a consequence of Lemma 6.5, we see that a rank one point in t_0 is sent to a rank one point in g through A^{-1} . This will fix the ordering σ .

□

Finding the rank 1 points $\{a_1, \dots, a_n\}$ is an instance of MinRank, so we can use the algorithm given in [BBC⁺20] that is summarized in Section 6.1.2. In this particular context, we want to find n matrices with target rank 1. This means that our coefficient variable matrix, R , is simply a row matrix. Denote the entries in R as $r_1, \dots, r_n$. This leaves us with a bilinear system in the target variables, $\{x_i\}_{i=1}^n$, and the coefficient variables, $\{r_i\}_{i=1}^n$, where we will have $n^2(n-1)/2$ equations and $n(n-1)$ monomials. If the equations are not related by linear dependencies, we can solve the system by direct linearization. In our experiments, this was indeed the case.

Going forward, Lemma 6.19 shows that we can recover some matrix A_0^{-1} whose rows are given by $\{a_i\}_{i=1}^n$. Our findings from Section 6.2.3 tell us that neither the permutation of these rows, nor the scalar multiples of them, will affect our search for a suitable isomorphism satisfying Equation 6.3. In fact, fixing these rows and scalar multiples is a good idea as it reduces the number of solutions and variables in our search.

Once we have recovered some A_0^{-1} , we can then solve for B_0 and C_0 to complete the isomorphism by considering the system of n^3 linear equations in $2n^2$ variables given by $(I, B_0, I) \star t_0 = (A_0^{-1}, I, C_0^{-1}) \star g$. The system, as is, is very under-determined because of the size of the equivalence class of (A, B, C) via t_0 . To slim down the solution space, thanks to Lemma 6.12, we can similarly normalize the first row of B_0 , by setting the entries of this row to arbitrary values of $\mathbb{F}_q^*$.

There is a chance that this will not give a solution, in case one or more of the entries in the first row of B were 0. We argue in the proof of Theorem 6.20 that when q is large enough compared to n , the probability of this failure is very low. In the small chance of failure, however, we can apply a new known isomorphism, $(I, B_1, I) \in G$, and now consider $(I, B_1, I) \star ((A, B, C) \star t_0) = (I, B_1, I) \star g$ instead. This essentially “rerandomizes” our instance, in the hopes that the solution space for $B_1 \cdot B$ will contain an entry whose first row has no zeros. Alternatively, we can also normalize a randomly chosen element in each column of B .

These steps are summarized in Figure 6.2.

CTI solver in orbits of t_0

- 1: **Given:** $g \in \mathbf{V}$, isomorphic to t_0
- 2: $a_1, \dots, a_n \leftarrow \text{MinRank}(g, \text{target rank} = 1)$
- 3: $A_0^{-1} \leftarrow \text{Matrix}(a_1, \dots, a_n)$
- 4: $B_0, C_0 \leftarrow \text{Solutions}((I, B_0, I) \star t_0 = (A_0^{-1}, I, C_0^{-1}) \star g)$
 $\cap \text{Solutions}((B_0)_{1,i} = 1, i = 1, \dots, n)$
- 5: **return** (A_0, B_0, C_0)

Figure 6.2: Algorithm for solving Problem 6.17.

Theorem 6.20. *There exists a probabilistic algorithm that solves Problem 6.17 using $O(n^6)$ operations.*

Proof. We first show that the probability of failure of Algorithm 6.2 is negligible for large q . When q is small, in case of failure one can just apply a new isomorphism $(I, B_1, I) \in G$ to g , and repeat the algorithm with this new instance as described above.

We have a chance of failure when we normalize the first row of B_0 . Since we have already fixed A_0 , this fixes the permutation of columns of B_0 as well (see Lemma 6.13). Hence, we risk failure when one of these normalized columns necessarily has a leading zero. The chance of this happening is the complement probability to the columns having no leading zero. This gives us a total probability of failure of

$$P(\text{Alg 6.2 fails}) = 1 - \frac{(q-1)^n}{q^n},$$

which is negligible for large q .

We now prove the complexity of this algorithm. The dominating subroutines in Algorithm 6.2 include the MinRank computation, the tensor group action arithmetic, and solving the final system of equations (which can be done using Gaussian elimination). While the tensor arithmetic could be done in time $O(n^4)$, as noted in Section 6.2 of [TDJ⁺22], the other two subroutines require $O(n^6)$ operations. $\square$

Remark 6.21. An alternate approach to proving Theorem 6.20 could be to apply Lemma 6.19 over all three axis, allowing us to recover some A, B, C , each up to some stabilizer elements. The issue with this approach, however, is that those stabilizer elements will a priori not be the same for all three matrices, so at this point we can write a system of equations to recover these stabilizer elements and finish the attack. This approach, while correct, adds an extra step, making it neither conceptually simpler nor more efficient than what is outlined above. Additionally, the above approach has the advantage of directly using the partial knowledge we have about A when recovering B and C .

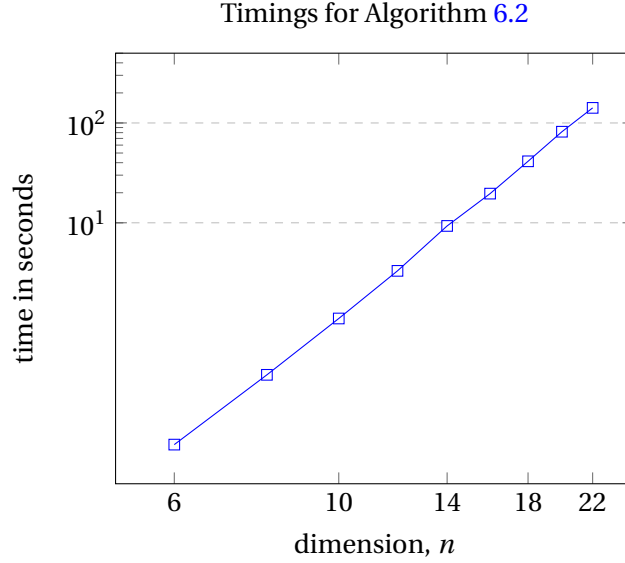
We test the attack from Theorem 6.20 on the parameters from Section 6.1.1 and give the results in seconds in Table 6.1. We consider some additional parameters in Figure 6.3 to further support our complexity claims. It corroborates our claim that Algorithm 6.2 has complexity $O(n^6)$, since the curve follows very closely to a line of slope 6.

6.2.5 Solving CTI in orbits of t_1

The approach of computing the set of rank 1 points via the MinRank algorithm from [BBC⁺20] will only work in the case the tensor has rank n (i.e. $b = 0$ in the commitment scheme). Any smaller rank would result in too many rank 1 points to compute. In the $b = 1$ case, however, we

n	q	time (s)
14	4093	9.3
22	4093	141.6
30	2039	858.9

Table 6.1: Timings in seconds for solving the CTI variant from Problem 6.17.

Figure 6.3: Timings of Algorithm 6.2 for $q = 4093$ and varying values of n on a log-log scale.

will see that we can edit MinRank to further filter distinct rank 1 points using our knowledge of the stabilizer group of t_1 .

We begin by computing the one rank 0 point in g (which is unique up to scalars). This point is easy to compute, as seen in Section 6.2.2. Denote it P_0 . Now, we know that any rank 1 point could have a rank 0 point added to it, and still remain a rank 1 point. In other words, we get an equivalence class on the set of rank 1 points where we say two rank 1 points, R_i, R_j , are equivalent if $R_i - R_j = \lambda P_0$ for some scalar λ . This means, for each $i \in [1, \dots, n-1]$, the equivalence class can be written as the subspace $\langle P_i, P_0 \rangle$, for some rank 1 point P_i , and the rank 0 point P_0 . Hence, we can obtain a representative from each equivalence class, P_i , by appropriately filtering the results returned from MinRank. More precisely, we normalize a component of the rank 1 points, thus filtering much of the equivalence class. The index of the component to be normalized corresponds to the index of the first non-zero entry of P_0 .

After these computations, we will be left with one rank 0 point and $n-1$ distinct rank 1 points. We are assured to get exactly $n-1$ linearly independent rank 1 points since each equivalence class, $\langle P_i, P_0 \rangle$, is the image under an isomorphism of the subspace $\langle e_i, e_n \rangle$. We can then follow a similar attack as that described in Section 6.2.4, Figure 6.2, to reconstruct an isomorphism

$$(A_0, B_0, C_0) \equiv_{t_1} (A, B, C).$$

First we need to be careful to place the rank 0 point in the final row of A_0^{-1} due to the restriction on permutation matrices in the stabilizer group of t_1 , as shown in Lemma 6.14. We can also use Lemma 6.15 to further filter out options by fixing the final columns of B_0, C_0 to random elements. To do so we slightly modify the equations we are solving to avoid high degree equations involving the inverse of C_0 . We consider the new equation $(I, B_0, C_0) \star t_1 = (A_0^{-1}, I, I) \star g$. This new equation is no longer linear since the left hand side will have quadratic terms in the variables of B_0 and C_0 . This means we cannot solve the system of equations linearly, but instead will require a Gröbner basis. Experiments show the run times of this attack to be very close to those of the computational attack on t_0 , which we give in Table 6.2. Thus, we also estimate the complexity of the overall attack to be polynomial.

n	q	time (s)
14	4093	8.7
22	4093	158.1
30	2039	1235.9

Table 6.2: Timings in seconds for solving CTI for t_1 .

6.3 Repairing the Asiacrypt 2023 scheme

In this section, we propose a new construction of a commitment scheme and include proofs of its security. This commitment is the first non-interactive commitment scheme from non-transitive group actions proposed; such group actions are less restrictive, and arise naturally. In particular, tensor-based group actions are such an example. At the moment isogenies have been among the only group actions receiving attention. By studying tensors in this application, and showing how they can be used in protocols, our hope is to contribute to the assumption diversification of the field.

The key components that we aim to preserve from [DFG23] are the following: a commitment scheme based on a non-transitive group action for which the group action should be uncertified^I, and the commitment scheme should be non-interactive.

We therefore propose a non-interactive bit commitment scheme from non-transitive uncertified group actions, described in Figure 6.4. It is statistically binding (Theorem 6.27) and computationally hiding (Theorem 6.29). We improve on [QSY19]’s construction, which is also statistically binding and computationally hiding but requires interaction. We lose the perfect binding property of [DFG23], however, this allows us to have a concrete instantiation that relies on standard problems. In practice, replacing perfect binding by statistical binding

^IAn *uncertified* group action is a group action for which checking that two elements are in the same orbit is hard.

does not affect the security of more advanced constructions that could be built from bit commitments. Note that we do not keep the framework of Problem 6.6, but rather introduce a slightly different and simpler one, which still does not introduce any new assumptions.

Definition 6.22 (Decisional group action framework). *Consider a group action $\star : G \times V \rightarrow V$ that has the following properties :*

1. *It is an Effective Group Action.*
2. *Given two elements v_0, v_1 such that $v_1 = g \star v_0$ for some $g \in G$, computing g is hard.*
3. *The probability, p , that two random elements are in different orbits is $p = 1 - \text{negl}(\lambda)$.*
4. *The Decisional Group Action Inversion Problem for $\star$ is hard (recall Definition 2.35).*

Note that the first three properties are described in [JQSY19].

Setup(1^λ)	Commit(b)	Verify($c, (b, g)$)
1: $v_0, v_1 \xleftarrow{\$} V$	1: $g \xleftarrow{\$} G$	1: return ($c = g \star v_b$)
2: return v_0, v_1	2: $c \leftarrow g \star v_b$	
	3: return c	

Figure 6.4: Commitment scheme

We can instantiate this framework using tensors through the group action described in Equation 6.2.3. The crucial difference with the construction of [DFG23] is that v_0, v_1 are now two **randomly** generated tensors and they do not possess any special structure. In particular, a random tensor will only have low rank points with negligible probability, as shown in Lemma 6.25.

Remark 6.23. One can extend the commitment scheme to accommodate a polynomial number of messages say $0, \dots, N$ by slightly modifying the Setup and Commit algorithms of Figure 6.4. Indeed, during Setup one can instead randomly sample N elements $v_0, \dots, v_N$ and output these. Then during Commit, to commit to a message $k \in \{0, \dots, N\}$, one samples $g \in G$ and computes $g \star v_k$.

Remark 6.24. Note that v_0, v_1 must be sampled randomly, in practice this can be done by sampling them as the output of a pseudo-random generator or a cryptographic hash function. What matters for our proofs is that they are indeed indistinguishable from elements sampled from a uniformly random distribution.

Lemma 6.25. *Given a random tensor $t \in \mathbf{V}$, the average number of rank $n - d$ points (up to scalar multiplication) is q^{-d^2+n-1} as q goes to infinity.*

Proof. Consider our tensor t as a list of n random $n \times n$ matrices, over $\mathbb{F}_q$, say $M_1, \dots, M_n$. A rank r point in t is $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{F}_q^n$, $\lambda \neq 0$, such that $\text{rank}(\lambda_1 M_1 + \dots + \lambda_n M_n) = r$.

First, we have that the probability that a random matrix M has rank $n - d$ for some integer d goes to q^{-d^2} as q goes to infinity. This result is a direct observation from [FG15]. Then we observe that there are about q^{n-1} possibilities projectively for λ , hence we get that the average number of rank $n - d$ points is q^{-d^2+n-1} .

□

This means that our distinguishing attack from Section 6.2.2 is not applicable on a randomly generated instance. Indeed, this lemma shows that it is unlikely for a random tensor to have points of small rank. For points of larger rank, the MinRank attack described in Section 6.1.2 becomes impractical.

Let us now check that all the desired properties from Definition 6.22 are satisfied: Property 1 has been studied in [QSY19] and Property 2 corresponds to the Computational Tensor Isomorphism Problem (Problem 2.73). Property 4 is introduced in [DFG23] and they discuss its applicability to tensors. While we show it is broken for their instantiation using unit tensors, t_b , it is still believed to be hard in the case of random tensors, which is precisely our case. Regarding Property 3, we must compute the probability, p , of two random tensors being in different orbits under $\star$. Note that [QSY19] assumes that this probability is high but they do not give any estimates.

Lemma 6.26. *The probability, p , of two random tensors being in distinct orbits under $\star$ is $p \geq 1 - \frac{1}{q^{n^3-3n^2}}$.*

Proof. Let $t \in \mathbf{V}$ be any tensor, then trivially

$$\#Orb(t) \leq \#G = (\#GL_n(q))^3 = O(q^{3n^2}).$$

Furthermore, we have $\#\mathbf{V} = q^{n^3}$. So the total number of orbits is at least $\#\mathbf{V}/(\#GL_n(q))^3 = O(q^{n^3-3n^2})$. The probability of two tensors being in the same orbit is therefore bounded by $\frac{1}{q^{n^3-3n^2}}$, hence $p \geq 1 - \frac{1}{q^{n^3-3n^2}}$. □

We have shown that all the properties from Definition 6.22 are satisfied. Now it remains to show that our proposed commitment scheme from Figure 6.4 is both hiding and binding.

Theorem 6.27 (Binding). *The commitment scheme described in Figure 6.4 is statistically binding.*

Proof. Consider G to be the binding security game from Section 2.1.2 applied to our scheme.

Taking the following probabilities over the randomness of the game we have :

$$\begin{aligned}
\Pr[G(\mathcal{A}) \rightarrow 1] &= \Pr[G(\mathcal{A}) \rightarrow 1 \wedge v_0, v_1 \text{ are in the same orbit}] \\
&\quad + \Pr[G(\mathcal{A}) \rightarrow 1 \wedge v_0, v_1 \text{ are in different orbits}] \\
&\leq \Pr[v_0, v_1 \text{ are in the same orbit}] \\
&\quad + \Pr[G(\mathcal{A}) \rightarrow 1 | v_0, v_1 \text{ are in different orbits}] \cdot p,
\end{aligned}$$

where p is the probability defined by Property 3, and computed in Lemma 6.26. We have $\Pr[v_0, v_1 \text{ are in the same orbit}] = 1 - p$ by definition of p . Furthermore, $\Pr[G(\mathcal{A}) \rightarrow 1 | v_0, v_1 \text{ are in different orbits}] = 0$ since otherwise if $\mathcal{A}$ wins G in that case, this means they return (A_0, B_0, C_0) and (A_1, B_1, C_1) such that

$$(A_0, B_0, C_0) \star v_0 = (A_1, B_1, C_1) \star v_1.$$

In particular this would mean that $(A_1^{-1} A_0, B_1^{-1} B_0, C_1^{-1} C_0) \star v_0 = v_1$ which is impossible since v_0, v_1 are in different orbits and thus no such isomorphism exists.

Overall we get

$$\Pr[G(\mathcal{A}) \rightarrow 1] \leq (1 - p) + 0 \cdot p = (1 - p) = \text{negl}(\lambda).$$

□

Remark 6.28. Notice that in the case where v_0, v_1 are indeed in the same orbit (which happens only with negligible probability), breaking the Binding problem amounts to solving the Computational Tensor Isomorphism problem (Problem 2.73) in a *random* orbit, hence remains plausibly hard. We could obtain a perfectly hiding and computationally binding scheme by sampling $v_1 = td \star v_0$ for a trapdoor $td \in G$ in the setup. But then the knowledge of the trapdoor allows to break the binding property. For this reason, we do not explore this version further.

Theorem 6.29 (Hiding). *Assuming the hardness of the dGA-IP problem (Definition 2.35), our commitment scheme is hiding.*

Proof. Consider G to be the hiding game from Section 2.1.2. Taking the following probabilities over the randomness of the game, we have :

$$\begin{aligned}
\Pr[G(\mathcal{A}) \rightarrow 1] &= \Pr[G(\mathcal{A}) \rightarrow 1 \wedge v_0, v_1 \text{ are in the same orbit}] \\
&\quad + \Pr[G(\mathcal{A}) \rightarrow 1 \wedge v_0, v_1 \text{ are in different orbits}] \\
&\leq \Pr[v_0, v_1 \text{ are in the same orbit}] \\
&\quad + \Pr[G(\mathcal{A}) \rightarrow 1 | v_0, v_1 \text{ are in different orbits}] \cdot p,
\end{aligned}$$

where p is the probability defined by Property 3. We have $\Pr[v_0, v_1 \text{ are in the same orbit}] = 1 - p = \text{negl}(\lambda)$.

In the case where v_0, v_1 are in different orbits, one can refer to [DFG23], Theorem 2. Roughly, given an adversary $\mathcal{B}$ against dGA-IP, that receives as input s, t , they instantiate two Hiding games with adversaries $\mathcal{A}_1, \mathcal{A}_2$ to which they respectively give s or t as input, and $\mathcal{B}$ returns 1 if the outputs of $\mathcal{A}_1$ and $\mathcal{A}_2$ are equal and 0 otherwise. They get that

$$\Pr[G(\mathcal{A}) \rightarrow 1 | v_0, v_1 \text{ are in different orbits}] \leq \frac{1}{2} + 2\varepsilon(\lambda)^2$$

where $\varepsilon(\lambda)$ is the advantage in the dGA-IP game.

So overall we have

$$\begin{aligned} \Pr[G(\mathcal{A}) \rightarrow 1] &\leq (1-p) + p \cdot \left(\frac{1}{2} + 2\varepsilon(\lambda)^2\right) \\ &= \frac{1}{2}p + (1-p) + 2p\varepsilon(\lambda)^2 \end{aligned}$$

and by assumption both $\varepsilon(\lambda)$ and $1-p$ are negligible. $\square$

Remark 6.30. Consider the extension of our commitment scheme to a message space of $\{0, \dots, N\}$ as described in Remark 6.23. Then the hiding and binding properties are derived in a straightforward way since $v_0, \dots, v_N$ are in different orbits with probability at least $1 - \frac{N}{q^{n^3-3n^2}}$.

6.3.1 Cost comparison

The cost of generating the commitment in [DFG23] is $O(n^3)$ multiplications and additions over $\mathbb{F}_q$ (the cost of matrix multiplication, ignoring asymptotic improvements). Due to our use of generic tensors, we have to perform $3n$ matrix multiplications to compute our commitment, bringing the cost to $O(n^4)$. The cost of multiplications over $\mathbb{F}_q$ is $O(\log q \log \log q)$. While this provides a slowdown compared to [DFG23], it is still reasonable when compared to other cryptographic operations. In particular, we benefit from the fact that matrix multiplication already has existing low-level code and algorithms, including parallel algorithms. It is also worth noting that our loss in efficiency only occurs during the actual computation of the commitment. For any operation on the commitment itself or involving it (e.g zero-knowledge proofs) the cost of manipulating it would be the same for both our construction and that of [DFG23] since they are both comprised of matrix multiplication on a (seemingly) random tensor.

MEDS [CNP⁺23b], a NIST signature candidate, only uses two matrices A, B , but introduces *systematic form computation* to recover a third matrix, C . This allows a trade-off between computation time and size. Therefore, the cost of a single group action on a matrix code is the cost of matrix multiplication plus the systematic form computation. This trade-off could also be applied to the case of our commitment scheme, along with their other (future) group action optimizations.

6.3.2 On proofs of knowledge

To use commitment schemes within more advanced protocols such as verifiable secret sharing or zero-knowledge proofs for generic statements, it is useful to have non-interactive zero-knowledge proofs of knowledge of commitments. More precisely, one wishes to prove that they know the value to which a commitment c corresponds without giving away any information about it.

Let us consider the commitment scheme of Figure 6.4, extended to the message space $\{0, \dots, N\}$. Given public parameters $\text{pp} = \{t_0, \dots, t_N\}$ consisting of $N+1$ random tensors from $\mathbf{V}$, we consider the relation

$$\mathcal{R} = \{(c, (i, (A, B, C))) \mid c = (A, B, C) \star t_i\},$$

where c is the statement and $(i, (A, B, C))$ is the witness. This means given c we wish to prove knowledge of the message i and randomness (A, B, C) to which c corresponds, without revealing anything about them. We briefly expose two ways of obtaining such a proof.

Using a variant of OR-proofs.

A proof for $\mathcal{R}$ can be obtained by combining a variant of an OR-proof [CDS94b] and a proof for $\mathcal{R}_i = \{((c, t_i), (A, B, C))) \mid c = (A, B, C) \star t_i\}$, to show that one knows an isomorphism mapping c to a particular choice of one of the t_i . A proof for $\mathcal{R}$ can be obtained by adapting, for example, the graph isomorphism proof [GMW91]. However, using OR-proofs increases the communication cost in the answer of the prover, and we will therefore prefer to build a direct proof for the full relation $\mathcal{R}$.

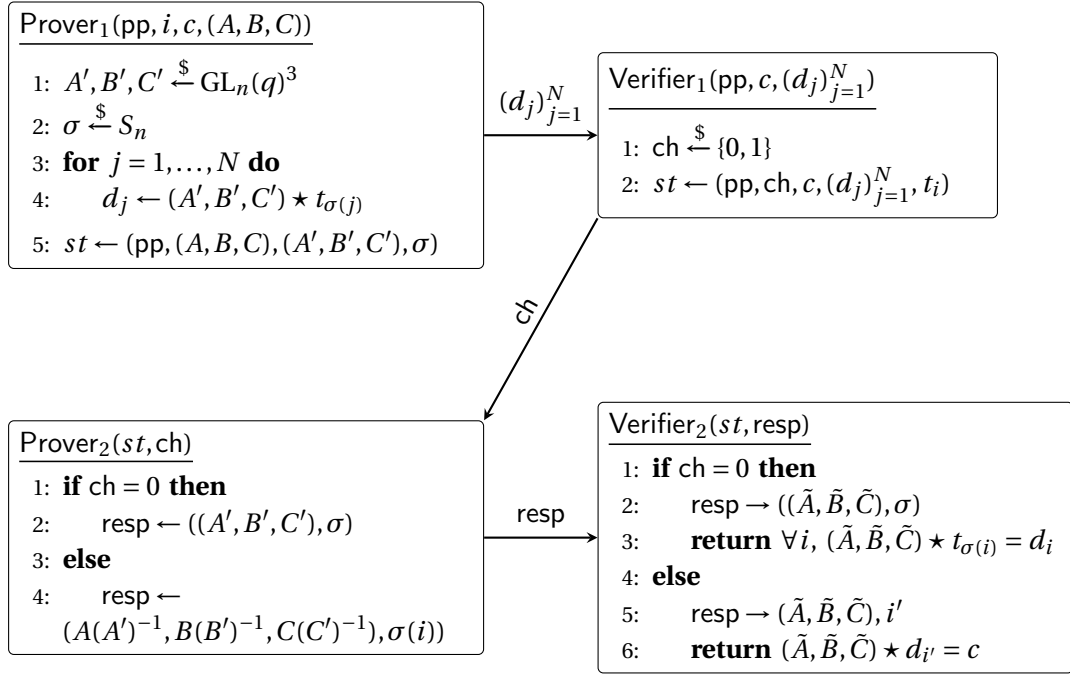
A direct proof for $\mathcal{R}$.

We give in Figure 6.5 a sigma protocol that directly gives a proof for $\mathcal{R}$.

We sketch the extractor for special soundness and simulator for honest-verifier zero-knowledge below.

Special-soundness. Suppose we have two accepting transcripts with different challenges, $((d_j)_{j=1}^N, \text{ch}_0, \text{resp}_0)$ and $((d_j)_{j=1}^N, \text{ch}_1, \text{resp}_1)$. Without loss of generality, we assume $\text{ch}_0 = 0$. Then we can build an extractor $\mathcal{E}$ which, given these transcripts as input, does the following : Recover (A', B', C') from resp_0 and $(A(A')^{-1}, B(B')^{-1}, C(C')^{-1})$ from resp_1 . Compute (A, B, C) by multiplying the latter by the former. Recover i' from resp_0 and recover the correct $i = \sigma^{-1}(i')$. Then $(i, (A, B, C))$ is the witness.

Honest-verifier zero-knowledge. We now sketch the simulator. In the case $\text{ch} = 0$, the simulator just follows the protocol honestly and will be producing an accepting transcript.

Figure 6.5: Proof system for $\mathcal{R}$.

In the $ch = 1$ case, the simulator samples (A', B', C') and σ as in the protocol, then picks some index j and sets $d_j = ((A')^{-1}, (B')^{-1}, (C')^{-1}) \star c$. The other d_i for $i \neq j$ are computed according to the protocol. The distribution will remain computationally indistinguishable since (A', B', C') is random. The answer then consists of j and (A', B', C') .

6.4 Conclusion

In this work, we study low rank points and stabilizers on unit tensors. We show how this information gives us the tools to break the hiding property of the bit commitment scheme proposed in [DFG23] at Asiacrypt 2023, that uses these unit tensors. Our attacks further allow to recover all the secret information used during the creation of the commitment. All of these algorithms run in polynomial time, which leaves the framework introduced in [DFG23] with no concrete instantiation. Prior to this work, there was no evidence in the literature suggesting that unit tensors were not secure for use in cryptography.

With these attacks in mind, we propose a slightly different framework as well as a construction that makes use of *random* tensors. This allows us to build a statistically binding, computationally hiding commitment from non-transitive group actions. Finally, we complete this work by proposing a zero-knowledge proof of opening for our new commitment scheme.

At the moment, as pointed out by [DFG23], the field of commitment schemes from non-transitive group actions is lacking protocols that are both post-quantum and non-interactive.

We hope our new construction helps to fill this gap, and that our attacks can help to inform future cryptography designers on some of the limits of working with unit tensors. We leave any further investigation into a commitment scheme using the framework proposed in [\[DFG23\]](#) as future work.

7 Analysis of Code-equivalence Problems and Applications to the Security of Two Blind Signature Constructions

This chapter is based on [GMPT25], published in IACR Communications in Cryptology, Volume 2, Issue 4.

The framework of cryptographic group actions allow to design protocols in an abstract setting to be later instantiated from concrete problems. Examples of these "cut and paste" group action frameworks include multi-signatures [DFMS25], commitment schemes [DFG23, JWL⁺25], threshold signatures [BBMP24, BBD⁺25], and blind signatures [DKQ⁺25]. Isogeny-based cryptography has commonly been considered the main candidate for such concrete instantiations, with the main caveats being that it tends to suffer from slow runtimes and extremely technical implementations.

Recently, more and more proposals using code-based problems have appeared in the community. Code-based protocols are attractive because they are more efficient than their isogeny counterparts, are easier to understand, and thus are also easier to implement correctly. On the other hand, they offer less compact protocols and the resulting group action is non-abelian, which can be restrictive for the design of more advanced protocols. The main problem considered is the *Code Equivalence problem*, which asks whether two codes belong to the same equivalence class under the action of some group, and, when it is the case, to compute the acting group element. This problem was first introduced and analyzed in [SS13], and then later found applications in the design of digital signature schemes (such as LESS [BMPS20], and then MEDS [CNP⁺23b]).

Following these initial proposals, the cryptographic community developed code equivalence problems in two orthogonal directions.

The first direction consists in varying the group that acts on the code itself. For example LESS ([BMPS20]) considers the action of the group of monomial matrices $\text{Mono}(n, q)$ on the set $[n, m]_q$ linear codes (See definition 2.74). This is called the Linear Code Equivalence problem (LCE, Problem 2.77). On the other hand, MEDS ([CNP⁺23b]) considers the action

Figure 7.1: Illustration of Inverse and t -sample Code Equivalence Problem

of $GL(n, q) \times GL(m, q)$ on the set of $[mn, k]$ matrix codes (or equivalently $[mn, k]_q$ linear codes), which results in the Matrix Code Equivalence Problem (MCE, Problem 2.83). These problems have further subdivided themselves into sub-categories. For instance, the action of a monomial matrix, as considered in LCE, can be decomposed as the product of a diagonal matrix by a permutation matrix. When setting the diagonal matrix to the identity, one therefore obtains the so-called Permutation Code Equivalence problem (PCE, not studied in this work). In the case of MCE, one can impose further condition on the acting matrices of $GL(n, q) \times GL(m, q)$, for example imposing their symmetry, or anti-symmetry, and considering the action of $Sym(m, q) \times Sym(n, q)$ instead. This observation underlines the MIMCE problem, which will be studied in section 7.2. Or, one can also consider the action of $GL(n, q) \times GL(m, q)$ on distinguished low-rank elements as suggested in [DFG23], although this has been proven insecure in [GMPT24].

The second direction in which these code-equivalence problems have evolved is that of considering inverse and multi-sample problems.

More specifically, inverse problems look into the following scenario: given three codes, where two of them are related to a third one by a group element g and its inverse respectively, how hard is it to compute g ? We illustrate this situation in Figure 7.1(a), where $\mathcal{C}_i$ are linear codes and $g \in G$ is a group element.

This gives rise to the Inverse Linear Code Equivalence problem ILCE (Problem 2.81), when the $\mathcal{C}_i$'s are linear codes and $G = \text{Mono}(n, q)$, and its analogous for matrix codes IMCE (Problem 2.84) when the $\mathcal{C}_i$'s are matrix codes and $G = GL(m, q) \times GL(n, q)$. ILCE has been shown to be polynomially solvable when $m = n/2$ by [BCD⁺24], but no such results exists for IMCE. To overcome the attack on ILCE, [DKQ⁺25] introduces the Diagonal-masked Inverse Linear Code Equivalence Problem (Problem 2.82), a variant of ILCE which will be studied in Section 7.1. Similarly, [KLP25] introduce the MIMCE problem, a variant of IMCE which instantiates our diagram with the $\mathcal{C}_i$'s as matrix codes and $G = \text{Sym}(m, q) \times \text{Sym}(n, q)$.

A natural extension of the inverse problem scenario, is the following: what if we give two pairs of codes related by the same group elements? Does that make the problem easier, and if so by how much? And what about the case where one were to give t such additional pairs of codes? This is what we call t -sample problems, which we illustrate in Figure 7.1(b).

[BCD⁺24] showed that the 2-LCE Problem (i.e $t = 2$ above, Problem 2.80) is solvable in polynomial-time when $n = m/2$. Additionally, [BCD⁺24] also gives bounds on the number of samples t needed to solve the t -MCE problem. We will investigate similar results for DmILCE and MIMCE.

We summarize this discussion in Table 7.1.

Code Type	Linear $[n, m]_q$		Matrix $[nm, k]_q$	
Group acting	Mono(n, q)	$S_n(q)$	$GL(n, q) \times GL(m, q)$	$Sym(n, q) \times Sym(m, q)$
Problem	LCE	DmILCE	MCE	MIMCE
Usage	Signature [BMPS20]	Blind signature [DKQ ⁺ 25]	Signature [CNP ⁺ 23b]	Blind signature [KLP25]
Hardness	Hard [BBPS23]	Section 7.1	Hard [CNP ⁺ 23a]	Section 7.2
Inverse Problem	ILCE, ring signatures [BBN ⁺ 22], broken for $n = m/2$	-	IMCE, hard [CNP ⁺ 23a]	-
t -sample Problem	2-LCE, Threshold signatures [BBMP24] broken for $n = m/2$	Section 7.1	Hard for small enough t [BCD ⁺ 24]	Section 7.3

Table 7.1: Overview of code equivalence problems

Our contributions In Section 7.1 we study a variant of the LCE problem. Namely, we will study the DmILCE problem which was introduced in Versions 1 and 2 of the blind signature scheme [DKQ⁺25]¹. We show that these problems are polynomial-time equivalent, strengthening confidence in this hardness assumption for future use in cryptography.

Then in Section 7.2 we turn our attention to the MIMCE problem, a variant of MCE. We focus our analysis on the parameters initially proposed by [KLP25], that scale the MEDS parameters linearly. We algebraically study its complexity, showing that these parameters do not reach the target security level. Note, the concurrent work [CD25] also analyzes the security of this problem. In this work, Chi-Domínguez presents an efficient attack on the problem, however, it does not affect the scaled MEDS parameters that we are targeting in this work. Thus, [CD25] and this work can be seen as orthogonal from each other.

Lastly, in Section 7.3, we consider a variant of the MIMCE problem where an adversary has

¹Shortly after the original version of our work was released, [DKQ⁺25] updated their own paper to alter the formulation of the DmILCE problem.

access to several samples. We first adapt the result of Budroni et al. [BCD⁺24] for this problem variant, showing that between 6-11 samples (depending on the code dimension) are needed to recover the secret in polynomial time. We improve upon this result by providing a polynomial-time algorithm that recovers the secret using only 2 samples, regardless of the dimension. This discourages attempts at building cryptographic primitives such as ring, group, or threshold signatures from the multiple sample version of this problem.

Related work

Algebraic modeling from MEDS

The digital signature MEDS [CNP⁺23a] relies on the MCE problem. To assess its security, in the original paper [CNP⁺23b], they introduce an algebraic modeling of the MCE problem (Problem 2.83) based on maximal minors modeling. This allows them to collect various equations, out of which a subset is independent. They then model the complexity of the polynomial system using Hilbert series and deduce an estimated bit complexity per parameter set. In the specification document [CNP⁺23a] submitted to the NIST signature scheme, they refined this modeling further. We will elaborate more on this modeling in Section 7.2.2.

Note that both [CNP⁺23b] and [CNP⁺23a] also evaluate further attacks on the original MCE problems, such as the Leon-like algorithm, but these attacks are vastly agnostic to the algebraic structure of the matrices acting on the code which is precisely what we wish to exploit in this work. We therefore do not consider them further.

Cryptanalysis techniques in the multiple-sample setting

In [BCD⁺24], the authors study variants of LCE and MCE in the setting when one has access to t -independent samples. They show that when $m = n/2$, 2-LCE is broken, as well as ILCE. They also give estimates on the number of independent samples needed to solve the MCE and LMCE problems with non-negligible probability in probabilistic polynomial time. They obtain their results by using a modeling based on the generator matrices of the codes and their parity check matrices. We develop their technique further and adapt them to the context of MIMCE in Section 7.3.1.

7.1 Reduction between LCE and DmILCE

The work of [DKQ⁺25] introduces a general framework for blind signatures from cryptographic group actions. They instantiate their framework using code-based group actions. In Versions 1 and 2 of their work, they presented the DmILCE problem (Problem 2.82). It was later updated, following the release of the paper [GMPT25] corresponding to this chapter to a newer version of the assumption to which the results below do not extend. In the following, we analyze this new assumption and show that it is in fact polynomial-time equivalent to the LCE problem,

a well-studied code-based assumption. We recall further reductions from LCE known in the literature to consolidate the confidence in this assumption.

We will make use of the following lemma.

Lemma 7.1. *Let $D \in D(n, q)$ and $P \in S_n$ then $PDP^{-1} \in D(n, q)$.*

Proof. Let $\{d_{i,i}\}_{i=1}^n$ denote the diagonal entries of D . Then left multiplication by a permutation matrix $P \in S_n$ will permute the rows of D by some permutation function π , thus it sends $d_{i,i}$ to $d_{\pi(i),i}$. On the other hand, right multiplication by the same permutation matrix P permutes the columns of D by π^{-1} . Thus it sends $d_{i,i}$ to $d_{i,\pi^{-1}(i)}$. Therefore left multiplication by P and right multiplication by P^{-1} sends $d_{i,i}$ to $d_{\pi(i),(\pi^{-1})^{-1}(i)} = d_{\pi(i),\pi(i)}$. This means that every diagonal entry in D remains on the diagonal. The other entries of D are identically zero, hence permuting them will not change the shape of the resulting matrix. Therefore, $PDP^{-1} \in D(n, q)$. $\square$

We now give polynomial-time algorithms that can solve LCE given access to a DmLLCE solver, and vice versa.

Theorem 7.2. *LCE and DmLLCE are polynomial-time equivalent.*

Proof. (LCE solves DmLLCE:)

Given the DmLLCE instance (G_0, G_1, G_2) , observe that the pair (G_0, G_1) forms an LCE instance. Thus we can use the LCE solver to recover $Q = D_0P$ such that $G_1 = A_0G_0D_0P$ for $A_0 \in GL(m, q)$, $D_0 \in D(n, q)$ and a permutation matrix $P \in S_n$. With Q , we can easily read off P . This trivially solves DmLLCE.

(DmLLCE solves LCE:)

Let (G_0, G_1) be an instance of LCE. Thus $G_1 = A_0G_0D_0P$ for some $A_0 \in GL(m, q)$, $D_0 \in D(n, q)$ and a permutation matrix $P \in S_n$. We are asked to recover all of A_0, D_0, P .

Consider an adversary $\mathcal{B}$ against DmLLCE. We want to build an adversary $\mathcal{A}$ that can solve LCE given access to $\mathcal{B}$.

The key insight of our reduction consists in observing that in the DmLLCE instance, the codes corresponding to G_0 and G_2 are in the same equivalence class, i.e. $[G_0]_D = [G_2]_D$. Indeed, recall that $[G_0]_D := \{AG_0D : A \in GL(m, q), D \in D(n, q)\}$. We have that

$$\begin{aligned} G_2 &= A_1G_1D_1P^{-1} \\ &= A_1(A_0G_0D_0P)D_1P^{-1} \\ &= (A_1A_0)G_0(D_0PD_1P^{-1}). \end{aligned}$$

Note that $(A_1 A_0) \in GL(m, q)$. Further, by Lemma 7.1 the matrix $PD_1 P^{-1}$ is diagonal, and thus so is $D_0 P D_1 P^{-1}$. This shows that $G_2 \in [G_0]_D$.

Using this insight, let us now describe the corresponding adversary, which is summarized in Algorithm 7.2.

$\mathcal{A}(G_0, G_1)$

- 1: $D \xleftarrow{\$} D(n, q)$
- 2: $A \xleftarrow{\$} GL(m, q)$
- 3: $G_2 \leftarrow A G_0 D$
- 4: $\mathcal{B}(G_0, G_1, G_2) \rightarrow P$
- 5: Solve $G_0 D_0 P H_1^T = 0$ to recover some D_0
- 6: Solve $G_1 = A_0 G_0 D_0 P$ to recover A_0
- 7: **return** (A_0, D_0, P)

Figure 7.2: Adversary $\mathcal{A}$ against LCE.

Given G_0, G_1 of rank m , $\mathcal{A}$ will start by selecting a random $D' \in D(n, q)$ and $A' \in GL(m, q)$. Then it computes G_2 in the following way:

$$\begin{aligned} G_2 &:= A' G_0 D' \\ &= A' (A_0 G_1 P^{-1} D_0^{-1}) D' \end{aligned}$$

Let $A_1 := A' A_0 \in GL(m, q)$. Since A' is sampled from a uniformly random distribution, then A_1 will be randomly distributed as well. Further, note that G_2 will always have rank m since all of A' , G_0 , and D' will have full rank by definition.

Define $D_1 := P^{-1} D_0^{-1} D' P$. Then, by Lemma 7.1, D_1 is diagonal and we can rewrite our equation as

$$G_2 = A_1 G_1 D_1 P^{-1}.$$

As before, since D' is sampled uniformly within the set of $n \times n$ diagonal matrices over $\mathbb{F}_q$, it follows that D_1 will also be uniformly random within that same set. Hence the G_2 built here follows the same distribution as a DmLCE instance.

It follows that the tuple (G_0, G_1, G_2) constitutes a full instance of DmLCE. After passing it to the DmLCE solver $\mathcal{B}$, we are provided P .

From here, we may recover D_0 using the parity check matrix of G_1 , denote it H_1 . Observe that

$$\begin{aligned} G_1 H_1^T &= 0 \\ \iff A_0 G_0 D_0 P H_1^T &= 0 \\ \iff G_0 D_0 P H_1^T &= 0 \end{aligned}$$

This gives a system of mn linear equations and only n variables, hence we can recover D_0 using

Gaussian elimination in $O(n^3)$ elementary operations.

From here, we can recover A_0 by solving the linear system $G_1 = A_0 G_0 D_0 P$. This computation also requires $O(n^3)$ elementary operations.

Suppose that the adversary $\mathcal{B}$ solves the DmLCE problem in polynomial time, then by construction the adversary $\mathcal{A}$ will solve the LCE problem in polynomial time too. $\square$

This reduction gives theoretical support for the hardness of the blind signature scheme from [DKQ⁺25]. Additionally, in [BW25, Corr. 5.2] Bennett and Win give a polynomial-time reduction from LCE (over a field whose order is polynomially-bounded) to the Lattice Isomorphism Problem (LIP). Therefore, [BW25] together with Theorem 7.2, proves the existence of a reduction from DmLCE to two well-studied problems: LCE itself and LIP.

Remark 7.3. Note that the 2-sample version of LCE is broken in polynomial time by the work of [BCD⁺24]. Thus our equivalence result suggests against using several-sample versions of DmLCE too.

7.2 (One sample) MIMCE

In what follows we will consider the pure MIMCE problem described in Problem 2.85. This problem was introduced in [KLP25] to prove the one-more unforgeability property of the blind signature they introduce, but its analysis was left as an open problem. This section and the next one aim at filling this gap and providing an in-depth analysis of this new code-based problem. We proceed to study this problem algebraically by modeling it as a system of polynomial equations, in the same spirit as [CNP⁺23a] and [RST23]. We will model the problem using a bilinear system of equations. Note, for the sake of conciseness, we will assume that the matrices A, B are symmetric, though the same approach will apply in the anti-symmetric case.

7.2.1 Bilinear system

Recall that we are working with generator matrices $G_0, G_1, G_2 \in \mathbb{F}_q^{k \times mn}$ such that

$$\begin{cases} G_1 = S_0 G_0 (A \otimes B) \\ G_2 = S_1 G_0 (A \otimes B)^{-1} \end{cases} \quad (7.1)$$

for some $S_0, S_1 \in GL(k, q)$, and $A \in GL(m, q), B \in GL(n, q)$ symmetric.

Remark 7.4. Throughout the attack, we will be decomposing G_0, G_1 into m blocks of size $k \times n$ and n blocks of size $k \times m$. Thus, in order for the notion of invertibility to be well-defined, we require $n = m = k$.

Relations depending on only A and B

Just as in [CNP⁺23a] and [CNP⁺23b], we would like to eliminate S_0 and S_1 to obtain a set of equations only depending on A and B . We start from the equations

$$\begin{cases} S_0^{-1} G_1 = G_0(A \otimes B) \\ S_1 G_0 = G_2(A \otimes B) \end{cases} \quad (7.2)$$

We write the right-hand side as a list of n blocks of size $k \times m$. We denote a block of index i for a matrix M as $[M]_i$. Hence we are interested in $[G_j(A \otimes B)]_i$, with $j = 0, 2$ and $i = 1, \dots, n$.

We use the same block decomposition on G_0, G_1 and similarly denote the blocks as $[G_1]_i$, respectively $[G_0]_i$. Assuming the inverses are well-defined, we get that $S_0^{-1} = [G_0(A \otimes B)]_i [G_1]_i^{-1}$ and $S_1 = [G_2(A \otimes B)]_i [G_0]_i^{-1}$ for all $i = 1, \dots, n$.

Remark 7.5. The probability that any one block M has rank $n - d$ for some integer d goes to q^{-d^2} as q goes to infinity. This result is a direct observation from [FG15]. To ensure invertibility, we are interested in full rank matrices i.e. $d = 0$, hence as q grows, this probability tends to 1. For small values of q this could still be an issue. However, when a block is not full rank, one can add a linear combination of the other blocks to itself to make it invertible.

Taking pairwise differences, this gives $2(n - 1)$ sets of mk equations of the form:

$$\begin{cases} [G_0(A \otimes B)]_1 [G_1]_1^{-1} - [G_0(A \otimes B)]_i [G_1]_i^{-1} = 0 \\ [G_2(A \otimes B)]_1 [G_0]_1^{-1} - [G_2(A \otimes B)]_i [G_0]_i^{-1} = 0 \end{cases} \quad (7.3)$$

In total, we obtain $2(n - 1)mk$ bilinear equations in $n(n + 1)/2 + m(m + 1)/2$ variables. We experimentally verified that these equations are linearly independent. Further, note that the system itself is very structured, and this can be exploited while solving the system of equations. This structure is described in the following lemma.

Lemma 7.6. *Each $k \times n$ block $[G_0(A \otimes B)]_i$ (respectively $[G_2(A \otimes B)]_i$) has $n(n + 1)/2$ variables from B but only m variables from A .*

Proof. This can be seen since

$$G_0(A \otimes B) = G_0 \begin{pmatrix} a_{1,1} \cdot B & \dots & a_{1,m} \cdot B \\ \dots & \dots & \dots \\ a_{m,1} \cdot B & \dots & a_{m,m} \cdot B \end{pmatrix} = [[G_0]_1, \dots, [G_0]_m] \begin{pmatrix} a_{1,1} \cdot B & \dots & a_{1,m} \cdot B \\ \dots & \dots & \dots \\ a_{m,1} \cdot B & \dots & a_{m,m} \cdot B \end{pmatrix}$$

where $a_{i,j}$ denotes the $(i, j)^{th}$ coefficient of the matrix A , and $[G_0]_i$ denotes the i^{th} $k \times n$ block of G_0 . Thus, the i^{th} block $[G_0(A \otimes B)]_i$ will be of the form $[G_0(A \otimes B)]_i = \sum_{j=1}^m [G_0]_j a_{i,j} B$ which uses all the variables of (the symmetric matrix) B but only m from A . And then, each entry of the block matrices involves at most mn distinct monomials. $\square$

Remark 7.7. If we denote by H_1 the parity check matrix for G_1 , we can also see that $G_0(A \otimes B)H_1^T = 0$. At first sight this gives an additional k^2 equations, however, this matrix equation is linearly dependent with the above system. Indeed, the rows of H_1 span the right kernel of G_1 , which itself is obtained from G_0 through the linear transformation defined by (A, B, S_0) . Hence H_1 depends linearly on S_0^{-T} , and S_1^{-1} also depends linearly on S_0 . Consequently, the matrix equation $G_0(A \otimes B)H_1^T = 0$ does not impose any new independent constraint on A and B .

The following lemma shows that the system can be explicitly linearized up to $n = 5$.

Lemma 7.8. *Assuming the above set of equations are linearly independent, linearization is possible up to $n = 5$.*

Proof. By counting equations and monomials, we can give a lower bound on when linearization will be possible. We have $2(n-1)mk$ equations in $nm(n+1)(m+1)/4$ monomials. Therefore, when $n = m = k$ (which is the main case of interest in [KLP25]), we have $2(n-1)n^2$ equations for $\frac{n^2(n+1)^2}{4}$ monomials. Further, we can normalize one variable in each of A and B , which immediately gives an extra two linear equations. We can multiply these linear equations by every other variable to get an additional $n(n+1)$ equations. This allows for direct linearization up to $n = 5$. $\square$

Experimentally, the system can be solved for higher n values using a Groebner basis. We list the average runtimes for small values of n in Table 7.2 using the computer algebra software Magma [BCP97] run on a laptop. The code for these experiments is included at <https://github.com/vgilchri/blind-sigs-mimce>.

n	5	6	7	8	9
Time to recover MIMCE secret keys	0.02s	0.3s	5.9s	2.0 min	1.38 hr

Table 7.2: Average runtime (over 10 runs) of solving the bilinear system described in Section 7.2.1.

7.2.2 MEDS modeling

In the original MEDS paper [CNP⁺23b], as well as in the specifications [CNP⁺23a], an algebraic modeling of the MCE problem is proposed. We proceed to describe it in detail, as it is the main point of comparison with our work.

Maximal minors modeling from [CNP⁺23b] We start with the improved modeling technique of [CNP⁺23b, Sect. 6.2]. Let us consider the secret matrices $A \in GL_n(\mathbb{F}_q)$, $B \in GL_m(\mathbb{F}_q)$ to be variables. The MCE problem is modeled using the generator matrix interpretation, i.e. we are given $G_1, G_0 \in \mathbb{F}_q^{k \times mn}$ such that $G_1 = S_0 G_0 (A \otimes B^T)$ for $S_0 \in GL_k(q)$ and we are solving for A, B .

Let $\tilde{G}_1 = G_0(A \otimes B^T)$. A key observation of [CNP⁺23b], is that augmenting G_1 with any row i of $\tilde{G}_1$ results in a matrix G'_i that is not full rank, since G_1 and $\tilde{G}_1$ are in the same equivalence class of codes.

Using this observation, they then derive bilinear equations on A, B by setting the maximal minors of these augmented matrices G'_i to be zero. This gives $\binom{mn}{k+1}$ bilinear equations in $m^2 + n^2$ variables, but only $(mn - k)k$ of these are linearly independent. Indeed, for a single minor, since the rank of the matrix is k , each column can be written as a linear combination of k columns, resulting in $(mn - k)$ independent equations for each of the k minors, thus $(mn - k)k$ equations in total.

They also consider another modeling, using the parity check matrix of G_1 , say H_1 and observing that $H_1 \cdot \tilde{G}_1^T = 0$, but they observe it only gives redundant equations with respect to the modeling described above.

The modeling of [CNP⁺23b] gives the same number of independent equations as derived in Section 7.2.1. However, [CNP⁺23b] collects a total of $\binom{mn}{k+1}$ (or k^2 for the parity check one) equations out of which only $(mn - k)k$ are independent whilst Section 7.2.1 generates directly a set of independent equations. This gives no asymptotic improvement but it may improve the runtime of a practical attack.

Extended modeling from [CNP⁺23a] The modeling described above is extended in [CNP⁺23a, Section 4.3.1] by using the fact that matrix codes can be viewed as trilinear maps. Recall, a given matrix code $\mathcal{C}$ can be represented as $\mathcal{C}(x, y, z) = \sum_{i,j,s=1}^{m,n,k} C_{ij}^{(s)} x_i y_j z_s$. One can then reformulate the MCE problem as $\mathcal{C}_0(Ax, By, S_0 z) = \mathcal{C}_1(x, y, z)$. This framework allows them to generalize the techniques of [CNP⁺23b]. More specifically, rewriting the equation as $\mathcal{C}_0(Ax, By, z) = \mathcal{C}_1(x, y, S_0^{-1} z)$, the modeling presented above allows to remove S_0^{-1} as done by [CNP⁺23b] and in Section 7.2.1. They observe that one can rewrite the original equation in two more ways (moving either A^{-1} or B^{-1} to $\mathcal{C}_1$), and use a similar technique to remove A^{-1} and B^{-1} respectively.^{II} This allows them to collect a total of $k(nm - k) + m(nk - m) + n(mk - n)$ bilinear equations in the variables of A, B and S_0 . Taking into account syzygies, they are then able to deduce a tri-graded Hilbert series to estimate the concrete complexity of the attack.

Going forward, since the modeling from [CNP⁺23b] aligns more closely with the analysis from Section 7.2.1 (in that it only removes S_0^{-1}), we will use [CNP⁺23b] as a baseline for comparison.

7.2.3 Theoretical analysis

Let us start by taking a theoretical detour and establishing the necessary background on the analysis of polynomial systems of equations.

^{II}Note that this manipulation implicitly assumes that $n = m = k$ so that the three transformations A, B, S_0 act on the same space and their inverses can be consistently moved between the trilinear forms.

Polynomial systems of equations

Most of our cryptanalysis results involve solving non-linear systems of equations. We briefly recall some techniques that are used to solve such systems. The general idea consists of collecting combinations of the original polynomial equations, until one can solve it by solving a linear system with respect to some monomials. The precise way of doing so varies depending on the technique, but generally it involves some variant of the Gröbner basis algorithm and estimating the precise number of combinations needed is non-trivial.

One main tool is the use of so-called Macaulay matrices.

Definition 7.9 (Macaulay matrix). *The Macaulay matrix $\mathcal{M}_d$ of degree d of a polynomial system $(p_1, \dots, p_m) \subset \mathbb{F}_q[x_1, \dots, x_n]$ is a matrix whose columns are labeled by the monomials m of $\mathbb{F}_q[x_1, \dots, x_n]$ up to degree d and whose rows are labeled by the products $r \cdot p_i$ for $r \in \mathbb{F}_q[x_1, \dots, x_n]$ such that $\deg(r) \leq d - \deg(p_i)$. Then the entry $(r \cdot p_i, m)$ consists of the coefficient of m in $r \cdot p_i$.*

Observe that the rowspace of $\mathcal{M}_d$ is precisely the space of linear combinations of the polynomial equations up to a certain degree d . By echelonizing $\mathcal{M}_d$ we can see whether we obtain linear relations in the monomials and solve the system. We are interested in several metrics that characterize the complexity of solving a system of polynomial equations.

- The first one is the number of *syzygies* at degree d , each of which capture the linear dependencies between the rows of the corresponding Macaulay matrix $\mathcal{M}_d$.
- The second one is the *solving degree* which is the value of d at which $\mathcal{M}_d$ has corank 1 (in the case of a 0-dimensional ideal with unique solution). In general, the solving degree is the maximum degree of the polynomials that appear in the Gröbner basis computation (See [CG21, Definition 1.1]).
- The third one is *linearization degree*, which estimates when the number of independent equations is greater than the number of monomials, and one can solve a system by direct linearization.

Another important tool to understand the complexity of polynomial systems of equations is their associated Hilbert series.

Definition 7.10 (Hilbert series). *Let $P = K[x_1, \dots, x_n]$ be a polynomial ring over a field K , and I be an homogeneous ideal of P . Then P/I is a graded vector space in the sense that it is obtained as a direct sum of the K -vector space of homogeneous polynomials of degree d , say V_d , for $d = 0, 1, \dots$. For such a graded vector space P/I , the Hilbert series is given by*

$$\mathcal{H}(t) = \sum_{d \geq 0} \dim(V_d) t^d$$

which can also be written as a rational function of t .

Computing the expression of the Hilbert series as a rational function of t lets us deduce information about the dimension of the V_d , which then allows us to compute the solving degree of our polynomial system.

For example, if $f \in P \setminus \{0\}$ is an homogeneous polynomial of degree d , then $\mathcal{H}_{P/\langle f \rangle}(t) = \frac{1-t^d}{(1-t)^n}$. Or if $I = \langle t_1, \dots, t_r \rangle$ such that the t_i 's are pairwise coprime monomials in P of degree d_i , then $\mathcal{H}_{P/I}(t) = \frac{\prod_{i=1}^r (1-t)^{d_i}}{(1-t)^n}$.

We now consider a system of polynomial equations with two types of variables. We can model it as a bi-graded ring where we separate variables into two types, and the number of variables of each type is denoted by n_1 and n_2 respectively. This means we work over $\mathbb{F}_q[x_1, \dots, x_{n_1+n_2}]$ graded by

$$\begin{aligned} \deg x_1 = \dots = \deg x_{n_1} &= (1, 0) \\ \deg x_{n_1+1} = \dots = \deg x_{n_1+n_2} &= (0, 1) \end{aligned}$$

where $\deg(h) = d \in \mathbb{N}^2$ if $h \in \mathbb{F}_q[x_1, \dots, x_{n_1+n_2}]_d$.

Following the formalism of [PS20, Lemma 2] and [NIW⁺20], if we identify $x_1, \dots, x_{n_1}$ to r , $x_{n_1+1}, \dots, x_{n_1+n_2}$ to s , we obtain the following result on the number of monomials of a given bi-degree (α, β) .

Lemma 7.11. *[PS20, Lemma 2] The number of monomials of bi-degree (α, β) is given exactly by the coefficient of $s^\alpha t^\beta$ in the series*

$$\mathcal{M}(s, t) = \frac{1}{(1-s)^{n_1} \cdot (1-t)^{n_2}}.$$

We consider a system of polynomial equations where all the equations are bilinear i.e. their bi-degree is $(1, 1)$.

The work of [PS20] and [NIW⁺20] shows that the bi-graded Hilbert series of such a system can then be conjectured as

$$\mathcal{H}(s, t) = \mathcal{M}(r, s, t) \cdot (1-st)^{\#\text{independent eqs. of bi-degree}(1,1)}$$

One can then approximate the solving degree of our system by looking at the coefficients of this series. Let $[s^\alpha t^\beta] \mathcal{H}$ denote the coefficient of $s^\alpha t^\beta$ in the Hilbert series $\mathcal{H}(s, t)$. Then, [NIW⁺20], [PS20] show that the minimal value of the following set $\{\alpha + \beta \mid [s^\alpha t^\beta] \mathcal{H} < 0\}$ is a good approximation for the solving degree.

One can then deduce an estimate for the complexity of the system by plugging the estimated value of the solving degree into a complexity estimator for system-solving algorithms. As in [CNP⁺23a] and [RST23], in this work we will estimate the complexity of using the block

Wiedemann XL algorithm as:

$$\min_{\substack{(\alpha, \beta) < (n, m) \\ [s^\alpha t^\beta] \mathcal{H} \leq 0}} 3 \cdot ([s^\alpha t^\beta] \mathcal{M})^2 \cdot \text{density}.$$

where $[s^\alpha t^\beta] \mathcal{M}$ denotes the coefficient of the $s^\alpha t^\beta$ term in $\mathcal{M}$, density is the density of the system (i.e. the average number of distinct monomials per equations), and 3 is a constant of the algorithm.

Complexity of our attack

Using the theory we just exposed, we can evaluate the complexity of our attack by modeling our system within a bi-graded polynomial ring, with two types of variables, those for A ($m(m+1)/2$ of them) and those for B ($n(n+1)/2$ of them). We can then write the generating function of monomials as follows:

$$\mathcal{M}(s, t) = \frac{1}{(1-s)^{m(m+1)/2} \cdot (1-t)^{n(n+1)/2}}.$$

Recall, we have $2(n-1)mk$ independent bilinear equations and have verified their independence at degree $(1, 1)$ experimentally. Therefore we can conjecture the Hilbert series to be of the following form:

$$\mathcal{H}(s, t) = \mathcal{M}(s, t) \cdot (1-st)^{2(n-1)mk}.$$

This allows us, as outlined in Section 7.2.3, to estimate the solving degree and deduce the bit complexity of using the block Wiedemann XL algorithm as:

$$\min_{\substack{(\alpha, \beta) < (n, m) \\ [s^\alpha t^\beta] \mathcal{H} \leq 0}} 3 \log_2(q) \cdot ([s^\alpha t^\beta] \mathcal{M})^2 \cdot \text{density}.$$

where 3 is a constant of the algorithm and $\log_2(q)$ accounts for the cost of field arithmetic. To estimate density, one needs to estimate the number of distinct monomials appearing in our equations. From Lemma 7.6, we get that we have on average mn distinct monomials per equation in each block, so when we subtract the blocks to obtain our equations, we obtain at most $2mn$ distinct monomials. We therefore take $\text{density} = 2mn$.

We compare our estimates for MIMCE against similar metrics from IMCE, and MCE to give an intuition on the related complexities of these problems. To estimate the complexity of MCE and IMCE, we use the modeling of MEDS [CNP⁺23b]. We have n^2 , resp. m^2 variables in s , resp. t , since A, B are no longer symmetric, and the modeling gives $2(mn-k)k$ independent equations in the case of IMCE versus $(mn-k)k$ for MCE, the density is now $nm(k+1)$ and we can similarly deduce a conjectured Hilbert series.

We report our findings in Table 7.3.

n	MIMCE		IMCE		MCE	
	Solving degree	Complexity	Solving degree	Complexity	Solving degree	Complexity
5	3	33	5	50	9	68
13	4	61	7	106	12	155
14	4	63	8	117	13	169
15	4	64	8	121	13	176
16	4	66	8	127	14	191
17	4	67	8	130	14	197
18	4	69	9	144	15	211
19	4	70	9	147	16	226
20	4	72	9	151	16	233

Table 7.3: Comparison of solving degree and complexity for MIMCE (Problem 2.85), IMCE (Problem 2.84, based on [CNP⁺23b]), and MCE (Problem 2.83, based on [CNP⁺23b]) across different values of n . The NIST Level I parameters proposed in MEDS ($n = 14$) and MIMCE ($n = 20$) are highlighted. The size of the prime used is 13 bits, as done in [CNP⁺23b].

Predicted complexity of MCE, IMCE, MIMCE

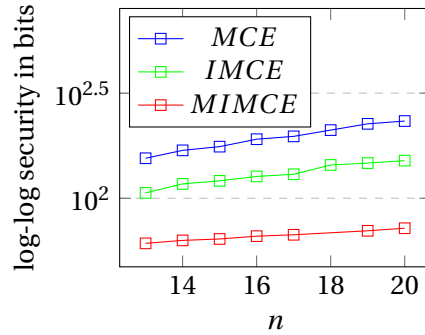


Figure 7.3: Values reported in Table 7.3.

In [KLP25], the authors acknowledged that the use of (anti)symmetric matrices would decrease the security of the MIMCE problem when compared to MCE. As a countermeasure, they suggested to increase the dimension of the matrices n by a factor of $\sqrt{2}$. Table 2., clearly shows that [KLP25] heavily underestimated the impact of the symmetry of matrices on the hardness of the problems. For example, in the case of NIST level 1 security that targets 126 bits of security, their proposed countermeasure only seems to achieve about 75 bits of security. Further, the fact that the solving degree does not increase much among the increasing values of n may indicate that the security of the parameters does not scale at the same rate as that of MCE. This makes it unclear how to select appropriate parameters for the blind signature.

7.3 Solving 2-(M)IMCE

We now consider a variant of (M)IMCE (Problem 2.85), where an adversary has access to more than one instance using the same secret matrices A, B . Such a formulation would be of interest in the context of multi-party protocols, such as threshold or ring signatures. First, we establish the state-of-the-art for this problem using the findings from Budroni et al. [BCD⁺24]. Later, we present our attack and show that actually, in the case of MIMCE, two samples are sufficient to recover A, B in polynomial time.

7.3.1 t -symIMCE

We follow the analysis of [BCD⁺24] to study the MIMCE variants with more samples, i.e. we look at the variants of Problems 2.83 and 2.84 when the matrices involved are symmetric and two samples are provided.

In Corollaries 1 and 2 of [BCD⁺24], the authors give an estimate on the number of samples needed to solve MCE, respectively IMCE, with non negligible probability in time $O((mn)^{2\omega})$, where ω is the linear algebra constant. We follow their modeling to deduce a similar bound in the context of the blind signature from [KLP25] whose hardness relies on MIMCE.

The IMCE problem can be expressed using the generator matrices of the underlying code. Given G_0 , a generator matrix of $\mathcal{C}_0$, we get that $G_1 = SG_0Q$ is the generator matrix of $\mathcal{C}_1$, with $Q = A^T \otimes B^T$ in $GL(mn, q)$.

One can then use the equation that stems from [BCD⁺24, Prop. 2] to write

$$G_1 H_1^T = 0 \iff SG_0 Q H_1^T = 0 \iff G_0 Q H_1^T = 0 \iff (G_0 \otimes H_1^T) \text{vec}(Q) = 0$$

[BCD⁺24, Prop. 2] shows that this expression gives $k(mn - k)$ linear equations in $(mn)^2$ variables (the entries of $Q = A^T \otimes B^T$). However, in our case A, B are symmetric. That is, $A = A^T$ and $B = B^T$ so in particular $Q = A^T \otimes B^T = A \otimes B$ is also symmetric by the properties of the Kronecker product $((A \otimes B)^T = (A^T \otimes B^T))$. So we can instead model Q using $mn(mn + 1)/2$ variables. By following their reasoning, we can adapt their results to obtain the following two corollaries.

Corollary 7.12. For $t \geq \lfloor \frac{mn(mn+1)/2}{k(mn-k)} \rfloor + 1$ samples, the t -symMCE problem is solvable with non-negligible probability in time $O((mn)^{2\omega})$.

Corollary 7.13. For $t \geq \lfloor \frac{mn(mn+1)/2}{2k(mn-k)} \rfloor + 1$ samples, the t -symIMCE problem is solvable with non-negligible probability in time $O((mn)^{2\omega})$.

Following the NIST Level 1 MEDS parameters, we fix $n = m = k$. We compute in Table 7.4 the number of samples t needed according to the results of [BCD⁺24] and above.

NIST level	n	t_{MCE}	t_{IMCE}	t_{symMCE}	t_{symIMCE}
1	14	15	8	8	4
3	22	23	12	12	6
5	30	31	16	16	8

Table 7.4: Parameters from MEDS spec and t -sample estimates necessary to recover the secret, derived from [BCD⁺24].

This shows that the symmetric version of the MCE and IMCE does not reach the same security level against the t -sample attack as the original version. In fact, the number of samples needed to run the attack is divided by two for all parameter sets.

In [KLP25], they suggest increasing the MEDS parameters by a factor $\sqrt{2}$ to ensure security against the basic algebraic brute-force attack. In Table 7.5, we report the same metrics for these new parameters. It shows that this increase would not be enough to preserve the same security level against these t -sample attacks.

NIST level	n	t_{symMCE}	t_{symIMCE}
1	20	11	6
3	31	17	9
5	42	22	11

Table 7.5: Increased MEDS parameters as suggested by [KLP25] and corresponding t -sample estimates, derived from [BCD⁺24].

In the following section, we improve upon the results derived from [BCD⁺24] by recovering the secret inputs using only 2 samples instead of 6-11 samples, regardless of the dimension of the underlying matrix code.

7.3.2 Attack on 2-(M)IMCE

We state below the 2 sample version of the symIMCE problem in terms of generator matrices. Recall from Lemma 2.88, that this is equivalent to the 2 sample MIMCE problem.

Problem 7.14. [2-symIMCE] Suppose we are given two tuples of three codes C_0, C_1, C_2 and C'_0, C'_1, C'_2 with associated generator matrices G_0, G_1, G_2 and F_0, F_1, F_2 forming two symIMCE instances.

They verify

$$\begin{cases} G_1 = S_0 G_0 (A \otimes B) \\ G_2 = S_1 G_0 (A^{-1} \otimes B^{-1}) \\ F_1 = S_2 F_0 (A \otimes B) \\ F_2 = S_3 F_0 (A^{-1} \otimes B^{-1}) \end{cases} \quad (7.4)$$

with $A \in GL(m, q)$, $B \in GL(n, q)$ (anti)symmetric and $S_i \in GL(k, q)$, $i = 0, \dots, 3$. Recover A, B (up to equivalence).

Recall that applying the results from [BCD⁺24] would require extra samples in order to recover the secret inputs to 2-symIMCE. We show below that the 2 sample instance can in fact already be solved in polynomial time.

Theorem 7.15. *The 2-symIMCE problem can be solved in $O(k^{2\omega} + (m^2 n)^\omega + n^{2\omega})$ operations. When $n = m = k$, this gives a dominating complexity of $O(n^{3\omega})$ operations.*

Proof. In what follows we outline an attack on the 2-symIMCE problem when the secret matrices are symmetric. Note that the antisymmetric case follows analogously.

Recovering S_0 . The first step consists in enumerating the possibilities for the isomorphism S_0 . We will make use of the symmetry of the underlying matrices. Observe that $G_1 G_2^T = S_0 G_0 (A \otimes B) (A \otimes B)^{-T} G_0^T S_1^T$. Since A, B are symmetric, so is $A \otimes B$, we have that $(A \otimes B)^{-T} = (A \otimes B)^{-1}$. Thus the equation does not involve A or B , and we obtain $G_1 G_2^T = S_0 G_0 G_0^T S_1^T$. We can repeat this for our second instance, as well as for cross terms to obtain the following set of equations.

$$\begin{cases} G_1 G_2^T S_1^{-T} = S_0 G_0 G_0^T \\ F_1 F_2^T S_3^{-T} = S_2 F_0 F_0^T \\ G_1 F_2^T S_3^{-T} = S_0 G_0 F_0^T \\ F_1 G_2^T S_1^{-T} = S_2 F_0 G_0^T \end{cases} \quad (7.5)$$

This gives us a system of $4k^2$ linear equations in $4k^2$ variables (the S_i matrices). However, solving naively gives a large number of solutions due to redundancies in the system. We provide an explicit description of the solution space of (7.5) in the following lemma.

Lemma 7.16. *Let $V := (G_0 G_0^T)(F_0 G_0^T)^{-1}(F_0 F_0^T)(G_0 F_0^T)^{-1}$. Assume that the minimal polynomial of V has degree n (i.e., it coincides with the characteristic polynomial of V). Let S'_0 be a solution for S_0 in System (7.5). Then any other solution for S_0 in the system (7.5) will be of the form $S'_0 P_i$ where $P_i := \sum_{j=0}^{k-1} \lambda_j^i V^j$ for some $\lambda_j^i \in \mathbb{F}_q$.*

Proof. Let $(S'_i)_{i=0}^3$ be a fixed solution for (7.5), and let $(S_i)_{i=0}^3$ be another arbitrary solution. Since both solutions verify (7.5), we have

$$S_0 G_0 G_0^T S_1^T = G_1 G_2^T = S'_0 G_0 G_0^T S'_1{}^T$$

In particular, this gives us

$$S_0^{-1} S'_0 G_0 G_0^T S'_1{}^T S_1^{-T} = G_0 G_0^T$$

The other equations of the system in 7.5 follow similarly. Let us denote $P_i = S_i^{-1} S'_i$, then we obtain the following system.

$$\begin{cases} P_0^{-1} G_0 G_0^T P_1 &= G_0 G_0^T \\ P_2^{-1} F_0 F_0^T P_3 &= F_0 F_0^T \\ P_0^{-1} G_0 F_0^T P_3 &= G_0 F_0^T \\ P_2^{-1} F_0 G_0^T P_1 &= F_0 G_0^T \end{cases}$$

We would like to characterize the matrices P_i appearing in this system.

If we express P_1, P_2, P_3 in terms of P_0 , we obtain the following equations:

$$\begin{cases} P_1 &= (G_0 G_0^T)^{-1} P_0 (G_0 G_0^T) \\ P_3 &= (F_0 F_0^T)^{-1} P_2 (F_0 F_0^T)^T \\ P_3 &= (G_0 F_0^T)^{-1} P_0 (G_0 F_0^T) \\ P_2 &= (F_0 G_0^T) P_1 (F_0 G_0^T)^{-1} \end{cases}$$

Replacing the expression for P_1 in P_2 , we get $P_2 = (F_0 G_0^T) (G_0 G_0^T)^{-1} P_0 (G_0 G_0^T) (F_0 G_0^T)^{-1}$ Which we can now plug into the first expression for P_3 and get

$$P_3 = (F_0 F_0^T)^{-1} (F_0 G_0^T) (G_0 G_0^T)^{-1} P_0 (G_0 G_0^T) (F_0 G_0^T)^{-1} (F_0 F_0^T)^T$$

We now have two expressions for P_3 in terms of P_0 which we can set equal to obtain

$$P_0 = (G_0 F_0^T) (F_0 F_0^T)^{-1} (F_0 G_0^T) (G_0 G_0^T)^{-1} P_0 (G_0 G_0^T) (F_0 G_0^T)^{-1} (F_0 F_0^T)^T (G_0 F_0^T)^{-1}$$

Then, if we set $V := (G_0 G_0^T) (F_0 G_0^T)^{-1} (F_0 F_0^T) (G_0 F_0^T)^{-1}$, we observe that $P_0 = V^{-1} P_0 V$. In other words, P_0 and V are commuting matrices.

By our assumption that the minimal and characteristic polynomials of V coincide, we conclude that P_0 can be written as a polynomial in V [Cur84, Sect. 25, Exercise 5]. Thus we can write P_0 as a polynomial in V as desired and $S'_0 P_0$ will be a solution by construction. $\square$

Remark 7.17. For a random matrix over a finite field $\mathbb{F}_q$, [NP95] tells us that the minimal and characteristic polynomial will coincide with probability greater than 99% for our choices of q . This probability was corroborated experimentally for actual V matrices of the form described above. In fact, when checking over several hundred runs, we never encountered a failure case. This shows that our assumption on V is not restrictive in general.

Thus, we can solve System (7.5) while normalizing n entries to obtain a unique solution S'_0 , requiring $O(k^{2\omega})$ elementary operations, where ω is the linear algebra constant. This will work so long as the normalized entries do not correspond to zero entries in the original matrix S_0 that we are solving for. The probability of this occurring is $1 - (\frac{q-1}{q})^n$, i.e. the complement probability to having all the entries be non-zero. In such a case, we can repeat by normalizing

a different set of entries.

Lemma 7.16 allows us to characterize the possible solutions for S_0 based on S'_0 .

Recovering A and B. Observe that

$$G_1 = S_0 G_0 (A \otimes B) \iff G_1 (I_m \otimes B^{-1}) = S_0 G_0 (A \otimes I_n). \quad (7.6)$$

If we plug in the characterization of S_0 deduced from Lemma 7.16, we get the following:

$$G_1 = S'_0 P_0 G_0 (A \otimes B) \iff G_1 (I_m \otimes B^{-1}) = S'_0 P_0 G_0 (A \otimes I_n). \quad (7.7)$$

This gives a linear expression for B depending bilinearly on A and P_0 .

From here, we would like to recover all of A ($m(m+1)/2$ variables), B ($n(n+1)/2$ variables), and P_0 (k variables). We will use a similar approach to that of Section 7.2 to construct bilinear equations. First, write G_1 as n blocks of size $k \times m$. This gives

$$G_1 = [[G_1]_1, \dots, [G_1]_n].$$

Then from Eq. 7.7 we have

$$G_1 (I_m \otimes B^{-1}) = [[G_1]_1 B^{-1}, \dots, [G_1]_n B^{-1}].$$

Similarly, we can write the rest of Eq. 7.7 in terms of blocks of the form

$$S'_0 P_0 G_0 (A \otimes I_n) = [\mathcal{B}_0(A, P_0) \cdots \mathcal{B}_n(A, P_0)],$$

where the $\mathcal{B}_i(A, P_0)$ are $k \times m$ blocks whose entries are bilinear equations depending on the entries of A and P_0 .

Now, for any $i \in [n]$, we have that

$$[G_1]_i^{-1} [G_1]_i B^{-1} = [G_1]_i^{-1} \mathcal{B}_i(A, P_0).$$

Note that sometimes the G_1^i are not invertible. However as explained in Remark 7.4, the probability of them being invertible tends to 1 for large q .

Now consider all pairs $(1, j)$ for $j \in [2, n]$, and subtract the corresponding equations:

$$0 = B^{-1} - B^{-1} = [G_1]_1^{-1} \mathcal{B}_0(A, P_0) - [G_1]_j^{-1} \mathcal{B}_j(A, P_0).$$

Any other relations derived in this way would be linear combinations of each other, so we only consider these $n - 1$ relations.

NIST level	n	q	Thm. 7.15 runtime
1	20	4093	35.5 s
3	31	4093	25.3 min
5	42	2039	6.58 hr

Table 7.6: Average runtime of our attack on 2-MIMCE taken from 10 runs. We use the MEDS parameters but with n scaled by a factor of $\sqrt{2}$, as suggested by [KLP25]. We run the attack using Magma [BCP97] on a laptop.

In total this gives a system of $n - 1$ bilinear matrix equations depending on the entries of A and the k coefficients in P_0 . Each of these relations contain $m \cdot k$ homogenous equations. In total, there are $(n - 1)mk$ equations, and $\frac{m(m+1)}{2} + k$ variables. Further, since the variables of A are being multiplied by those of P_0 , we get $\frac{m(m+1)}{2}k$ monomials in our bilinear system. Experiments verify that these equations are indeed linearly independent from each other. This gives a system that can be efficiently linearized. With this number of monomials, we use direct linearization which should require roughly $O((m^2n)^\omega)$ elementary operations, where ω is the linear algebra constant.

After having recovered A and P_0 , we may solve for B directly using the relation $G_1 = S_0 G_0(A \otimes B)$. This relation contains a total of kmn linear equations, and $\frac{n(n+1)}{2}$ unknowns. Gaussian elimination will suffice here, which uses $O(n^{2\omega})$ elementary operations. $\square$

Experimental results. The authors of [KLP25] suggested using MEDS [CNP⁺23b] parameter sets, and scaling them up to a small factor (around $\sqrt{2}$). In particular, this means that $n = m = k$. The original parameters from MEDS propose vector dimensions and field size, (n, q) , of $(14, 4093)$, $(22, 4093)$, $(30, 2039)$. We run our attack against the modified MEDS parameters, and report on the timings in Table 7.6. The code used to obtain these timings can be found at <https://github.com/vgilchri/blind-sigs-mimce>.

7.4 Conclusion

Code-based problems have emerged as candidates for post-quantum cryptography, and received particular attention as a new option for a cryptographic group action. While the core LCE and MCE problems have received (and continue to receive) extensive cryptanalysis, variants of these problems have begun to appear in the literature that also require analysis. These variants are often developed with the purpose of achieving some new functionalities and advanced protocols.

In this work, we study two variants of LCE and MCE that were introduced in the context of blind signatures. We show a polynomial-time equivalence between LCE and its DmLCE variant. This provides strong theoretical confidence in the hardness of this new problem, and gives confidence in the security of the blind signature scheme of [DKQ⁺25]. With new

confidence in the DmLCE problem, it is an interesting question to see which further primitives could be built from it.

We then study the hardness of the MCE variant called MIMCE. This variant is polynomial-time equivalent to the IMCE problem with (anti)symmetric matrices. However, our analysis shows that it is strictly easier than the original MCE and IMCE problems. We give an in-depth analysis through algebraic modeling and show that the suggested parameter sets are far from reaching their claimed security. Our analysis shows that selecting appropriate parameter sizes in order to reach desired security levels for the blind signature scheme of [KLP25] is not trivial, and that appropriately scaled security parameters may hinder the practicality of the scheme.

Finally, multiple samples problems have been used in the past to build multi-user primitives such as [BBMP24] which builds threshold signatures from 2-LCE. We anticipate such future attempts and analyze the multiple sample version of the MIMCE problem. We show that two samples are enough to break the problem, regardless of the dimensions of the code, suggesting that this problem is not suitable for building multi-user primitives.

8 Conclusion

In this thesis, we explored the design and analysis of post-quantum primitives beyond basic encryption. Our study followed two main directions, emphasizing the *diversity* of post-quantum assumptions and of the primitives they enable. On one hand, we examined several isogeny-based assumptions (Chapters 3 to 5) and multiple code-based problems (Chapters 6 and 7). On the other, we investigated how these assumptions support different cryptographic functionalities, including commitment schemes, digital signatures, proof systems, and updatable public-key encryption.

Through this thesis, I wish to convey two main takeaways to drive further research on the topics we explored. First, both isogeny-based and code-based cryptography are sufficiently expressive to support a broad range of cryptographic constructions. We presented three distinct applications of isogeny-based techniques: digital signatures (Chapter 3), updatable public-key encryption (Chapter 4), and zero-knowledge proofs (Chapter 5). Each construction relies on a different isogeny-based assumption, and we provided a detailed study of their security and efficiency. These results show that isogeny-based cryptography can yield practical and secure schemes for varied applications. Notably, an isogeny-based signature scheme, SQI-Sign, is currently a contender in the NIST digital signature additional standardization process [AAA⁺24]. Similarly, code-based assumptions can support primitives such as commitment schemes (Chapter 6) and blind signatures (Chapter 7). Together, these results demonstrate that competitive post-quantum primitives can be built without relying on structured lattices. In some cases, alternative assumptions even enable constructions that appear unattainable in the lattice setting, such as updatable public-key encryption from isogenies. Moreover, in a scenario where lattice assumptions were compromised, these other hardness assumptions would provide viable foundations for a diverse ecosystem of cryptographic primitives. Therefore, it is essential to explore further applications of isogeny-based and code-based cryptography, particularly in areas where structured lattices seem to underperform.

The second conclusion of this thesis concerns the state of post-quantum cryptanalysis. Protocol designers today have access to several families of assumptions, yet the security of many of these problems remains far less understood than that of classical foundations such

as factoring or discrete logarithms. This work, which began in late 2021, was shaped by the SIDH attacks of 2022. These attacks revealed two important lessons: first, that isogeny-based cryptography, despite over a decade of study, was far less well understood than expected; and second, that major cryptanalytic breakthroughs *need not always mark the end of a field*, but can instead open new directions and tools. Other areas of post-quantum cryptography could experience similar shifts, and rather than fearing them, the community should welcome cryptanalysis as a driver of progress. In this spirit, we contributed new cryptanalytic results on several code-based problems (Chapters 6 and 7) and hope they will encourage further exploration of these questions. Recent years have shown that isogeny-based cryptography continues to evolve at a fast pace, driven by new techniques coming from higher-dimensional isogeny computations, and it is essential that cryptanalysts should not fall too far behind protocol designers in order to maintain the confidence in the hardness of the underlying problems. Similarly, variants of code-based problems appear to design new primitives and should be analysed in-depth.

Both of these lessons should be kept in mind as NIST announces another standardization project for threshold cryptography¹. Only through a better understanding of the security of post-quantum assumptions can we build secure and efficient cryptographic schemes for a post-quantum future.

¹<https://csrc.nist.gov/projects/threshold-cryptography>

A Representing computations for proof systems

The goal of this appendix is to give a brief overview on the different ways to represent computations for proof systems. We cover two main representations: Arithmetic circuits and Rank 1 Constraint Systems (R1CS) as they are used in Chapter 5.

The key insight behind these representations is that they allow us to express computations in a way that is more amenable to the construction of efficient proof systems. Each representation has its own strengths and weaknesses, and the choice of which one to use often depends on the specific requirements of the proof system being designed.

Most of the material is adapted from Vitalik's Buterin's blogpost¹.

A.1 Arithmetic circuits

We start with the conceptually simplest representation: Arithmetic circuits. It is a mathematical model used to represent computations over a finite field. It is similar to Boolean circuits, but instead of using logical gates (AND, OR, NOT), it uses arithmetic operations (addition and multiplication) over a field. So it is represented as a directed acyclic graph (DAG) where the inputs are the leaves, the internal nodes are the arithmetic operations, and the output is the root of the graph. Each edge in the graph represents the flow of data between the nodes.

To prove a computation using arithmetic circuits, we need to show that we know the inputs to the circuit that produce a specific output. We can subdivide this task. First, we must ensure that the values computed at each internal node of the circuit are consistent with the arithmetic operations defined at that node. This involves verifying that, for each addition or multiplication gate, the output indeed equals the sum or product of its inputs. This can be done by first committing to the input and output values and then using a proof system to show that the committed values satisfy the arithmetic operations defined by the circuit. Next, we need to ensure that the final output of the circuit matches the expected output. This involves

¹<https://vitalik.eth.limo/general/2016/12/10/qap.html>

verifying that the output value computed at the root of the circuit is equal to the expected output value. This can be done by committing to the output value and then using a proof system to show that the committed output value matches the expected output value.

A.2 Rank 1 Constraint Systems (R1CS)

A more efficient representation than arithmetic circuits is a Rank 1 Constraint Systems (R1CS). An R1CS is a system of equations that can be used to represent computations in a more compact way than arithmetic circuits. An R1CS is a sequence of groups of three matrices (A, B, C) , as well as a solution z such that

$$(Az) \circ (Bz) = Cz \quad (\text{A.1})$$

where $\circ$ denotes the dot product. Each line in the matrices represents a constraint for a gate that must be satisfied by the solution z . Typically, z is of the form $(1, v, w)$ where v represents the circuit's public inputs and w the private inputs and wire values.

To use R1CS within proof systems, we therefore need to prove knowledge of z such that Equation A.1 holds. Modern proof systems typically divide the verification process into two subprotocols: **RowCheck** and **LinCheck**.

In the first step of the protocol, the prover commits to z, Az, Bz, Cz using a commitment scheme of their choice, and obtains the corresponding values C_z, C_A, C_B, C_C .

The **RowCheck** protocol ensures the structural consistency of the R1CS instance, i.e. it verifies that the committed vectors are consistent with the structure of the instance. This means that given a statement $(A, B, C, C_z, C_A, C_B, C_C)$ containing the constraints and committed values, and a witness z, r_Z, r_A, r_B, r_C where the r_i stand for the randomnesses used by the commitment scheme to generate C_i , it needs to generate a proof for the relation

$$C_A = \text{Commit}(Az; r_A), \quad C_B = \text{Commit}(Bz; r_B), \quad C_C = \text{Commit}(Cz; r_C), \quad C_z = \text{Commit}(z; r_z), \quad (\text{A.2})$$

This can be done using any suitable proof system.

The **LinCheck** protocol ensure that the committed vector satisfies the R1CS equation.

Let y_A, y_B, y_C represent the vectors within C_A, C_B, C_C respectively. The **LinCheck** protocol must verify that $y_A \circ y_B = y_C$ holds, without revealing any information about the committed vectors. This means that given a statement (C_A, C_B, C_C) containing the constraints, and a witness $y_A, y_B, y_C, r_A, r_B, r_C$ where the r_i stand for the randomnesses used by the commitment scheme to generate C_i , it needs to generate a proof for the relation

$$C_A = \text{Commit}(y_A; r_A), \quad C_B = \text{Commit}(y_B; r_B), \quad C_C = \text{Commit}(y_C; r_C), \quad y_C = y_A \circ y_B, \quad (\text{A.3})$$

This can be done using any suitable proof system.

At the end of the protocol, the verifier checks both proofs to ensure that the R1CS instance is valid and that the prover knows a valid solution z .

B Computational proofs for Chapter 6

The proofs included here are done via a very computational approach, using core definitions. We encourage the reader who wants to familiarize themselves with these definitions to do them as exercise.

Lemma 6.11. *Let $\lambda_a, \lambda_b, \lambda_c \in \mathbb{F}_q$ be such that $\lambda_a \lambda_b \lambda_c = 1$. Then for all $v \in \mathbf{V}$, we have that $(\lambda_a I_n, \lambda_b I_n, \lambda_c I_n) \star v = v$.*

Proof. Let us write $v = \sum_{i,j,k=1}^n v(i,j,k) e_i \otimes e_j \otimes e_k$. We have

$$\begin{aligned} (\lambda_a I_n, \lambda_b I_n, \lambda_c I_n) \star v &= \sum_{i,j,k=1}^n v(i,j,k) (\lambda_a I_n) e_i \otimes (\lambda_b I_n) e_j \otimes (\lambda_c I_n) e_k \\ &= \sum_{i,j,k=1}^n v(i,j,k) \lambda_a \lambda_b \lambda_c e_i \otimes e_j \otimes e_k \\ &= \sum_{i,j,k=1}^n v(i,j,k) e_i \otimes e_j \otimes e_k \text{ since } \lambda_a \lambda_b \lambda_c = 1 \\ &= v \end{aligned}$$

□

Lemma 6.13. *Let $\sigma \in S_n$, then there exists P_σ such that $(P_\sigma, P_\sigma, P_\sigma) \star t_0 = t_0$, given by $P_\sigma = (p_{ij})_{i,j=1}^n$ and $p_{ij} = \mathbf{1}_{i=\sigma(j)}$.*

Proof. Let $t_0 = \sum_{s=1}^n e_s \otimes e_s \otimes e_s$, then we can express the action $(A, B, C) \star t_0$ (from (6.2.3)) as

$$(A, B, C) * (\sum_{s=1}^n e_s \otimes e_s \otimes e_s) = \sum_{i,j,k}^n A_{is} B_{js} C_{ks} e_i \otimes e_j \otimes e_k.$$

$$\begin{aligned} (P_\sigma, P_\sigma, P_\sigma) \star t_0 &= \sum_{i,j,k=1}^n \sum_{s=1}^n (P_\sigma)_{is} (P_\sigma)_{js} (P_\sigma)_{ks} e_i \otimes e_j \otimes e_k \\ &= \sum_{i,j,k=1}^n \sum_{s=1}^n \mathbf{1}_{i=\sigma(s)} \mathbf{1}_{j=\sigma(s)} \mathbf{1}_{k=\sigma(s)} e_i \otimes e_j \otimes e_k \\ &= \sum_{i,j,k=1}^n \sum_{s=1}^n \mathbf{1}_{i=j=k} \mathbf{1}_{i=\sigma(s)} e_i \otimes e_j \otimes e_k \\ &= \sum_{m=1}^n \sum_{s=1}^n \mathbf{1}_{m=\sigma(s)} e_m \otimes e_m \otimes e_m \\ &= \sum_{m=1}^n e_m \otimes e_m \otimes e_m = t_0 \end{aligned}$$

since σ is a bijection. □

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EDUCATION

SEP 2021- ONGOING	PhD in Computer Science, EPFL, Switzerland <i>Working on cryptography in the LASEC lab. Co-supervised by Pr. Serge Vaudenay (EPFL) and Pr. Christophe Petit (ULB).</i>
SEP 2020- SEP 2021	Master in Mathematics and Computer Science, University of Geneva, Switzerland <i>Specialized in Cryptography, Final Grade 5.9/6.</i>
OCT 2019- JUL 2020	MASt in Pure Mathematics (Part III), University of Cambridge, United Kingdom <i>Specialized in Number Theory, Final Grade 84/100. No class awarded due to COVID.</i>
SEP 2016-JUN 2019	Bachelor in Mathematics, University of Geneva, Switzerland <i>Final grade 5.5/6</i>
SEP 2018-DEC 2018	Exchange semester, University of Toronto, Canada

EXPERIENCE

JUN 2025- AUG 2025	PhD Research Intern, Nokia Bell Labs, Stuttgart, Germany <i>Conducting research on zero-knowledge proofs for data-processing pipelines. Patent in submission.</i>
JUL 2024- JAN 2025	Visiting PhD Student, University of Auckland, New Zealand <i>Research on the computation of modular polynomials supervised by Pr. Steven Galbraith.</i>
FEB 2022- ONGOING	Graduate Teaching Assistant, EPFL, Lausanne, Switzerland <i>Spring 2022: Analysis, Fall 2022, 2023: Cryptography and Security, Spring 2023: Security Protocols and Applications, Spring 2024, 2025: Advanced Cryptography.</i>
OCT 2020-JAN 2022 JAN 2019-JUN 2019	Substitute Teacher, Public Instruction Department, Geneva, Switzerland <i>Long-term contract (5 months) and on-call for Mathematics and English.</i>
JAN 2017-JUN 2019	Tutor, ARA, Geneva, Switzerland <i>Tutoring students 15-19 years old in Mathematics.</i>

PUBLICATIONS

Updatable Public-key Encryption from FESTA. Andrea Basso, Tako Boris Fouotsa, Fatna Kouider, Péter Kutas, Luciano Maino *Laurane Marco*. In submission.

On the security of two blind signatures from code equivalence problems. Valerie Gilchrist, *Laurane Marco*, Christophe Petit, Gang Tang. IACR CiC, Volume 2, Issue 4.

Non-Transferable Anonymous Tokens by Secret Binding. F.Betül Durak, *Laurane Marco*, Abdullah Talayhan, Serge Vaudenay. CCS 2024.

Solving the Tensor Isomorphism Problem for special orbits with low rank points: Cryptanalysis and repair of an Asiacrypt 2023 commitment scheme. Valerie Gilchrist, *Laurane Marco*, Christophe Petit, Gang Tang. CRYPTO 2024.

Exploring SIDH-based Signatures Parameters. Andrea Basso, Mingjie Chen, Tako Boris Fou-

osta, Péter Kutas, Abel Laval, *Laurane Marco*, Gustave Tchoffo Saah. ACNS 2024.

Malleable Commitments from Group Actions and Zero-Knowledge Proofs for Circuits based on Isogenies. Mingjie Chen, *Laurane Marco*, Yi Fu Lai, Abel Laval, Christophe Petit. IN-DOCRYPT 2023.

Making Classical Signatures Post-Quantum for Single Use on a Public Ledger. *Laurane Marco*, Abdullah Talayhan, Serge Vaudenay. IWSEC 2023.

STUDENT SUPERVISION

IMPLEMENTATION OF PSIDH	Jacob Prud'homme, Semester Project, EPFL, Fall 2022
PROJECT MINT: BUILDING CYBERSECURITY COMPETENCE THROUGH EDUCATION	Dario Naepfer, Semester Project, EPFL, Fall 2023
POST-QUANTUM ANONYMOUS TOKENS	Sebastien Hauri, Master Thesis, EPFL, Fall 2023
BEYOND SIDH: A SURVEY ON COUNTERMEASURES AND NEW CONSTRUCTIONS	Valerio Ardizio, Semester Project, EPFL, Spring 2024
CRYPTANALYSIS OF A PUBLIC KEY ENCRYPTION SCHEME FROM EMBEDDINGS OF $SL(2, N)$	Bianca Rechsteiner, Semester Project, EPFL, Spring 2025
ALGEBRAIC STUDY OF THE 3-TENSOR ISOMORPHISM PROBLEM	David Pulido Cornejo, Semester Project, EPFL, Spring 2025

SKILLS

LANGUAGES:	French (Fluent), English (Fluent), Spanish (Intermediate) German (Advanced beginner)
PROGRAMMING:	Python, Sagemath and Magma for Cryptography

AWARDS AND GRANTS

EPFL EXCELLENT TEACHING ASSISTANT AWARD 2025	For excellence in teaching.
EPFL DocMOBILITY GRANT 2024	Grant for EPFL doctoral student who wish to conduct a research stay in an academic institution abroad.
IC DISTINGUISHED SERVICE AWARD 2023	For outstanding contributions to the EPFL EPIC Association through the Women+ in IC initiative.
EDIC FELLOWSHIP 2021-2022	One year funding fellowship awarded for first-year PhD students at EPFL.
FNRS ASPIRANT FELLOWSHIP	Declined. Fully-funded PhD grant awarded with grade "A+ Outstanding" by the Belgium National Fund for Scientific Research (FNRS).

SERVICE

EPFL EPIC:	Committee member of the EPFL PhD in Computer Science association since 2022 (EPIC, www.epfl.ch/campus/associations/list/epic/).
EPFL WOMEN+ IN IC:	Founder and president (2022-2024) of the Women+ in IC initiative (https://wic-epfl.github.io) and member (2024-now).
WOMEN IN CRYPTOGRAPHY:	Seminar and workshop organization for the women in cryptography community (www.womenincryptography.com).
SUMMERSCHOOL MINT:	Lecturer for the EPFL MINT summer school for high school students from Köniz and Lebermatt gymnasium (2022, 2023).
OUTREACH TALKS:	EPFL Open Days 2023, Thementag Gymnasium Thun 2022.